Statistical Models and Application 2025

Egle Caudullo

Contents

I Gaussian Linear Model **11**

1 Model definition **13**
1.1 Assumption . . . 13
1.2 Model Parameters and Conditional Sample Space . . . 16
1.2.1 Probability Density Function . . . 17
1.3 Matrix Representation . . . 17
1.4 Model Definition with ε_i . . . 18
1.4.1 Assumption . . . 18

2 Maximum Likelihood Estimation **19**
2.1 Parameters . . . 19
2.2 Maximum Likelihood Estimation for $\boldsymbol{\beta}$ . . . 19
2.2.1 Properties . . . 20
2.3 MLE for σ^2 . . . 20
2.3.1 Properties of $\boldsymbol{\varepsilon}$ and $\hat{\sigma}^2_{\text{ML}}$ . . . 21
2.3.2 Restricted Maximum Likelihood for σ^2 . . . 21
2.3.3 Standardised Residuals . . . 22

3 Hypothesis Testing **23**
3.1 Examples when $k = 3$ . . . 23
3.2 Solutions - Nested Models . . . 23
3.3 Likelihood Ratio Tests- known σ^2 . . . 24
3.3.1 Constrained Maximum Likelihood Estimation: . . . 24
3.3.2 Distribution . . . 25
3.4 Likelihood Ratio Tests - unknown σ^2 . . . 25
3.5 Wald Test Statistics . . . 26
3.5.1 Special Cases . . . 26
3.5.2 Confidence Intervals . . . 26
3.6 Linear independence hypothesis . . . 27
3.7 H_0: linear independence between X_j and Y: $\beta_j = 0 \; \forall \; j$ . . . 27

4 Categorical regressor **29**
4.1 Coding scheme: indicator/dummy variables . . . 29
4.1.1 Linear Hypothesis . . . 30
4.1.2 Results . . . 30
4.1.3 F test . . . 31
4.2 Coding scheme: Exclusion of the intercept . . . 32
4.2.1 Results . . . 32
4.2.2 F test . . . 33
4.3 Coding scheme: Incremental/split coding . . . 34
4.3.1 R implementation . . . 34
4.3.1.1 Interpretation . . . 35

4.3.2 Linear trend hypothesis . . . 35
4.3.3 Linear constraints . . . 36
4.3.4 F test for the linear trend hypothesis . . . 36
4.4 Changing reference category of the regressor . . . 39
4.4.1 Contrast (NON CONSIGLIATO) . . . 40
4.4.2 Relevel (CONSIGLIATO) . . . 41
4.5 Models . . . 42
4.5.1 Modello con dist: rural categoria di riferimento . . . 42
4.5.2 Modello con new.dist: major cities categoria di riferimento . . . 43

5 Model selection **45**
5.1 Nested models . . . 46
5.1.1 F statistic with anova function . . . 47
5.2 Non nested models . . . 48
5.3 Adjusted R^2 . . . 49
5.3.1 Correspondence R^2 and $\bar{R}^2$ . . . 49
5.4 Cross validation . . . 50
5.4.1 Leove-one-out . . . 50
5.4.2 CV for Gaussian linear regression models . . . 50
5.5 Akaike information Criteria . . . 51
5.5.0.1 Definition . . . 51
5.5.1 How does it work . . . 52
5.5.2 AIC for Gaussian linear models . . . 52
5.6 Bayesian Information Criterion (BIC) . . . 53
5.6.0.1 Definition . . . 53
5.6.1 BIC for Gaussian linear models . . . 54
5.7 AIC or BIC? . . . 54
5.8 Non nested models R . . . 55

II Gaussian Linear Mixed Models **57**

1 Introduction **58**
1.1 Fixed or random factors . . . 58
1.1.1 Modelling implications . . . 58
1.2 Hierarchical structure . . . 59
1.3 Regressors for multilevel data . . . 59
1.4 Examples . . . 59
1.4.1 Clustered data . . . 59
1.4.2 Longitudinal data–Growth model . . . 60

2 Exploratory Analysis **62**
2.0.1 Dati . . . 62
2.0.2 Pre-analysis . . . 65
2.0.2.1 Graphical comparison between observed and fitted values 66

3 Model with random intercept **69**
3.1 Notation . . . 69
3.1.1 Example (longitudinal data) . . . 69
3.2 Model definition . . . 70
3.3 Model assumptions . . . 70
3.4 Properties . . . 71

4 Model with random intercept and slope **72**
4.1 Model definition . . . 72
4.2 Model assumptions . . . 73
4.3 Properties . . . 73

5 Models for units in the i–th group **75**
5.1 Notation . . . 75
5.2 Model Definition . . . 76
5.3 Assumptions . . . 76
5.3.1 Possible choices for $\boldsymbol{Q}$ . . . 76
5.4 Conditional heteroscedasticity and dependence . . . 77
5.4.1 Marginal formulation . . . 77
5.4.2 Conditional formulation . . . 77

6 Global model (for all sample units) **79**
6.1 Notation . . . 79
6.2 Assumptions . . . 80
6.3 Properties . . . 80
6.4 Conditional heteroscedasticity and dependence . . . 81
6.4.1 Conditioning on level 2 units . . . 81
6.5 Extended/general Gaussian linear mixed models . . . 82
6.5.1 Heteroscedastic errors . . . 82
6.5.2 Correlated errors . . . 83
6.5.3 A note about extended Gaussian linear mixed models . . . 83
6.5.4 Marginal representation . . . 83

7 Maximum Likelihood Estimation **84**
7.1 Maximum likelihood estimation for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ ($\boldsymbol{\vartheta}$ known): . . . 85
7.1.1 Generalised/weighted least squares . . . 85
7.2 Maximum likelihood estimation for $\boldsymbol{\vartheta}$ . . . 86
7.2.1 Profile maximum likelihood estimation for $\boldsymbol{\vartheta}$ . . . 86
7.2.2 Restricted MLE for $\boldsymbol{\vartheta}$ . . . 87

8 Linear hypotheses on fixed effects **88**
8.1 Wald test statistic . . . 88
8.1.1 Wald test statistic $\boldsymbol{\vartheta}$ known . . . 88
8.1.2 Wald test statistic $\boldsymbol{\vartheta}$ uknown . . . 89
8.1.3 Special case . . . 89
8.2 Likelihood ratio test statistics . . . 90

9 Hypotheses on random effects **91**
9.1 Identification of the constraints . . . 91
9.2 Test statistics . . . 92
9.2.1 An approximation for the true p-value . . . 92

III Polynomial regression **93**

1 Polynomial regression **95**
1.1 Model Definition . . . 95
1.1.1 Matrix notation . . . 95
1.2 Model fitting . . . 96
1.2.1 Linear basis expansions . . . 96
1.3 Maximum Likelihood estimation . . . 100
1.4 Output- polynomial of orther 5 . . . 100
1.5 Hypothesis testing . . . 102
1.5.1 Example on R . . . 103
1.6 Some cautionary remarks . . . 106

IV Regression splines **108**

1 Regression splines **109**
1.1 Spline function . . . 109
1.2 Linear Splines . . . 110
1.3 Cubic splines . . . 110
1.4 Number of free parameters of a spline function . . . 111

2 Basis expantion **112**
2.1 Linear basis expansion for spline functions . . . 112
2.2 Truncated power basis for spline functions . . . 112
2.3 B-spline basis functions . . . 113
2.4 Matrix notation . . . 114

3 Knots selection **116**
3.1 Inference for regression spline models . . . 116
3.2 Knots Location . . . 116
3.3 Knots selection on R . . . 117
3.4 Polynomials and spline function . . . 121

4 P-splines **124**
4.1 Penalized log-likelihood function . . . 124
4.1.1 Penalty terms for P-splines . . . 125
4.1.2 Matrix formulation: . . . 125
4.2 Penalized maximum likelihood estimation . . . 126
4.2.1 Linear smoothers and smoother matrices . . . 127
4.3 Choice of the smoothing parameter . . . 127
4.4 Inference for P-splines . . . 128
4.5 Hypothesis testing for P-splines . . . 129

4.5.1 Crash test data – P-splines – R output 129

V Binary regression 132

1 Beyond Gaussian models 133

2 Model Structure 134
2.1 Binary regression 134
2.2 Logit model 135
2.3 Choice of the smoothing parameter 135
2.4 Inference for P-splines 136
2.5 Hypothesis testing for P-splines 137
2.5.1 Crash test data – P-splines – R output 137

VI Binary regression 140

1 Beyond Gaussian models 141

2 Model Structure 142
2.1 Binary regression 142
2.2 Logit model 143
2.3 Probit model 144
2.4 Complementary log-log model 144
2.5 A latent utility formulation 145

3 Grouped data 147
3.0.1 Dataset example: Forbes100 147
3.0.2 su R 148
3.0.3 Residuals: visual inspection 151
3.1 Overdispersion 154
3.2 Model Comparison 155
3.2.1 Akaike criterion 155
3.2.2 Regarding Test and statistics 157

4 Maximum likelihood estimation 158
4.1 Individual data 158
4.1.1 Iterative computation for MLE 159
4.1.2 Asymptotic properties of ML estimator 159

5 Hypotheses testing 160
5.0.1 R implementation 160
5.0.2 Tests: LRT, Wald, Score 162
5.1 Grouped data 164
5.1.1 Model fit 164
5.1.1.1 su R 164
5.1.2 Estimating overdispersion 165
5.2 Individual data 166
5.2.1 Residuals: visual inspection 167

5.2.2 Model fit . . . 168
5.2.3 On the deviance and Pearson statistic: Hosmer-Lemeshow Test . 168
5.3 Binary regression issues . . . 170

VII Count data regression 171

6 Models for count data 172
6.1 Maximum Likelihood . . . 172
6.2 Model fit, model choice, quasi-likelihood . . . 173
6.2.1 Modello con dist: rural categoria di riferimento . . . 173
6.3 Goodness of fit . . . 175
6.3.1 Deviance . . . 175
6.3.2 Pearson . . . 176
6.3.3 Graphical inspection of the Residual . . . 176
6.4 Pratical Session . . . 177
6.4.1 Fitting a `quasipoisson` . . . 177

7 Offset 178
7.1 Modelling with an offset . . . 178
7.2 Interpretation of the Offset Term . . . 179
7.3 Warnings and Limitations . . . 179
7.4 Offset Exclusion . . . 179
7.5 Using offset as a regressor . . . 181
7.5.1 Hypothesis testing su H0: $\beta_{offset} = 1$ . . . 182

VIII Nonnegative continuous data 184

8 Intro 185
8.1 Log-Normal Model . . . 186
8.1.1 Problems . . . 186

9 Gamma regression 187
9.0.1 Gamma GLM Output Example . . . 187

10 R nonneg 189

IX A general model definition 201
10.1 Univariate exponential family (EF) . . . 202
10.1.1 GLMs and EFs . . . 203
10.2 Likelihood inference in GLMs . . . 203
10.2.1 Fisher scoring algorithm as IWLS . . . 204
10.2.2 Asymptotic properties of ML estimates . . . 206
10.2.3 Overdispersion parameter . . . 206
10.3 Model fit and selection for GLMs . . . 207
10.4 Quasi-likelihood for EFs . . . 207
10.4.1 Quasi-ML estimator and asymptotics . . . 207

10.5 Exam possible question 208

X Categorical regression 210

1 Model definition 211
1.1 From categorical to dummy 211
1.2 Distributional assumption 212
1.3 Group data 214

2 Models for unordered categories 215
2.1 Generalization of logit model to *categorical (multinomial)* logit 215
2.1.1 Linear predictor 215
2.2 Latent utility formulation 216
2.3 Level-specific effects 216
2.4 Dataset example -output 217

3 R 218

4 Models for ordered categories 224
4.1 Cumulative model 224
4.1.1 Definition 224
4.1.2 The effect of β 225
4.2 Example Dataset 225
4.3 Proportional Odds Linear Regression 226
4.3.0.1 Threshold interpretation 227
4.3.1 Coeffients interpretation 228
4.3.1.1 Different paramterization on R 229
4.4 Model comparison (come prima) 232
4.5 proportional hazards model 232

5 Appendix 233
5.1 ML estimator 234
5.2 Numerical computation of $\hat{\beta}$ 234
5.2.1 Model connection 236

XI Generalized Linear Mixed Models 237

1 Generalized Linear Mixed Models 238
1.1 Model structure: 238
1.2 Matrics Formulation 239

2 Mixed binary and count 240
2.1 Mixed Logit/Probit models 240
2.2 Mixed Poisson models 240

3 Mixed models for categorical data **241**
3.1 Mixed multinomial logit model . . . 241
3.2 Conditional and marginal models . . . 241
3.2.0.1 In log-linear Poisson random intercept model ($n_i = 1$) . . 242

4 Maximum Likelihood Estimation **243**
4.1 Known variance-covariance parameters . . . 243
4.2 Unknown variance-covariance parameters . . . 244

5 R mixed models **245**
5.1 Dati longitudinali (conteggi) individuali . . . 245
5.1.0.1 Dati longitudinali (conteggi) clustered . . . 248
5.1.0.2 Dati Binari clustered . . . 250
5.1.1 Count Data . . . 252
5.1.1.1 Modello con `time` . . . 252
5.1.1.2 Modello senza `time` . . . 254
5.1.1.3 Modello semplice senza random effects . . . 256
5.1.1.4 Modello con `subjID` come fattore . . . 257
5.1.2 Cluster Data . . . 258
5.1.2.1 Modello con intercetta sia fissa che random e X fisse . . 258
5.1.2.2 Modello con solamente l'intercetta random . . . 260
5.1.2.3 Modello con anche random slope . . . 261
5.1.2.4 Modello con variabile cluster come predittore categorico 264

XII Generalized Additive Models **266**

1 Generalized Additive Models **267**
1.1 Model definition . . . 267
1.2 Estimation of $f_j(z_j)$ . . . 268
1.2.1 Problem of Identifiability . . . 268
1.2.2 Smoothness . . . 269
1.3 Matrix representation . . . 269
1.4 STAR Models . . . 269

2 Generalized STAR and GAMs **270**
2.1 Model definition . . . 270
2.2 Inference in STAR models . . . 271
2.3 Inference in Generalized STAR models and GAMs . . . 271
2.4 Summary . . . 272
2.5 Generalised Additive Models . . . 273
2.5.0.1 Visualizzazione delle relazioni e introduzione dei tensor product smooth . . . 274
2.5.1 Linear regression . . . 275
2.5.2 Spline . . . 276
2.5.2.1 Spline con basi tp . . . 276
2.5.2.2 Spline con basi br . . . 280
2.5.2.3 Spline con Edu lineare ed Income ed Edu con basi tp . . 281
2.5.2.4 Spline con Income ed Edu con basi tp . . . 282

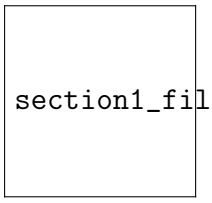
2.5.3 Tensor 284
2.5.3.1 Plot 285

Part I

Gaussian Linear Model

- **regressori numerici** : scatter plots con la variabile dipendente sull'asse Y e i regressori sull'asse X
- **regressori categorici** : multiple box plots, cioè box plot specific per ogni categoria del regressore

```
par(mfrow=c(1, 2))
plot(bwt~lwt,data=dati)
plot(bwt~race,data=dati)
```



Nota: plot è una funzione generica, che restituisce plot diversi in base alla natura di ciò che mettiamo dentro. Nella formula abbiamo un lato sinistro e un lato destro:

- a sinistra mettiamo la **variabile dipendente**
- a destra i **regressori**

Queste sono separate da una tilde.
Possiamo notare che, per esempio, i bambini nati da madre bianca sembrano avere un peso medio alla nascita di bambini nati da madre nera o altra etnia.
Vediamo inoltre la presenza di overlapping tra i boxplots, cosa che ci fa capire che dobbiamo controllare se le differenze siano significative.

1. Model definition

$$f(y_i, x_{1i}, x_{2i}, x_{3i}) = f(y_i \mid x_{1i}, x_{2i}, x_{3i}) \cdot f(x_{1i}, x_{2i}, x_{3i}) \qquad i = 1, \ldots, 2032$$

The specification of a family of conditional distribution is needed to define the parametric models. When doing this analysis you are interested in the analysis of the **conditional distribution** of the dependent variable given the covariates.

1.1 Assumption

Y_i Random variable that describes the value for the dependent variable observed on the i–th sample unit $(i = 1, \ldots, n)$

$x_{1i}, x_{2i}, \ldots, x_{ki}$ values of the regressors/covariates for the i–th sample unit *(covariate pattern)*: a specific combination of values of the regressor, typically each units has its own pattern.

A) $\mathbb{E}[Y_i \mid x_{1i}, \ldots, x_{ki}] = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} \quad \forall i$

B) $\mathrm{Var}[Y_i \mid x_{1i}, \ldots, x_{ki}] = \sigma^2 \quad \forall i$

C) $\mathrm{Cov}[Y_i \mid x_{1i}, \ldots, x_{ki},\ Y_h \mid x_{1h}, \ldots, x_{kh}] = 0 \quad \forall i \neq h$

D) $Y_i \mid x_{1i}, \ldots, x_{ki} \sim \mathcal{N}(\mu_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki},\ \sigma^2)$ (cond. independent) $\forall i$

A) **linearity assumption**, here we specify the functional form of the relation

$$\mathbb{E}[Y_i \mid x_{1i}, x_{2i}, x_{3i}] = \mu_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i}$$

The expression is linear in 2 sense

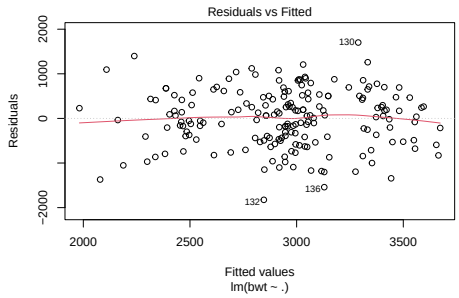
1. with respect to the **regressors**: since the impact of each regressor is constant on the dependent variable
2. with respect to the **parameter** (μ_i): since it is a linear combination of regressors

Notice that
β_i is the measure of the impact of the *ith* regressor, recall that it takes into account also the effect of other regressor, which are considered constant while measuring the x_i's effect. For example β_1 gives the impact of excercise after controlling for the presence of any risk factor.

$$\beta_0 + \beta_1 \mathbf{1}\{\text{exercise}_i = \text{yes}\} + \beta_2 \mathbf{1}\{\text{drinkany}_i = \text{yes}\} + \beta_3 \text{BMI}_i, \quad \forall i$$

Residual analysis

```
plot(model1, which=1)
```



Abbiamo le **y stimate** sull'asse X e i **raw residuals** sull'asse Y.

Se l'ipotesi di linearità è adeguata, dovremmo osservare dei raw residuals distribuiti uniformemente introno a 0, assumendo valori positivi e negativi che tendono a cancellarsi a vicenda in termini di grandezza.

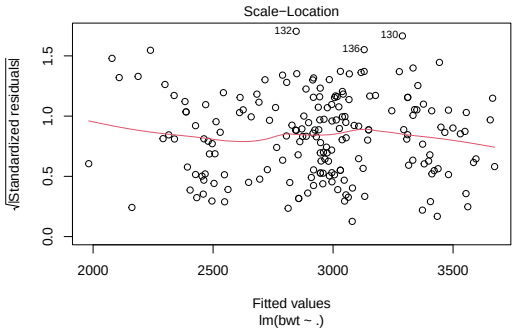
Se ciò accade allora possiamo dire che **non c'è una differenza sistematica tra i valori osservati e quelli fitted** quando ci focalizzizamo su unità specifiche che condividono lo stesso covariate pattern.

La linea rossa presente nel grafico è la local avarege per i residui: quando è orizzontale e vicina allo 0 possiamo dire che la l'ipotesi di linearità è adeguata.

B) homoschedasticity assumption (constant conditional variance assumption) within different groups σ^2 is the same.

$$\text{Var}[Y_i \mid x_{1i}, x_{2i}, x_{3i}] = \sigma^2 \quad \forall i$$

```
plot(model1, which=3)
```



Nel grafico abbiamo in X i **fitted values** in orizzontale, in Y abbiamo la **radice quadrata del valore assoluto dei residui standardizzati**. Il plot si basa quindi

sull'ordine di grandezza dei residui, non sul segno, infatti prendiamo il valore assoluto. La teoria ci dice infatti che quando le assunzioni sono adeguate gli standardize residuals sono omoschedastici e hanno una varianza pari a 1.

Se hanno la stessa variabilità allora devono avere anche la stessa grandezza, quindi la media in valore assoluto dei residui deve essere simile.

Nota: stiamo usando nuovamente una local average, che possiamo vogliamo sia vicina ad 1. Questo perché quando la varianza doveva essere vicino a 1 lo è anche la sua radice.

In esempio vediamo che non c'è un cambiamneto nella variabilità degli standardize residuals se compariamo unità con different fitted values. La local average è un po' meno orrizzontale ma costante, quindi possiamo dire che non ci sono evidenze contro l'assunzione di omoschedasticità.

Se vediamo comportamenti strani, per esempio che i bound dell'asse y sono sempre più larghi al crescere dei fitted values, allora abbiamo un segno tipico di variabilità condizionale non è costante tra le unità. Questa infatti ma tende a crescre al crescere del valore atteso di y.

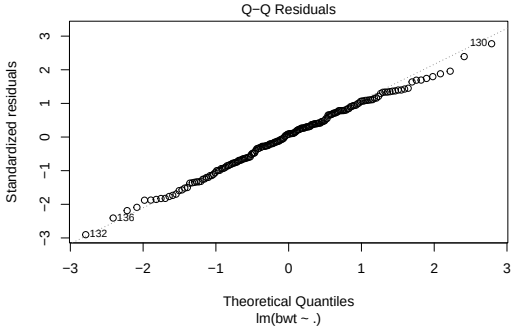
C) conditional independence assumption this is true when units are selected from a random sample. It means: there is no linear correlation across the predictors, when belonging to different groups.

$$\text{Cov}[Y_i \mid x_{1i}, x_{2i}, x_{3i}, Y_h \mid x_{1h}, x_{2h}, x_{3h}] = 0 \quad \forall i \neq h$$

D) distribution assumption the dependent random variable is assumed to have a gaussian distribution when conditioned on the covariates (gaussian linear models)

$$y_i \mid x_{1i}, x_{2i}, x_{3i} \sim \mathcal{N}(\mu_i, \sigma^2) \quad \forall i$$

```
plot(model1,which=2)
```



con questo graficio compariamo coppie di distribuizioni plottando i rispettivi quantili, sapendo che due possiamo definire due distrbuzioni uguali se queste hanno uguali quantili.

In **X** abbiamo i **quantili teorici associati a una gaussian distribution**, in **Y** abbiamo i **quantili estratti dall'empirical distribution degli standardize residuals**.

Se quindi l'assunzioni fatte sono adeguate i residui dovrebbero avere una distribuzione normale.

Guardando il grafico se i punti sono allineati lungo la bisettrice possiamo dire che non ci sono differenze sistematiche tra la distrbuzione empirica degli standardize residuals e la distribuzione teoretica per una random variable gaussian.

Se invece troviamo dei pattern strani, questa potrebbe essere un'evidenza contro l'adeguatezza delle assunzioni. (per esempio se sono sulle code i punti sono sistematicamente più grandi o più piccoli della bisettrice)
Nell'esempio sono allineati perfettamente alla bisettrice, quindi vanno bene.

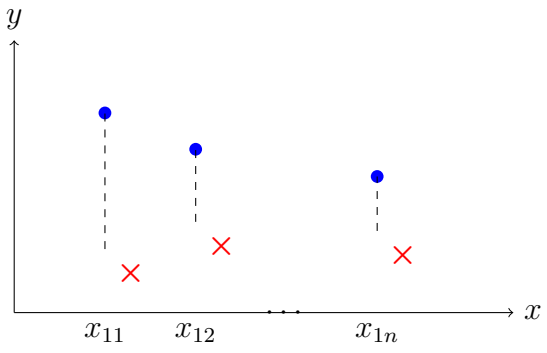
1.2 Model Parameters and Conditional Sample Space

It includes regression coefficient and σ^2

- $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_k)^\top \in \mathbb{R}^p$, where $p = (k+1)$
- $\sigma^2 \in \mathbb{R}^+$ is a positive scalar value
- $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \sigma^2)^\top \in \Theta = \mathbb{R}^p \times \mathbb{R}^+$

Conditional sample space: a set that contains n covariates pattern, which is the cartesian product of the real line representing the y_i : it is the realization of y_i for that specifc covariates pattern.

$$\mathbb{R} \times \left\{(x_{1i}, \ldots, x_{ki})^\top,\ i = 1, \ldots, n\right\}$$



red x do not belongs to conditional sample space since the covariates pattern has not been drown while blue one dows

1.2.1 Probability Density Function

The consequence on the model's assumption of independence allow us to compute the **joint** conditional density function as the product of each conditional density function of y_i due to conditional independence.

$$f(y_1, \ldots, y_n \mid x_{11}, \ldots, x_{1n}, \ldots, x_{kn}) = \prod_{i=1}^{n} f(y_i \mid x_{1i}, \ldots, x_{ki}) \qquad \text{(conditional independence)}$$

1.3 Matrix Representation

- $\boldsymbol{Y} = (Y_1, \ldots, Y_n)^\top$: n-dimensional random variable for the dependent variable
- $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$: observed sample values
- $\boldsymbol{x}_i = (x_{0i}, x_{1i}, \ldots, x_{ki})^\top$: covariates for i-th unit, value of the regressor for a single unit
- $\boldsymbol{x}_{[j]} = (x_{j1}, \ldots, x_{jn})^\top$: values for the j-th regressor across units $j = 0, \ldots, k$
- $\boldsymbol{x}_{[0]} = (1, 1, \ldots, 1)^\top$ for intercept

Design matrix $\boldsymbol{X}$ $(n \times p)$ with $p = k + 1$:

$$\boldsymbol{X} = \begin{bmatrix} 1 & x_{11} & \ldots & x_{k1} \\ 1 & x_{12} & \ldots & x_{k2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1n} & \ldots & x_{kn} \end{bmatrix} = \left[\begin{array}{c} \boldsymbol{x}_1^\top \\ \hline \boldsymbol{x}_2^\top \\ \hline \vdots \\ \hline \boldsymbol{x}_n^\top \end{array}\right] = [\boldsymbol{x}_{[0]}\ \boldsymbol{x}_{[1]} \cdots \boldsymbol{x}_{[p]}]$$

Row interpretation: each one presents the value observed for each units
Column interpretation: each columns present the value for a specific covariate

Then the conditional expected values

$$\mathbb{E}[Y_i \mid x_{1i}, \ldots, x_{pi}] = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} = \boldsymbol{x}_i^\top \boldsymbol{\beta}, \quad \forall i$$

In compact form:

$$\boldsymbol{X}\boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{x}_1^\top \boldsymbol{\beta} \\ \boldsymbol{x}_2^\top \boldsymbol{\beta} \\ \vdots \\ \boldsymbol{x}_n^\top \boldsymbol{\beta} \end{bmatrix}$$

$$\boldsymbol{Y} \mid \boldsymbol{X} \sim \mathcal{N}_n(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n)$$

1.4 Model Definition with ε_i

$$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$
$$\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^\top$$

- deterministic component not affected by sample variability
- random error affected by sample variability

This means that we can observed different y_i values for the same given pattern of covariates due to ε_i

1.4.1 Assumption

A) $\mathbb{E}[\varepsilon_i \mid x_{1i}, \ldots, x_{ki}] = 0 \quad \forall i$

B) $\mathrm{Var}[\varepsilon_i \mid x_{1i}, \ldots, x_{ki}] = \sigma^2 \quad \forall i$ (Homoschedasticity)

C) $\mathrm{Cov}[\varepsilon_i, \varepsilon_h \mid x_{1i}, \ldots, x_{ki}, x_{1h}, \ldots, x_{kh}] = 0 \quad \forall i \neq h$

D) $\varepsilon_i \mid x_{1i}, \ldots, x_{ki} \sim \mathcal{N}(0, \sigma^2) \quad \forall i$

$$\boldsymbol{\varepsilon} \mid \boldsymbol{X} \sim \mathcal{N}_n(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n)$$

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Hence, from properties of multivariate Gaussians:

$$\boldsymbol{Y} \mid \boldsymbol{X} \sim \mathcal{N}_n(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n)$$

Observation
When there is no sample variability (in regressors) you do control the value of the regressor, which mean that data can be treated as deterministic, this allow to work directly on the marginal without conditioning, when this not happens by conditioning on $\boldsymbol{X}$ allow us to ignore the randomness of the covariates, this implies that all the conclusion are conditional to $\boldsymbol{X}$ (conditional inference).

2. Maximum Likelihood Estimation

2.1 Parameters

The unkown parameter in a Gaussian linear regression models are

- $\boldsymbol{\beta}$: regression coefficients
- σ^2: the conditional variance

Respectvely the *likelihood* and $log - likelihood$ can be defined as

$$L(\boldsymbol{\beta}, \sigma^2) = L(\boldsymbol{\beta}, \sigma^2, \boldsymbol{y} \mid \boldsymbol{X}) = \left\{ \frac{1}{\sqrt{2\pi\sigma^2}} \right\}^n \exp\left[-\frac{1}{2\sigma^2} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) \right]$$

$$\ell(\boldsymbol{\beta}, \sigma^2) = \log L(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \ln 2\pi - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})$$

Note: $-\frac{n}{2} \ln 2\pi$ is a constant and can be ignored

MLE estimates

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

$$\hat{\sigma}^2_{ML} = \frac{1}{n} (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})^\top (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})$$

$$\hat{\sigma}^2 = \frac{1}{n-p} (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})^\top (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})$$

2.2 Maximum Likelihood Estimation for $\boldsymbol{\beta}$

The vector $\hat{\boldsymbol{\beta}}$ is the maximum likelihood estimator for $\boldsymbol{\beta}$ if and only if:

$$\ell(\hat{\boldsymbol{\beta}}, \sigma^2) = \max_{\boldsymbol{\beta} \in \mathbb{R}^p} \ell(\boldsymbol{\beta}, \sigma^2) \quad \text{or} \quad \hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta} \in \mathbb{R}^p} \ell(\boldsymbol{\beta}, \sigma^2)$$

- $\frac{\partial}{\partial \boldsymbol{\beta}} \ell(\boldsymbol{\beta}, \sigma^2) \Big|_{\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}} = \mathbf{0}_p$
- $\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} \ell(\boldsymbol{\beta}, \sigma^2) \Big|_{\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}}$ is negative definite

Recall: A matrix $\boldsymbol{A}$ is negative definite if $\boldsymbol{z}^\top \boldsymbol{A} \boldsymbol{z} < 0$ for all nonzero $\boldsymbol{z}$.

It is obtained

- $\frac{\partial}{\partial \boldsymbol{\beta}} \ell(\boldsymbol{\beta}, \sigma^2) = \frac{\boldsymbol{X}^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})}{\sigma^2}$
- $\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} \ell(\boldsymbol{\beta}, \sigma^2) = -\frac{\boldsymbol{X}^\top \boldsymbol{X}}{\sigma^2}$

If $\boldsymbol{X}$ has full column rank the second order partial derivatives is always negative defined ($\forall \beta \in \mathbb{R}^p$):

$$\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \mathbf{0}_p \quad \Leftrightarrow \quad \boldsymbol{\beta} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$$

Note:
For Gaussian linear models, maximizing $\ell(\boldsymbol{\beta}, \sigma^2)$ is equivalent to minimizing the following expression (ordinary least square estimation for Gaussian linear models):

$$(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_k x_{ki})^2$$

2.2.1 Properties

Depends on

- $\boldsymbol{X}$ regressors values
- $\boldsymbol{y_i}$ realization of $\boldsymbol{Y}$

$\hat{\boldsymbol{\beta}}$ is a linear transformation of $\boldsymbol{Y}$:

$$\hat{\boldsymbol{\beta}} = \boldsymbol{D}\boldsymbol{Y} + \boldsymbol{d}, \quad \text{with} \quad \boldsymbol{D} = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top, \quad \text{and} \quad \boldsymbol{d} = \mathbf{0}_p$$

According to the Gaussian linear model assumption $\boldsymbol{Y} \mid \boldsymbol{X} \sim \mathcal{N}_n(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2\boldsymbol{I}_n)$:

$$\Rightarrow \mathbb{E}[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \boldsymbol{\beta}, \quad \mathrm{Cov}[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$$

$$\hat{\boldsymbol{\beta}} \mid \boldsymbol{X} \sim \mathcal{N}_p(\boldsymbol{\beta}, \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1})$$

Which implies that the estimator is unbiased and efficient according to the Rao-Cramer lowee bound

2.3 MLE for σ^2

Due to equivalence with the OLS procedure

$$\hat{\sigma}^2_{\mathrm{ML}} = \arg\max_{\sigma^2 \in \mathbb{R}^+} \ell(\hat{\boldsymbol{\beta}}, \sigma^2) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \boldsymbol{x}_i^\top\hat{\boldsymbol{\beta}})^2 =$$

$$= \frac{1}{n}\sum_{i=1}^{n}\hat{\varepsilon}^2 \; = \frac{\hat{\varepsilon}^\top\hat{\varepsilon}}{n} = \frac{\mathrm{SSE}}{n}$$

Where:

$$\varepsilon_i = y_i - \boldsymbol{x}_i^\top\hat{\boldsymbol{\beta}} \quad \text{raw residual}, \quad \mathrm{SSE} = \sum_{i=1}^{n}\varepsilon_i^2 \quad \text{residual sum of squares}$$

Note: residual are still a linear transformation of y_i

$$\boldsymbol{\varepsilon} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$$

The Hat matrix is obtained buy substiuing in the formula the MLE for $\hat{\boldsymbol{\beta}}$

$$\hat{\boldsymbol{Y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}$$

$$\hat{\boldsymbol{Y}} = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})\boldsymbol{X}^\top\boldsymbol{Y}$$
$$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top$$

2.3.1 Properties of $\boldsymbol{\varepsilon}$ and $\hat{\sigma}^2_{\mathbf{ML}}$

Assuming Gaussian linear model:

$$\hat{\boldsymbol{\varepsilon}} \mid \boldsymbol{X} \sim \mathcal{N}_n(\boldsymbol{0}, \sigma^2(\boldsymbol{I}_n - \boldsymbol{H}))$$

Now raw resisual are not linarly independent: by summing them you do not obtain n degree of freedome, you loose some

$$\text{SSE} \mid \boldsymbol{X} \sim \sigma^2\chi^2(n-p)$$
$$\frac{\text{SSE}}{\sigma^2} \mid \boldsymbol{X} \sim \chi^2(n-p)$$

$$\Rightarrow \quad \mathbb{E}[\text{SSE} \mid \boldsymbol{X}] = \sigma^2(n-p), \quad \mathbb{E}[\hat{\sigma}^2 \mid \boldsymbol{X}] = \sigma^2$$

$$\Rightarrow \quad \mathbb{E}[\hat{\sigma}^2_{\text{ML}} \mid \boldsymbol{X}] = \sigma^2 \cdot \tfrac{n-p}{n} \neq \sigma^2$$

$$\Rightarrow \quad \mathbb{E}[\hat{\sigma}^2_{\text{ML}} \mid \boldsymbol{X}] \xrightarrow{n\to\infty} \sigma^2$$

An unbiased estimator for σ^2 is given by $\hat{\sigma}^2 = \frac{\text{SSE}}{n-p}$
It in a Biased estimator but works for large n Unbiased estimator for σ^2:

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-p}$$

Note: since $\frac{\text{SSE}}{\sigma^2}$ is independent of $\hat{\boldsymbol{\beta}} \Rightarrow \hat{\sigma}^2_{\text{ML}}$ and $\hat{\sigma}^2$ are independent of $\hat{\boldsymbol{\beta}}$.

2.3.2 Restricted Maximum Likelihood for σ^2

REML is obtained by integrating out $\boldsymbol{\beta}$:

$$L(\sigma^2) = \int \ell(\boldsymbol{\beta}, \sigma^2)\, d\boldsymbol{\beta}$$

- REML estimates are generally less biased than ML estimates.
- For Gaussian linear models, REML is unbiased.

2.3.3 Standardised Residuals

Assuming:

$$\hat{\boldsymbol{\varepsilon}} \mid \boldsymbol{X} \sim \mathcal{N}_n(\mathbf{0_n}, \sigma^2(\boldsymbol{I}_n - \boldsymbol{H}))$$

The diagonal elements of $(\boldsymbol{I}_n - \boldsymbol{H})$ differ across i:

$$1 - H_{ii} = 1 - \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i$$

which means that the $\boldsymbol{H}$ matrix is not invertible. This is why estimated row residual are computed:

$$r_i = \frac{e_i}{\sqrt{\hat{\sigma}^2(1 - H_{ii})}} \sim \mathcal{N}(0, 1)$$

Standardised residuals r_i are approximately standard normal:

$$r = (r_1, \ldots, r_n)^\top \sim \mathcal{N}_n(\mathbf{0}, \boldsymbol{I}_n)$$

3. Hypothesis Testing

Hypthesis formulation involves a linear system defined on the parameter of interested

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \qquad p = (k+1) \times 1 \text{ parameter vector}$$

- $\boldsymbol{C}$: $r \times p$ matrix composed of known constants with full row rank $r \leq p$ (indicates to which β will be constrained)
- $\boldsymbol{d}$: $r \times 1$ vector composed of known constants
- $H_0 : \boldsymbol{C\beta} = \boldsymbol{d}$ is the system of linear hypotheses on $\boldsymbol{\beta}$
- r number of constraint

3.1 Examples when $k = 3$

(A) $\boldsymbol{C} = [0\ 1\ 0\ 0], \quad \boldsymbol{d} = 0 \quad \Rightarrow \quad H_0 : \beta_1 = 0$

(B) $\boldsymbol{C} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{d} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow H_0 : \begin{cases} \beta_1 = 0 \\ \beta_3 = 0 \end{cases} \quad \text{or} \quad \beta_1 = \beta_3 = 0$

(C) $\boldsymbol{C} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{d} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow H_0 : \begin{cases} \beta_1 = 0 \\ \beta_2 = 0 \\ \beta_3 = 0 \end{cases}$

$\Rightarrow \quad \beta_1 = \beta_2 = \beta_3 = 0$ (linear independence)

(D) $\boldsymbol{C} = [0\ 1\ 0\ -1], \quad \boldsymbol{d} = 0 \quad \Rightarrow \quad H_0 : \beta_1 = \beta_3$

(E) $\boldsymbol{C} = [0\ 1\ 0\ 0], \quad \boldsymbol{d} = 3 \quad \Rightarrow \quad H_0 : \beta_1 = 3$

Note: the row's order is irrelevant

3.2 Solutions - Nested Models

Linear Hypotheses lead to a Gaussian linear models that can be obtained by removing some regressors from the starting model:
Starting model: $\mathbb{E}[Y_i \mid x_{1i}, x_{2i}, x_{3i}] = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i}$

(A) $\mathbb{E}_{H_0}[Y_i \mid x_{1i}, x_{2i}, x_{3i}] = \beta_0 + \beta_2 x_{2i} + \beta_3 x_{3i} = \mathbb{E}[Y_i \mid x_{2i}, x_{3i}]$

(B) $\mathbb{E}_{H_0}[Y_i \mid x_{1i}, x_{2i}, x_{3i}] = \beta_0 + \beta_2 x_{2i} = \mathbb{E}[Y_i \mid x_{2i}]$

(C) $\mathbb{E}_{H_0}[Y_i \mid x_{1i}, x_{2i}, x_{3i}] = \beta_0 = \mathbb{E}[Y_i]$

Note: $\mathbb{E}_{H_0}[\cdot]$ denotes the expected value under the null hypothesis H_0- under the restriction imposed by the linear hypothesis.

3.3 Likelihood Ratio Tests- known σ^2

It is the ratio beetween the maximesed *loglikelihood* without any restriction on the parameter and the maximesed *loglikelihood* with restriction on the parameter space, to the values that respect the linear Hp.

$$LR = \frac{L(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2)}{L(\hat{\boldsymbol{\beta}}_{H_0}, \hat{\sigma}^2)} \quad \text{or, equivalently,} \quad lr = 2\ln\left(\frac{L(\hat{\boldsymbol{\beta}}, \hat{\sigma}^2)}{L(\hat{\boldsymbol{\beta}}_{H_0}, \hat{\sigma}^2)}\right)$$

$$\begin{aligned} lr &= -n\ln 2\pi\sigma^2 - \frac{1}{\sigma^2}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}) \\ &\quad + n\ln 2\pi\sigma^2 + \frac{1}{\sigma^2}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{H_0})^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{H_0}) \\ &= \frac{\boldsymbol{\varepsilon}_{H_0}^\top\boldsymbol{\varepsilon}_{H_0} - \boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon}}{\sigma^2} = \frac{\mathrm{SSE}_{H_0} - \mathrm{SSE}}{\sigma^2} = \frac{\Delta\mathrm{SSE}}{\sigma^2} \end{aligned}$$

Where:

$$\hat{\boldsymbol{\beta}}_{H_0} = \arg\max_{\{\boldsymbol{\beta}:\boldsymbol{C}\boldsymbol{\beta}=\boldsymbol{d}\}\subset\mathbb{R}^p} \ell(\boldsymbol{\beta}, \sigma^2), \quad \boldsymbol{\varepsilon}_{H_0} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}_{H_0}, \quad \mathrm{SSE}_{H_0} = \boldsymbol{\varepsilon}_{H_0}^\top\boldsymbol{\varepsilon}_{H_0}$$

3.3.1 Constrained Maximum Likelihood Estimation:

Under the constraint $\boldsymbol{C}\boldsymbol{\beta} = \boldsymbol{d}$, we have:

$$\hat{\boldsymbol{\beta}}_{H_0} = \hat{\boldsymbol{\beta}} - (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\left[\boldsymbol{C}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\right]^{-1}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})$$

$$\mathrm{SSE}_{H_0} = \mathrm{SSE} + (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{H_0})^\top\boldsymbol{X}^\top\boldsymbol{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{H_0})$$

No matter who $\boldsymbol{C}$ is we will obtain the constraint MLE estimates. If $\boldsymbol{X}$ has full column rank, then $\boldsymbol{X}^\top\boldsymbol{X}$ is positive definite:

$$(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{H_0})^\top\boldsymbol{X}^\top\boldsymbol{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{H_0}) > 0 \quad \forall\hat{\boldsymbol{\beta}} \neq \hat{\boldsymbol{\beta}}_{H_0} \Rightarrow \mathrm{SSE}_{H_0} > \mathrm{SSE}$$

The estimates of $\boldsymbol{\beta}$ under H_0 always lead to an increas in the SSE unless the two estimates coincide. Because when you introduce linear restriction you loose goodness of fit

$$\Delta\mathrm{SSE} = (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{H_0})^\top\boldsymbol{X}^\top\boldsymbol{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{H_0}) = (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})^\top\left[\boldsymbol{C}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\right]^{-1}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})$$

Note: ΔSSE can be computed without knowing $\hat{\boldsymbol{\beta}}_{H_0}$ (this is true for GLM)

$$\begin{aligned} LRT &= 2[\ell(\hat{\boldsymbol{\beta}}, \hat{\vartheta}) - \ell(\hat{\boldsymbol{\beta}}_{H_0}, \hat{\vartheta}_{H_0})] \sim \chi^2_r \\ RLRT &= 2[\ell_R(\hat{\boldsymbol{\beta}}, \hat{\vartheta}_{REML}) - \ell_R(\hat{\boldsymbol{\beta}}_{H_0}, \hat{\vartheta}_{REML,H_0})] \sim \chi^2_r \end{aligned}$$

3.3.2 Distribution

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p\left(\boldsymbol{\beta}, \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\right)$$

⇓ **Linear transformation:**

$$\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d} \sim \mathcal{N}_r\left(\boldsymbol{C}\boldsymbol{\beta} - \boldsymbol{d},\ \sigma^2\boldsymbol{C}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\right)$$

⇓ **If H_0 is true:**

$$\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d} \mid H_0 \sim \mathcal{N}_r\left(\boldsymbol{0},\ \sigma^2\boldsymbol{C}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\right)$$

⇓ **Standardisation:**

$$\boldsymbol{z} = \frac{1}{\sigma}\left[\boldsymbol{C}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\right]^{-1/2}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d}) \quad \mid H_0 \sim \mathcal{N}_r(\boldsymbol{0}, \boldsymbol{I}_r)$$

⇓ **Quadratic form (sum of squares):**

$$\boldsymbol{z}^\top\boldsymbol{z} = \frac{(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})^\top\left[\boldsymbol{C}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{C}^\top\right]^{-1}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})}{\sigma^2} = lr \mid H_0 \sim \chi^2_r$$

Note: Due to the formula of the lr the larger the observed difference in the SSE of the two models the stronger will be the evidence against the null hypotheses

3.4 Likelihood Ratio Tests - unknown σ^2

we do replace the unknown value of σ^2 with the MLE estimate.
Properties of raw residuals:

$$\frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-p} \quad \text{and independence between } \hat{\boldsymbol{\beta}} \text{ and } \hat{\sigma}^2$$

If H_0 is true:!!

$$\begin{aligned} F &= \frac{lr}{r}\cdot\frac{n-p}{\text{SSE}/\sigma^2} = \frac{\frac{\Delta SSE}{\sigma^2}}{\frac{SSE}{\sigma^2}}\cdot\frac{n-p}{r} = \\ &= \frac{\Delta\text{SSE}}{r}\cdot\frac{n-p}{\text{SSE}} \\ &= \frac{\Delta\text{SSE}/r}{\text{SSE}/(n-p)} = \frac{\frac{\chi^2_r}{r}}{\frac{\chi^2_{n-p}}{n-p}} \Rightarrow F \mid H_0 \sim F_{r,n-p} \end{aligned}$$

Where r = degree of freedom of the numerator and $n-p$ = degree of freedom of the denumerator.

Note: One chi-squared divided by its degrees of freedom is not an F, but two of them in ratio form is.

3.5 Wald Test Statistics

It is an approximation of the Likelihood ratio test, but in the special case of GLM they do coincide. The F test statistic can be rewritten as:

$$F = \frac{\Delta\text{SSE}}{r} \cdot \frac{1}{\hat{\sigma}^2} = \frac{1}{r} \frac{(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})^\top \left[\boldsymbol{C}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{C}^\top\right]^{-1} (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})}{\hat{\sigma}^2}$$

with $\hat{\sigma}^2 = \frac{\text{SSE}}{n-p}$

$$= \frac{1}{r}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})^\top \left[\hat{\sigma}^2\boldsymbol{C}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{C}^\top\right]^{-1} (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d}) = \frac{1}{r}W$$

$$= \frac{1}{r}(\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d})^\top \left[\hat{\text{Cov}}(\boldsymbol{C}\hat{\boldsymbol{\beta}})\right]^{-1} (\boldsymbol{C}\hat{\boldsymbol{\beta}} - \boldsymbol{d}) = \frac{1}{r}W$$

Where:

- $\hat{\text{Cov}}(\boldsymbol{C}\hat{\boldsymbol{\beta}})$ is the estimates for $\text{Cov}(\boldsymbol{C}\hat{\boldsymbol{\beta}}) = \sigma^2\boldsymbol{C}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{C}^\top$
- W is the Wald test statistic, representing distance between $\boldsymbol{C}\hat{\boldsymbol{\beta}}$ and the hypothesized value $\boldsymbol{d}$, weighted by its covariance. (considering $\boldsymbol{d}$ the value of $\boldsymbol{C}\hat{\boldsymbol{\beta}}$ when the Hp is true)

3.5.1 Special Cases

If $H_0 : \beta_j = 0$, then:

$$F = \left(\frac{\hat{\beta}_j}{\sqrt{\text{Var}(\hat{\beta}_j)}}\right)^2 = t_j^2 \Rightarrow \frac{\hat{\beta}_j}{\sqrt{\text{Var}(\hat{\beta}_j)}} \sim t_{n-p}$$

Note: Link between t and F: $t_{n-p}^2 = F_{1,n-p}$

3.5.2 Confidence Intervals

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{Var}(\hat{\beta}_j)}} \sim t_{n-p} \quad \text{(pivotal quantity for} \quad \hat{\beta}_j)$$

Student-t confidence interval at $(1-\alpha)$ confidence level:

$$\left[\hat{\beta}_j - t_{n-p,1-\alpha/2}\sqrt{\text{Var}(\hat{\beta}_j)}, \quad \hat{\beta}_j + t_{n-p,1-\alpha/2}\sqrt{\text{Var}(\hat{\beta}_j)}\right]$$

3.6 Linear independence hypothesis

Defined as: $H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$
it occurs when none of the regressor have a signficicant imapct on the dependent variable.

Then F reduces to:

$$F = \frac{R^2}{1 - R^2} \cdot \frac{n - p}{k} \quad \text{where} \quad R^2 = 1 - \frac{\text{SSE}}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Note that: R^2 quantifies the goodness of fit, and it represent the fraction of the sample variability explained by the regressor

Recall that:

$0 \leq \text{SSE} \leq \sum_{i=1}^{n}(y_i - \bar{y})^2 \Rightarrow R^2 \in [0, 1]$

$R^2 = 0 \Leftrightarrow \text{SSE} = \sum_{i=1}^{n}(y_i - \bar{y})^2$ (no explanatory power)

$R^2 = 1 \Leftrightarrow \text{SSE} = 0$ (perfect fit)

3.7 H_0: linear independence between X_j and Y: $\beta_j = 0 \ \forall \ j$

Partendo dall'ipotesi di indipendenza lineare dobbiamo per prima cosa definire una matrice C, in cui abbiamo in questo caso 9 righe (number of linear constraints) e 10 colonne (number of constrainable parameters). La prima colonna sarà di soli 0 perché l'ipotesi di indipendenza lineare non rguiarda l'intercetta, le rimanenti colonne conterranno degli 1.

```
C1<-cbind(rep(0,9),diag(9)) # matrix of Constraints
C1
```

```
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
[1,]    0    1    0    0    0    0    0    0    0     0
[2,]    0    0    1    0    0    0    0    0    0     0
[3,]    0    0    0    1    0    0    0    0    0     0
[4,]    0    0    0    0    1    0    0    0    0     0
[5,]    0    0    0    0    0    1    0    0    0     0
[6,]    0    0    0    0    0    0    1    0    0     0
[7,]    0    0    0    0    0    0    0    1    0     0
[8,]    0    0    0    0    0    0    0    0    1     0
[9,]    0    0    0    0    0    0    0    0    0     1
```

```
d1<-rep(0,9)
d1
```

```
[1] 0 0 0 0 0 0 0 0 0
```

L'ordine delle righe è irrilevante, le righe devono essere linearmente indipendenti, la matrice deve essere quindi a rango di righe pieno.

```
lht(model1,C1,d1)
```

```
  Linear hypothesis test:
  age = 0
  lwt = 0
  raceblack = 0
  raceother = 0
  smokeyes = 0
  ptl = 0
  htyes = 0
  uiyes = 0
  ftv = 0

  Model 1: restricted model
  Model 2: bwt ~ age + lwt + race + smoke + ptl + ht + ui + ftv

    Res.Df      RSS Df Sum of Sq      F    Pr(>F)
  1    188 99969656
  2    179 75702317  9  24267339 6.3756 7.891e-08 ***
  ---
  Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Le funzioni `lht` e `linearHypothesis` sono equivalenti.

- Il primo argomento è il modello fittato.
- Il secondo argomento, `hypothesis.matrix`, è la matrice C delle ipotesi:
- Ha q righe e p colonne.
- Il terzo argomento, `rhs`, è il vettore d:
- `rhs` sta per "right-hand side" del sistema di ipotesi lineari che vogliamo specificare.

Nella funzione è possibile specificare il test da eseguire.

- un test χ^2 se la varianza σ^2 è nota;
- un test F di Fisher se la varianza non è nota.

Nota:

- se il vettore d1 è di soli 0 possiamo ignorarlo nella funzione `lht`
- l'output di `summary(model1)` restituiva già questo test test F (ultima riga), quindi non è necesario utilizzare questa funzione per fare questo tipo di test. Lo usiamo per testare altri tipi d'ipotesi.

4. Categorical regressor

	X_{MLi}	X_{SLi}	X_{AAi}	X_{SMi}	X_{MMi}
much less active	1	0	0	0	0
somewhat less active	0	1	0	0	0
about as active	0	0	1	0	0
somewhat more active	0	0	0	1	0
much more active	0	0	0	0	1

- 5 indicator variables allow to code a qualitative regressor with 5 categories

- These 5 indicator variables sum up to 1, for any sample unit:

$$X_{MLi} + X_{SLi} + X_{AAi} + X_{SMi} + X_{MMi} = 1 \quad \forall i$$

  if they are included in a linear model along with an intercept term, the corresponding regressor matrix $\mathbf{X}$ will not have full column rank.

$\rightarrow$ it is necessary to exclude one of the indicator variables. The corresponding category is termed *baseline/reference category*.

4.1 Coding scheme: indicator/dummy variables

	X_{Bi}	X_{Ci}	X_{Di}	X_{Ei}
much less active	0	0	0	0
somewhat less active	1	0	0	0
about as active	0	1	0	0
somewhat more active	0	0	1	0
much more active	0	0	0	1

The reference category choice leads to different interpretation of the model and the meaning of regressor coefficient

Reference category: `physact`$_i$ `= much less active`

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i] = \beta_0 + \beta_1 X_{SLi} + \beta_2 X_{AAi} + \beta_3 X_{SMi} + \beta_4 X_{MMi} \qquad i = 1, \ldots, 2032$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{much less active}] = \beta_0$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{somewhat less active}] = \beta_0 + \beta_1$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{about as active}] = \beta_0 + \beta_2$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{somewhat more active}] = \beta_0 + \beta_3$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{much more active}] = \beta_0 + \beta_4$$

$\hat{\boldsymbol{\beta}}_j$ interpretation:
Each regression coefficient represents the difference between the conditional expected value given the corresponding category and the conditional expected value given the reference category. *For example* observing $\hat{\boldsymbol{\beta}}_j < 0$ means that the units in this category have lower y_i with respectc to the reference category.

4.1.1 Linear Hypothesis

Absence of significant differences in the average glucose level among women characterized by different physical activity levels

$$\begin{aligned} H_0: \quad & \mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{much less active}] \\ =& \mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{somewhat less active}] \\ =& \mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{about as active}] \\ =& \mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{somewhat more active}] \\ =& \mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{much more active}] \end{aligned}$$

Hypothesis of interest (linear independent hypothesis)

$$\Rightarrow H_0 : \begin{cases} \beta_0 = \beta_0 + \beta_1 \\ \beta_0 = \beta_0 + \beta_2 \\ \beta_0 = \beta_0 + \beta_3 \\ \beta_0 = \beta_0 + \beta_4 \end{cases} \quad \text{or, equivalently} \quad H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

```
> C1
```

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

```
> d1
[1] 0 0 0 0
```

The regression coefficients for the four indicator variables are simultaneously not significantly different from 0
$\Rightarrow$ when only a categorical regressor is considered, this hypothesis coincides with the linear independence hypothesis

4.1.2 Results

In order to evaluate the significance of the categorical regressor you have to simultaneusly set equal to zero more than 1 coefficient

```
> physact1 <- lm(glucose ~ physact, data = hers.nod)
> summary(physact1)
```

```
Estimate   Std. Error  t value   Pr(>|t|)
(Intercept)                    98.421     0.939     104.784  < 2e-16
physactsomewhat less active   -0.858     1.084     -0.792   0.429
physactabout as active        -1.226     1.011     -1.213   0.225
physactsomewhat more active   -2.434     1.011     -2.408   0.016
physactmuch more active       -3.278     1.121     -2.924   0.003


Residual standard error: 9.716 on 2027 degrees of freedom
Multiple R-squared: 0.008668 Adjusted R-squared: 0.006712
F-statistic: 4.431 on 4 and 2027 DF, p-value: 0.001441
```

the relevant test statistic is the F test statistic
t test statistics allow to evaluate differences between each category and the reference category

Note: here it happens to coincide with the tests for linear independence, but just because it is the only one regressor in this model.

4.1.3 F test

```
> linearHypothesis(physact1, C1, d1, test = "F")
Linear hypothesis test
Hypothesis:

physactsomewhat less active = 0
physactabout as active = 0
physactsomewhat more active = 0
physactmuch more active = 0

Model 1:  restricted model
Model 2:  glucose   physact

Res.Df    RSS       Df  Sum of Sq     F       Pr(>F)
2031   193017.70
2027   191344.61    4     1673.09    4.43    0.0014
```

where:

- `2027`$= n - p$
- `4`$= r$
- `Sum of Sq`$= \Delta\text{SS}$

4.2 Coding scheme: Exclusion of the intercept

If one consider a regression model without intercept, it is possible to include all the 5 indicator variables (without choosing a reference category)

	X_{Ai}	X_{Bi}	X_{Ci}	X_{Di}	X_{Ei}
much less active	1	0	0	0	0
somewhat less active	0	1	0	0	0
about as active	0	0	1	0	0
somewhat more active	0	0	0	1	0
much more active	0	0	0	0	1

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i] = \mu_1 X_{MLi} + \mu_2 X_{SLi} + \mu_3 X_{AAi} + \mu_4 X_{SMi} + \mu_5 X_{MMi} \qquad i = 1, \ldots, 2032$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{much less active}] = \mu_1$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{somewhat less active}] = \mu_2$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{about as active}] = \mu_3$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{somewhat more active}] = \mu_4$$

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i = \text{much more active}] = \mu_5$$

Hypothesis of interest (Linear hypothesis)

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$$

$$H_0 : \begin{cases} \mu_1 = \mu_2 \\ \mu_1 = \mu_3 \\ \mu_1 = \mu_4 \\ \mu_1 = \mu_5 \end{cases} \quad \Leftrightarrow \quad H_0 : \begin{cases} \mu_1 - \mu_2 = 0 \\ \mu_1 - \mu_3 = 0 \\ \mu_1 - \mu_4 = 0 \\ \mu_1 - \mu_5 = 0 \end{cases} \quad \text{or, equivalently, } H_0 : \begin{cases} \mu_1 - \mu_2 = 0 \\ \mu_2 - \mu_3 = 0 \\ \mu_3 - \mu_4 = 0 \\ \mu_4 - \mu_5 = 0 \end{cases}$$

4.2.1 Results

```
> C2
```

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 & -1 \end{bmatrix} \quad \text{or, equivalently,} \quad \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

```
> d2
[1] 0 0 0 0
```

```
> physact2 <- lm(glucose ~ physact - 1, data = hers.nod)
> summary(physact2)

...

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
physactmuch less active     98.4206     0.9393  104.78  <2e-16 *
physactsomewhat less active 97.5621     0.5414  180.19  <2e-16 *
physactabout as active      97.1442     0.3742  259.71  <2e-16 *
physactsomewhat more active 95.9867     0.3734  257.05  <2e-16 *
physactmuch more active     95.1429     0.6120  155.45  <2e-16 *

Residual standard error: 9.716 on 2027 degrees of freedom
Multiple R-squared: 0.99, Adjusted R-squared: 0.99
F-statistic: 4.023e+04 on 5 and 2027 DF,  p-value: < 2.2e-16
```

WARNING:

- In this setting the function `lm` computes R^2 using $\sum_{i=1}^{n} y_i^2$ as denominator
- the F test statistic refers to $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = 0$, no longer to the linear independence hypothesis

4.2.2 F test

```
> linearHypothesis(physact2, C2, d2, test = "F")
Linear hypothesis test

Hypothesis:
physactmuch less active - physactsomewhat less active = 0
physactmuch less active - physactabout as active = 0
physactmuch less active - physactsomewhat more active = 0
physactmuch less active - physactmuch more active = 0

Model 1: restricted model
Model 2: glucose ~ physact


Res.Df    RSS         Df   Sum of Sq   F     Pr(>F)
2031    193017.70
2027    191344.61    4     1673.09    4.43  0.0014
```

Note:
The choice of the coding scheme is irrelevant in the test, the only things that change is the interpretation

4.3 Coding scheme: Incremental/split coding

An alternative coding scheme can be used if categories are ordered:

	X_{Bi}	X_{Ci}	X_{Di}	X_{Ei}
much less active	0	0	0	0
somewhat less active	1	0	0	0
about as active	1	1	0	0
somewhat more active	1	1	1	0
much more active	1	1	1	1

It is possible to show that these alternative indicator variables can be obtained by **summing subsets of the indicator variables introduced above**. $\rightarrow$ each of this dummy can be written as a sum of dummies, we still needd $k-1$ dummy (it is like if the 5-*th* is the intercept

$$\mathbb{E}[\text{glucose}_i \mid \text{physact}_i] = \beta_0 + \beta_B X_{Bi} + \beta_C X_{Ci} + \beta_D X_{Di} + \beta_E X_{Ei} \qquad i = 1, \ldots, 2032$$

$$\mathbb{E}[\text{glucose}_i \mid \text{much less active}] = \beta_0$$

$$\mathbb{E}[\text{glucose}_i \mid \text{somewhat less active}] = \beta_0 + \beta_B$$

$$\mathbb{E}[\text{glucose}_i \mid \text{about as active}] = \beta_0 + \beta_B + \beta_C$$

$$\mathbb{E}[\text{glucose}_i \mid \text{somewhat more active}] = \beta_0 + \beta_B + \beta_C + \beta_D$$

$$\mathbb{E}[\text{glucose}_i \mid \text{much more active}] = \beta_0 + \beta_B + \beta_C + \beta_D + \beta_E$$

Each regression coefficient represents the difference between the conditional expected values associated with two consecutive categories

Hypothesis of interest

$$H_0 : \begin{cases} \beta_0 = \beta_0 + \beta_B \\ \beta_0 = \beta_0 + \beta_B + \beta_C \\ \beta_0 = \beta_0 + \beta_B + \beta_C + \beta_D \\ \beta_0 = \beta_0 + \beta_B + \beta_C + \beta_D + \beta_E \end{cases} \qquad \text{or, equivalently, } H_0 : \beta_B = \beta_C = \beta_D = \beta_E = 0$$

4.3.1 R implementation

You need to create the matrix `splitcoding`

```
> splitcoding
  1 2 3 4
1 0 0 0 0
2 1 0 0 0
3 1 1 0 0
4 1 1 1 0
5 1 1 1 1
```

```
> contrasts(hers.nod$physact) <- splitcoding
```

```
> physact3 <- lm(glucose ~ physact, data = hers.nod)
> summary(physact3)

Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)   98.421    0.939     104.784   0.000
physactd1     -0.858    1.084     -0.792    0.429
physactd2     -0.368    0.658     -0.559    0.576
physactd3     -1.208    0.529     -2.284    0.022
physactd4     -0.844    0.717     -1.177    0.239

Residual standard error: 9.716 on 2027 degrees of freedom
Multiple R-squared: 0.008668, Adjusted R-squared: 0.006712
F-statistic: 4.431 on 4 and 2027 DF, p-value: 0.001441
```

4.3.1.1 Interpretation

Here the values for $\hat{\beta}_j$ is: the average difference in y level between j-th category and the $j-1$ category.
So here the p-values are comparing if the difference of the response is significant among the current category and the previous one, so just a subsection of all the possible comparison that can be done.
The results differ with the others code scheme

4.3.2 Linear trend hypothesis

Due to similar amount of reduction observed in y values when changing category (when dealing with ordered categorical regressor) $\rightarrow$ one can think of **linear trend**
This assumption implies a constant impact on the response regardless which pair of categories you are considering.

The introduction of suitable linear constraints in the regression coefficients associated with the incremental coding scheme allows to test for the existence of a linear trend in the conditional expected values:

$$H_0 : \beta_B = \beta_C = \beta_D = \beta_E = \beta\,(\neq 0)$$

$$\Rightarrow\ \mathbb{E}[\text{glucose}_i \mid \text{much less active}]_{H_0} = \beta_0 + 0 \cdot \beta = \beta_0$$

$$\Rightarrow\ \mathbb{E}[\text{glucose}_i \mid \text{somewhat less active}]_{H_0} = \beta_0 + 1 \cdot \beta$$

$$\Rightarrow\ \mathbb{E}[\text{glucose}_i \mid \text{about as active}]_{H_0} = \beta_0 + 2 \cdot \beta$$

$$\Rightarrow\ \mathbb{E}[\text{glucose}_i \mid \text{somewhat more active}]_{H_0} = \beta_0 + 3 \cdot \beta$$

$$\Rightarrow\ \mathbb{E}[\text{glucose}_i \mid \text{much more active}]_{H_0} = \beta_0 + 4 \cdot \beta$$

Rather than defining 5 dummies we assign a score to each category Testing the linear trend hypothesis is equivalent to testing the adequacy of a numeric coding based on integer scores from 0 to 4

4.3.3 Linear constraints

$$H_0 : \beta_B = \beta_C = \beta_D = \beta_E = \beta(\neq 0) \quad \Rightarrow \quad H_0 : \begin{cases} \beta_B = \beta_C \\ \beta_C = \beta_D \\ \beta_D = \beta_E \end{cases} \quad \text{or } H_0 : \begin{cases} \beta_B = \beta_C \\ \beta_B = \beta_D \\ \beta_B = \beta_E \end{cases}$$

```
> c.lin
```

$$\begin{bmatrix} 1 & 0 & 1 & -1 & 0 \\ 2 & 0 & 0 & 1 & -1 \\ 3 & 0 & 1 & 0 & -1 \end{bmatrix} \quad \text{or, equivalently,} \quad \begin{bmatrix} 1 & 0 & 1 & -1 & 0 \\ 2 & 0 & 1 & 0 & -1 \\ 3 & 0 & 0 & 1 & -1 \end{bmatrix}$$

```
> d.lin   [1] 0 0 0
```

4.3.4 F test for the linear trend hypothesis

```
> linearHypothesis(physact3, C.lin, d.lin, test="F")

    Linear hypothesis test
    Hypothesis:

    physactd1 - physactd2 = 0
    physactd2 - physactd3 = 0
    physactd3 - physactd4 = 0

Model 1:  restricted model
Model 2:  glucose ~ physact

\begin{verbatim}
Res.Df       RSS          Df    Sum of Sq     F       Pr(>F)
1     2030   191419.47
2     2027   191344.61    3     74.86       0.26     0.8511
```

The linear trend hypothesis is not rejected

$H_0 : \beta_{black} = \beta_{other}$ **(esempio di $\beta_j = \beta_h$ $j \neq k$)** Per esempio ci focalizziamo sulle due dummy variables associate con l'etnia. Abbiamo due coefficienti di regressione stimati che non sono esattamante uguali tra loro, ma se prendiamo in considerazione il loro estimated standard error possiamo sospettare che non ci siano delle differenze significative tra i due. In altri termini la differenza tra donne caucasiche e africane sembra essere la stessa che c'è fra donne caucasiche e di altre etnie. Questa tesi è supportata anche da uno dei primi grafici che abbiamo fatto (boxplot) in cui potevamo vedere che le medie tra i due gruppi sono molto simili.
Quindi invece che considerare questo regressore come categorico con tre categorie possiamo cercare di semplificare la situazione unendo `black` e `others`.
In questo modo riduciamo le categorie da 3 a 2. Per farlo però dobbiamo verificare che questo sia adeguato. In questo caso $j = 4, k = 5$.

```
C2<-C1[3,]
C2
```

```
 [1] 0 0 0 1 0 0 0 0 0 0
```

```
C2[5] <- -1
C2
```

```
 [1]  0  0  0  1 -1  0  0  0  0  0
```

```
d2 <- 0
```

Abbiamo preso prima la terza riga di `C1` (che ha un 1 in 4° posizione, proprio per $j = 4 = black$) e poi abbiamo assegnato -1 alla 5° posizione del vettore appena creato (corrispondente a $others$).
Questo vettore ci permette di prendere le differenze tra il β delle donne africane e quello di donne di altre etnie.

```
lht(model1,C2,d2)
```

```
Linear hypothesis test:
raceblack - raceother = 0

Model 1: restricted model
Model 2: bwt ~ age + lwt + race + smoke + ptl + ht + ui + ftv

  Res.Df      RSS Df Sum of Sq      F Pr(>F)
1    180 75998327
2    179 75702317  1    296010 0.6999 0.4039
```

Cediamo che l' F test ha un p-value grande, quindi non ci sono evidenze contro la nostra H_0. Possiamo allora semplificare il nostro modello accorpando le due categorie.

Modello con le due classi accorpate

```
       FALSE TRUE
white     96    0
black      0   26
other      0   67
```

```
model4<-update(model1,.~.-race+nowhite,data=dati)
```

Definiamo una nuova variabile (dummy) categorica che prende 1 per ogni donna con origine non caucasica e 0 per quelle caucasiche.
Se vogliamo fittare il modello sotto la restrizione che non ci siano differenze tra nere e altre, quello che possiamo fare è sostituire il regressore originale con quello appena creato. Aggiorniamo quindi il modello 1 con la stessa variabile dipendente, gli stessi regressori, ma da quelli originali sto togliamo `race` e aggiungiamo quello nuovo. Specifichiamo inoltre quale sia il dataframe che contiene queste variabili, dal momento che abbiamo aggiunto nuove variabili, in questo modo evitiamo problemi con l'uso della funzione `update`.

```
anova(model4,model1)
```

```
Analysis of Variance Table

Model 1: bwt ~ age + lwt + smoke + ptl + ht + ui + ftv + nowhite
Model 2: bwt ~ age + lwt + race + smoke + ptl + ht + ui + ftv
  Res.Df      RSS Df Sum of Sq      F Pr(>F)
1    180 75998327
2    179 75702317  1    296010 0.6999 0.4039
```

Vediamo che usando la funzione `anova` tra restricted e unrestricted models abbiamo test F come identico a quello ottenuto usando la funzione lht, cioè definendo la matrice C e il vettore d associati a questa restrizione lineare.

```
summary(model4)
```

```
Call:
lm(formula = bwt ~ age + lwt + smoke + ptl + ht + ui + ftv +
    nowhite, data = dati)

Residuals:
     Min       1Q   Median       3Q      Max
-1838.40  -443.44    12.04   467.77  1675.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2971.999    308.188   9.643  < 2e-16 ***
age           -2.833      9.572  -0.296 0.767613
lwt            3.946      1.664   2.371 0.018795 *
smokeyes    -366.181    105.039  -3.486 0.000616 ***
ptl          -47.525    101.881  -0.466 0.641437
htyes       -591.885    202.149  -2.928 0.003853 **
uiyes       -512.896    138.717  -3.697 0.000289 ***
ftv          -15.276     46.406  -0.329 0.742405
nowhiteTRUE -397.275    102.988  -3.857 0.000159 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 649.8 on 180 degrees of freedom
Multiple R-squared:  0.2398, Adjusted R-squared:  0.206
F-statistic: 7.097 on 8 and 180 DF,  p-value: 3.828e-08
```

```
summary(model1)
```

```
Call:
lm(formula = bwt ~ ., data = dati)
```

```
Residuals:
     Min       1Q   Median       3Q      Max
-1825.26  -435.21    55.91   473.46  1701.20

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 2927.962    312.904   9.357  < 2e-16 ***
age           -3.570      9.620  -0.371 0.711012
lwt            4.354      1.736   2.509 0.013007 *
raceblack   -488.428    149.985  -3.257 0.001349 **
raceother   -355.077    114.753  -3.094 0.002290 **
smokeyes    -352.045    106.476  -3.306 0.001142 **
ptl          -48.402    101.972  -0.475 0.635607
htyes       -592.827    202.321  -2.930 0.003830 **
uiyes       -516.081    138.885  -3.716 0.000271 ***
ftv          -14.058     46.468  -0.303 0.762598
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 650.3 on 179 degrees of freedom
Multiple R-squared:  0.2427, Adjusted R-squared:  0.2047
F-statistic: 6.376 on 9 and 179 DF,  p-value: 7.891e-08
```

Osserviamo come abbiamo adesso solo una dummy per race. Possiamo vedere come il coefficiente stimato (-397) è una sorta di media tra i due coefficienti di regressione che avevamo prima (-488 e -355).

4.4 Changing reference category of the regressor

```
load("C:/Users/eglew/Desktop/RSMA/claims.RData")
str(claims)
```

```
'data.frame':    64 obs. of  5 variables:
 $ n   : int  197 284 133 24 85 149 66 9 35 53 ...
 $ c   : int  38 63 19 4 22 25 14 4 5 10 ...
 $ age : Factor w/ 4 levels "<25","25-29",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ dist: Factor w/ 4 levels "rural","small towns",..: 1 1 1 1 2 2 2 2 3 3 ...
 $ car : Factor w/ 4 levels "<1","1-1.5","1.5-2",..: 1 2 3 4 1 2 3 4 1 2 ...
```

- age: arranged in classes (<25, 25-29, 30-35, >35)
- dist: type of area where the policyholders live (rural, small town, large towns, major cities)
- car: engine capacity (<1 liter, 1-1.5 liter, 1.5-2 liter, >2 liter)
- n: number of policyholders for each covariate pattern (that is, each data entry of the cross-table built on [age,dist,rural,car])
- c: number of claims for each covariate pattern

4.4.1 Contrast (NON CONSIGLIATO)

Lo possiamo fare usando il comando `contrasts` su dei regressori categorici. In questo modo vediamo come coding the qualitative regressors how categorical predictors are turned into binary variables for levels

```
claims$car
```

```
  [1] <1    1-1.5 1.5-2 >2    <1    1-1.5 1.5-2 >2    <1    1-1.5 1.5-2 >2
 Levels: <1 1-1.5 1.5-2 >2
```

```
claims$age
```

```
  [1] <25   <25   <25   <25   <25   <25   <25   <25   <25   <25   <25   <25
 Levels: <25 25-29 30-35 >35
```

Possiamo vedere dall'output come non ci sia un ordine nei livelli (cosa che teoricamente dovrebbe esserci). Lo capiamo dall'ultima riga dell'output (`Levels`):
`<1 1-1.5 1.5-2 >2` per `car` e `<25 25-29 30-35 >35` per `age`.

```
order.car <- factor(claims$car, ordered = T)
order.car
```

```
  [1] <1    1-1.5 1.5-2 >2    <1    1-1.5 1.5-2 >2    <1    1-1.5 1.5-2 >2
 Levels: <1 < 1-1.5 < 1.5-2 < >2
```

Se ci fosse stato un ordine avremmo avuto (come nella nuova variabile `order.car`) avremmo visualizzato `<1 < 1-1.5 < 1.5-2 < >2`. Si può vedere infatti come fra ogni livello ci sia un `<`, cosa che ci fa capire che i livelli sono stati ordinati.

```
contrasts(claims$car)
```

```
      1-1.5 1.5-2 >2
 <1       0     0  0
 1-1.5    1     0  0
 1.5-2    0     1  0
 >2       0     0  1
```

```
contrasts(claims$age)
```

```
      25-29 30-35 >35
 <25      0     0   0
 25-29    1     0   0
 30-35    0     1   0
 >35      0     0   1
```

```
contrasts(claims$dist)
```

```
              small towns large towns major cities
 rural                  0           0            0
 small towns            1           0            0
 large towns            0           1            0
 major cities           0           0            1
```

Utilizzando il comando `contrasts` possiamo vedere come vengono codificate le nostre variabili categoriche e quali livelli di queste vengono scelti come categorie di riferimento (sono quelli a cui è associato un vettore di 0)

```
contrasts(order.car)
```

```
              .L   .Q         .C
 [1,] -0.6708204  0.5 -0.2236068
 [2,] -0.2236068 -0.5  0.6708204
 [3,]  0.2236068 -0.5 -0.6708204
 [4,]  0.6708204  0.5  0.2236068
```

Vediamo come applicare il comando ad una variabile ordinata ci porti ad un problema di interpretazione. Infatti se la variabile è ordinata abbiamo dei parametri associati alle categorie che si riferiscono alla parte lineare, quadratica e cubica della nostra variabile. Questi parametri NON sono facili da interpretare!
Devi stare quindi attento a capire su che variabile la usi.

```
# ?contr.treatment -> Return a matrix of contrasts
```

Per cambiare la categoria di riferimento quindi possiamo fare nel seguente modo:

```
contrasts(claims$dist) <- contr.treatment(4 ,base=4)
contrasts(claims$dist)
```

Specifichiamo quindi quale sia la nuova categoria di riferimento utilizzando il comando `contr.treatment`. In questo caso stiamo dicendo che la variabile ha 4 livelli e che come categoria di riferimento consideriamo proprio il quarto.

```
              1 2 3
 rural        1 0 0
 small towns  0 1 0
 large towns  0 0 1
 major cities 0 0 0
```

Possiamo vedere infatti come la categoria di riferimento sia diventata `major cities` a cui ora è associato il vettore `(0, 0, 0)`.

4.4.2 Relevel (CONSIGLIATO)

In genere il comando potremmo anche usarlo direttamente sulla variabile del dataset, ma è sconsigliato. Meglio creare una nuova variabile e modificare quella.

```
# claims£dist
contrasts(claims$dist)
```

```
             1 2 3
 rural       1 0 0
 small towns 0 1 0
 large towns 0 0 1
 major cities 0 0 0
```

Lo usiamo sulla variabile `dist` che ha 4 livelli: rural; small towns; large towns; major cities. Vediamo come di dafault la reference category sia rural (a cui corrisponde un vettore di 0).
Andiamo a creare quini la nuova variabile `new.dist`:

```
new.dist <- relevel(claims$dist, "major cities")
# new.dist
contrasts(new.dist)
```

```
             rural small towns large towns
 major cities     0           0           0
 rural            1           0           0
 small towns      0           1           0
 large towns      0           0           1
```

In questo modo stiamo dicendo che della variabile dist in claims, la categoria di riferimento diventa `major cities`, cosa che vediamo lanciando il comando `contrasts` sulla nuova variabile: il vettore di 0 è infatti associato a `major cities`.

4.5 Models

4.5.1 Modello con dist: rural categoria di riferimento

```
# ?glm
```

```
m1<-glm(c~offset(log(n))+car+dist+age, data=claims, family=poisson)
# non c'è overdispersion perchè non c'è quasipossion
```

```
summary(m1)
```

```
 Call:
 glm(formula = c ~ offset(log(n)) + car + dist + age, family = poisson,
     data = claims)

 Coefficients:
             Estimate Std. Error z value Pr(>|z|)
 (Intercept) -1.58753    0.09276 -17.115  < 2e-16 ***
```

```
car1-1.5     0.16134    0.05053   3.193 0.001409 **
car1.5-2     0.39281    0.05500   7.142 9.18e-13 ***
car>2        0.56341    0.07232   7.791 6.65e-15 ***
dist1       -0.23421    0.06167  -3.798 0.000146 ***
dist2       -0.20834    0.06476  -3.217 0.001294 **
dist3       -0.19568    0.06984  -2.802 0.005081 **
age25-29    -0.19101    0.08286  -2.305 0.021149 *
age30-35    -0.34495    0.08137  -4.239 2.24e-05 ***
age>35      -0.53667    0.06996  -7.672 1.70e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 236.26  on 63  degrees of freedom
Residual deviance:  51.42  on 54  degrees of freedom
AIC: 388.74

Number of Fisher Scoring iterations: 4
```

Coefficient Interpretation
Per interpretare correttamente i coefficienti di un modello che ha come link function il logaritmo (per esempio Poisson), è necessario prendere l'esponenziale dei coefficienti. Il segno del coefficiente ci fornisce un'indicazione sull'effetto della variabile: un segno positivo implica un aumento del tasso di eventi, uno negativo una diminuzione.

Nel nostro caso, solo una delle categorie della variabile dist risulta significativa. Questo NON implica automaticamente che possiamo eliminare i livelli non significativi: potremmo considerare di collassare alcune categorie, ma ciò richiede cautela e una motivazione chiara, quindi non è sempre consigliabile.

Se almeno un livello di una variabile categorica è significativo, è buona norma mantenere tutta la variabile nel modello. Un'alternativa utile può essere modificare la categoria di riferimento, perché la **significatività** dei coefficienti **può dipendere** da quale livello è utilizzato come **categoria di riferimento**.

Nota: i coefficienti dei singoli livelli variano (in grandezza, segno e significatività) a seconda della categoria di riferimento scelta.

4.5.2 Modello con new.dist: major cities categoria di riferimento

```
m1.b <- update(m1, . ~ offset(log(n))+car+new.dist+age)
summary(m1.b)
```

```
Call:
glm(formula = c ~ car + new.dist + age + offset(log(n)), family = poisson,
```

```
    data = claims)

Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)         -1.58753    0.09276 -17.115  < 2e-16 ***
car1-1.5             0.16134    0.05053   3.193 0.001409 **
car1.5-2             0.39281    0.05500   7.142 9.18e-13 ***
car>2                0.56341    0.07232   7.791 6.65e-15 ***
new.distrural       -0.23421    0.06167  -3.798 0.000146 ***
new.distsmall towns -0.20834    0.06476  -3.217 0.001294 **
new.distlarge towns -0.19568    0.06984  -2.802 0.005081 **
age25-29            -0.19101    0.08286  -2.305 0.021149 *
age30-35            -0.34495    0.08137  -4.239 2.24e-05 ***
age>35              -0.53667    0.06996  -7.672 1.70e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 236.26  on 63  degrees of freedom
Residual deviance:  51.42  on 54  degrees of freedom
AIC: 388.74

Number of Fisher Scoring iterations: 4
```

Vediamo come l'overall fit del modello non cambia. Varia solamente la categoria di riferimento, cosa che ha fatto diventare tutti i coefficienti significativi.

5. Model selection

A crucial step is represented by the choice of the regressors to be included in the model among those in the set of candidate regressors.*(including the intercept and any set of dummy variables needed to code categorical regressors that are available in the data set).*

- $\{x_0, x_1, \ldots, x_k\}$ set of candidate regressors
- $M \subseteq \{0, 1, \ldots, k\}$ index set identifying the regressors to be included in a Gaussian linear models
- $\boldsymbol{X}_M$ regressor matrix obtained considering the candidate regressors indexed in M

Parsimony vs. Accuracy, a trade-off

When selecting the regressors to be included in a Gaussian linear model, two contrasting requirements need to be addressed:

- **parsimony**: models with fewer regressors are simpler to handle and interpret
  - $\rightarrow$ measured by the number of model parameters *(dimension of the parameter space)*
  - $\rightarrow$ for a Gaussian linear model, the total number of parameters is given by $|M|+1$, where $|M|$ is the cardinality of the index set M and corresponds to the number of columns of $\boldsymbol{X}_M$
- **accuracy**: models with many regressors are more accurate
  - $\rightarrow$ for a Gaussian linear model, a common choice to measure the adequacy (*goodness of fit*) are based on functions of SSE such as the coefficient of linear determination

$$R^2 = 1 - \frac{\text{SSE}}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Impact of relevant vs. irrelevant regressors

Under the assumptions characterizing Gaussian linear models, it is possible to prove that:

- excluding a **relevant regressor** (i.e., a regressor whose regression coefficient differ from 0) can lead to a bias in $\hat{\boldsymbol{\beta}}$ *(the ML estimator for the regressors included in the model)*
- including an **irrelevant regressor** (i.e., a regressor whose regression coefficient is equal to 0) leads to an increase in the variability of $\hat{\boldsymbol{\beta}}$

$\Rightarrow$ *What if the distinction between relevant and irrelevant regressors is not known?*
Some solutuions are based on recasting the problem of regressor selection as a model comparison problem
Note: Detailed proof on the book
Which is the most adequate Gaussian linear model for a given random sample $\boldsymbol{Y}$?

Procedure

Having

$$A \subseteq \{0, 1, \dots, k\} \Rightarrow \boldsymbol{Y} \mid \boldsymbol{X}_A \sim \mathcal{N}_n(\boldsymbol{X}_A\boldsymbol{\beta}_A, \sigma^2\boldsymbol{I}_n), \quad \boldsymbol{\beta}_A \in \mathbb{R}^{|A|}$$
$$B \subseteq \{0, 1, \dots, k\} \Rightarrow \boldsymbol{Y} \mid \boldsymbol{X}_B \sim \mathcal{N}_n(\boldsymbol{X}_B\boldsymbol{\beta}_B, \sigma^2\boldsymbol{I}_n), \quad \boldsymbol{\beta}_B \in \mathbb{R}^{|B|}$$

The two models differ because they are characterized by different sets of regressors: $\boldsymbol{X}_A \neq \boldsymbol{X}_B$ (without loss of generality: $|A| > |B|$).

Note: it is better to focus on the dimension of the X matrix, in case of the presence of categorical regressor.

5.1 Nested models

B is a proper subset of $A \rightarrow$ likelihood ratio test

If $\boldsymbol{X}_B$ can be obtained by removing one or more that one columns from $\boldsymbol{X}_A$

$\Rightarrow$ Model B can be obtained by introducing suitable linear constraints on the parameters of model A

$$\left.\begin{array}{l} \mathbf{Y} \mid \mathbf{X}_A \sim \mathcal{N}_n\left(\mathbf{X}_A\boldsymbol{\beta}_A, \sigma^2\mathbf{I}_n\right) \\ H_0 : \mathbf{C}_B\boldsymbol{\beta}_A = \mathbf{d}_B \end{array}\right\} \quad \Rightarrow \quad \mathbf{Y} \mid \mathbf{X}_B \sim \mathcal{N}_n\left(\mathbf{X}\boldsymbol{\beta}_B, \sigma^2\mathbf{I}_n\right)$$

Where:

- $\boldsymbol{C}_B$: $q \times |A|$ matrix
- $\boldsymbol{d}_B = \mathbf{0}_q$

Likelihood Ratio Test

In particular, such test can be expressed as a function of the two corresponding residual sums of squares. By removing irrelevant column, the LRT test helps us in the choice of rejecting or not the null hypothesis.

$$lr = 2\ln\left[\frac{L\left(\hat{\boldsymbol{\beta}}_A, \sigma^2\right)}{L\left(\hat{\boldsymbol{\beta}}_A \mid H_0, \sigma^2\right)}\right] = 2\ln\left[\frac{L\left(\hat{\boldsymbol{\beta}}_A, \sigma^2\right)}{L\left(\hat{\boldsymbol{\beta}}_B, \sigma^2\right)}\right] = \frac{\boldsymbol{\varepsilon}_B^\top\boldsymbol{\varepsilon}_B - \boldsymbol{\varepsilon}_A^\top\boldsymbol{\varepsilon}_A}{\sigma^2} = \frac{\mathrm{SSE}_B - \mathrm{SSE}_A}{\sigma^2}$$

Furthermore,

$$F = \frac{\mathrm{SSE}_B - \mathrm{SSE}_A}{\mathrm{SSE}_A} \cdot \frac{n - |\mathcal{A}|}{q} = \frac{R_A^2 - R_B^2}{1 - R_A^2} \cdot \frac{n - |A|}{q}$$

Where:

- $q = |A| - |B|$ number of elements excluded from A to obtain B
- $\frac{n-|A|}{q}$ are the degrees of freedom: the numbers of columns of the larger models considering the removed column
- $\frac{R^2_A - R^2_B}{1-R^2_A}$ significance in the difference

what is the model testing:
R^2 is structured such that, when a column is removed, it automatically decreases. The test statistic checks whether this reduction is significantly different from zero. If it is, the larger model is preferred, as this indicates that the null hypothesis is rejected. *A significant difference from zero implies that we are rejecting the null hypothesis. and the larger model is accepted*

If H_0 is true – if model B is as "adequate" as A:

$$\Rightarrow \quad lr| \text{ Model } B \sim \chi^2_q$$

$$\Rightarrow \quad F| \text{ Model } B \sim F_{q,(n-|A|)}$$

5.1.1 F statistic with anova function

per prima cosa costruiamo un nested model, che sarà il modello che otteniamo quando l'ipotesi di indipendenza lineare è vera. In questo caso è il modello con solamente l'intercetta. Questo lo possiamo creare riusando la funzione lm di prima o aggiornando un modello già esistente con la funzione update cambiando la formula.

Può essere utile per rifittare un modello dopo aver cambiato qualcosa. Possiamo rifittare lo stesso modello su un campione diverso, si può creare un modello con gli stessi regressori ma con diversa variabile dipendente o invece cambaire i regressori lasciando la stessa variabile dipendente.

```
model2<-update(model1,.~1)
```

Aggiorniamo il `model1`, e nel secondo argomento mettiamo la nuova formula.

Con il punto a sinistra sto dicendo che voglio la stessa parte sinistra della formula precedente, ma una nuova parte destra. Specificando solamente 1 sto dicendo che voglio un modello con solo l'intercetta. Questo sarebbe il modello (nested/reduced model) ottenuto quando l'ipotesi di indipendenza lineare è vera.

```
summary(model2)
```

```
Call:
lm(formula = bwt ~ 1, data = dati)

Residuals:
```

```
     Min       1Q   Median       3Q      Max
-2235.59  -530.59    32.41   542.41  2045.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2944.59      53.04   55.51   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 729.2 on 188 degrees of freedom
```

anova function La usiamo quando stiamo lavorando con i fitted model. Un modo per usarla è metterle dentro due modelli di R: il primo sarà è il reduced/nested model, il secondo invece il full model. In questo modo otteniamo un F test statistic per comparare questi due modelli

```
anova(model2,model1)
```

```
Analysis of Variance Table

Model 1: bwt ~ 1
Model 2: bwt ~ age + lwt + race + smoke + ptl + ht + ui + ftv
  Res.Df      RSS Df Sum of Sq      F    Pr(>F)
1    188 99969656
2    179 75702317  9  24267339 6.3756 7.891e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# summary(model1)
```

5.2 Non nested models

B is not a proper subset of A

> If $\mathbf{X}_B$ cannot be obtained by removing one or more than one columns from $\mathbf{X}_A$,
>
> $\Rightarrow$ Model B cannot be expressed trough a restriction on the parameter of model A.
>
> $\Rightarrow$ Model B can be obtained by simultaneously excluding some (or all) regressors in model A and adding some new regressors to model A.
>
> $\Rightarrow$The two models are characterised by two sets of regressors that are only partially overlapping, or non-overlapping.
>
> $\Rightarrow$ The differences between SSE_B and SSE_A does not have a known random distribution, and thus a likelihood ratio test cannot be used to choose between the two models

R^2 is not informative due to the difference in the numbers of columns

5.3 Adjusted R^2

Problems with R^2

- When $\mathbf{X}_A$ and $\mathbf{X}_B$ have different numbers of columns, R^2 could favour the model whose design matrix has the largest number of columns.
- It is possible to prove that SSE never increases after adding a regressor to a Gaussian linear model,even if the regressor is irrelevant

Solution with the adjusted R^2

$$\bar{R}^2 = 1 - \frac{n-1}{n-|M|}\left(1-R^2\right) = 1 - \left(\frac{n-1}{n-|M|} \cdot \frac{\text{SSE}}{\sum_{i=1}^{n}(y_i-\bar{y})^2}\right)$$

- $\frac{n-1}{n-|M|}$ a **correcting factor that consider model complexity** (the cardinality of the index set), it balanced the reduction observed in SSE due to introduction of an irrelevant regressor, (it balanced the increase in SSE due to removing a regressor)
- Adjusted R^2 does not necessarly decrease: it reach a maximum and then decrease when the increase in goodness of fit due to added columns is neglible
- It can also be negative, dependeing on how large $\frac{n-1}{n-|M|}$ is
- Differently from R^2, $\bar{R}^2$ consider the trade-off between parsimony and adequacy
- in case of same $\bar{R}^2$, the choice must be done according to parsimony

5.3.1 Correspondence R^2 and $\bar{R}^2$

Consider two models A and B same number of columns in $\mathbf{X}_A$ and $\mathbf{X}_B$). Then there is no Bias in the comparison trough R^2

$$\begin{aligned}
\bar{R}^2(A) > \bar{R}^2(B) &\Leftrightarrow 1 - \left(\frac{n-1}{n-p} \cdot \frac{\text{SSE}_A}{\sum_{i=1}^{n}(y_i-\bar{y})^2}\right) > 1 - \left(\frac{n-1}{n-p} \cdot \frac{\text{SSE}_B}{\sum_{i=1}^{n}(y_i-\bar{y})^2}\right) \\
&\Leftrightarrow \frac{n-1}{n-p} \cdot \frac{\text{SSE}_A}{\sum_{i=1}^{n}(y_i-\bar{y})^2} < \frac{n-1}{n-p} \cdot \frac{\text{SSE}_B}{\sum_{i=1}^{n}(y_i-\bar{y})^2} \\
&\Leftrightarrow \text{SSE}_A < \text{SSE}_B \\
&\Leftrightarrow \bar{R}^2(A) > \bar{R}^2(B) \\
&\Leftrightarrow \quad \bar{R}^2(A) > \bar{R}^2(B)
\end{aligned}$$

5.4 Cross validation

Differently from SSE, CV may increase if an irrelevant regressor is added to the model.
$\Rightarrow$ The best model is the one achieving the **minimum value for CV** (among all the considered models).

5.4.1 Leove-one-out

Between two regression models, choose the one with the smallest prediction error.
Trough C-V we check the goodness of fit on an independent test set

- $\hat{y}_{iM}^{-i}$ predicted value for y_i obtained using model M after excluding the i-th unit from the observed sample (independent of the i-th unit)
- $CV = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_{iM}^{-i}\right)^2$ Unbiased estimate of the prediction error

In order to compute CV for a given regression model, the estimation procedure should be repeated n times after omitting each sample unit. Then, each fitted model is used to compute a prediction for the corresponding omitted sample unit. The quantities $y_i - \hat{y}_{iM}^{-i}$ are also referred to as *deleted residuals*. Some authors/software use the acronym PRESS (PRedictive Error Sum of Square) to denote CV.

5.4.2 CV for Gaussian linear regression models

CV for Gaussian linear regression models can be computed without repeating the fitting process n times.

- $\hat{\boldsymbol{\beta}}_M^{-i}$ ML estimate of $\boldsymbol{\beta}_M$ obtained after excluding the i-th unit from the observed sample (independent of the i-th unit)
- $\hat{y}_{iM}^{-i} = \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}_M^{-i}$ predicted value for y_i using the regressors in M after excluding the i-th unit from the observed sample (independent of the i-th unit)

It is possible to prove that

$$y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}_M^{-i} = \frac{y_i - \hat{y}_{iM}}{1 - h_{iiM}} = \frac{e_i}{1 - h_{iiM}}$$

Where h_{iiM} is the i-th diagonal element of the matrix

$$\boldsymbol{H}_M = \boldsymbol{X}_M \left(\boldsymbol{X}_M^\top \boldsymbol{X}_M\right)^{-1} \boldsymbol{X}_M^\top$$

5.5 Akaike information Criteria

It is useful beyond linear regression, because it can be applied to different regressors' structure.

5.5.0.1 Definition

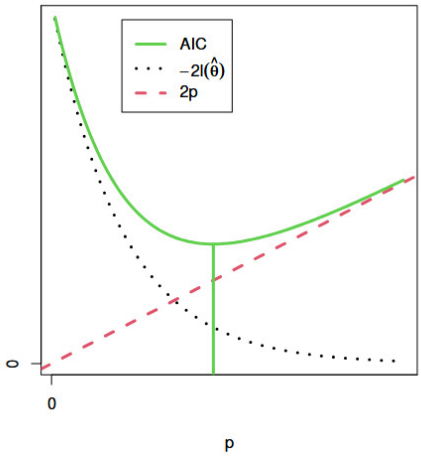
- Y random sample with unknown probability/density $f_0(y)$
- $\mathcal{F}_A = \{f_A(\cdot;\theta_A), \theta_A \in \Theta_A \subseteq \mathbb{R}^{p_A}\}$ **parametric statistical model** A
- $\mathcal{F}_B = \{f_B(\cdot;\theta_B), \theta_B \in \Theta_B \subseteq \mathbb{R}^{p_B}\}$ **parametric statistical model** B

parametric statistical model refers to a set of probability distribution among which there is the true one.

It is possible to prove that, under suitable regularity conditions that for a generic parametric statistical model:

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p$$

As you can see the $-2\ell(\hat{\theta})$ decrease as the number of predictors increase



- $-2\ell(\hat{\theta})$ (twice the MLE estimates ?) measures the accuracy/goodness of fit of a statistical model to the data.
  In general, this quantity decreases as the number of parameters increases.
- $2p$ measures the complexity (= the opposite of parsimony) of a statistical model.
  It increases as the number of the parameters increases.

This way it is establishing a formulation of the trade-off. By choosing according to the AIC means choosing the *Best trade-off between goodness of fit and accuracy/parsimony* → The best model is the one achieving the **minimum value for AIC** (among all the considered models)

5.5.1 How does it work

Between two statistical models, choose the one that contains the element that is the most "similar" to $f_0(\cdot)$, select the one that contains element as close as possible to the true distribution of the radom sample.

1. Define a measure for the distance between the true one $f_0(\cdot)$ and the estimated one.

   Kullback-Leibler divergence is the selected one. It measurees the amount of information that is lost when $f_0(\cdot)$ is approximated with $f_A(\cdot;\theta_A) \in \mathcal{F}_A$

   $$\mathcal{K}(f_A, f_0) = \mathbb{E}\left[\ln \frac{f_0(Y)}{f_A(Y;\theta_A)}\right]$$

2. check where the distance's measure is minumum

   It is possible to prove that, under suitable regularity conditions,

   $$\min_{\mathcal{F}_A} \mathcal{K}(f_A, f_0) < \min_{\mathcal{F}_B} \mathcal{K}(f_B, f_0) \quad \Leftrightarrow \quad -2\ell_A(\hat{\theta}_A) + 2p_A < -2\ell_B(\hat{\theta}_B) + 2p_B$$

5.5.2 AIC for Gaussian linear models

- Maximum likelihood estimates for $\boldsymbol{\beta}$ and σ^2 are considered (even if $\hat{\sigma}^2_{ML}$ is biased).

  $$-2\ell\left(\hat{\boldsymbol{\beta}}_M, \hat{\sigma}^2_{ML}\right) = 2\left(\tfrac{n}{2}\ln 2\pi - \tfrac{n}{2}\ln\tfrac{\mathrm{SSE}_M}{n} - \tfrac{1}{2\cdot\frac{\mathrm{SSE}_M}{n}}\mathrm{SSE}_M\right)$$
  $$= n\ln 2\pi + n + n\ln\hat{\sigma}^2_{ML}$$
  $$\Rightarrow \quad \mathrm{AIC} = n\ln 2\pi + n + n\ln(\hat{\sigma}^2_{ML}) + 2(|M|+1)$$

  Note: $\hat{\sigma}^2_{ML} = \frac{\mathrm{SSE}_M}{n}$

- When all the candidate models are Gaussian linear models, the following simplified formula can be used:

  $$\mathrm{AIC} = n\ln(\hat{\sigma}^2_M) + 2(|M|+1)$$

- When $|A| = |B|$ (same number of columns in $\boldsymbol{X_A}$ and $\boldsymbol{X_B}$):

  $$\mathrm{AIC}(A) < \mathrm{AIC}(B) \Leftrightarrow \mathrm{SSE}_A < \mathrm{SSE}_B \Leftrightarrow R^2(A) > R^2(B)$$

5.6 Bayesian Information Criterion (BIC)

5.6.0.1 Definition

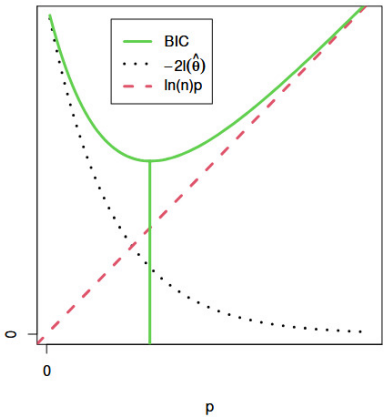
- $\boldsymbol{y}$: observed sample
- $\mathcal{F}_A$: parametric statistical model A
- $\mathcal{F}_B$: parametric statistical model B

In the Bayesian framework, each element in $\mathcal{F}_A$ and in $\mathcal{F}_B$ has an a priori probability of being the true distribution that generated $\boldsymbol{y}$:

- $g_A(\theta_A)$: a priori probability/density function for distributions belonging to $\mathcal{F}_A$
- $g_B(\theta_B)$: a priori probability/density function for distributions belonging to $\mathcal{F}_B$

For a generic parametric statistical model:

$$\text{BIC} = -2\ell(\hat{\theta}) + \ln(n)p$$



- $-2\ell(\hat{\theta})$ measures the goodness of fit of a statistical model to the data.
  In general, this quantity decreases as the number of parameters increases.
- $\ln(n)p$ measures the complexity (= the opposite of parsimony) of a statistical model.
  It increases as the number of the parameters increases.

This way it is establishing a formulation of the trade-off. By choosing according to the BIC means choosing the *Best trade-off between goodness of fit and complexity/parsimony* → The best model is the one achieving the **minimum value for BIC** (among all the considered models)

It represent the probability that the data has been generated from a specific model and then it choose the model with the largest probability of having generated the data.

$$\Pr(\boldsymbol{y}|\mathcal{F}_A) = \int g_A(\theta_A) f_A(\boldsymbol{y};\theta_A) d\theta_A$$

It is possible to prove that, under suitable regularity conditions,choosing according to the BIC is equivalent of choosing accorrding to the AIC:

$$\Pr(\boldsymbol{y}|\mathcal{F}_A) > \Pr(\boldsymbol{y}|\mathcal{F}_B) \quad \Leftrightarrow \quad -2\ell_A(\hat{\theta}_A) + \ln(n)p_A < -2\ell_B(\hat{\theta}_B) + \ln(n)p_B$$

5.6.1 BIC for Gaussian linear models

$$-2\ell\left(\hat{\boldsymbol{\beta}}_M, \hat{\sigma}^2_{ML}\right) = 2 \cdot \left(\frac{n}{2}\ln 2\pi - \frac{n}{2}\ln\frac{\mathrm{SSE}_M}{n} - \frac{1}{2 \cdot \frac{\mathrm{SSE}_M}{n}} \cdot \mathrm{SSE}_M\right)$$

$$= n\ln 2\pi + n + n\ln\hat{\sigma}^2_{ML} \quad \Rightarrow \quad \mathrm{BIC} = n\ln 2\pi + n + n\ln(\hat{\sigma}^2_{ML}) + 2\ln(n)(|M| + 1)$$

- Maximum likelihood estimates for $\boldsymbol{\beta}$ and σ^2 are considered
- When all the candidate models are Gaussian linear models, the following simplified formula can be used:

$$\mathrm{BIC} = n\ln(\hat{\sigma}^2_{ML}) + \ln(n)(|M| + 1)$$

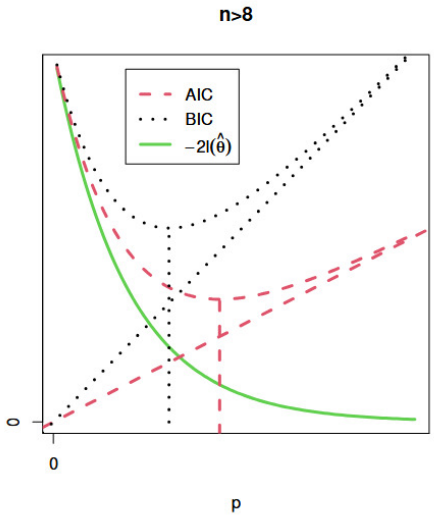
- When $|A| = |B|$ (same number of columns in $\boldsymbol{X}_A$ and $\boldsymbol{X}_B$):

$$\mathrm{BIC}(A) < \mathrm{BIC}(B) \Leftrightarrow \mathrm{SSE}_A < \mathrm{SSE}_B \Leftrightarrow R^2(A) > R^2(B)$$

5.7 AIC or BIC?

Although derived within two completely different frameworks, *AIC* and *BIC* have a very similar functional form. The main "practical" difference is the way in which model complexity is weighted:

- In general, **BIC** puts more weight on model complexity, it will tend to favor simpler model more than AIC:
$$n > 8 \quad \Rightarrow \quad \ln(n) > 2$$
- For a given observed sample, **BIC** tends to favour less complex models (than those selected according to *AIC*)
- Under suitable conditions, both criteria are consistent (when the sample size is large, they select or are able to select the "best" model – according to the corresponding conceptual framework)



- There is not any test to evaluate the significance of the difference among *AIC* (or *BIC*) values

Note: there is not a way to evauluate the significance of the difference observed in AIC or BIC between different models
Notice the different weight that each criteria gives to complexity

5.8 Non nested models R

```
model3 <-update(model1,.~.-age-ftv)
```

stiamo togliendo dal `model1` i regressori `age` e `ftv`
Allora `model3` e `model4` non sono nested: non c'è modo di ottenere uno di questi modelli imponendo restrizioni lineari all'altro.

```
summary(model3)$adj.r.squared
  [1] 0.2122369

summary(model4)$adj.r.squared
  [1] 0.2059988
```

possiamo compararli usando l'adjusted R2, che troviamo nel summary. In questo caso preferiamo il 3 al 4, perchè ha valore più alto.
*AIC*Abbiamo due funzioni in R per calolcare l'AIC: `AIC` e `extractAIC`

```
AIC(model3)
  [1] 2992.861

AIC(model4)
  [1] 2995.305
```

```
extractAIC(model3)
  [1]    8.000 2454.502

extractAIC(model4)
  [1]    9.000 2456.946

AIC(model3) - extractAIC(model3)[2]
  [1] 538.3588

AIC(model4) - extractAIC(model4)[2]
  [1] 538.3588
```

I valori che queste ci restituiscono sono leggermente diversi anche se sono applicate allo stesso modello.

Comparando la differenza tra queste due applicate allo stesso modello ci accorgiamo che queste sono costanti, quindi non dipendono dal fitted model ma dalla funzione.
Ciò è infatti legato al fatto di avere una formula semplificata per calcolare AIC quando consideriamo i gaussian linear models.
Il **miglior modello** in ogni caso sarà quello con **AIC più piccolo**, che è sempre il model3.

Abbiamo quindi sia AIC che $Adjusted\ R^2$ ci consigliano di selezionare il modello 3, i model selection criteria quindi concordano.

Part II

Gaussian Linear Mixed Models

1. Introduction

Mixed beacuse include both random and fixed effect.
Through Gaussian Linear Mixed Models it is possible to relax on the assumption of:

- **conditional homoscedasticity** → we can model heteroschedasticity. We need to introduce a parametric structure for $\text{Cov}[\mathbf{Y}|\mathbf{X}] \neq \sigma^2\mathbf{I}_n$, which cannot be no longer diagonal, so we will need new parameters

- **conditional independence** → pecific dependence structures can be introduced when the data are characterised by a **multilevel structure**

Gaussian linear mixed models use **random factors** to introduce conditional heteroscedasticity and dependence structures.

1.1 Fixed or random factors

- **Fixed factor:** all the categories of interest have been observed. *All the category*(of interest) *have been observed*

- **Random factor:** *only a random subset of the categories* of interest is observed. The category of interest are also those not observed.

The distinction between fixed and random factors depends on:

  ◦ the sampling scheme

  ◦ the aim of the analysis

Note: In some cases, a categorical variable can be interpreted either as a fixed or a random factor.Generally, a random factor can be treated as fixed, but **a fixed factor should never be treated as random**.

1.1.1 Modelling implications

- **Fixed factors:** modeled via numeric coding (indicator variables).
  → As many parameters as the number of categories (minus 1 for identifiability).

- **Random factors:** introduction of **random effects**(to exploit the factor's randomness), parameters defined for each (observed or unobserved) category (*also for level 2 units not observed in the sample.*
  → These require assumptions on their probability distribution to handle unobserved categories.

1.2 Hierarchical structure

Clustered data: instead of random sample it involves a two-stage sampling scheme.
Any "Cluster" (two-stage) sampling scheme leads to samples with a (two-level) hierarchical structure.
Statistical units are divided in two random sample:

1. m **level 2** units are drown at **random** –*you have m groups*
   → your sample will only have a subset of all the possible groups, not all the possible groups will be in the sample

2. n_i **level 1** units are drown from each level 2 units selected at stage 1 with $i = 1, \cdots, m$

*Note that the observed sample is composed of statistical units that are split according to a **random subset** of the categories of the factor used at Stage 1*

Note:

- Each level 2 unit is composed of some level 1 units
- Any level 1 unit belongs to a specific level 2 unit *(it cannot belong to more than one level 2 unit)*

There can be data with structures with more than two levels

1.3 Regressors for multilevel data

The multilevel structure of an observed sample has an impact also on the regressors observed on each sample unit:

- **level 1 regressor** specific feature of each level 1 unit
  Example: gender of a rat pup (male/female)
- **level 2 regressor** common feature for all level 1 units belonging to the same level 2 unit, constant inside clusters, it might change when you compare level l1 units belonging to different level 2 units
  Example: Dosage of a drug given to the rat mother during pregnancy

1.4 Examples

1.4.1 Clustered data

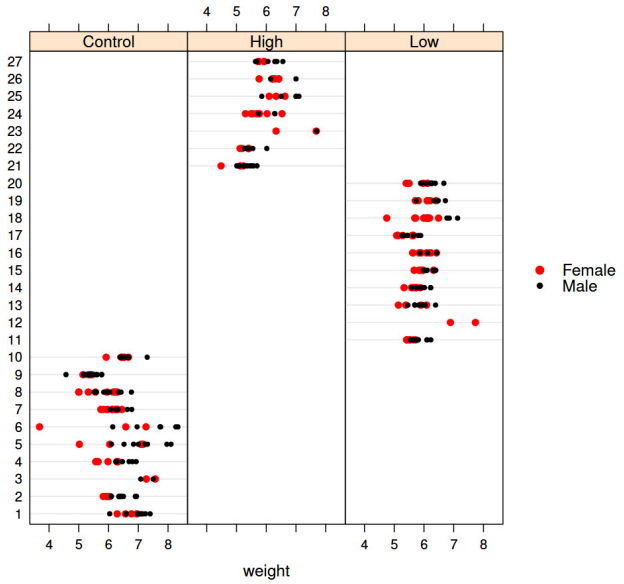
- Birth weight, statistical phenomenon of interest (dependent variable)
- Rat pup, statistical unit
- Litter factor, (categorical variable) that splits the statistical units into groups

Two-stage sampling scheme:

1. m litters
2. n_i rat pup belonging to each litter selected at Stage 1 are examined

   level 1 regressor → gender of a rat pup (male/female)

   level 2 regressor → Dosage of a drug given to the rat mother during pregnancy



1.4.2 Longitudinal data–Growth model

Two-level hierarchical structure:

- Level 1: value at a given time point on the i-th individual
- Level 2: all the values on the i-th individual

Example:

- Distance between the pituitary gland and the pterygomaxillary fissure, Statistical phenomenon of interest (dependent variable)
- Distance at a given time point,value at a given time point
- Child, factor (categorical variable) that splits time points into groups

Two-stage/cluster sampling scheme:

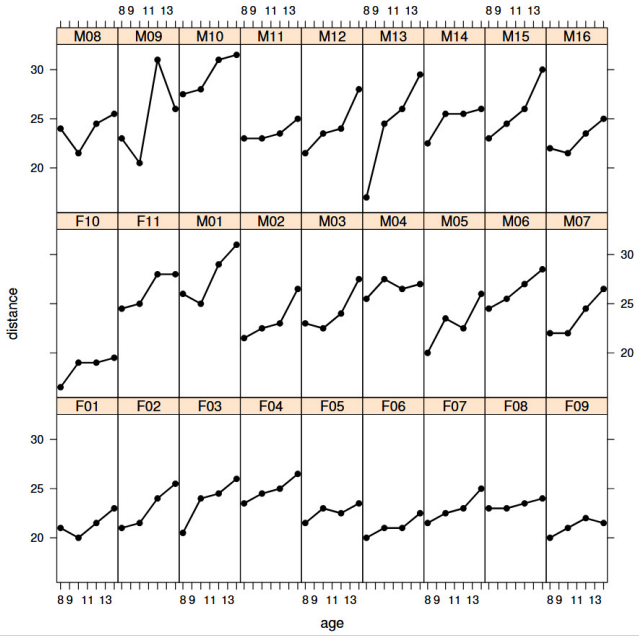
1. m children are randomly selected from a population of interest
2. n_i time points $t_{i1}, t_{i2}, \ldots, t_{in_i}$ are considered for each child selected at stage 1 ($i = 1, \ldots, n$)

the individual can be interpreted as a factor, that allows to split the values at all the time points

Regressors:

level 1 regressor → Age at a given time point

level 2 regressor → Gender (Male/Female)



2. Exploratory Analysis

Pacchetti e Librerie

```
library(nlme)
library(lattice)
```

2.0.1 Dati

*

```
ortho <- read.table("Orthodont.txt", header=TRUE, sep=";")
```

```
str(ortho)
```

```
'data.frame':    108 obs. of  4 variables:
 $ distance: num  26 25 29 31 21.5 22.5 23 26.5 23 22.5 ...
 $ age     : int  8 10 12 14 8 10 12 14 8 10 ...
 $ Subject : chr  "M01" "M01" "M01" "M01" ...
 $ Sex     : chr  "Male" "Male" "Male" "Male" ...
```

Abbiamo osservazioni che si riferiescono a dei bambini organizzate verticalmente: le prime 4 righe (1-4) sono per il primo bambino, le seconde 4 (5-8) per il secondo bambino e così via.

- distance: distance between the pituitary gland and the pterygomaxillary fissure (taken from x-rays of the children's skulls). Questa è monitorata nel tempo per controllare la crescita dei bambini
- age: age at x-ray
- Subject: child identifier (categorical variable with 27 categories). Nello studio ci sono27 bambini diversi.
- Sex: gender (male, female)

```
ortho$Subject <- factor(ortho$Subject)
ortho$Sex <- factor(ortho$Sex)
```

Entrambe sono variabili qualitative (fattori) e devono essere trattate in quanto tali.
Centrare la variabile age

```
summary(ortho$age)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    8.0     9.5    11.0    11.0    12.5    14.0
```

Possiamo vedere come la media dell'età sia 11. Le osservazioni sono infatti rilevate ogni due anni: a 8-10-12-14 anni

```
ortho$age <- ortho$age - 11
```

in questo modo stiamo centrando l'età, lo stiamo facendo perché vogliamo che l'intercetta del modello abbia un'iterpretazione rappresenta la distanza media (la misurazione media) presa a 11 anni il regressore age prenderà valore 0 quando il bambino ha 11 anni É solo un modo per dare un significato all'intercetta

Preliminary data management

Nel pacchetto `nlme` non ci sono solo funzioni che permettono di fittare un GLMM, ma anche funzioni che permettono di maneggiare dei dati con una multilevel structure.
Una di queste è la funzione `groupedData`
Questa crea un `groupedData object`, che è una versione migliorata del dataframe che stiamo utilizzando. É utile quando lavoriamo con dei multilevel dataframe perché possiamo attaccare al dataframe una formula in cui diciamo esplicitamente a R quali sono i random factor, cioè le variabili che utilizziamo per identificare le level 1 units, in questo caso le misurazioni associate ad ogni bambino.
La formula deve contere i level 1 regressors. I level 2 regressors possono essere aggiunti usando l'argomento `outer`
Nota: è consigliabile aggiungere solamente regressori categorici in `outer`

```
# ?groupedData
ortho1 <- groupedData(distance ~ age | Subject, outer = ~ Sex,
                      data = ortho)
```

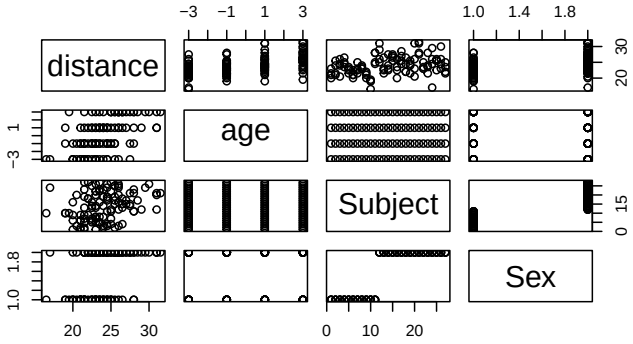
Usiamo la formula tipica: y ~ level 1 regr | group

- `Subject` è quindi la variabile che identifica i gruppi di unità
- `outer = ~ Sex` in questo modo specifichiamo i level 2 regressors, tipicamente ci mettiamo categorical level 2 regressors

Nota: è utile creare questo oggetto (con dati ma anche formula) perché in questo modo quando facciamo i grafici R riconosce la formula nell'oggetto e la usa per produrre grafici.

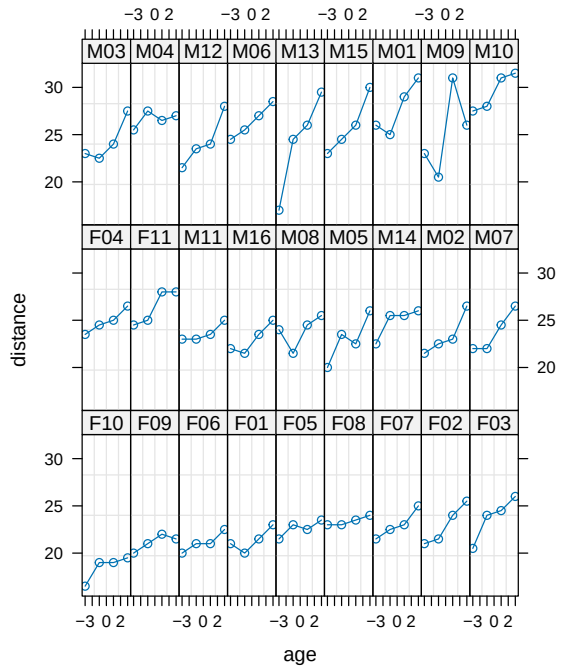
Explorative analysis - graphical displays

```
# ?plot
plot(ortho)
```



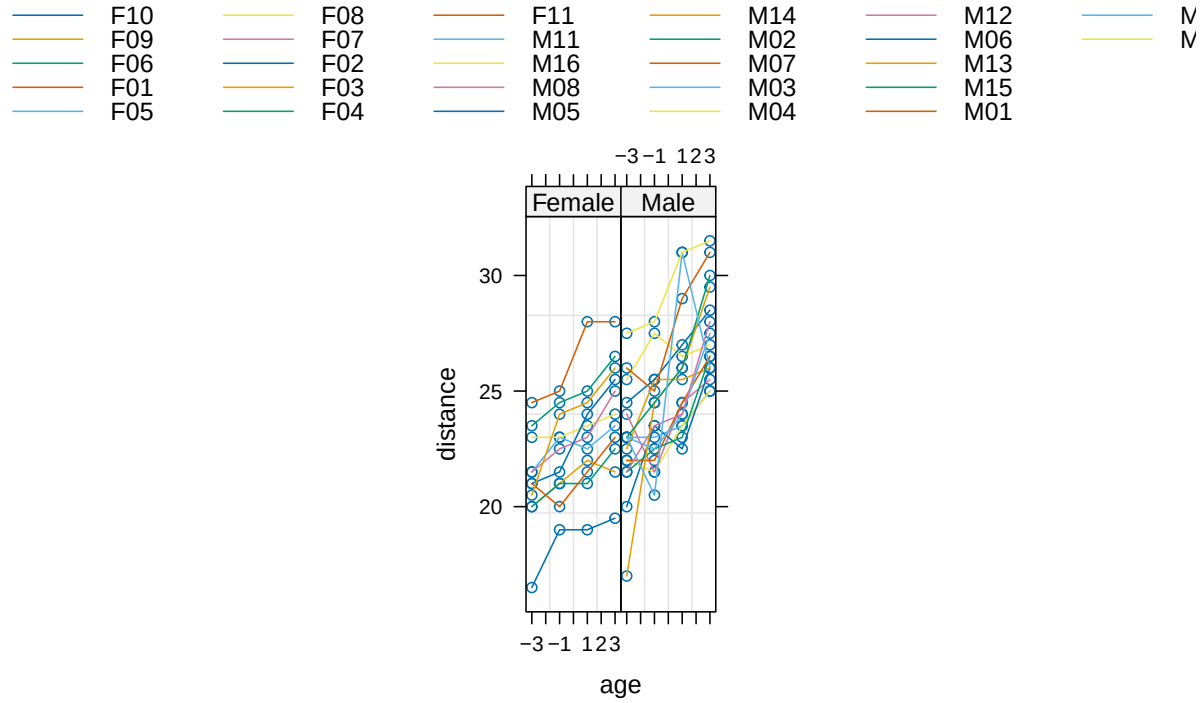
Quello che otteniamo utilizzando il dataset "normale" è una scatterplot matrix che contiene tutti gli scatterplot per ogni coppia di variabili.

```
plot(ortho1)
```



dal momento che R riconosce che questo dataframe è di tipo `groupedData`, sapendo che al suo interno c'è una formula che specifica i random factors, ci restituisce un plot molto più informativo.
Abbiamo quindi un plot per ogni bambino.
Quello che visualizziamo è quindi la distanza in base all'età per ognuno dei 27 bambini.

```
plot(ortho1,outer=TRUE)
```



Aggreghiamo le informazioni di ogi bambino. Abbiamo le femmine a sinistra e i maschi a destra. In questo modo possiamo vedere come all'inizio dello studio la media della distanza per femmine e maschi fosse diversa (per F era più piccola), come ci siano dei pattern di crescita diversi tra soggetti diversi e come comunque alla fine dello studio i maschi tendano ad avere una distanza media più grande delle femmine.
Possiamo quindi affermare la presenza di un effetto di genere sulla distanza e sul pattern di crescita.

2.0.2 Pre-analysis

Andiamo a fittare un GLM diverso per ogni soggetto. Nel dataset abbiamo 4 osservazioni per bambino. Per farlo usaimo la funzione `lmList`.
Questa analisi può dare dei suggerimenti in merito a quale random effect includere nei GLMM.
La funzione `lmList` divide i dati in gruppi in accordo con le variabili che sono usate per identificare i random factor e poi fitta un modello lineare individuale per ognuno di questi gruppi. In questo modo creaimo quindi una lista di GLM in cui `age` è usata come regressore.

```
# ?lmList
lmList1 <- lmList(distance~age|Subject,data=ortho1)
```

Dobbiamo innanzitutto specificare una formula come prima e i dati che inseriamo devono essere di tipo `grupedData`.

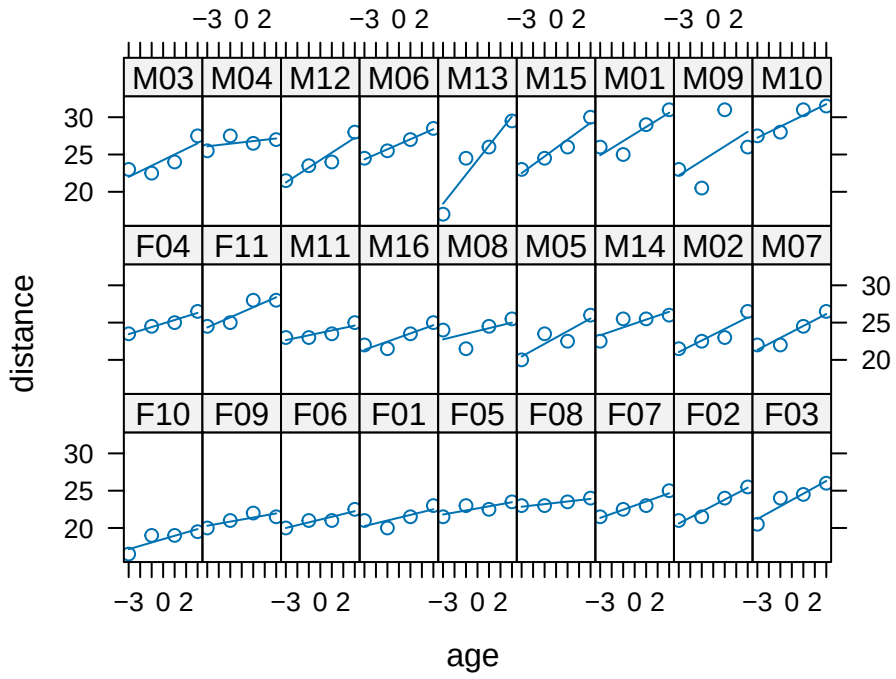
```
lmList2<-lmList(ortho1)
```

Se però la formula è la stessa presente che nel `groupedData object`, possiamo anche non specificarla, questo ci restituisce gli stessi risultati.

2.0.2.1 Graphical comparison between observed and fitted values

La funzione `augPred` può essere applicata ad un oggetto creato con la funzione `lmList`.

```
# ?augPred
plot(augPred(lmList1),layout=c(9,3))
```



Questo plot compara le misure osservate per la variabile dipendente con quelle ottenute, cioè le predette, fittando un singolo GLM per ogni bambino. Specificando `layout=c(9,3)` posizioniamo 9 grafici in ogni riga e abbiamo 3 righe.
Ogni bambino ha quindi le 4 osservazioni e le corrispettiva retta di regressione. Notiamo come ognuno abbia intercetta e pendenza differente.
Nota: `age` è l'unico level 1 regressor.
Non possiamo includere anche `sex` nell'analisi esplorativa perché prende valori costanti dentro ogni level 2 unit (la matrice X avrebbe quindi delle colonne linearmente dipendenti)

```
head(coefficients(lmList1))
```

```
    (Intercept)   age
F10      18.500 0.450
F09      21.125 0.275
```

Per ogni bambino possiamo visualizzare l'intercetta stimata, che è la distanza media a 11 anni. Abbiamo anche il coefficiente di regressione per l'età, il quale aumenterà la distanza per ogni anno aggiuntivo di età. Abbiamo quindi intercette diverse e coefficienti di regressione diversi per ogni bambini diversi. Abbiamo quindi bambini con dimensioni diverse a 11 anni e che hanno un diverso tasso di crescita.

```
head(intervals(lmList1))
```

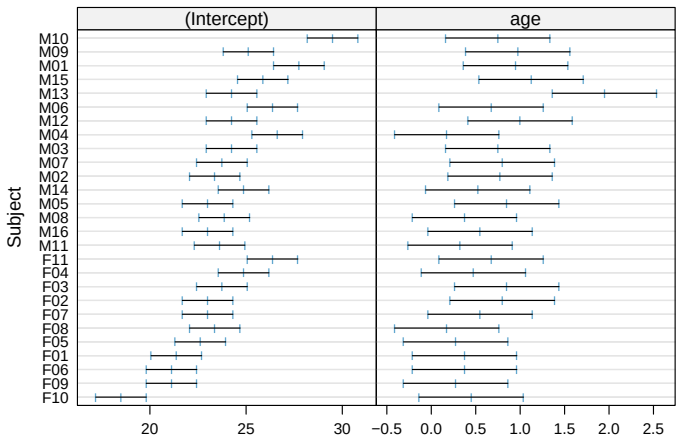
```
, , (Intercept)

       lower   est.    upper
F10 17.18676 18.500 19.81324


, , age
```

Possiamo anche visualizzare gli intervalli di confidenza associati ai parametri stimati per ogni bambino.

```
plot(intervals(lmList1))
```



Plottando gli intervalli di confidenza possiamo vedere come questi per l'intercetta non siano sovrapposti, ci sono quindi alcuni bambini con intercetta sostanzialmente differente dagli altri. Abbiiamo anch un grado di non sovrapposizione in termini di effetti stimati per `age`. Per l'intercetta è quindi evidente, mentre per age è debole.
Guardando questo grafico possiamo e gli intervalli di confidenza possiamo quindi avere degli hints circa quale possano essere i random effect da includere nel GLMM. In questo caso è ragionevole includere un random effect per entrambi.

```
# lmList(distance~age+Sex|Subject,data=ortho1) -> da errore
```

Se proviamo nella formula ad aggiungere `sex` la funzione ci restituisce un messaggio di errore, questo perché `sex` è costante.
Per esempio le 4 osservazioni per `M01` avranno lo stesso valore per gender, questo ci da un problema nel fittare il modello perché oltre l'intercetta abbaimo un'altra colonna costante.

```
head(coefficients(lmList(distance~age+Sex|Subject,data=ortho1)))
```

```
Warning: 27 volte con gli stessi errori in ‘contrasts<-‘(‘*tmp*‘, value =
contr.funs[1 + isOF[nn]]): i contrasti si possono applicare solo a variabili
factor con 2 o più livelli

$F10
NULL

$F09
NULL
```

⋮

Se proviamo a ottenere i coefficienti vediamo come ci sia `NULL` per ogni bambino incluso nello studio.
Nota: finita l'explorative analysis, possiamo passare al FIT del modello GLMM. Nel farlo però dobbiamo tenere a mente che durante questa fase appena conclusa abbiamo fittato molti modelli, ottenendo quindi molti parametri fittati. Abbiamo quindi overparametrizzato il problema (avremo tanti modelli e tanti parametri), che è una cosa che vorremmo evitare.

3. Model with random intercept

3.1 Notation

m number of groups/clusters/level 2 units/observed categories of the random factor

n_i (level 1) units in the i-th group/level 2 unit, showing the i-th category of the random factor $(i = 1, \dots, m)$

Y_{ij} Random variable that describes the value for the dependent variable observed on the j-th unit in the i-th group $(i = 1, \dots, m,\ j = 1, \dots, n_i)$

x_{ij} value of the **level 1** regressor X on the j-th unit belonging to the i-group $(i = 1, \dots, m,\ j = 1, \dots, n_i)$

$\boldsymbol{x}_i$ vector containing the values of X on units belonging to the i-th group $(i = 1, \dots, m)$

3.1.1 Example (longitudinal data)

$m = 11$ girls selected at random from a population of the random factor

$n_i = 4$ time points considered for the i-th girl $(i = 1, \dots, 11)$

Y_{ij} distance between the pituitary gland and the pterygomaxillary fissure observed on the i-th girl at the j-th time point

x_{ij} age of the i-th girl at the j-th time point quad *(centered at 11)*

Aims of the study:

- evaluate the impact of age on the growth pattern
- assess the variation in the growth pattern among girls *(between-girl variability)*

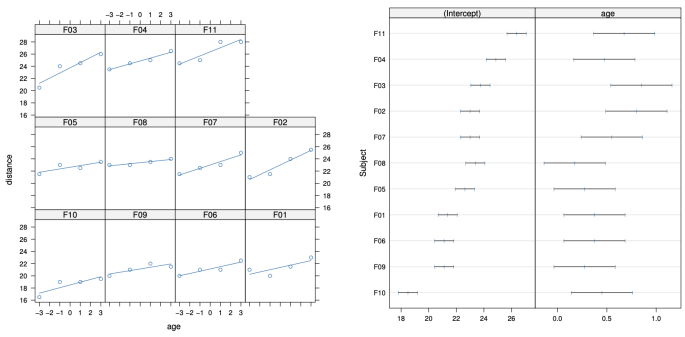


Figure 3.1: Gaussian linear models fitted on each girl separately

A clear linear pattern seem to emerge by considering the 11 (girl-specific) Gaussian linear models these 11 models seem characterised by similar slopes, but different intercepts. **HOWEVER**, in this example the interest is not limited to a comparison among the 11 observed girls, but it extends also to all girls belonging to the same population

3.2 Model definition

$$Y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \varepsilon_{ij} \qquad i = 1, \ldots, m,\ j = 1, \ldots, n_i$$

where:

$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$ fixed effect vector (unknown parameters – common to all units

γ_{0i} random effect (– one for each group/level 2 unit) $(i = 1, \ldots, m)$

ε_{ij} random error associated with the j-th unit of the i-th group/level 2 unit

$(i = 1, \ldots, m;\ j = 1, \ldots, n_i)$

Source of randomness:

ε_{ij} regards level 1 units

γ_{0i} regards level 2 units

3.3 Model assumptions

- $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ (conditionally) independent and identically distributed $\forall i,\ \forall j$
- $\gamma_{0i} \sim \mathcal{N}(0, \tau_0^2)$ (conditionally) independent and identically distributed $\forall i$
- $\text{Cov}[\varepsilon_{ij}, \gamma_{0i}] = 0 \quad \forall i, \forall j$

$\Rightarrow$ *(conditional) independence and homoscedasticity among random errors and random effects*

Conditioning on γ_{0i} allow us to threath the random factor as a fixed factor ***N.B.***: *for the sake of simplicity, the conditioning on the values of the regressor observed in the sample has been removed from the formulas*

Note A model with just the random intercept allow us to relax (and to account for) conditional independence among level 1 units hypothesis, non the homoschedasticity one.

3.4 Properties

It is possible to prove that:

$$\mathbb{E}[Y_{ij}] = \beta_0 + \beta_1 x_{ij} \quad \begin{cases} \beta_0 & \text{population intercept} \\ \beta_1 & \text{population slope} \end{cases} \quad \text{common to all level 1 units}$$

$$\mathbb{E}[Y_{ij} \mid \gamma_{0i}] = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} = \beta_{0i} + \beta_1 x_{ij} \quad \text{with } \beta_{0i} = \beta_0 + \gamma_{0i}$$

$$\rightarrow \textbf{specific intercept for the } i\textbf{-th group/level 2 unit}$$

Remarks:

- conditioning on γ_{0i} (on the random effects) is equivalent to treat the random factor as fixed
- the population intercept can be interpreted as the expected value (over all possible groups/level 2 units) of the group-specific intercepts:

  $$\mathbb{E}[\beta_{0i}] = \beta_0$$

  furthermore:

  $$\mathbb{E}\left[\mathbb{E}[Y_{ij} \mid \gamma_{0i}]\right] = \mathbb{E}[Y_{ij}]$$

It is possible to prove that:

$$\mathrm{Var}[Y_{ij}] = \tau_0^2 + \sigma^2 \quad \Rightarrow \quad \textbf{homoscedasticity}$$

$$\mathrm{Cov}[Y_{ij}, Y_{lh}] = \begin{cases} 0 & \text{if } i \neq l \\ \tau_0^2 & \text{if } i = l \end{cases} \quad \Rightarrow \quad \textbf{within-group correlation}$$

Remarks:

$\Rightarrow \mathrm{Cov}[Y_{ij} \mid \gamma_{0i}, Y_{lh} \mid \gamma_{0l}] = 0 \quad \forall i, l$

- *The correlation structure among level 1 units belonging to the same level 2 unit is entirely due to the randomness of the factor (to the presence of a random effect)*
- means that each girl is random, and so independent $\Rightarrow$ there is no correlation between level 1 units selected from different level 2 group

  Intraclass correlation coefficient:

  $$\mathrm{Cor}[Y_{ij}, Y_{lh}] = \frac{\mathrm{Cov}[Y_{ij}, Y_{lh}]}{\sqrt{\mathrm{Var}[Y_{ij}]\,\mathrm{Var}[Y_{lh}]}} = \begin{cases} 0 & \text{if } i \neq l \\ \dfrac{\tau_0^2}{\tau_0^2 + \sigma^2} & \text{if } i = l \end{cases}$$

4. Model with random intercept and slope

4.1 Model definition

$$Y_{ij} = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \gamma_{1i} x_{ij} + \varepsilon_{ij} \qquad i = 1, \ldots, m, \quad j = 1, \ldots, n_i$$

where:

$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$ fixed effect vector (unknown parameters – common to all units)

$\boldsymbol{\gamma}_i = \begin{bmatrix} \gamma_{0i} \\ \gamma_{1i} \end{bmatrix}$ random effect vectors (one for each group/level 2 unit)

$(i = 1, \ldots, m,\ j = 1, \ldots, n_i)$

Source of randomness:

ε_{ij} random error associated with the j-th unit of the i-th group/level 2 unit *(level 1 units specific)*

γ_{0i} random intercept, accounts for **dependence** among level 1 units. *(level 2 units specific)*

γ_{1i} random slope accounts for **heteroschedasticity** *(level 2 units specific)*

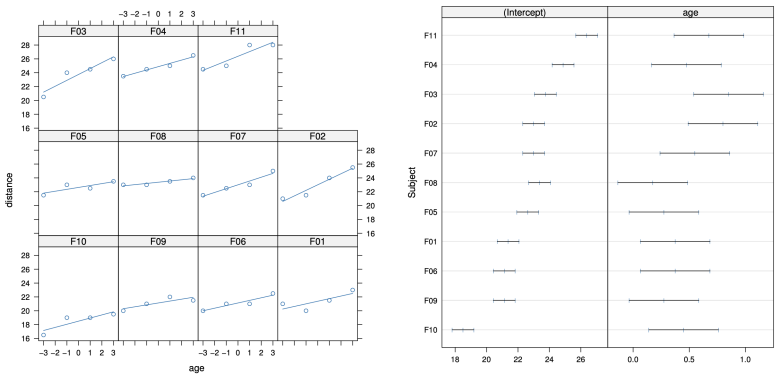


Figure 4.1: Gaussian linear models fitted on each girl separately

4.2 Model assumptions

- $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ (conditionally) independent and identically distributed $\forall i,\ \forall j$
- $\boldsymbol{\gamma}_i \sim \mathcal{N}_2\left(\begin{bmatrix}0\\0\end{bmatrix},\ \mathbf{Q} = \begin{bmatrix}\tau_0^2 & \tau_{01}\\ \tau_{01} & \tau_1^2\end{bmatrix}\right)$
  (conditionally) independent and identically distributed $\forall i$
  τ_{01} refers to the possible correlation between γ_{0i} and

$$\gamma_{1i}$$

- $\text{Cov}[\boldsymbol{\gamma}_i, \varepsilon_{ij}] = \begin{bmatrix}0\\0\end{bmatrix} \quad \forall i,\ \forall j$

$\Rightarrow$ *(conditional) independence among random errors and random effects*

N.B.:
For the sake of simplicity, the conditioning on the values of the regressor observed in the sample has been removed from the formulas

4.3 Properties

It is possible to prove that:

$$\mathbb{E}[Y_{ij}] = \beta_0 + \beta_1 x_{ij} b \quad \rightarrow \quad \text{just fixed effect}$$

$$\mathbb{E}[Y_{ij} \mid \boldsymbol{\gamma}_i] = \beta_0 + \beta_1 x_{ij} + \gamma_{0i} + \gamma_{1i} x_{ij}$$

$$= \underbrace{(\beta_0 + \gamma_{0i})}_{\beta_{0i}} + \underbrace{(\beta_1 + \gamma_{1i})}_{\beta_{1i}} x_{ij}$$

With

$\beta_{0i} = \beta_0 + \gamma_{0i} \quad \rightarrow \quad$ *specific intercept for the i-th group/level 2 unit*

$\beta_{1i} = \beta_1 + \gamma_{1i} \quad \rightarrow \quad$ *specific slope for the i-th group/level 2 unit*

Regarding the parametric structure of the var-cov matrix induced by the random effect

the population intercept and slope can be interpreted as the expected value (over the groups/level 2 units) of the group-specific intercepts and slopes, respectively

$$\mathbb{E}[\beta_{0i}] = \beta_0, \qquad \mathbb{E}[\beta_{1i}] = \beta_1$$

It is possible to prove that:

$$\text{Var}[Y_{ij}] = \tau_0^2 + 2\tau_{01}x_{ij} + \tau_1^2 x_{ij}^2 + \sigma^2 \quad \Rightarrow \quad \textit{heteroscedasticity}$$

$$\text{Cov}[Y_{ij}, Y_{lh}] = \begin{cases} 0 & \text{if } i \neq l \\ \tau_0^2 + \tau_{01}x_{ij} + \tau_{01}x_{lh} + \tau_1^2 x_{ij}x_{lh} & \text{if } i = l \end{cases} \quad \Rightarrow \quad \textit{within-group correlation}$$

Remarks:

The heteroscedasticity and the correlation structure among level 1 units belonging to the same level 2 unit are entirely due to the randomness of the factor (to the presence of the random effects)

- $\text{Var}[Y_{ij} \mid \boldsymbol{\gamma}_i] = \sigma^2$
- $\text{Cov}[Y_{ij} \mid \boldsymbol{\gamma}_i,\ Y_{lh} \mid \boldsymbol{\gamma}_l] = 0 \quad \forall i, l$

5. Models for units in the i–th group

5.1 Notation

Here we will represent the information singularly for each n_i groups of units, belonging to the i-th group/level 2 unit. So for each of these vector/matrix we will have m of them.

$i = 1, \dots, m \rightarrow$ it does not vary inside each matrix, refers to how many of matrix/vectors we will observe

$$Y_i = \begin{bmatrix} Y_{i1} \\ Y_{i2} \\ \vdots \\ Y_{in_i} \end{bmatrix} \qquad \varepsilon_i = \begin{bmatrix} \varepsilon_{i1} \\ \varepsilon_{i2} \\ \vdots \\ \varepsilon_{in_i} \end{bmatrix} \qquad (n_i \times 1\ )$$

$k \rightarrow$ number of predictors with *fixed effect*

$p = 1 + k$

$$\boldsymbol{X_i} = \begin{bmatrix} 1 & x_{1_i 1} & \cdots & x_{k_i 1} \\ 1 & x_{1_i 2} & \cdots & x_{k_i 2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{1_i n_1} & \cdots & x_{k_i n_1} \end{bmatrix} \quad (n_i \times p) \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad (p \times 1 \text{ vector})$$

The introduction of random effects requires the definition of two new quantities:

$q \rightarrow$ number of predictors for which we want to introduce the *random effect*

$$U_i = \begin{bmatrix} u_{1_i 1} & \cdots & u_{q_i 1} \\ u_{1_i 2} & \cdots & u_{q_i 2} \\ \vdots & \ddots & \vdots \\ u_{1_i n_i} & \cdots & u_{q_i n_i} \end{bmatrix} \quad n_i \times (q+1) \text{ matrix}$$

$$\gamma_i = \begin{bmatrix} \gamma_{i0} \\ \gamma_{i1} \\ \vdots \\ \gamma_{iq} \end{bmatrix} \quad (q+1) \times 1 \text{ vector}$$

The matrix $\boldsymbol{U_i}$ contains those regressors (and the intercept) with a random effect (effects that change as the group/level 2 unit changes); generally the same regressor can have both fixed and random slope.

- not necessarily $\boldsymbol{X_i} = \boldsymbol{U_i}$ (usually $\boldsymbol{U_i}$ contains a subset of the columns of $\boldsymbol{X_i}$)
- $\boldsymbol{U_i}$ can contain only level 1 regressors

5.2 Model Definition

$$Y_{ij} = \beta_0 + \beta_1 x_{ij1} + \cdots + \beta_k x_{ijk} + \gamma_{i0} u_{ij0} + \cdots + \gamma_{iq} u_{ijq} + \varepsilon_{ij} \quad j = 1, \ldots, n_i$$

or, equivalently:

$$Y_i = \boldsymbol{X_i}\boldsymbol{\beta} + \boldsymbol{U_i}\boldsymbol{\gamma}_i + \boldsymbol{\varepsilon_i}$$

where

- $X_i\boldsymbol{\beta}$ is cocmmon to all level 2 units
- $U_i\boldsymbol{\gamma}_i$ specific for each level two units

N.B.: For the sake of simplicity, the conditioning on the values of the regressors in X_i and U_i has been removed from the formulas.

5.3 Assumptions

- $\boldsymbol{\varepsilon_i} \sim \mathcal{N}_{n_i}(\mathbf{0}_{n_i}, \sigma^2 \boldsymbol{I}_{n_i})$
- $\boldsymbol{\gamma_i} \sim \mathcal{N}_{q+1}(\mathbf{0}_{q+1}, \boldsymbol{Q})$ → one fro each level 2 regressor (and intercept)
- $\mathrm{Cov}(\varepsilon_i, \gamma_i) = \mathbf{0}_{n_i \times (q+1)}$
  → *Conditional independence between error terms and random effects*

5.3.1 Possible choices for $\boldsymbol{Q}$

$\boldsymbol{Q}$ is an $(q+1) \times (q+1)$ symmetric and positive definite matrix.
→ which means that the **actual number of independent unique parameter** is lower then the dimensionality $(q+1) \times (q+1)$.

Some structures commonly used in Gaussian linear mixed models:

- **Proportional to the identity matrix** → 1 unknown parameter
  $\boldsymbol{Q} = \tau_0^2 \boldsymbol{I}_{q+1}$
- **Compound symmetry** → 2 unknown parameters
  which means having specific value on the diagonal and off–diagonal elements

$$Q = \tau_0^2 \boldsymbol{I}_{q+1} + \tau_1 \mathbf{1}_{q+1} \mathbf{1}_{q+1}^\top \quad \rightarrow \quad \begin{bmatrix} \tau_0^2 + \tau_1 & \tau_1 & \cdots & \tau_1 \\ \tau_1 & \tau_0^2 + \tau_1 & \cdots & \tau_1 \\ \vdots & \vdots & \ddots & \vdots \\ \tau_1 & \tau_1 & \cdots & \tau_0^2 + \tau_1 \end{bmatrix}$$

- **Diagonal** (variance components) → $q+1$ unknown parameters
  here each random effects has its own variance τ_q^2 plus the random intercept variance τ_0^2 and the off–diagonal elements , the covariances, are all zero.
- **Unstructured** → $\frac{(q+1)(q+2)}{2}$ unknown parameters
  here each random effects has its own variance τ_q^2 plus the random intercept variance τ_0^2 an the covariances are not zero, so each of them requires it's own parameter. (since upper matrix coincide with lower matrix you divide by 2)

5.4 Conditional heteroscedasticity and dependence

5.4.1 Marginal formulation

$$\boldsymbol{Y_i} \mid \boldsymbol{X_i} \sim \mathcal{N}_{n_i}(\boldsymbol{X_i\beta}, \sigma^2 \boldsymbol{I}_{n_i} + \boldsymbol{U_i} Q \boldsymbol{U_i}^\top)$$

The non diagonal structure of $\text{Cov}(\boldsymbol{Y_i} \mid \boldsymbol{X_i})$ let us relax the homoschedasticity assumption, such that we can introduce hetreoschedasticity.

- $\mathbb{E}[\boldsymbol{Y_i}] = \mathbb{E}[\boldsymbol{X_i\beta}] + \mathbb{E}[\boldsymbol{U_i\gamma}_i] + \mathbb{E}[\varepsilon_i] = \boldsymbol{X_i\beta}$

  Recall $\boldsymbol{\gamma_i} \sim \mathcal{N}_{q+1}(\boldsymbol{0}, \boldsymbol{Q}) \Rightarrow \mathbb{E}[\boldsymbol{U_i\gamma}_i]$ (*since* $\boldsymbol{U_i}$ *are known constant*)

- $\text{Cov}(\boldsymbol{Y_i}) = \boldsymbol{U_i} \underbrace{\text{Cov}(\boldsymbol{\gamma}_i)}_{\boldsymbol{Q}} \boldsymbol{U_i}^\top + \underbrace{\text{Cov}(\varepsilon_i)}_{\sigma^2 \boldsymbol{I}_{n_i}} + \underbrace{2\text{Cov}(\varepsilon_i, \boldsymbol{\gamma}_i)}_{0_{n_i \times n_i)}} = \boldsymbol{U_i Q U_i}^\top + \sigma^2 \boldsymbol{I}_{n_i}$

$\text{Cov}(\boldsymbol{Y_i})$ is a symmetric matrix with (no more than) $\dfrac{n_i(n_i+1)}{2}$ distinct elements obtained by combining:

- $n_i \times r$ known values contained in the matrix $\boldsymbol{U}_i$
- no more than $\dfrac{(q+1)(q+2)}{2} + 1$ unknown parameters (elements in $\boldsymbol{Q}$ plus σ^2)

Note:

1. The parameters of the marginal distribution of Y_i do not involve the random effects γ_i, but only the parameters of the distribution of γ_i contained in $\boldsymbol{Q}$. $\rightarrow$ the random effects do not appear explicitly.
2. The number of parameter needed to compute $\text{Cov}(\boldsymbol{Y_i})$ is quadratic in n_i (or q?)

5.4.2 Conditional formulation

Generative model for level 1 units, given the level 2 unit:

$$\boldsymbol{\gamma_i} \sim \mathcal{N}_{q+1}(\boldsymbol{0}, \boldsymbol{Q})$$

Generative model for level 2 units:

$$\boldsymbol{Y_i} \mid \boldsymbol{X_i}, \boldsymbol{\gamma}_i \sim \mathcal{N}_{n_i}(\boldsymbol{X_i\beta} + \boldsymbol{U_i\gamma}_i, \sigma^2 \boldsymbol{I}_{n_i})$$

Conditioning on the level 2 units means dealing with the random factor as if it was fixed:

- $\mathbb{E}[\boldsymbol{Y_i}|\boldsymbol{\gamma}_i] = \mathbb{E}[\boldsymbol{X_i\beta}|\boldsymbol{\gamma}_i] + \mathbb{E}[\boldsymbol{U_i\gamma}_i|\boldsymbol{\gamma}_i] + \mathbb{E}[\varepsilon_i] = \boldsymbol{X_i\beta} + \boldsymbol{U_i\gamma}_i$
- $\text{Cov}(\boldsymbol{Y_i}|\boldsymbol{\gamma}_i) = \sigma^2 \boldsymbol{I}_{n_i}$

The correlation structure among level 1 units belonging to the same level 2 unit is entirely due to the randomness of the factor (to the presence of random effects).

Note:

$$f(\gamma) \cdot f(y \mid \gamma) = f(y)$$

The presence of $\boldsymbol{\gamma_i}$ wasn't explicit in the marginal formulation but it is actually there. So the introduction of **heteroschedasticity**, is the result of the introduction of random effects.

6. Global model (for all sample units)

6.1 Notation

Information about the $n = \sum_i n_i$ units can be organised in matrices and vectors
$i = 1, \ldots, m \rightarrow$ number of level 2 units m

$$\mathbf{Y} = \begin{bmatrix} \mathbf{Y_1} \\ \hline \mathbf{Y_2} \\ \hline \vdots \\ \hline \mathbf{Y_m} \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \boldsymbol{\varepsilon_1} \\ \hline \boldsymbol{\varepsilon_2} \\ \hline \vdots \\ \hline \boldsymbol{\varepsilon_m} \end{bmatrix} \quad \text{a } n \times 1 \text{ vectors}$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{X_1} \\ \hline \mathbf{X_2} \\ \hline \vdots \\ \hline \mathbf{X_m} \end{bmatrix} \quad (n \times p \text{ matrix}) \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad (p \times 1 \text{ vector})$$

These quantities can be obtained by stacking vectors $\mathbf{Y}_i$ and ε_i, and matrices $\mathbf{X}_i$, $(i = 1, \ldots, m)$, and contain all the level 1 units in the sample considering on the level 2 units which they belong.

$\mathbf{U} = \text{blockdiag}(\underline{\mathbf{U}}_1, \underline{\mathbf{U}}_2, \ldots, \underline{\mathbf{U}}_m) \quad n \times m \cdot (q+1) \rightarrow$ number of column of eah matrix $q+1$ times the number of level 2 units m

$$\mathbf{U} = \left[\begin{array}{c|c|c|c} \mathbf{U}_1 & \mathbf{0}_{n_1 \times (g+1)} & \cdots & \mathbf{0}_{n_1 \times (g+1)} \\ \hline \mathbf{0}_{n_2 \times (g+1)} & \mathbf{U}_2 & \cdots & \mathbf{0}_{n_2 \times (g+1)} \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline \mathbf{0}_{n_m \times (g+1)} & \mathbf{0}_{n_m \times (g+1)} & \cdots & \mathbf{U}_m \end{array}\right] \quad n \times m \cdot (q+1) \text{ matrix}$$

$$\boldsymbol{\gamma} = \begin{bmatrix} \boldsymbol{\gamma}_1 \\ \hline \boldsymbol{\gamma}_2 \\ \hline \vdots \\ \hline \boldsymbol{\gamma}_m \end{bmatrix} \quad m \cdot (q+1) \times 1 \text{ vector} \quad \mathbf{U}\boldsymbol{\gamma} = \begin{bmatrix} \mathbf{U}_1\boldsymbol{\gamma}_1 \\ \hline \mathbf{U}_2\boldsymbol{\gamma}_2 \\ \hline \vdots \\ \hline \mathbf{U}_m\boldsymbol{\gamma}_m \end{bmatrix}$$

Note:

1. The matrix $\mathbf{U}$ can be obtained by diagonally concatenating matrices $\underline{\mathbf{U}}_i$, while the vector $\boldsymbol{\gamma}$ can be obtained by stacking vectors $\boldsymbol{\gamma}_i$, $(i = 1, \ldots, m)$
2. The vector $\mathbf{U}\boldsymbol{\gamma}$ can be obtained by stacking vectors $\underline{\mathbf{U}}_i\boldsymbol{\gamma}_i$, $(i = 1, \ldots, m)$
3. Trough this notation you can also write the model with only the random intercept

6.2 Assumptions

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}\boldsymbol{\gamma} + \boldsymbol{\epsilon}$$

with

$$\boldsymbol{\epsilon} \sim \mathcal{N}_n(\mathbf{0}_n, \sigma^2\mathbf{I}_n) \quad \boldsymbol{\gamma} \sim \mathcal{N}_{m\cdot(q+1)}(\mathbf{0}_{m\cdot(q+1)}, \mathbf{G}) \quad \boldsymbol{\epsilon} \perp \boldsymbol{\gamma}$$

$$\mathbf{G} = \text{blockdiag}(\mathbf{Q}_{(1)}, \ldots, \mathbf{Q}_{(m)}) \quad \text{an } m(g+1) \times m(g+1) \text{ matrix}$$

$$\left[\begin{array}{c|c|c|c} \mathbf{Q} & \mathbf{0}_{(q+1)\times(q+1)} & \cdots & \mathbf{0}_{(q+1)\times(q+1)} \\ \hline \mathbf{0}_{(q+1)\times(q+1)} & \mathbf{Q} & \cdots & \mathbf{0}_{(q+1)\times(q+1)} \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline \mathbf{0}_{(q+1)\times(q+1)} & \mathbf{0}_{(q+1)\times(q+1)} & \cdots & \mathbf{Q} \end{array}\right]$$

Level 2 units are assumed to be independent and identically distributed, for this reason $\boldsymbol{G}$ is a block–diagonal matrix, also it reflects the random effect structure.
$\mathbf{G}$ symmetric, positive definite, block-diagonal (each block can be diagonal).

$$\Rightarrow \quad \text{Cov}\left[\boldsymbol{\varepsilon}, \boldsymbol{\gamma}\right] = \mathbf{0}_{n\times m\cdot(q+1)} \quad \Leftrightarrow \quad \begin{bmatrix}\boldsymbol{\gamma} \\ \boldsymbol{\varepsilon}\end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix}\mathbf{0} \\ \mathbf{0}\end{bmatrix}, \begin{bmatrix}\mathbf{G} & \mathbf{0} \\ \mathbf{0} & \sigma^2\mathbf{I}\end{bmatrix}\right)$$

Note:
It is possible to observe correlatioin among the vector $\boldsymbol{\gamma_i}$, but it is always independent.

6.3 Properties

$$\begin{aligned}\mathbb{E}[\mathbf{Y}] &= \mathbb{E}[\mathbf{X}\boldsymbol{\beta}] + \underbrace{\mathbb{E}[\mathbf{U}\boldsymbol{\gamma}]}_{0} + \underbrace{\mathbb{E}[\boldsymbol{\varepsilon}]}_{0} \\ &= \mathbf{X}\boldsymbol{\beta}\end{aligned}$$

$$\begin{aligned}\text{Cov}[\mathbf{Y}] &= \mathbf{U}\underbrace{\text{Cov}[\boldsymbol{\gamma}]}_{\mathbf{G}}\mathbf{U}^\top + \underbrace{\text{Cov}[\boldsymbol{\varepsilon}]}_{\sigma^2\mathbf{I}_n} + \underbrace{2\text{Cov}[\boldsymbol{\varepsilon}, \boldsymbol{\gamma}]}_{\mathbf{0}_{n\times n}}\mathbf{U}^\top \quad \rightarrow \\ &= \mathbf{U}\mathbf{G}\mathbf{U}^\top + \sigma^2\mathbf{I}_n\end{aligned}$$

$$\mathbf{Y} \sim \mathcal{N}_n\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n + \mathbf{U}\mathbf{G}\mathbf{U}^\top\right)$$

- $\underbrace{2\text{Cov}[\boldsymbol{\varepsilon}, \boldsymbol{\gamma}]}_{\mathbf{0}_{n\times n}}\mathbf{U}^\top \quad \rightarrow$ represent the contribution of the random effect

- Recall that both $\boldsymbol{U}$ and $\boldsymbol{G}$ have a block diagonal structure

6.4 Conditional heteroscedasticity and dependence

Exploiting the block-diagonal structure of $\mathbf{G}$, it is possible to prove that:

$$\text{Cov}[\mathbf{Y}] = \sigma^2\mathbf{I}_n + \mathbf{U}\mathbf{G}\mathbf{U}^\top = \text{blockdiag}(\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_m)$$

$$= \left[\begin{array}{c|c|c|c} \mathbf{V}_1 & \mathbf{0}_{n_1\times n_2} & \cdots & \mathbf{0}_{n_1\times n_m} \\ \hline \mathbf{0}_{n_2\times n_1} & \mathbf{V}_2 & \cdots & \mathbf{0}_{n_2\times n_m} \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline \mathbf{0}_{n_m\times n_1} & \mathbf{0}_{n_m\times n_2} & \cdots & \mathbf{V}_m \end{array}\right] = \mathbf{V}$$

The conditional independence assumption among random effects γ_i $(i = 1, \ldots, m)$ (and in the sampling scheme) leads to a block-diagonal structure in $\mathbf{V}$.

- $\mathbf{V}$ can be obtained by diagonally concatenating matrices $\mathbf{V}_i$ $(i = 1, \ldots, m)$
- $\mathbf{V}$ is symmetric, block-diagonal, with (no more than) $\sum_{i=1}^{m} \frac{n_i(n_i+1)}{2}$ distinct elements obtained from:
  - $n \times (q + 1)$ known values in the matrices $\mathbf{U}_i$ $(i = 1, \ldots, m)$
  - (no more than) $\dfrac{(q+1)(q+2)}{2} + 1$ unknown parameters (unknown elements of $\mathbf{Q}$ plus σ^2)
- The parameters characterising $\mathbf{V}$ can be collected in the vector $\boldsymbol{\vartheta}$

6.4.1 Conditioning on level 2 units

- $\mathbb{E}[\mathbf{Y} \mid \boldsymbol{\gamma}] = \mathbb{E}[\mathbf{X}\boldsymbol{\beta}] + \underbrace{\mathbb{E}[\mathbf{U}\boldsymbol{\gamma} \mid \boldsymbol{\gamma}]}_{\mathbf{U}\boldsymbol{\gamma}} + \underbrace{\mathbb{E}[\boldsymbol{\varepsilon} \mid \boldsymbol{\gamma}]}_{\mathbf{0}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}\boldsymbol{\gamma} \rightarrow$ level 2 units specific
- $\text{Cov}[\mathbf{Y} \mid \boldsymbol{\gamma}] = \sigma^2\mathbf{I}_n$

The previously observed dependence structure is justified by γ

$\rightarrow$ The correlation structure among level 1 units belonging to the same level 2 units is entirely due to the randomness of the observed categories for the random factor (to the presence of random effects)

Conditional formulation:

- Joint generative model for level 2 units in the observed sample:

$$\boldsymbol{\gamma} \sim \mathcal{N}_{m\cdot(q+1)}(\mathbf{0}_{m\cdot(q+1)}, \mathbf{G})$$

- Joint generative model for level 1 units, given the level 2 units in the observed sample:

$$\mathbf{Y} \mid \boldsymbol{\gamma} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{U}\boldsymbol{\gamma}, \sigma^2\mathbf{I}_n)$$

6.5 Extended/general Gaussian linear mixed models

*Overcoming the **homoschedasticity assumption** on the random error ε_i*

The class of Gaussian linear mixed models can be extended by removing the within-group homoschedasticity and independence assumptions for the error term $\boldsymbol{\varepsilon}$

$$\textbf{level 1 units} \quad \varepsilon_i \sim \mathcal{N}_{n_i}(\mathbf{0}_{n_i}, \mathbf{R}_i) \quad \Rightarrow \quad \mathbf{V}_i = \mathrm{Cov}[\mathbf{Y}] = \mathbf{R}_i + \mathbf{U}_i \mathbf{G} \mathbf{U}_i^\top$$

$\mathbf{R}_i$: $n_i \times n_i$ symmetric and positive definite matrix, $i = 1, \ldots, m$

$$\textbf{level 2 units} \quad \boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \mathbf{R}) \quad \Rightarrow \quad \mathbf{V} = \mathrm{Cov}[\mathbf{Y}] = \mathbf{R} + \mathbf{U}\mathbf{G}\mathbf{U}^\top$$

$$\mathbf{R} = \mathrm{blockdiag}(\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_m)$$

$$= \left[\begin{array}{c|c|c|c} \mathbf{R}_1 & \mathbf{0}_{n_1 \times n_2} & \cdots & \mathbf{0}_{n_1 \times n_m} \\ \hline \mathbf{0}_{n_2 \times n_1} & \mathbf{R}_2 & \cdots & \mathbf{0}_{n_2 \times n_m} \\ \hline \vdots & \vdots & \ddots & \vdots \\ \hline \mathbf{0}_{n_m \times n_1} & \mathbf{0}_{n_m \times n_2} & \cdots & \mathbf{R}_m \end{array}\right]$$

$\mathbf{V}$ *retains its block-diagonal structure (independence among groups)*

6.5.1 Heteroscedastic errors

still related to the sampling scheme at **level 1**
$\rightarrow$ Allowing different error's variance between different **level 2 units**.
Some examples:

- **between-group heteroscedasticity**
  each **level 1 unit** has a different error variance, it could be accounted for by the introduction of a new regressor

$$\mathbf{R}_i = \sigma_i^2 \mathbf{I}_{n_i} \quad \Rightarrow \quad m \text{ unknown parameters}$$

- **between-strata heteroschedasticity**

  S: stratifying variable (qualitative regressor) with T categories $\{A_1, \ldots, A_T\}$

$$\mathrm{Var}[\varepsilon_{ij}] = \sigma_t^2 \iff s_{ij} = A_t \Rightarrow T \text{ unknown parameters}$$

When you want to account for heteroschedasticity trhough injecting a categorical regressor

- S level 1 regressor $\Rightarrow$ **within-group** heteroscedasticity
- S level 2 regressor $\Rightarrow$ **between-group** heteroscedasticity

6.5.2 Correlated errors

Some examples:

- **Constant correlation within groups**
  ($\mathbf{R}_i$ with compound symmetric structure) $\Rightarrow$ 2 (or $2m$) unknown parameters
- **Autocorrelation structures** based on ARMA models (typically for longitudinal data)

6.5.3 A note about extended Gaussian linear mixed models

$$\mathbf{V}_i = \text{Cov}[\mathbf{Y_i}] = \mathbf{R}_i + \mathbf{U}_i\mathbf{Q}\mathbf{U}_i^\top, \quad i = 1, \ldots, m$$

When dealing with extended Gauddian Linear Mixed models you are given two ingredients to describe the dependence structure and heteroscedasticity:

- $\mathbf{U}_i\mathbf{Q}\mathbf{U}_i^\top$: *"random effect" component*$\rightarrow$ here the random effect is not explicitly present, it comes inside the equation trough $\boldsymbol{G}$ which contains the parameter associated with the distribution of the random effect γ_i
- $\mathbf{R}_i$: *"within-group" component* $\rightarrow$ the independence among group remain while it is relaxed inside groups

In many practical applications similar results can be obtained combining:

- a complex $\mathbf{U}_i\mathbf{Q}\mathbf{U}_i^\top$ component with a simple $\mathbf{R}_i$ component (such as $\mathbf{R}_i = \sigma^2\mathbf{I}_{n_i}$)
- a complex $\mathbf{R}_i$ component with a simple $\mathbf{U}_i\mathbf{Q}\mathbf{U}_i^\top$ component (such as $\tau_0^2\mathbf{1}\mathbf{1}^\top$)

6.5.4 Marginal representation

Until now we have seen only conditional representation, here the idea is to combine both your source of randomness.

Consider all level 1 units belonging to the same level 2 unit:

$$\varepsilon_i^* = \mathbf{U}_i\boldsymbol{\gamma}_i + \boldsymbol{\varepsilon}_i \quad \Rightarrow \quad \varepsilon_i^* \sim \mathcal{N}_{n_i}(\mathbf{0}_{n_i}, \mathbf{V}_i)$$

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \varepsilon_i^* \quad \Rightarrow \quad \mathbf{Y}_i \sim \mathcal{N}_{n_i}(\mathbf{X}_i\boldsymbol{\beta}, \mathbf{V}_i)$$

When $\mathbf{V}_i$ is not diagonal then you are accounting for heteroschedasticity within group
$\rightarrow$ recalling $\mathbf{V}_i$ structure it contains the parameter associated with the distribution of the random effect.

Consider all level 1 units:

$$\varepsilon^* = \mathbf{U}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} \quad \Rightarrow \quad \varepsilon^* \sim \mathcal{N}_n(\mathbf{0}_n, \mathbf{V})$$

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \varepsilon^* \quad \Rightarrow \quad \mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$$

Any Gaussian linear mixed model can be seen as a **Gaussian linear model containing only fixed effects** coupled with correlated errors

$\Rightarrow$ implicit presence of random effects, thanks to the particular structure of the matrices $\mathbf{V}_i$

7. Maximum Likelihood Estimation

The unknown quantities that characterise a Gaussian linear mixed model are:

- $\boldsymbol{\beta}$ fixed effects (**parameters**)
- $\boldsymbol{\gamma}$ random effects (**form an inferential point of view $\boldsymbol{\gamma}$ elements are not parameter, but are unobservable realisations of random variables, the difference stays in the fact that a parameter (according to classical inference) is a fixed quantity which does not vary**)
- $\boldsymbol{\vartheta}$ variances (and covariances) of the random effects and of the error terms (**parameters**)

The question is which Lieklihood function should be used?

Both the conditional and the marginal formulation can be exploited as starting point to obtain maximum likelihood estimates for $\boldsymbol{\beta}$.

- the conditional formulation can be exploited as starting point to obtain predicted values for $\boldsymbol{\gamma}$, since $\boldsymbol{\gamma}$ is involved explicitly
- the marginal formulation is usually exploited as starting point to obtain maximum likelihood estimates for $\boldsymbol{\vartheta}$, where $\boldsymbol{\gamma}$ is involved implicitly

Marginal formulation ($\boldsymbol{\vartheta}$ known):

$$L(\boldsymbol{\beta}) = \frac{\exp\left\{-\frac{1}{2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})\right\}}{(2\pi)^{n/2}|\mathbf{V}|^{1/2}}$$

$$\ell(\boldsymbol{\beta}) = \ln L(\boldsymbol{\beta}) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\mathbf{V}| - \frac{1}{2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})$$

Note that the random effects $\boldsymbol{\gamma}$ do not appear explicitly in these expressions.

Conditional formulation ($\boldsymbol{\vartheta}$ known):

$$L(\boldsymbol{\beta},\boldsymbol{\gamma}) = \frac{\exp\left\{-\frac{1}{2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}-\mathbf{U}\boldsymbol{\gamma})^\top \mathbf{R}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}-\mathbf{U}\boldsymbol{\gamma})\right\}}{(2\pi)^{n/2}|\mathbf{R}|^{1/2}} \cdot \frac{\exp\left\{-\frac{1}{2}\boldsymbol{\gamma}^\top \mathbf{G}^{-1}\boldsymbol{\gamma}\right\}}{(2\pi)^{m(q+1)/2}|\mathbf{G}|^{1/2}}$$

$$\ell(\boldsymbol{\beta},\boldsymbol{\gamma}) = \ln L(\boldsymbol{\beta},\boldsymbol{\vartheta}) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\mathbf{R}| - \frac{m(q+1)}{2}\ln 2\pi - \frac{1}{2}\ln|\mathbf{G}|$$

$\rightarrow$ The function you should maximize is the following:

$$-\frac{1}{2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}-\mathbf{U}\boldsymbol{\gamma})^\top \mathbf{R}^{-1}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}-\mathbf{U}\boldsymbol{\gamma}) - \frac{1}{2}\boldsymbol{\gamma}^\top \mathbf{G}^{-1}\boldsymbol{\gamma}$$

7.1 Maximum likelihood estimation for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ ($\boldsymbol{\vartheta}$ known):

It is possible to prove that, when $\mathbf{X}$ has full column rank:

$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}$
$\rightarrow$ corresponds to the expression in simple GLM where $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$

$\hat{\boldsymbol{\gamma}} = \mathbf{G}\mathbf{U}^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$

Note:
Here beside in the simple Gaussian Linear Model, information regarding the variance is needed, and they are summarized by $\boldsymbol{V}$

- $\hat{\boldsymbol{\beta}}$ maximises $\ell(\boldsymbol{\beta})$ (marginal formulation)
- $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$ simultaneously maximise $\ell(\boldsymbol{\beta}, \boldsymbol{\gamma})$ (conditional formulation)

$\hat{\boldsymbol{\beta}}$ is the Best Linear Unbiased Estimator (BLUE) for $\boldsymbol{\beta}$ and $\hat{\boldsymbol{\gamma}}$ is the Best Linear Unbiased **Predictor** (not parameter) (BLUP) for $\boldsymbol{\gamma}$

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}\left(\boldsymbol{\beta}, (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1}\right)$$

Note:
Knowledge regarding $\boldsymbol{\vartheta}$ includes knowledge about $\mathbf{G}$ and $\mathbf{V}$. They are both needed in the estimation of $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$

7.1.1 Generalised/weighted least squares

$\hat{\boldsymbol{\beta}}$ is also the **generalised/weighted least squares** estimator for $\boldsymbol{\beta}$

$\Rightarrow$ $\ell(\boldsymbol{\beta})$ is maximised if and only if the generalised/weighted least squares criterion

$$GLS(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

is minimised.

$GLS(\boldsymbol{\beta})$ is an extension of the ordinary least squares criterion that takes into account the covariance structure characterising the sample units.

Difference with the OLS:

Here the distance within observed and fitted value is wighted by the var-cov matrix structure that characterize the sample units, in order to compute the GLS you need V

differences between observed and fitted values are downweighted for units characterised by a large variance and/or for pairs of units that are highly correlated

7.2 Maximum likelihood estimation for $\boldsymbol{\vartheta}$

In order to obtain the actual estimates for $\boldsymbol{\beta}$ and the actual predicted values for $\boldsymbol{\gamma}$ an estimate for $\boldsymbol{\vartheta}$ must be plugged in the expressions for $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$

$\rightarrow$ the **marginal formulation** of Gaussian linear mixed models is usually exploited:

$$\ell(\boldsymbol{\beta}, \boldsymbol{\vartheta}) = -\frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|\mathbf{V}(\boldsymbol{\vartheta})| - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

two possible strategies:

- **Profile maximum likelihood**
- **Restricted maximum likelihood**

once a maximum likelihood estimate $\hat{\boldsymbol{\vartheta}}$ is obtained, estimates for the covariance matrices characterising a Gaussian linear mixed models can be obtained:

$$\hat{\mathbf{R}} = \mathbf{R}(\hat{\boldsymbol{\vartheta}}) \qquad \hat{\mathbf{G}} = \mathbf{G}(\hat{\boldsymbol{\vartheta}}) \qquad \hat{\mathbf{V}} = \mathbf{V}(\hat{\boldsymbol{\vartheta}}) = \hat{\mathbf{R}} + \mathbf{U}\hat{\mathbf{G}}\mathbf{U}^\top$$

7.2.1 Profile maximum likelihood estimation for $\boldsymbol{\vartheta}$

To maximize the profile maximum likelihood you need the MLE of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, which to be obtained need the estimates of $\boldsymbol{\vartheta} \Rightarrow$ you loose properties of the estimates for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ for finite sample size, but they remain asymptotically.

The profile log-likelihood for $\boldsymbol{\vartheta}$ is obtained by replacing $\boldsymbol{\beta}$ in $\ell(\boldsymbol{\beta}, \boldsymbol{\vartheta})$ using

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}) = (\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1}\mathbf{X})^{-1}\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1}\mathbf{y} \quad \rightarrow \text{with } \mathbf{V}(\boldsymbol{\vartheta}) \text{ unknown}$$

$$\ell_P(\boldsymbol{\vartheta}) = \ell(\hat{\boldsymbol{\beta}}(\boldsymbol{\vartheta}), \boldsymbol{\vartheta}) =$$

$$= -\frac{1}{2}\ln|\mathbf{V}(\boldsymbol{\vartheta})| - \frac{1}{2}\mathbf{r}(\boldsymbol{\vartheta})^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1}\mathbf{r}(\boldsymbol{\vartheta})$$

where

$\mathbf{r}(\boldsymbol{\vartheta}) = \left[\mathbf{I}_n - \mathbf{X}(\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1}\mathbf{X})^{-1}\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta})^{-1}\right]\mathbf{y}$ $\rightarrow$ still ϑ is the unknown quantity

$$\hat{\boldsymbol{\vartheta}}_{ML} = \arg\max \ell_P(\boldsymbol{\vartheta})$$

Apart from some simple situations (e.g. random intercept models), the maximisation of $\ell_P(\boldsymbol{\vartheta})$ requires iterative numerical approximation techniques $\rightarrow \vartheta$ contains our choice for representing the dependency an heteroscedasticity in the data

$\rightarrow$ it is possible to prove that $\hat{\boldsymbol{\vartheta}}_{ML}$ is a biased estimator for $\boldsymbol{\vartheta}$
as for Gaussian Linear Model $\hat{\boldsymbol{\vartheta}}$ was biased for $\boldsymbol{\vartheta}$

7.2.2 Restricted MLE for $\boldsymbol{\vartheta}$

The **restricted log-likelihood** for $\boldsymbol{\vartheta}$ is obtained by integrating out $\boldsymbol{\beta}$ from $L(\boldsymbol{\beta}, \boldsymbol{\vartheta})$:

$$\ell_R(\boldsymbol{\vartheta}) = \ln \left(\int_{\mathbb{R}^p} L(\boldsymbol{\beta}, \boldsymbol{\vartheta}) d\boldsymbol{\beta} \right)$$

$$\hat{\boldsymbol{\vartheta}}_{REML} = \arg\max \ell_R(\boldsymbol{\vartheta})$$

Apart from some simple situations (e.g. random intercept models), the maximisation of $\ell_R(\boldsymbol{\vartheta})$ requires iterative numerical approximation techniques.

it is possible to prove that

- $\ell_R(\boldsymbol{\vartheta}) = \ell_P(\boldsymbol{\vartheta}) - \frac{1}{2} \ln |\mathbf{X}^\top \mathbf{V}(\boldsymbol{\vartheta}) \mathbf{X}|$ $\quad \rightarrow$ connection with profile likelihood
- $\hat{\boldsymbol{\vartheta}}_{REML}$ is an unbiased estimator for $\boldsymbol{\vartheta}$

Note:
Here we do have a close form expression fo $\boldsymbol{\vartheta}$, but there is a price to pay: problems in Hypothesis testing and test statistics.

8. Linear hypotheses on fixed effects

Recall: *an hypotheses is linear if it can be expressed trough a linear restriction*

$$\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{R} + \mathbf{U}\mathbf{G}\mathbf{U}^\top) \qquad \text{(general) Gaussian linear mixed model}$$

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \qquad p = (k+1) \times 1 \text{ fixed effects}$$

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

- $\mathbf{C}$: $r \times p$ matrix composed of known constants with full row rank $r \leq p$
- $\mathbf{d}$: $r \times 1$ vector composed of known constants

8.1 Wald test statistic

8.1.1 Wald test statistic ϑ known

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p \left(\boldsymbol{\beta}, \left[\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \right]^{-1} \right)$$

$$\Downarrow \quad \textit{Linear transformation}$$

$$\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d} \sim \mathcal{N}_q \left(\mathbf{C}\boldsymbol{\beta} - \mathbf{d}, \mathbf{C} \left(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \right)^{-1} \mathbf{C}^\top \right)$$

$$\Downarrow \quad \text{if } H_0 \text{ is true}$$

$$\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d} \mid H_0 \sim \text{MVN}_q \left(\mathbf{0}_q, \mathbf{C} \left(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \right)^{-1} \mathbf{C}^\top \right)$$

$$\left(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d} \right)^\top \left[\mathbf{C} \left(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \right)^{-1} \mathbf{C}^\top \right]^{-1} \left(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d} \right) \Big|_{H_0} \sim \chi^2_r$$

$$\text{it is equivalent to} \quad \frac{\text{Var}(\mathbf{C}\hat{\boldsymbol{\beta}})}{\text{Var}(\hat{\boldsymbol{\beta}})}$$

8.1.2 Wald test statistic ϑ uknown

Here results are not as good as GLM, so the approximation of the **Wald test statistics** with a χ^2 is poor, the problem is that this approximation is true only asymptotically.

$\mathbf{V}$ is replaced with estimate $\hat{\mathbf{V}}$ (obtained using either $\hat{\boldsymbol{\vartheta}}_{ML}$ or $\hat{\boldsymbol{\vartheta}}_{REML}$)

$$\mathbf{W} = \left(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}\right)^{\top} \left[\mathbf{C}\left(\mathbf{X}^{\top}\hat{\mathbf{V}}^{-1}\mathbf{X}\right)^{-1}\mathbf{C}^{\top}\right]^{-1} \left(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}\right) \quad \Big|_{H_0} \xrightarrow{d} \chi^2_r$$

$$F = \frac{1}{r}\mathbf{W} \quad \Big|_{H_0} \xrightarrow{d} F_{r,g_2}$$

The exact value of g_2 (degrees of freedom of the denominator) is not known. Some authors suggest the following approximations:

- if H_0 involves only level 1 regressors (and/or the intercept β_0):

$$g_2 \approx n - m - k_1$$

  k_1 = number of fixed effects associated with level 1 regressors

- if H_0 involves only level 2 regressors:

$$g_2 \approx m - k_2 - 1$$

  $k_2 = k_1 + k_2$ is the number of fixed effects associated with level 2 regressors

8.1.3 Special case

$H_0 : \beta_j = 0$
In order to derive the t-test in which a single fixed effect is setted on zero you need to find the MLE and divide it for the estimated variance.

$$H_0 : \beta_j = 0$$

Both W and F reduce to

$$\frac{\hat{\beta}_j^2}{\mathrm{Var}[\hat{\beta}_j]} = \left\{\frac{\hat{\beta}_j}{\sqrt{\mathrm{Var}[\hat{\beta}_j]}}\right\}^2 = \{t_j\}^2$$

$\mathrm{Var}[\hat{\beta}_j]$*: element on the main diagonal of*

$$\left(\mathbf{X}^{\top}\hat{\mathbf{V}}^{-1}\mathbf{X}\right)^{-1} \quad \text{corresponding to } \hat{\beta}_j$$

$$\Rightarrow \quad \frac{\hat{\beta}_j}{\sqrt{\mathrm{Var}[\hat{\beta}_j]}} \quad \Big| H_0 \xrightarrow{d} t_{g_2} \ (\approx \mathcal{N}(0,1))$$

8.2 Likelihood ratio test statistics

While it was not required for **Wald test statistics** the **Likelihood ratio test statistics** requires the estimates on the restricted model **Wilks theorem**

- Likelihood ratio test statistic (*marginal formulation*) → arising from the profile maximum likelihood

$$LRT = 2\left[\ell\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}}_{ML}\right) - \ell\left(\hat{\boldsymbol{\beta}}_{H_0}, \hat{\boldsymbol{\vartheta}}_{ML,H_0}\right)\right] \quad \Big|_{H_0} \xrightarrow{d} \chi^2_r$$

- Restricted likelihood ratio test statistic → arising from the restricted maximum likelihood

$$RLRT = 2\left[\ell_R\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}}_{REML}\right) - \ell_R\left(\hat{\boldsymbol{\beta}}_{H_0}, \hat{\boldsymbol{\vartheta}}_{REML,H_0}\right)\right] \quad \Big|_{H_0} \xrightarrow{d} \chi^2_r$$

  where

$$\ell_R\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}}_{REML}\right) = \ell\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\vartheta}}_{REML}\right) - \frac{1}{2}\ln\left|\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\vartheta}}_{REML})\mathbf{X}\right|$$

Some authors suggest avoiding the use of RLRT when comparing nested models, due to a lack of invariance of the restricted likelihood function to one-to-one reparameterisations of the fixed effects (e.g. change in the reference category for categorical regressors)

Problems:
The restricted likelihood is not invariant on the parametrization of $\hat{\boldsymbol{\beta}}$, which means that something is changed in the designed matrix $\boldsymbol{X}$ (*for example the reference cathegory* the restricted MLE will change. → use Profile LRT

9. Hypotheses on random effects

Do we actually need to insert a random effect?

$$\boldsymbol{\gamma}_i \sim \mathcal{N}_{q+1}(\mathbf{0}_{q+1}, \mathbf{Q}) \qquad \boldsymbol{\gamma} \sim \mathcal{N}_{m(q+1)}(\mathbf{0}_{m(q+1)}, \mathbf{G})$$

In order to exclude a random effect from a Gaussian linear mixed model, **its variance must be set to 0** $\Rightarrow$ Hypotheses involves the unknown parameters in matrix $\mathbf{Q}$ (variances and covariances of random effects)

9.1 Identification of the constraints

When H_0 excludes a random effect from a Gaussian linear mixed model, it is necessary to correctly identify the number of constraints introduced by H_0.

$\rightarrow$ **when matrix Q is diagonal** it sufficies to set one parameter equal to zero (variance)

$\rightarrow$ **when matrix Q is not diagonal**, the exclusion of a random effect implies also the removal of all the covariances among that random effect and all the other remaining random effects

Example:

$$q+1=3, \qquad \boldsymbol{\gamma}_i = \begin{bmatrix} \gamma_{0i} \\ \gamma_{1i} \\ \gamma_{2i} \end{bmatrix} \qquad H_0 : \tau_1^2 = \tau_2^2 = 0 \iff \gamma_{1i} = \gamma_{2i} = 0 \quad \forall i$$

- if $\mathbf{Q} = \begin{bmatrix} \tau_0^2 & \tau_{01} & \tau_{02} \\ \tau_{01} & \tau_1^2 & \tau_{12} \\ \tau_{02} & \tau_{12} & \tau_2^2 \end{bmatrix}$, then $r = q+1+q = 3+2 = 5$

  $\rightarrow$ removing 2 random effect equals to imposing 5 restriction

- if $\mathbf{Q} = \begin{bmatrix} \tau_0^2 & 0 & 0 \\ 0 & \tau_1^2 & 0 \\ 0 & 0 & \tau_2^2 \end{bmatrix}$, then $r = 2$

9.2 Test statistics

(SI REML) Here there is no problem with the lack of invariance (as when dealing with random effect, both tests are good).

- $LRT = 2\left[\ell_P(\hat{\boldsymbol{\vartheta}}_{ML}) - \ell_P(\hat{\boldsymbol{\vartheta}}_{ML,H_0})\right]$
- $RLRT = 2\left[\ell_R(\hat{\boldsymbol{\vartheta}}_{REML}) - \ell_R(\hat{\boldsymbol{\vartheta}}_{REML,H_0})\right]$

Warning:
Since variances are always strictly positive, when H_0 leads to the exclusion of one (or more than one) random effect, the resulting constrained parameter space contains one (or more than one) boundary point of the unconstrained parameter space, *in other word* since the parameter space for the variance can't be negative setting a variance equal to zero means choosing a point on the boundary of the parameter space.

- **the regularity conditions required by the Wilks theorem** are not met (*in particular the one that state that the intervall must be included in the parameter space* , and the convergence of the distribution of the likelihood ratio test statistic to a χ^2 is not guarantee,
  when H_0 is true

  $\rightarrow$The distribution of the $LRT/RLRT$ statistic (when H_0 is true) **is not known** (even asymptotically)

Solution $\rightarrow$ a mixed χ^2 distribution.

9.2.1 An approximation for the true *p*-value

Some authors suggest the use of a $\chi^2_1 : \chi^2_0$ mixture to approximate the true distribution of the (R)LRT statistic:

$$\Pr\left((R)LRT \geq (R)LRT_{\text{obs}} \mid H_0\right)$$
$$\approx \frac{\Pr\left((R)LRT \geq (R)LRT_{\text{obs}} \mid \chi^2_{r-1}\right) + \Pr\left((R)LRT \geq (R)LRT_{\text{obs}} \mid \chi^2_r\right)}{2}$$

Note:we also loose the strict connection beetween the Wald and F test statistics

Special case: Exclusion on one random effect uncorrelated with any other random effect / with diagonal **Q**

- $r = 1$
- $\Pr\left((R)LRT \geq (R)LRT_{\text{obs}} \mid \chi^2_0\right) \stackrel{\text{def}}{=} 0$ *(degenerate distribution, point mass at 0)*
- $\Pr\left((R)LRT \geq (R)LRT_{\text{obs}} \mid H_0\right) \approx \frac{\Pr\left((R)LRT \geq (R)LRT_{\text{obs}} \mid \chi^2_1\right)}{2}$

Part III

Polynomial regression

Example

For simplicity we are gonna see the situation when there is only one regressor. In presence of a single regressor, a Gaussian nonlinear model can be defined by replacing the linearity assumption

$$\mathbb{E}[Y_i \mid z_i] = \beta_0 + \beta_1 z_i$$

with the following assumption

$$\mathbb{E}[Y_i \mid z_i] = f(z_i; \gamma_0, \ldots, \gamma_l)$$

where $f(\cdot\,; \gamma_0, \ldots, \gamma_l)$ is assumed to have a **known** functional form with $d = l+1$ unknown parameters ($l \geq 1$).

Depending on the choice of the functional form of $f(\cdot\,; \gamma_0, \ldots, \gamma_l)$

Different departures from linearity can be accommodated.

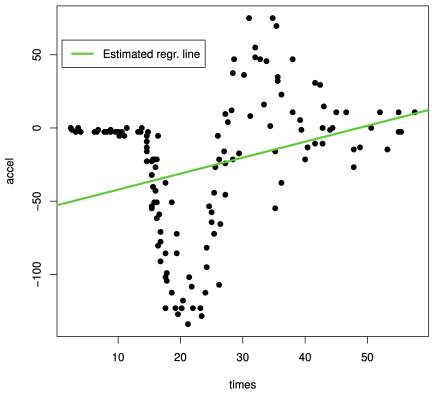


Figure 1

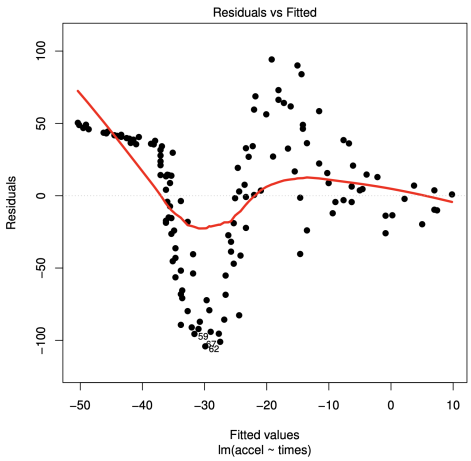


Figure 2

```
Coefficients:
             Estimate   Std.  Error   t value   Pr(>|t|)
  (Intercept)  -53.008          8.712    -6.084      0.000
        times    1.091          0.307     3.552      0.001

Multiple R-squared:  0.08785, Adjusted R-squared:  0.08089
F-statistic:  12.62 on 1 and 131 DF, p-value:  0.0005318
```

As shown in the summary the model can explain only a small fraction of the data → Multiple R-squared: 0.08785
The inadeguacy of the linearity assumption is evident from the residual plot.

1. Polynomial regression

1.1 Model Definition

- Y_i: Random variable that describes the value for the dependent variable observed on the i-th sample unit ($i = 1, \ldots, n$)
- z_i: value of the regressor for the i-th sample unit

$$f(z_i; \gamma_0, \ldots, \gamma_l) = \sum_{j=0}^{l} \gamma_j z_i^j$$

Is a nonlinear function in z, but it is **still linear in the unknown parameters** $\gamma_0, \ldots, \gamma_l$.This allow us to represent our expected value using basic matrix notation, treating each power function of the regressor as a new regressor.

Properties

(A) $\mathbb{E}[Y_i \mid z_i] = \gamma_0 + \gamma_1 z_i + \gamma_2 z_i^2 + \ldots + \gamma_l z_i^l = \sum_{j=0}^{l} \gamma_j z_i^j \quad \forall i$

(B) $\text{Var}[Y_i \mid z_i] = \sigma^2 \quad \forall i$

(C) $\text{Cor}[Y_i, Y_h] = 0 \quad \forall i \neq h$

(D) $Y_i \mid z_i \sim \mathcal{N}\left(\sum_{j=0}^{l} \gamma_j z_i^j, \sigma^2\right) \quad \forall i$

1.1.1 Matrix notation

Definition of regressor vectors:

$\left(1 = z_i^0, z_i^1, \ldots, z_i^l\right)^\top$ **row interpretation** powers of the regressor value for the i-th sample unit

$\mathbf{z}_{[j]} = \left(z_1^j, z_2^j, \ldots, z_n^j\right)^\top$ **column interpretation** powers of order j of all the n regressor values ($j = 0, \ldots, l$)

$\mathbf{z}_{[0]} = (1, 1, \ldots, 1)^\top$ intercept

Regressor matrix $n \times d = (l + 1)$**:**

$$\mathbf{Z} = \begin{bmatrix} z_1^0 & z_1^1 & \cdots & z_1^l \\ z_2^0 & z_2^1 & \cdots & z_2^l \\ \vdots & \vdots & \ddots & \vdots \\ z_n^0 & z_n^1 & \cdots & z_n^l \end{bmatrix} = \begin{bmatrix} \underline{\mathbf{z}_1^\top} \\ \underline{\mathbf{z}_2^\top} \\ \vdots \\ \underline{\mathbf{z}_n^\top} \end{bmatrix} = \left[\mathbf{z}_{[0]} \mid \mathbf{z}_{[1]} \mid \cdots \mid \mathbf{z}_{[l]}\right]$$

Regression coefficients:

$$\boldsymbol{\gamma} = \left[\gamma_0, \gamma_1, \ldots, \gamma_l\right]^\top \in \mathbb{R}^d$$

Conditional expected values in compact form:

$$\mathbb{E}[Y_i \mid z_i] = \sum_{j=0}^{l} \gamma_j z_i^j = \mathbf{z}_i^\top \boldsymbol{\gamma} \quad \forall i$$

$$\mathbb{E}[\mathbf{Y} \mid \mathbf{Z}] = \mathbf{Z}\boldsymbol{\gamma}$$

1.2 Model fitting

```
# ?poly
```

Questa funzione crea un matrice di regressione Z che contiene le potenze di uno specifico regressore. I suoi argomenti sono:

- `x` : vettore numerico che contiene i valori dei regressori
- `degree`: specifica il grado del polinomio.
  - Con `degree = 1` otteniamo le costanti e i regressori stessi,
  - Con `degree = 2` otteniamo i regressori stessi e i regressori al quadrato
  - ...
- `raw` : argomento per decidere che tipo di matrice vogliamo
  - `raw = F` : matrice di regressione associata con ai **polinomi ortogonali**
  - `raw = T` : matrice di regressione contente le **potenze** dei regressori.
  - Quindi se vogliamo la matrice Z che contiente le potenze dei regressori dobbiamo mettere `raw = TRUE`. Specificando `raw = FALSE` otteniamo allora la matrice $\tilde{Z}$.

1.2.1 Linear basis expansions

It is a transformation of the non linear functions $f(z_i; \gamma_0, \ldots, \gamma_l) = \sum_{j=0}^{l} \gamma_j z_i^j$ in a linear combination of basis. The functions $B_j(z)$ $(j = 0, \ldots, l)$ are called *basis*.
They are **nonlinear transformations** of z with a known functional form and without unknown parameters.

$$f(z_i; \gamma_0, \ldots, \gamma_l) = \sum_{j=0}^{l} \gamma_j z_i^j = \sum_{j=0}^{l} \gamma_j B_j(z_i)$$

In polynomial regression models, $B_j(z) = z^j \quad (j = 0, \ldots, l)$
The selection of proper basis is needed to represent the polynomial
Each plot in Figure2.1 represents $B_j(z) = z^j \quad (j = 0, \ldots, 5)$

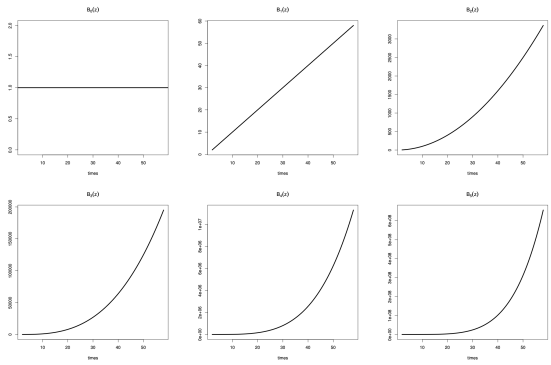


Figure 1.1: Polynomial basis 5

```
# ortogonal basis
poly12ort <- lm(accel ~ poly(times,degree = 12),data = mcycle)

# power basis
poly12 <- lm(accel ~ poly(times, degree = 12, raw = TRUE), data = mcycle)
```

Possiamo usare la funzione `lm` con la formula classica. La differenza è che adesso invece di scrivere solamente il regressore, specifichiamo che vogliamo considerare un polinomio di grado 12 associato ad esso.
Se lanciamo solamente il comando `poly`: otteniamo un matrice che ha tante colonne quanto il grado del polinomio.
La prima colonna è il regressore stesso, la seconda ha i quadrati del regressore, e così via.
Quindi usando la funzione `poly` function nella formula avremo che:
Verranno calcolate tutte le potenze del regressore, verranno costruite le basi necessarie per rappresentare il polinomio di grado scelto e queste basi verranno messe nella formula.

```
summary(poly12)
```

```
Call:
lm(formula = accel ~ poly(times, degree = 12, raw = TRUE), data = mcycle)

Residuals:
 Min      1Q  Median      3Q     Max
-75.781 -12.284   1.046  11.995  50.613

Coefficients:
                                      Estimate Std. Error t value Pr(>|t|)
(Intercept)                          2.386e+02  3.768e+02   0.633  0.52777
poly(times, degree = 12, raw = TRUE)1 -2.950e+02  3.488e+02  -0.846  0.39932
poly(times, degree = 12, raw = TRUE)2  1.440e+02  1.285e+02   1.121  0.26445
poly(times, degree = 12, raw = TRUE)3 -3.674e+01  2.543e+01  -1.445  0.15106
poly(times, degree = 12, raw = TRUE)4  5.485e+00  3.063e+00   1.791  0.07589
poly(times, degree = 12, raw = TRUE)5 -5.106e-01  2.401e-01  -2.126  0.03552  *
poly(times, degree = 12, raw = TRUE)6  3.085e-02  1.271e-02   2.426  0.01673  *
```

```
poly(times, degree = 12, raw = TRUE)7  -1.239e-03  4.627e-04  -2.677  0.00847  **
poly(times, degree = 12, raw = TRUE)8   3.332e-05  1.159e-05   2.875  0.00478  **
poly(times, degree = 12, raw = TRUE)9  -5.930e-07  1.961e-07  -3.024  0.00305  **
poly(times, degree = 12, raw = TRUE)10  6.701e-09  2.140e-09   3.131  0.00219  **
poly(times, degree = 12, raw = TRUE)11 -4.353e-11  1.359e-11  -3.202  0.00175  **
poly(times, degree = 12, raw = TRUE)12  1.239e-13  3.817e-14   3.245  0.00152  **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.67 on 120 degrees of freedom
Multiple R-squared:  0.7998, Adjusted R-squared:  0.7798
F-statistic: 39.96 on 12 and 120 DF,  p-value: < 2.2e-16
```

Se guardiamo il `summary` vediamo che di default viene inclusa l'intercetta (prima base) e poi un coefficiente per ogni colonna della matrice che è ottenuta usando la funzione `poly`.

Quindi un polinomio di grado 12 ha 13 coefficienti :

- l'intercetta (parametro associato con i regressori alla potenza di 0)
- tutte le altre potenze dei regressori

```
summary(poly12ort) # stime ottenute con le orthogonal basis
```

```
Call:
lm(formula = accel ~ poly(times, degree = 12), data = mcycle)

Residuals:
    Min      1Q  Median      3Q     Max
-75.781 -12.284   1.046  11.995  50.613

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                 -25.546      1.966 -12.993  < 2e-16 ***
poly(times, degree = 12)1   164.557     22.674   7.257 4.26e-11 ***
poly(times, degree = 12)2   131.227     22.674   5.788 5.82e-08 ***
poly(times, degree = 12)3  -239.790     22.674 -10.576  < 2e-16 ***
poly(times, degree = 12)4    -6.738     22.674  -0.297 0.766859
poly(times, degree = 12)5   245.799     22.674  10.841  < 2e-16 ***
poly(times, degree = 12)6   -83.906     22.674  -3.701 0.000326 ***
poly(times, degree = 12)7  -153.596     22.674  -6.774 4.94e-10 ***
poly(times, degree = 12)8   163.064     22.674   7.192 5.97e-11 ***
poly(times, degree = 12)9    31.879     22.674   1.406 0.162319
poly(times, degree = 12)10 -141.518     22.674  -6.241 6.77e-09 ***
poly(times, degree = 12)11   24.240     22.674   1.069 0.287191
poly(times, degree = 12)12   73.586     22.674   3.245 0.001520 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 22.67 on 120 degrees of freedom
Multiple R-squared:  0.7998, Adjusted R-squared:  0.7798
F-statistic: 39.96 on 12 and 120 DF,  p-value: < 2.2e-16
```

Anche utilizzando le basi ortogonali abbiamo 13 coefficienti, i cui valori però sono diversi da prima perché stiamo usando un diverso insieme di basi.
Confrontando però le altre statistiche notiamo che i due modelli sono equivalenti:

- Hanno lo stesso `Residual standard error` con gli stessi degree of freedom, stesso R^2 e $adj - R^2$
- Hanno la stessa F test statistic che ci permette di testare la presenza di una struttra di dipendenza polinomiale:

  → Stiamo comparando un modello completo (polinomio di ordine 12) con un modello ridotto (polinomio di ordine 0, cioè un termine costante). Rifiutando quindi quest'ipotesi andiamo a rifiutare l'ipotesi di indipendenza, cioè che non ci sia impatto dei regressori sulla variabile dipendente. Questo impatto in questo caso è quantificato usando un polinomio di grado 12.

1.3 Maximum Likelihood estimation

$$\hat{\gamma} = \left(\mathbf{Z}^\top \mathbf{Z}\right)^{-1} \mathbf{Z}^\top \mathbf{y}$$

However, the use of a basis representation gives rise to a specific **issue**:
Given their particular structure, the columns of $\mathbf{Z}$ tend to be **highly correlated** (nearly linearly dependent), we do still have full column rank (if $l < n$).

→ numerical instability due to the fact that $\mathbf{Z}^\top \mathbf{Z}$ is nearly singular,

→ possible inflation in the standard error estimates (especially when d is large compared with n)

These issues can be overcome by resorting to orthogonal polynomials: the matrix $\mathbf{Z}$ is transformed into a matrix $\tilde{\mathbf{Z}}$ such that

- $\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{I}_d$
- *for each* $\boldsymbol{\gamma} \in \mathbb{R}^d$ *there exists a unique* $\tilde{\boldsymbol{\gamma}} \in \mathbb{R}^d$ *ensuring that* $\mathbf{Z}\boldsymbol{\gamma} = \tilde{\mathbf{Z}}\tilde{\boldsymbol{\gamma}}$

(the actual recursive formula to be applied at each column of $\mathbf{Z}$ *to obtain* $\tilde{\mathbf{Z}}$ *is omitted)*

1.4 Output- polynomial of orther 5

These table reports the high correlation observed in the new fake regressors.

Sample correlation matrix among powers of the regressor

	times1	**times2**	**times3**	**times4**	**times5**
times1	1.0000	0.9688	0.9112	0.8499	0.7928
times2	0.9688	1.0000	0.9833	0.9479	0.9066
times3	0.9112	0.9833	1.0000	0.9895	0.9662
times4	0.8499	0.9479	0.9895	1.0000	0.9931
times5	0.7928	0.9066	0.9662	0.9931	1.0000

R^2 measure the linear dependence of each power on the other ones. Here z^1 is used as the dependent variable

R_j^2	**times1**	**times2**	**times3**	**times4**	**times5**
	0.9995298	0.9999860	0.9999972	0.9999971	0.9999776

Parameter estimates

```
Coefficients:
              Estimate   Std.  Error   t value   Pr(>|t|)
 (Intercept)  -105.8767       34.9883   -3.0261     0.0030
      times1    49.7816       10.3625    4.8040     0.0000
      times2    -6.3588        1.0149   -6.2655     0.0000
      times3     0.2969        0.0425    6.9819     0.0000
      times4    -0.0057        0.0008   -7.2385     0.0000
      times5     0.0000        0.0000    7.2504     0.0000
```

Figure 1.2: Original polynomial

```
Coefficients:
              Estimate   Std.  Error   t value   Pr(>|t|)
 (Intercept)   -25.5459        2.9396   -8.6902     0.0000
   times1ort   164.5566       33.9014    4.8540     0.0000
   times2ort   131.2271       33.9014    3.8708     0.0002
   times3ort  -239.7898       33.9014   -7.0732     0.0000
   times4ort    -6.7378       33.9014   -0.1987     0.8428
   times5ort   245.7987       33.9014    7.2504     0.0000
```

Figure 1.3: Orthogonal polynomial

Summary

```
Residual standard error:  33.9 on 127 degrees of freedom
Multiple R-squared:  0.5264, Adjusted R-squared:  0.5078
F-statistic:  28.24 on 5 and 127 DF, p-value:  < 2.2e-16
```

Figure 1.4: Original polynomial

```
Residual standard error:  33.9 on 127 degrees of freedom
Multiple R-squared:  0.5264, Adjusted R-squared:  0.5078
F-statistic:  28.24 on 5 and 127 DF, p-value:  < 2.2e-16
```

Figure 1.5: Orthogonal polynomial

As you can see from the table there are same fitted values for different set of basis

1.5 Hypothesis testing

Here comes an additional issue: *choosing the right degree for the polynomious* $\rightarrow$ **Model selection problem** in case of nested models.

Here we are setting a comparison between a polynomial of degree $l-r$ and one of degree l

$$\left.\begin{array}{l} H_0 : \gamma_{l-r+1} = \gamma_{l-r+2} = \cdots = \gamma_l = 0 \quad (r < d) \\ \mathbb{E}[Y_i \mid z_i] = \sum_{j=0}^{l} \gamma_j z_i^j \end{array}\right\} \quad \Rightarrow \quad \mathbb{E}_{H_0}[Y_i \mid z_i] = \sum_{j=0}^{l-r} \gamma_j z_i^j$$

If H_0 is true – *if the polynomial of degree $l-r$ is as "adequate" as the polynomial of degree l*:

$$\Rightarrow \quad F = \frac{\Delta \text{SSE}}{\text{SSE}} \cdot \frac{n-d}{r} \quad \Big| \quad H_0 \sim F_{(r,n-d)}$$

Example

$$H_0 : \boldsymbol{C}\boldsymbol{\gamma} = d$$
$$H_0 : \gamma_{13} = \gamma_{15} = 0$$

```
> c
   1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
1  0 0 0 0 0 0 0 0 0  0  0  0  0  1  0
2  0 0 0 0 0 0 0 0 0  0  0  0  0  0  1

> d
[1] 0 0

> linearHypothesis(poly14, C, d, test = "F")

Linear hypothesis test

Hypothesis:
poly(times, 14)13 = 0
poly(times, 14)14 = 0

Model 1: restricted model
Model 2: accel ~ poly(times, 14)

  Res.Df     RSS Df Sum of Sq      F  Pr(>F)
1    120 61693.46
2    118 61442.12  2     251.33  0.24  0.7860
```

The polynomial of degree 14 is not significantly better than the polynomial of degree 12.

1.5.1 Example on R

Nested Models → comparing a poly off order 12 with one of order 9

```
poly9ort <-lm(accel~poly(times,degree=9),data=mcycle) # orthog basis
poly9 <-lm(accel~poly(times,degree=9, raw=TRUE),data=mcycle) # power basis
```

Usando la funzione `anova` calcoliamo F test stastisc per comparare i due modelli.

Nota: dobbiamo mantenere lo stesso tipo di base anche nel modello che usiamo per il confronto.

Alternativa all'`anova`:
Questi test possono essere svolti usando la funzione `linearHypothesis (lht)` nel pacchetto `car`.
In quel caso però dobbiamo definire un'adeguata matrice C e un adeguato vettore d , quindi una matrice C con 3 righe e 13 colonne (perchè per avere un polinomio di ordine 9 dobbiamo rimuovere 3 basi dal polinomio di ordine 12).
Se non vogliamo usare la funzione `linearHypothesis (lht)`, possiamo fittare un modello completo e uno ridotto ed usare la funzione `anova`per avere F test statistic che cerchiamo.

```
anova(poly9ort,poly12ort)
```

```
Analysis of Variance Table

Model 1: accel ~ poly(times, degree = 9)
Model 2: accel ~ poly(times, degree = 12)
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1    123 87723
2    120 61693  3     26030 16.877 3.281e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(poly9,poly12)
```

```
Analysis of Variance Table

Model 1: accel ~ poly(times, degree = 9, raw = TRUE)
Model 2: accel ~ poly(times, degree = 12, raw = TRUE)
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1    123 87723
2    120 61693  3     26030 16.877 3.281e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Indipendentemente dalle basi che usiamo, il risultato è uguale.

In questo caso ctiamo settando tre parametri a 0 per ottenere un polinomio di ordine 9 da uno di ordine 12. Vediamo che l'F test statistic, ottenuto partendo dalla differenza nell'error sum of square, ha un p-value molto piccolo.
In questo caso allora l'ipotesi che un polinomio di ordine 9 non sia significativamente peggiore di un polinomio di ordine 12 deve essere rifiutata ⇒ Il polinomio di ordine 12 è significativamente migliore del polinomio di ordine 9.
Alternativa all'`anova`:
Questi test possono essere svolti usando la funzione `linearHypothesis (lht)` nel pacchetto `car`.
In quel caso però dobbiamo definire un'adeguata matrice C e un adeguato vettore d , quindi una matrice C con 3 righe e 13 colonne (perchè per avere un polinomio di ordine 9 dobbiamo rimuovere 3 basi dal polinomio di ordine 12).
Se non vogliamo usare la funzione `linearHypothesis (lht)`, possiamo fittare un modello completo e uno ridotto ed usare la funzione `anova`per avere F test statistic che cerchiamo.

Coefficienti in basi ortogonali e di potenze

What happens to coefficients estimates when changing the polynomial degree? Ci focalizziamo sul polinomio fittato usando le basi ortogonali.
Facciamo una comparazione dei primi 10 coefficienti di regressione stimati.

orthogonal basis

→ comparando le stime ottenute dal polinomio di ordine 9 con le stime ottenuto dal polinomio di ordine 12 per le prime 10 basi (cioè l'intercetta e le prime 9 ottenute per rappresentare il polinomio), vediamo che otteniamo esattamente gli stessi risultati. Questo significa che **quando lavoriamo con le basi ortogonali quando aumentiamo il grado del polinomio aggiungiamo parametri al modello con grado minore** ⇒ le stime dei coeffcienti che sono già nel modello rimangono invariate.. Il contributo delle basi sarà sempre lo stesso.

Con **contributo** si intende: l'impatto di una base sullafunzione polinomial che descrive la connessione tra Y e Z
Il comportamento appena descritto è tipico di polinomio rappresentati via basi ortogonali. Quindi usando polinomi con **basi orthogonali**, **l'esclusione di alcune basi non altera le stime dei coefficienti di regressione associati con le rimanenti basi**.

power basis

Se invece compariamo le stime dei coefficienti ottenute usando le power transformation notiamo che cambiando il grado del polinomio abbiamo un cambiamento anche nei coefficienti che erano già nel modello.
Per un polinomio di ordine 9 la stima del primo coefficiente (associato alla prima potenza) è -228.7551
Per un polinomio di ordine 9 la stima del primo coefficiente (associato alla prima potenza) è -294.9920. Quindi se usiamo le **basi di potenze** il comportamento peculiare che avevamo con le basi ortogonali non vale più. Abbiamo **stime diverse per i coefficienti in base al grado del polinomio**.

Alternativa alla funzione poly

I PRM con basi di potenze possono essere fittati anche elencando esplicitamentecan tutte le basi richieste nelle formula (quindi evitando di usare la funzione `poly`)
Per esempio, se vogliamo fittare un polinomio di ordine 3, utilizzando la funzione `poly` scriveremmo:

```
poly3<-lm(accel~poly(times,degree=3, raw=TRUE),data=mcycle)
```

In alternativa possiamo scrivere una formula in cui elenchiamo esplicitamente tutte le basi di potenza che vogliamo includere:

```
poly3b<-lm(accel ~ times + I(times ^ 2) + I(times ^ 3),data = mcycle)
```

Dove `times + I(times ^ 2) + I(times ^ 3)` significa $times + times^2 + times^3$
La funzione `I` è conveninete da usare quando definiamo formule che contengono trasformazioni dei regressori. In questo caso abbiamo 2 trasformazioni del regressore times, per evitare qualsiasi possibile confusione in R è meglio definire queste trasformazioni usando la funzione `I`. In questo modo R capisce che prima di includere il secondo regressore nel modello deve prendere times ed elevarlo al quadrato, e farà poi la stessa cosa con il cubo.
In linea di principio quindi possiamo fittare PRM senza usare la fuhnzione `poly`. Quando abbiamo formule molto lunghe però questo approccio non è conveniente
Se compariamo il `summary` dei due modelli vediamo:

```
summary(poly3)
```

```
   Call:
   lm(formula = accel ~ poly(times, degree = 3, raw = TRUE), data = mcycle)

   Residuals:
       Min      1Q  Median      3Q     Max
   -83.083 -28.772   2.935  31.004  94.108

   Coefficients:
                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                              78.582514  20.836118   3.771 0.000246 ***
poly(times, degree = 3, raw = TRUE)1 -17.113715   2.691521  -6.358 3.24e-09 ***
poly(times, degree = 3, raw = TRUE)2   0.667766   0.103098   6.477 1.79e-09 ***
poly(times, degree = 3, raw = TRUE)3  -0.007012   0.001170  -5.994 1.92e-08 ***
   ---
   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

   Residual standard error: 40 on 129 degrees of freedom
   Multiple R-squared:  0.3303, Adjusted R-squared:  0.3147
   F-statistic: 21.21 on 3 and 129 DF,  p-value: 3.134e-11
```

```
summary(poly3b)
```

```
Call:
lm(formula = accel ~ times + I(times^2) + I(times^3), data = mcycle)

Residuals:
    Min      1Q  Median      3Q     Max
-83.083 -28.772   2.935  31.004  94.108

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  78.582514  20.836118   3.771 0.000246 ***
times       -17.113715   2.691521  -6.358 3.24e-09 ***
I(times^2)    0.667766   0.103098   6.477 1.79e-09 ***
I(times^3)   -0.007012   0.001170  -5.994 1.92e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40 on 129 degrees of freedom
Multiple R-squared:  0.3303, Adjusted R-squared:  0.3147
F-statistic: 21.21 on 3 and 129 DF,  p-value: 3.134e-11
```

A parte differenze nelle label associate ad ogni coefficiente, vediamo che le stime per il model parameter sono le stesse. Usare quindi la funzione `poly` è più conveniente perché possiamo fittare polinomio complessi con un'istruzione molto corta, invece di dover elencare tutte le potenze nella formula o dover espandere il dataframe aggiungendo le colonne corrispondendi alle potenze dei regressori.
Nota: non c'è un modo simile che ci permette di fittare un polinomio con basi ortogonali. Nessuna funzione ci permette di creare ogni singola base ortogonale di cui abbiamo bisogno per fittare un polinomio usando le basi ortogonali. In quel caso quindi dobbiamo costruire le nostre basi usando la funzione poly e poi usarle.

1.6 Some cautionary remarks

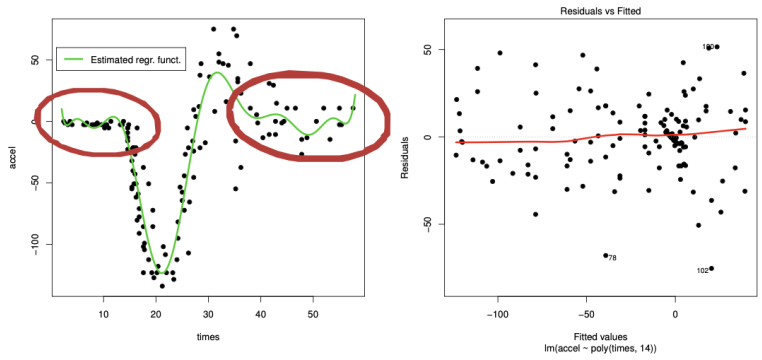


Figure 1.6: Polynomial of order 14

The plots suggest that a polynomial of degree 14 could be adequate to describe the effect of time on accelaration (no clear pattern in the average value of the residuals).
Moreover it is also evident that the polynomial is pretty rough in the extremes where on the other side data are pretty flat.

ML estimates of the coefficients of a polynomial regression function are **global function**
→ they are affected by each observation in the sample.
⇒ undesirable side effect

In particular, a small change in one observed value of the dependent variable can lead to a dramatic change in the fitted function(even for values of the regressors that are far from that observation).

This erratic/unstable behaviour is exacerbated:

- near the boundaries of the regressor domain
- when the degree of the polynomial is large (compared with the sample size)

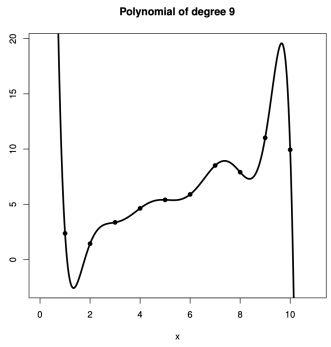


Figure 1.7: Polynomial basis 5 – Part 1

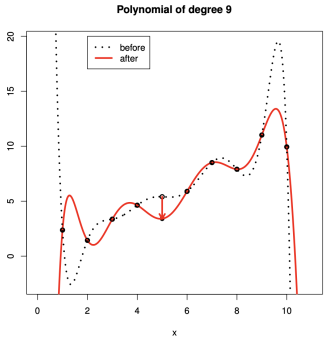


Figure 1.8: Polynomial basis 5 – Part 2

Note: the effect on the fitted regression function due to a small change in a single observation. → It acts good but when the degree of the polynomial is low.

Part IV

Regression splines

1. Regression splines

1.1 Spline function

A function $f(z) : [a, b] \to \mathbb{R}$ is said to be a r spline fuction if it is a spline function of degree l with knots $a = \kappa_1 < \kappa_2 < \ldots < \kappa_m = b$

Knots
Are a set of point selected in the range of the domain of the regressor. Those are needed because each spline function can be represented by a polynomial if order l between 2 knots $\to$ we divide the regressor domain in sub intervals and assume that within each interval there will be a specific polynomial, each of them will have different coefficient for the regressor:

with m knots you define

- $m - 1$ intervals
- $m - 2$ internal knots
- $m - 1$ polynomial of order l,
- $l + 1$ coefficients in each polynomial

$f(z)$ **is** :

- $l1$ times continuously differentiable (its partial derivatives with respect to z are continuous up to the order $l1$)
- is a polynomial of degree l on the $m1$ intervals $[_{j,j+1}), = 1, ..., m1$ defined by the knots
- is continuous but not differentiable

$$f(z) = \begin{cases} \sum_{h=0}^{l} \theta_{h1} z^h & k_1 = a \leq z < \kappa_2 \\ \vdots & \vdots \\ \sum_{h=0}^{l} \theta_{hj} z^h & \kappa_j \leq z < \kappa_{j+1} \quad (j = 2, \ldots, m-2) \\ \vdots & \vdots \\ \sum_{h=0}^{l} \beta_{h(m-1)} z^h & \kappa_{m-1} \leq z < b = k_m \end{cases}$$

1.2 Linear Splines

A function $f(z) : [a, b] \to \mathbb{R}$ is said to be a linear spline if it is a spline function of degree 1 with knots $a = \kappa_1 < \kappa_2 < \ldots < \kappa_m = b$

(its first partial derivative with respect to z does not exist at the internal knots $\kappa_2 < \ldots < \kappa_{m-1}$).

$\to f(z)$ is linear on the $m-1$ intervals $[\kappa_j, \kappa_{j+1})$, for $j = 1, \ldots, m-1$, defined by the knots.

$\to$ continuous and differentiable up to order 0

$$f(z) = \begin{cases} \theta_{01} + \theta_{11} z & a \leq x < \kappa_2 \\ \vdots & \vdots \\ \theta_{0j} + \theta_{1j} z & \kappa_j \leq z < \kappa_{j+1} \quad (j = 2, \ldots, m-2) \\ \vdots & \vdots \\ \theta_{0(m-1)} + \theta_{1(m-1)} z & \kappa_{m-1} \leq x < b \end{cases}$$

Each interval has its own intercept and slope

$\Rightarrow$ Linear splines are also known as **continuous piecewise linear functions.**

1.3 Cubic splines

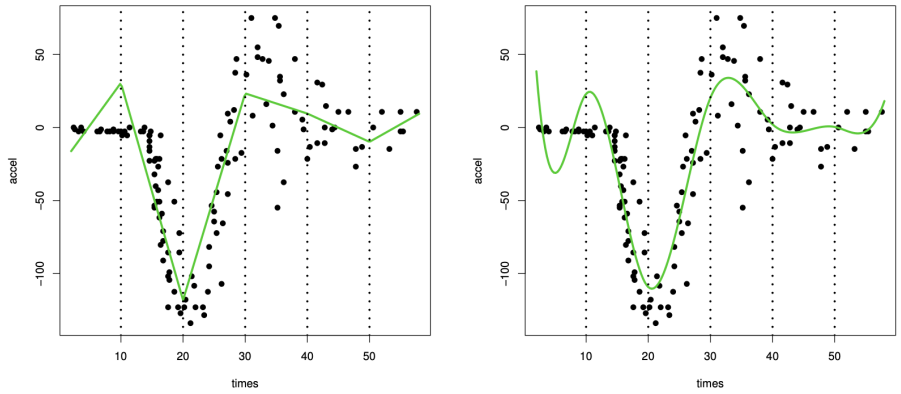
A function $f(z) : [a, b] \to \mathbb{R}$ is said to be a cubic spline if it is a spline function of degree 3 with knots $a = \kappa_1 < \kappa_2 < \ldots < \kappa_m = b$:

- $f(z)$ is continuous and differentiable up to the second order *(its third partial derivative with respect to z does not exist at the internal knots $\kappa_2 < \ldots < \kappa_{m-1}$).*
- $f(z)$ is cubic on the $m-1$ intervals $[\kappa_j, \kappa_{j+1})$, for $j = 1, \ldots, m-1$, defined by the knots.

$$f(z) = \begin{cases} \theta_{01} + \theta_{11} z + \theta_{21} z^2 + \theta_{31} z^3 & a \leq x < \kappa_2 \\ \vdots & \vdots \\ \theta_{0j} + \theta_{1j} z + \theta_{2j} z^2 + \theta_{3j} z^3 & \kappa_j \leq z < \kappa_{j+1} \quad (j = 2, \ldots, m-2) \\ \vdots & \vdots \\ \theta_{0(m-1)} + \theta_{1(m-1)} z + \theta_{2(m-1)} z^2 + \theta_{3(m-1)} z^3 & \kappa_{m-1} \leq x < b \end{cases}$$

It is claimed that cubic splines are the lowest order splines for which the discontinuity in the partial derivatives at the knots cannot be noticed by the human eye.

As you can see in the linear splines function the first partial derivative will not be continuous in the knots but the function is continuous.



The dashed vertical lines denote the location of the internal knots $_2 < ... <_6$

1.4 Number of free parameters of a spline function

Any spline function of degree l with knots $a = \kappa_1 < \kappa_2 < \ldots < \kappa_m = b$ has a total of:

- $(l+1)(m-1)$ total parameter
- $d = m + l - 1$ free parameters
- $(m-1)$ polynomials, each with $(l+1)$ coefficients $\Rightarrow$ *free parameter* $= (l+1)(m+1) - l \cdot (m-2)$

l restrictions must be imposed at each of the $(m-2)$ internal knots $\kappa_2 < \ldots < \kappa_{m-1}$, in order **to guarantee that its partial derivatives** with respect to z up to order $l-1$ are continuous

So the total amount of restriction that must be imposed is:
$l \cdot (m-2)$

linear splines have $d = m$ free parameters
cubic splines have $d = m + 2$ free parameters

Key feature of a spline function

1. it is a polynomial degree of order l
2. with $(m-2)$ intervals
3. allow linear basis expansion with $m + l - 1$ basis $\rightarrow$ this transformation allow us to estimates the parameter $\boldsymbol{\gamma}$ without setting any restriction

2. Basis expantion

2.1 Linear basis expansion for spline functions

Splines do admit basis, the number of which at maximum can be equal to the number of free parameter.
It is possible to prove that, regardless of m and l:

- any spline function of degree l with fixed knots $a = \kappa_1 < \kappa_2 < \ldots < \kappa_m = b$ can be represented using a linear basis expansion:

$$f(z) = \sum_{j=1}^{d} \gamma_j B_j(z)$$

- this linear basis expansion is not unique: several possible choices for the basis functions $B_j(\cdot)$. (while it was unique for polynomials)

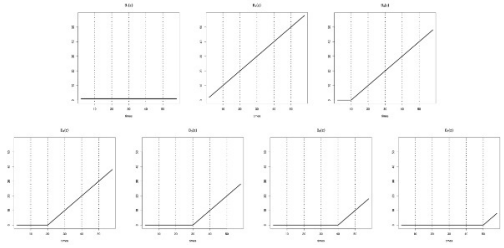
2.2 Truncated power basis for spline functions

$$B_j(z) = \begin{cases} z^{j-1} & j = 1, \ldots, l+1 \quad \text{global polynomial of degree } l \text{ for the first } l+1 \text{ basis} \\ (z - \kappa_{j-l})_+^l & j = l+2, \ldots, d \quad m - l - 2 \text{ additional bases for each internal knots} \end{cases}$$

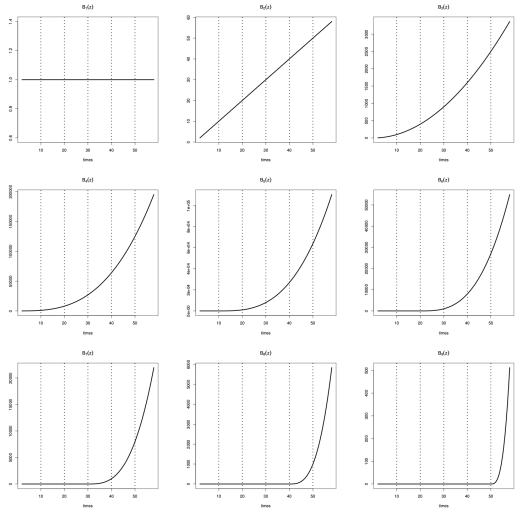
where one considers only the positive part:

$$(z - \kappa_{j-l})_+^l = \begin{cases} (z - \kappa_{j-l})^l & z \geq \kappa_{j-l} \\ 0 & z < \kappa_{j-l} \end{cases}$$

The basis is 0 for all the previous knots, this allows you to estimate free parameters without caring for the restriction.



Example of truncated linear basis



Example of truncated cubic basis

1. $B_1(z) = 0$
2. $B_2(z) = z$
3. $B_3(z) = (z - 10)_+$
4. $B_4(z) = (z - 20)_+$
5. ...
6. $B_7(z) = (z - 70)_+$

1. $B_1(z) = 1$
2. $B_2(z) = z$
3. $B_3(z) = z^2$
4. $B_4(z) = z^3$
5. $B_5(z) = (z - 10)^3$
6. $B_6(z) = (z - 20)^3$
7. ...
8. $B_7(z) = (z - 70)^3$

Figure 2.1: Examples of truncated linear and cubic bases

2.3 B-spline basis functions

Despite their simple and intuitive structure, truncated power bases are rarely used in practice (they are actually not implemented in R). This is because the columns of the corresponding matrix $\mathbf{Z}$ tend to be h**ighly correlated (nearly linearly dependent)**, which leads to near singularity of $\mathbf{Z}^\top\mathbf{Z}$ and numerical instability in the estimation process.

An alternative linear basis expansion that avoids these issues is the **B-spline basis**. These basis functions are defined using a recursive formula *(omitted)*. Each B-spline basis **takes positive values only between a pair of knots** otherwise is null (the interval depends on l), otherwise is zero. **at any point** $z \in [a, b]$**:**

- only $l + 1$ B-spline bases take positive values

$\Rightarrow \sum_{j=1}^{d} B_j(z) = 1$ simultaneusly for a given z

These are still **local basis** in the sense are 0 in others interval when it happens that the basis are simultaneous positive note that: if one is increasing the other one is decreasing Here each

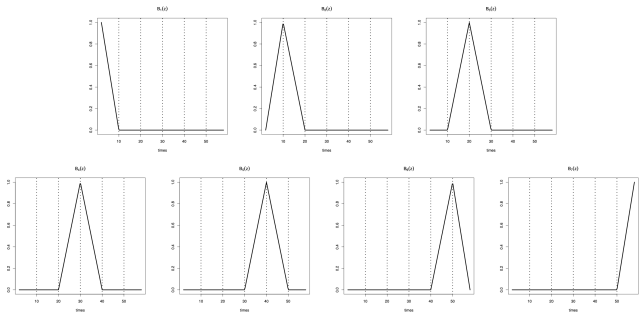


Figure 2.2: examples of B-spline bases - linear splines

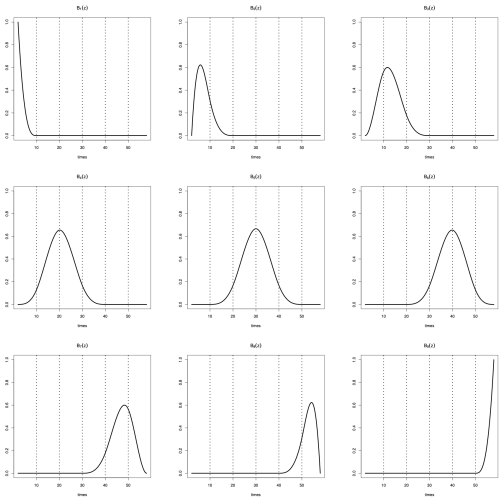


Figure 2.3: examples of B-spline bases - cubic splines

basis has a different shape and slope to satisfy the *sum-up-to-one* condition

2.4 Matrix notation

For a given choice of the basis functions: **Regressor matrix** $n \times d = (m + l - 1)$:

$$\mathbf{Z} = \begin{bmatrix} B_1(z_1) & B_2(z_1) & \cdots & B_{m+l-1}(z_1) \\ B_1(z_2) & B_2(z_2) & \cdots & B_{m+l-1}(z_2) \\ \vdots & \vdots & \ddots & \vdots \\ B_1(z_n) & B_2(z_n) & \cdots & B_{m+l-1}(z_n) \end{bmatrix}$$

Regression coefficients:

$$\boldsymbol{\gamma} = [\gamma_1, \gamma_2, \ldots, \gamma_{m+l-1}]^\top \in \mathbb{R}^d$$

Conditional expected values in compact form:

$$\mathbb{E}[\mathbf{Y} \mid \mathbf{Z}] = \mathbf{Z}\boldsymbol{\gamma}$$

```
# ?bs
# bs(x, df = NULL, knots = NULL,degree = 3,
intercept = FALSE, Boundary.knots = range(x),warn.outside = TRUE)
```

La funzione `bs` genera la matrice delle basi B-spline per un polinomio spline:

- `x`: è il regressore con cui vogliamo lavorare
- `degree`: ci fa settare il grado della funzione spline, cioè il valore di l
- `intercept`: quando è `FALSE`

`intercept = FALSE` la funzione `bs` costruisce un numero di basi che sarà uguale al numero *parametri liberi* -1. Questo ci permette di adattare l'espressione per calcolare le $l + m - 1$ basi, in modo tal da permettere che la prima base sia un valore costante.
Da un punto di vista tecnico lasciando quindi `intercept = FALSE`, non otteniamo le basi viste in classe nei grafici, ma delle basi leggermente diverse.
Dal momento però che vogliamo usare queste basi nella funzione `lm` è conventiente lasciare `intercept = F`, in questo modo non dobbiamo specificare una formula in cui dobbiamo rimuovere in modo esplicito l'intercetta.

`intercept = TRUE` Se invece vogliamo inserire l'effetto dell'intercetta nelle basi, sapendo però che quando andiamo a definire la formula nella funzione `lm` dobbiamo rimuoverla. Altrimenti dobbiamo lasciare `intercept = FALSE`, costruire le $l + m - 2$ basi e poi aggiungere tutte queste nella formula.

Nota di default se inseriamo una formula nella funzione `lm` viene inclusa l'intercetta.
Di default `bs` costruisce basi usando come primo knot k_1 e ultimo knot k_m il minimo e il massimo del regressore. Quello che dobbiamo fare per ottenere il set di basi che descrivono la spline function è definire gli **internal knots** cioè quelli da k_2 a k_{m-1}

3. Knots selection

Choosing knots mean choosing the optimal complexity In order to find the best model you have to compare.

3.1 Inference for regression spline models

In order to find the best model you have to compare models with different numbers of knots.

- Maximum likelihood estimates of the parameters $\gamma_1, \ldots, \gamma_d$ for a given linear basis expansion can be obtained using standard tools for Gaussian regression models with regression functions that are linear in the unknown parameters:

$$\Rightarrow \hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{y}$$

Model comparison

- *Nested models* can be obtained by adding/removing knots, if **truncated power bases** are used to represent the spline function

  $\Rightarrow$ use a l**ikelihood ratio test**

- A **change in the location** of one or more knots creates a *non-nested model*
  Note: It works only for truncated power basis not for B-splines, for example, here (due to the summation constraint) when changing one knot you have to change all the others.

  $\Rightarrow$ use **model selection criteria**

3.2 Knots Location

Can impact the ability of spline function to adapt to the data.

STRATEGY

1. subjective choice
2. equidistant knots
3. internal knots placed at quantiles of the regressor: $\rightarrow$ this guarantees an approximately constant number of sample units within each interval.

Internal knots for spline functions located at the quantiles of the regressor
For Example: to define 4 intervals I need 3 internal knots, such that 0.25 of the sample units fall in each interval $\rightarrow$ This way you give at each local polynomial the sam amount of information.

Due to $d = m + l - 1$ (*number of free parameters*

- the polynomial start at 0
- the linear splines start at 2,
- the cubic splines start at 5

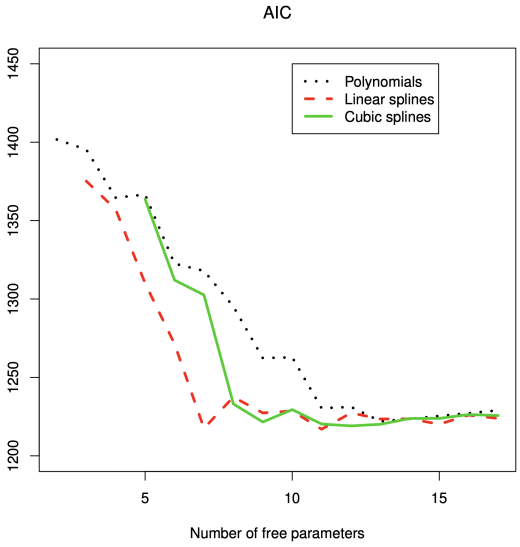


Figure 3.1: comparison among alternative models

3.3 Knots selection on R

Ci sono due modi possibili per scegliere la **location** e il **numero dei knots**:

- specificando `df` (degrees of freedom). Cioè inserendo il numero di parametri liberi, cioè $l + m - 1$
- Con l'argomento `knots` possiamo inserire un vettore che contiene le location esatte in cui vogliamo i nostri knots. Questo argomento si riferisce a $m - 2$ internal knots

Vediamo ora come possiamo usare la funzione `bs` per creare basi per una funzione spline con i knots in una location prespecificata.

Knots on regregresor's quantile

Quello che dobbiamo fare è include nell'argomento knots un vettore contente la location degli internal knots. Ricordiamoci che per esempio con 9 **internal knots** abbiamo un totale di 10 **intervalli**. Quindi per scegliere i quantili per una distribuzione dei regressori tale che finiamo con 10 intervalli dobbiamo scegliere 9 quantili.
Esempio: spline lineare con $m - 2 = 9$ *internal knots ai quantili:*

```
K.l <- 9
```

1. Iniziamo creando il vettore dei quantili utilizzando la funzione `quantile`. Definiamo il parametro `probs` come `(1:K.l)/(K.l+1)`, che rappresenta l'insieme delle probabilità corrispondenti ai quantili desiderati.

```
probs <- (1:K.l)/(K.l+1)
probs
```

```
[1] 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9
```

2. Applichiamo quindi la funzione `quantile` ai valori della variabile `times` per individuare le soglie corrispondenti alle probabilità specificate. In questo modo otteniamo un insieme ordinato di valori che suddivide i dati in intervalli contenenti ciascuno circa il 10

```
knots.l<-quantile(mcycle$times,probs=(1:K.l)/(K.l+1))
knots.l
```

```
   10%   20%   30%   40%   50%   60%   70%   80%   90%
10.04 14.68 16.20 18.44 23.40 26.52 31.52 36.20 43.80
```

→ Utilizzando questi quantili come posizioni dei knots interni, otteniamo una suddivisione dei dati tale da garantire che, approssimativamente, ciascun intervallo contenga il 10

3. Una volta definita la posizione dei knots, possiamo utilizzare la funzione `bs`, in modo analogo a quanto si fa con `poly`, per generare le basi spline. Si procede infine alla stima del modello tramite la funzione `lm`.

```
lspline<-lm(accel~bs(times,knots=knots.l, degree=1),data=mcycle)
```

Nella parte destra della formula abbiamo quindi incluso le basi che abbiamo ottenuto applicando la funzione bs al regressore times, posizionando gli internal knots ai valori che sono specificati nel vettrore knots.l. Specificando un `degree=1` otteniamo una **linear spline**.
Nota: Se non vogliamo internal knots ai quantili ma **equidistanti nel range del regressore**, possiamo usare l'argomento `df`, specificando *num tot basi* − 1. In questo modo la funzione `s` creerà una sequenza di internal knots tali che gli intervalli saranno tutti equidistanti.

```
summary(lspline)
```

```
Call:
lm(formula = accel ~ bs(times, knots = knots.l, degree = 1),
    data = mcycle)

Residuals:
    Min      1Q  Median      3Q     Max
-73.344 -11.889  -0.575  11.153  53.567

Coefficients:
                                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                                    -1.5222    10.5720  -0.144 0.885747
bs(times, knots = knots.l, degree = 1)1        -0.4519    15.1657  -0.030 0.976278
bs(times, knots = knots.l, degree = 1)2        -9.0326    12.5879  -0.718 0.474397
bs(times, knots = knots.l, degree = 1)3       -48.7028    12.6269  -3.857 0.000185 ***
bs(times, knots = knots.l, degree = 1)4      -102.4070    13.3444  -7.674 4.57e-12 ***
bs(times, knots = knots.l, degree = 1)5      -115.9931    14.0102  -8.279 1.83e-13 ***
bs(times, knots = knots.l, degree = 1)6       -23.8462    12.8728  -1.852 0.066378
```

```
bs(times, knots = knots.l, degree = 1)7    54.0932    14.2669   3.792 0.000234 ***
bs(times, knots = knots.l, degree = 1)8    10.6931    13.1640   0.812 0.418203
bs(times, knots = knots.l, degree = 1)9     1.5024    12.9866   0.116 0.908090
bs(times, knots = knots.l, degree = 1)10    0.9684    15.4128   0.063 0.950003
   ---
   Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

   Residual standard error: 22.4 on 122 degrees of freedom
   Multiple R-squared:  0.8014, Adjusted R-squared:  0.7851
   F-statistic: 49.22 on 10 and 122 DF,  p-value: < 2.2e-16
```

- Il termine `intercept` è aggiunto di default nel modello, anche se non è incluso direttamente nelle basi generate dalla funzione `bs`. Infatti, nella chiamata `bs(times, knots = knots.l, degree = 1)` il parametro `intercept` è impostato per default su `FALSE`.
- Otteniamo quindi 10 coefficienti associati alle basi spline generate dalla funzione `bs`, ai quali si aggiunge l'intercetta.
- Considerando che abbiamo scelto 9 *internal knots*, possiamo calcolare m come segue: $m - 2 = 9 \Rightarrow m = 11$.
- Poiché stiamo lavorando con spline lineari (cioè grado $l = 1$), il numero totale di parametri da stimare risulta essere $l + m - 1 = 1 + 11 - 1 = 11$.

In conclusione, nel modello stimiamo complessivamente 11 parametri: l'intercetta e i coefficienti associati alle 10 basi spline.
Se vogliamo utilizzare le basi nella stessa forma mostrata nelle slide, dobbiamo rifittare il modello utilizzando nuovamente la funzione `bs`. Tuttavia, questa volta è necessario
.

△ All'interno della funzione `bs` → `intercept = TRUE`

△ Nella formula del modello lineare → aggiungere `-1` per evitare doppia intercetta

```
lspline1 <- lm(accel~bs(times,knots=knots.l, degree=1, intercept= TRUE) - 1 ,
               data=mcycle)

summary(lspline1)
```

```
   Call:
   lm(formula = accel ~ bs(times, knots = knots.l, degree = 1, intercept = TRUE) -
       1, data = mcycle)

   Residuals:
       Min      1Q  Median      3Q     Max
   -73.344 -11.889  -0.575  11.153  53.567

   Coefficients:
    Estimate     Std. Error   t value      Pr(>|t|)
    1    -1.52225    10.57201   -0.144   0.885747
    2    -1.97414    8.18618   -0.241    0.809840
    3   -10.55488    7.26903   -1.452    0.149059
```

```
4   -50.22500    6.81826   -7.366    2.28e-11 ***
5  -103.92921    8.16317  -12.731     < 2e-16 ***
6  -117.51533    9.18759  -12.791     < 2e-16 ***
7   -25.36843    7.34609   -3.453    0.000762 ***
8    52.57096    9.57955    5.488    2.25e-07 ***
9     9.17081    7.84378    1.169    0.244609
10   -0.01986    7.54213   -0.003    0.997904
11   -0.55383   11.21545   -0.049    0.960697
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.4 on 122 degrees of freedom
Multiple R-squared:  0.845,  Adjusted R-squared:  0.831
F-statistic: 60.47 on 11 and 122 DF,  p-value: < 2.2e-16
```

Osservando il summary, notiamo che ora abbiamo 11 basi generate direttamente dalla funzione `bs`. Le stime dei coefficienti risultano diverse rispetto al modello precedente, ma le statistiche riassuntive del modello sono identiche. Questo conferma che entrambi gli approcci portano allo stesso risultato. È quindi possibile utilizzare entrambe le formulazioni, anche se si preferisce la prima per chiarezza.

Esempio: spline cubica con $m - 2 = 8$ internal knot ai quantili

```
K.c <- 8
probs <- (1:K.c)/(K.c+1)
probs
```

```
[1] 0.1111111 0.2222222 0.3333333 0.4444444 0.5555556 0.6666667 0.7777778
[8] 0.8888889
```

Avendo cambiato il numero di internal knot dobbiamo considerare dei quantili diversi. Allora è cambiata la location degli internal knots:

```
knots.c<-quantile(mcycle$times,probs=(1:K.c)/(K.c+1))
knots.c
```

```
11.11111% 22.22222% 33.33333% 44.44444% 55.55556% 66.66667% 77.77778% 88.88889%
10.46667  15.40000  16.80000  20.00000  25.40000  28.60000  35.53333  42.86667
```

Creato il vettore che ha le location degli internal knots, possiamo usare di nuovo la funzione `bs` per avere le basi e fittare la cubic spline. Se usiamo quindi i quantili per trovare le location dei knots, ovviamente queste posizioni per una spline con 9 intervalli saranno diversa da quella con 10 intervalli.

Nota: Non c'è alcun tipo di relazione annidata tra la location dei quantili per le due diverse splines usando un numero diverso di knots.

```
cspline<-lm(accel~bs(times,knots=knots.c, degree=3),data=mcycle)
```

Rispetto a prima abbiamo solamente cambiato l'argomento di knots e il grado della spline.

```
summary(cspline)
```

```
Call:
lm(formula = accel ~ bs(times, knots = knots.c, degree = 3),
    data = mcycle)

Residuals:
    Min      1Q  Median      3Q     Max
-74.384 -11.040  -0.056  11.560  52.830

Coefficients:
                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                                 -1.749     14.625  -0.120 0.905000
bs(times, knots = knots.c, degree = 3)1      4.228     37.658   0.112 0.910789
bs(times, knots = knots.c, degree = 3)2    -11.359     24.404  -0.465 0.642434
bs(times, knots = knots.c, degree = 3)3     13.580     20.002   0.679 0.498488
bs(times, knots = knots.c, degree = 3)4    -84.830     17.155  -4.945 2.48e-06  ***
bs(times, knots = knots.c, degree = 3)5   -132.878     22.153  -5.998 2.13e-08  ***
bs(times, knots = knots.c, degree = 3)6    -94.175     18.835  -5.000 1.96e-06  ***
bs(times, knots = knots.c, degree = 3)7     74.754     20.195   3.702 0.000324  ***
bs(times, knots = knots.c, degree = 3)8      2.639     19.839   0.133 0.894388
bs(times, knots = knots.c, degree = 3)9     10.840     24.738   0.438 0.662025
bs(times, knots = knots.c, degree = 3)10   -19.858     29.590  -0.671 0.503421
bs(times, knots = knots.c, degree = 3)11    15.108     23.786   0.635 0.526526
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.5 on 121 degrees of freedom
Multiple R-squared:  0.8012, Adjusted R-squared:  0.7831
F-statistic: 44.33 on 11 and 121 DF,  p-value: < 2.2e-16
```

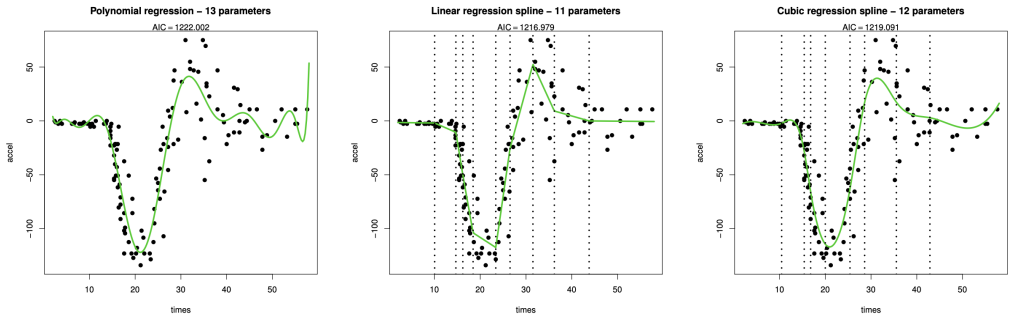
Nota: quando lavoriamo con polinomi e spline function non siamo direttamente interessati ai valori per le stime associati con ogni base, al **comportamento globale della funzione fittata**. Quindi guardando il `summary` mettiamo attenzione sulle parti più importanti: `Residual standard error`, `Multiple R-squared`, `Adjusted R-squared`, `F-statistic`.

Rispetto a prima nella cubic spline con 8 internal knots abbiamo un `Adjusted R-squared``=0.7831`, mentre nella linear spline con 9 internal knots `Adjusted R-squared=``0.831`. Quindi in questo caso apparentemente la linear con 9 internal knots sempra essere meglio della cubic con 8 internal knots.

Nota: il modello con la cubic spline ha un numero di parametri che è $l+m-1 = 3+10-1 = 12$, mentre la linear spline ne aveva 11.

3.4 Polynomials and spline function

- they are structured as local polynomials defined on non-overlapping intervals
- a change in one observed value for the dependent variable affects only some of the polynomials that compose the spline functions (the ones close to the interval to which the observation belongs), but leaves the other polynomials unchanged



Note:
where data are more sparse the knots are closer

Figure 3.2: best models

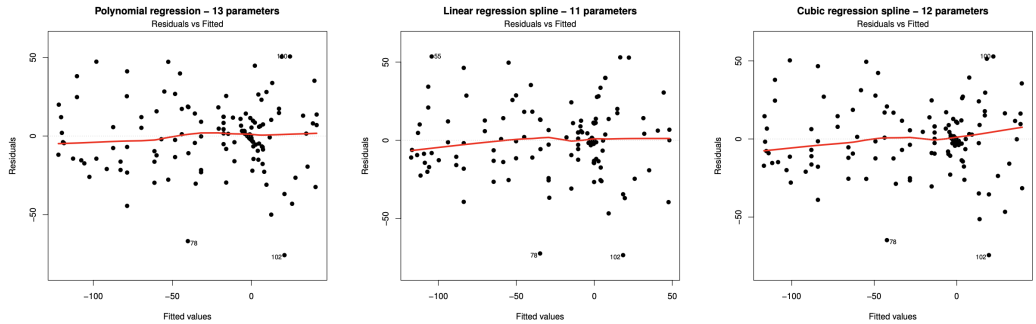


Figure 3.3: residuals

Differently from polynomials, spline functions have a local nature:

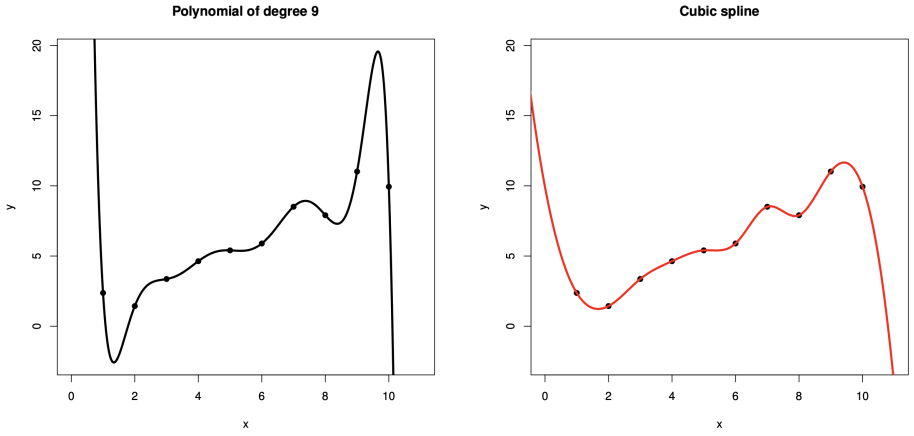


Figure 3.4
with splines the change is local, due to knots structure the impact will concern just closer intervals

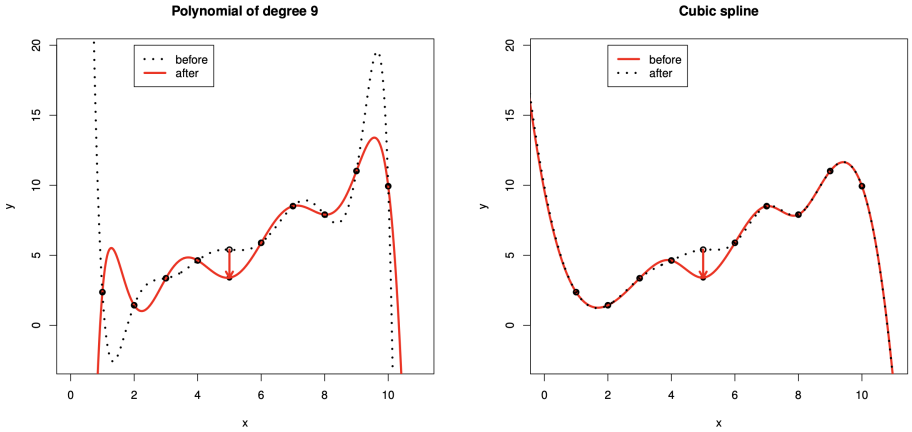

Polynomial of degree 9
before
after
y
x
0
5
10
15
20
0
2
4
6
8
10
Cubic spline
before
after
y
x
0
5
10
15
20
0
2
4
6
8
10

Figure 3.5

4. P-splines

Difficulties in using regression splines
Regression splines are an attractive solution to enhance the flexibility of Gaussian regression models.
However, they suffer from a major shortcoming: the choice of the number and of the locations of the knots

- it is basically not possible to completely avoid any amount of subjectivity and to make this choice in a systematic/objective manner
- considering equidistant knots or placing knots at quantiles are suboptimal strategies, leading to complications when performing model comparisons (models with different numbers of knots are not nested)

Basic idea

Introduction of a penalty therm:
In the context of Gaussian models with regression splines, an alternative strategy to overcome problems related to the selection of the number and location of the m knots (especially the $m-2$ internal knots) is:

1. choosing a relatively **large number of equally spaced internal knots** (e.g. $m-2=20$ or 40), in order to guarantee high flexibility in the resulting spline function
2. subtracting a **penalty/regularization term** to the log-likelihood function to account for the roughness/wiggliness of the resulting spline function

4.1 Penalized log-likelihood function

$$p_\lambda(\boldsymbol{\gamma}, \sigma^2) = l(\boldsymbol{\gamma}, \sigma^2) - \frac{\lambda}{2} J(\boldsymbol{\gamma})$$

- the likelihood $l(\boldsymbol{\gamma}, \sigma^2)$ determines the goodness of fit to the data *(the larger, the better)* $\rightarrow$ measures the amount of roughness
- $J(\boldsymbol{\gamma}) \geq 0$ is used to penalize for roughness/wiggliness *(complexity) — the smaller, the better*
- $\lambda \geq 0$ is the **regularization/smoothing parameter** controlling the trade-off
- $\lambda = 0 \quad \Rightarrow \quad$ no penalization (standard regression spline)
- $\lambda \rightarrow +\infty \quad \Rightarrow \quad$ any spline with $J(\boldsymbol{\gamma}) \neq 0$ is excluded

4.1.1 Penalty terms for P-splines

Penalty terms $J(\boldsymbol{\gamma})$ can be:

- **First-order differences** (squared):
  → accounts for difference in slope between note

$$J_1(\boldsymbol{\gamma}) = \sum_{j=2}^{d} (\gamma_j - \gamma_{j-1})^2 \quad \Rightarrow \quad J_1(\boldsymbol{\gamma}) = 0 \Leftrightarrow \sum_j \gamma_j B_j(z) \text{ is constant in } z$$

  → provides that the degree of the spline funcion is at least 1
  When the function is constant $J_1(\boldsymbol{\gamma}) = 0 \Leftrightarrow \gamma_j = \gamma_{j-1}$

- **Second-order differences** (squared):

$$J_2(\boldsymbol{\gamma}) = \sum_{j=3}^{d} (\gamma_j - 2\gamma_{j-1} + \gamma_{j-2})^2 \quad \Rightarrow \quad J_2(\boldsymbol{\gamma}) = 0 \Leftrightarrow \sum_j \gamma_j B_j(z) \text{ is linear}$$

  → provided that the degree of the spline funcion is at least 2
  It is an approximation of the second partial derivatives which is equal to zero when the function is linear

Note:
The actual order of the difference depends on the degree of the local polynomial: the first one is preferred for linear splines, while both are good for cubic splines.

Advantage
Rather than choosing number of knots, we will just need to choose the penalization parameter .

The joint use of B-spline basis functions and penalty terms based on differences leads to the lead to the so-called **P-spline approach**.

4.1.2 Matrix formulation:

$$J_1(\boldsymbol{\gamma}) = \boldsymbol{\gamma}^\top \mathbf{D}_1^{(d)\top} \mathbf{D}_1^{(d)} \boldsymbol{\gamma} = \boldsymbol{\gamma}^\top \mathbf{K}_1 \boldsymbol{\gamma}$$

$$J_2(\boldsymbol{\gamma}) = \boldsymbol{\gamma}^\top \mathbf{D}_1^{(d-1)\top} \mathbf{D}_1^{(d-1)} \mathbf{D}_1^{(d)\top} \mathbf{D}_1^{(d)} \boldsymbol{\gamma} = \boldsymbol{\gamma}^\top \mathbf{K}_2 \boldsymbol{\gamma}$$

Where:

$$\mathbf{D}_1^{(d)} = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}_{(d-1)\times d} \qquad \mathbf{D}_1^{(d-1)} = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & -1 & 1 & \end{bmatrix}_{(d-2)\times(d-1)}$$

4.2 Penalized maximum likelihood estimation

$$p_\lambda(\boldsymbol{\gamma}, \sigma^2) = -\frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{Z}\boldsymbol{\gamma})^\top(\mathbf{y}-\mathbf{Z}\boldsymbol{\gamma}) - \frac{\lambda}{2}\boldsymbol{\gamma}^\top\mathbf{K}\boldsymbol{\gamma} + c$$

Maximizing $p_\lambda(\boldsymbol{\gamma}, \sigma^2)$ with respect to $\boldsymbol{\gamma}$ is equivalent to minimizing **penalized least square criterion** (true due to gaussianity):

$$PLS(\tilde{\boldsymbol{\gamma}}) = (\mathbf{y}-\mathbf{Z}\boldsymbol{\gamma})^\top(\mathbf{y}-\mathbf{Z}\boldsymbol{\gamma}) + \tilde{\lambda}\boldsymbol{\gamma}^\top\mathbf{K}\boldsymbol{\gamma}, \quad \text{with } \tilde{\lambda} = \lambda\sigma^2$$

$$\hat{\boldsymbol{\gamma}}_{\tilde{\lambda}} = \left(\mathbf{Z}^\top\mathbf{Z} + \tilde{\lambda}\mathbf{K}\right)^{-1}\mathbf{Z}^\top\mathbf{y}$$

Note: the estimates depends on the specific value of the smoothing parameter

Example

When $\tilde{\gamma} \to \infty$ the function became constant $m2 = 20$ (equally spaced) internal knots, $l = 3$ cubic splines penalty: $\boldsymbol{K}_1$ (squared first-order differences)

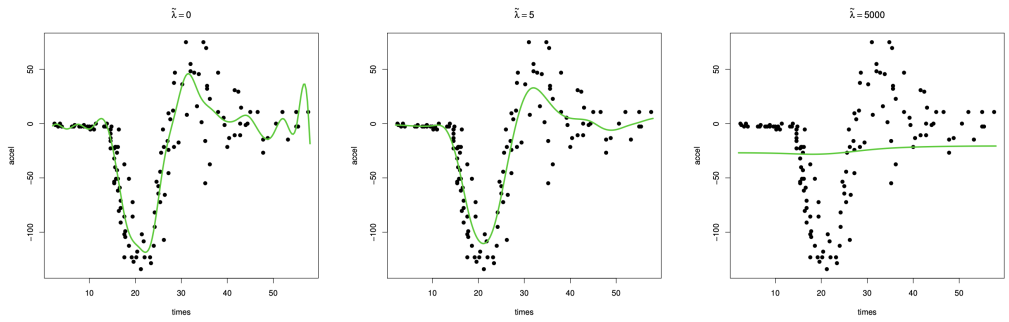


Figure 4.1: penalized ML estimation

$$\sum_j \hat{\gamma}_j B_j(z) \text{ approaches a constant as } \tilde{\lambda} \to +\infty$$

$\boldsymbol{K}_2$ (squared second-order differences)

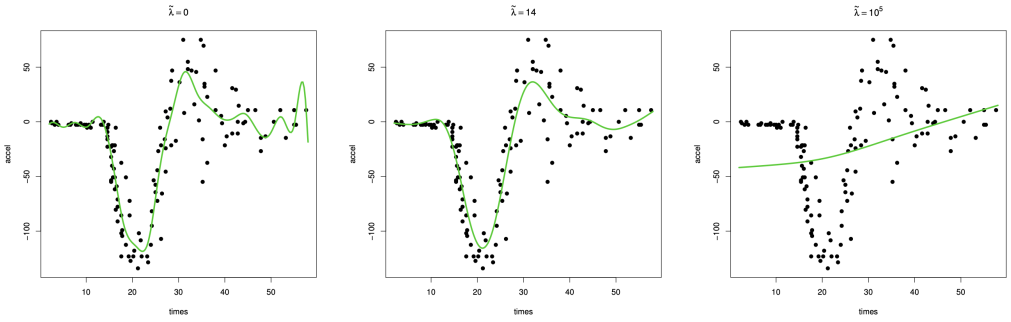


Figure 4.2: penalized ML estimation

$$\sum_j \hat{\gamma}_j B_j(z) \text{ approaches a linear function as } \tilde{\lambda} \to +\infty$$

4.2.1 Linear smoothers and smoother matrices

Fitted values (estimated conditional expected values) obtained by exploiting the linear structure:

$$\hat{f}_{\tilde{\lambda}} = \mathbf{Z}\tilde{\gamma} = \mathbf{Z}\left(\mathbf{Z}^\top\mathbf{Z} + \tilde{\lambda}\mathbf{K}\right)^{-1}\mathbf{Z}^\top\mathbf{y} = \mathbf{S}_{\tilde{\lambda}}\mathbf{y}$$

Where

- $\hat{f}_{\tilde{\lambda}}$ is a **linear smoother**, they are a smoothed version of the observed values. These are obtained by removing the random error ε from the determination of y_i, using the (symmetric) **smoother matrix** $\mathbf{S}_{\tilde{\lambda}}$
- $\mathbf{S}_{\tilde{\lambda}} = \mathbf{Z}\left(\mathbf{Z}^\top\mathbf{Z}\right)^{-1}\mathbf{Z}^\top$ is the equivalent of the hat matrix in GLM, it provides a quantification of the complexity of the model
- equivalent degrees of freedom/ effective number of parameters

$$\mathrm{df}(\mathbf{S}_{\tilde{\lambda}}) = \mathrm{trace}(\mathbf{S}_{\tilde{\lambda}})$$

Limiting cases:

- $\tilde{\lambda} = 0 \Rightarrow \quad \mathrm{trace}(\mathbf{Z}\left(\mathbf{Z}^\top\mathbf{Z}\right)^{-1}\mathbf{Z}^\top) = d = m + l - 1$
- $\tilde{\lambda} \to +\infty \Rightarrow \mathrm{trace}(\mathbf{Z}(\mathbf{Z}^\top\mathbf{Z} + \tilde{\lambda}\mathbf{K}_1)^{-1}\mathbf{Z}^\top) \to 1$ (number of free parameter in any constant function)
- $\tilde{\lambda} \to +\infty \Rightarrow \mathrm{trace}(\mathbf{Z}(\mathbf{Z}^\top\mathbf{Z} + \tilde{\lambda}\mathbf{K}_2)^{-1}\mathbf{Z}^\top) \to 2$

$$\hat{\sigma}^2 = \frac{\sum_i \left(y_i - \hat{f}_{\tilde{\lambda}}(z_i)\right)^2}{n - \mathrm{df}(\mathbf{S}_{\tilde{\lambda}})}$$

$\to \quad n - \mathrm{df}(\mathbf{S}_{\tilde{\lambda}})$ the equivalent of the degrees of freedome
Some authors suggest replacing the equivalent degrees of freedom $\mathrm{df}(\mathbf{S}_{\tilde{\lambda}})$

$$\mathrm{df}_{\mathrm{res}}(\mathbf{S}_{\tilde{\lambda}}) = \mathrm{trace}(2\mathbf{S}_{\tilde{\lambda}} - \mathbf{S}_{\tilde{\lambda}}\mathbf{S}_{\tilde{\lambda}}) \quad \text{or} \quad \mathrm{df}_{\mathrm{var}}(\mathbf{S}_{\tilde{\lambda}}) = \mathrm{trace}(\mathbf{S}_{\tilde{\lambda}}\mathbf{S}_{\tilde{\lambda}})$$

4.3 Choice of the smoothing parameter

In the P-spline approach the problem of choosing the number and location of knots is replaced by selecting the smoothing parameter $\tilde{\lambda}$ which is crucial for balancing goodness of fit vs. complexity. The selection of the optimal value for $\tilde{\lambda}$ can be based on a model selection criterion

- **Cross Validation** Leave one out:

$$CV = \frac{1}{n}\sum_i \left(\frac{y_i - \hat{f}_{\tilde{\lambda}}(x_i)}{1 - s_{\tilde{\lambda},ii}}\right)^2$$

  where $s_{\tilde{\lambda},ii}$ is the ith diagonal element of $\boldsymbol{S}_{\tilde{\lambda}}$ it is not necessary to refit the model n times

- **Generalized Cross Validation (GCV):**

$$GCV = \frac{1}{n}\sum_i \left(\frac{y_i - \hat{f}_{\tilde{\lambda}}(x_i)}{1 - \mathrm{df}(\mathbf{S}_{\tilde{\lambda}})/n}\right)^2$$

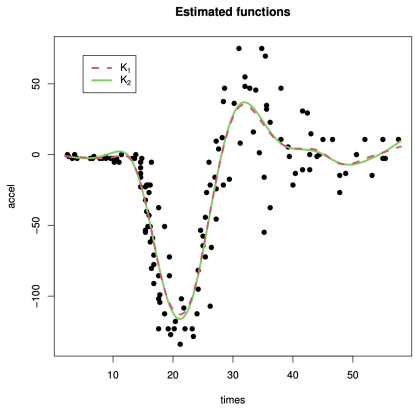


Figure 4.3: P-splines - optimal $\tilde{\lambda}$

Degree	Penalty	m	$\tilde{\lambda}$	df($\mathbf{S}_{\tilde{\lambda}}$)	GCV	AIC
cubic	first-order diff.	22	3.369	12.275	569.540	1222.092
cubic	second-order diff.	22	12.182	11.414	562.227	1220.542

Table 4.1: Comparison of penalties and model criteria for cubic splines

- **AIC or BIC:**
  - the maximum likelihood estimate for σ^2 is needed $\rightarrow \hat{\sigma}^2_{ML} = \frac{1}{n}\sum_i \left(y_i - \hat{f}_{\tilde{\lambda}}(x_i)\right)^2$
  - and number of parameters: $\text{df}(\mathbf{S}_{\tilde{\lambda}}) + 1$

Note: the number of parameter estimated trough df($\mathbf{S}_{\tilde{\lambda}}$) it's almost the same. Also, the actual value of γ is not that interesting, the linear smoothed estimates, these are interesting.

4.4 Inference for P-splines

Model assumptions:

$$\mathbf{Y} \mid \mathbf{Z} \sim \mathcal{N}_n(\mathbf{f}, \sigma^2\mathbf{I}_n) \quad \text{with } \mathbf{f} = (f(z_1), \ldots, f(z_n))^\top$$

where $\mathbf{f} = (f(z_1), \ldots, f(z_n))^\top$ is n-dimensional vector containing the (unknown) true conditional expected values $\mathbb{E}[Y_i \mid z_i]$

Note: here we are assuming conditional homoscedasticity and independence

Usually, the interest is in the function $f(\cdot)$, not in the individual parameters γ_j of the expansion

$$f(\cdot) \approx \sum_j \gamma_j B_j(\cdot)$$

Then:

$\hat{\mathbf{f}}_{\tilde{\lambda}} \mid \mathbf{Z} = \mathbf{S}_{\tilde{\lambda}}\mathbf{Y} \mid \mathbf{X} \sim \mathcal{N}_n(\mathbf{S}_{\tilde{\lambda}}\mathbf{f}, \sigma^2\mathbf{S}_{\tilde{\lambda}}\mathbf{S}_{\tilde{\lambda}}^\top) \rightarrow$ *here we do have the true vector multiplied by the smoothing matrix*

This introduce a BIAS:

$$\mathbb{E}\left[\hat{\mathbf{f}}_{\tilde{\lambda}} \mid \mathbf{Z}\right] = \mathbf{S}_{\tilde{\lambda}}\mathbf{f} \neq \mathbf{f}$$

- P-splines are biased estimators for the unknown function $f(\cdot)$: $\mathbf{S}_{\tilde{\lambda}}\mathbf{f} - \mathbf{f} \neq \mathbf{0}_n$

- This bias comes from penalty-induced constraints, and the fact that splines approximate $f(\cdot)$
- The distributional result can still be used for approximate inference on $f(\cdot)$

4.5 Hypothesis testing for P-splines

The approximate distributional result for $\hat{\mathbf{f}}_{\tilde{\lambda}}$ allows us to test hypotheses on $f(\cdot)$:

- **Independence assumption**:

$$H_0 : f(\cdot) \text{ is a constant function}$$

- **Linearity assumption**:

$$H_0 : f(\cdot) \text{ is a linear function}$$

Nested models test statistics: Since constants and linear functions are nested in cubic splines, one can use an approximate likelihood ratio or Wald test (typically approximated with an F-test).after expressing these hypothesis in terms of linear restrictions on the cubic spline coefficients (omitted)

Warning:he resulting p-values rely on several approximations and do not take into account the uncertainty related to the choice of the smoothing parameter. In particular, they are anticonservative, in the sense that they tend to underestimate the "true" p-values. → those p-values do not consider sample variability of the smoothing parameter
It is also possible to derive approximate pointwise confidence bands for $f(\cdot)$.

4.5.1 Crash test data – P-splines – R output

`gam` function (package: `mgcv`) – $l = 3, m = 22$, second-order difference penalty

Parametric coefficients:

```
Approximate significance of smooth terms:
edf Ref.df F p-value s(times) 10.414 12.438 37.198 0.000

R-sq.(adj) = 0.78, Deviance explained = 79.7GCV = 562.23, Scale est. = 513.98,
n = 133
```

The estimated function $\hat{f}_{\tilde{\lambda}}$ is decomposed into:

- the intercept ⇒ 1 degree of freedom
- a (nonconstant) function centred around 0⇒ df($\mathbf{S}_{\tilde{\lambda}}$) − 1equivalent degrees of freedom

⇒ *p-value suggests that* $\hat{f}_{\tilde{\lambda}}$ *according to the approximate p-value, one may conclude that the estimated function differs significantly from a constant one*
Approximate pointwise confidence bands can be associated with the centered estimated function.
Note:

- The horizontal line at 0 is not included in the band ⇒ Consistently confirms rejection of the constant function null hypothesis

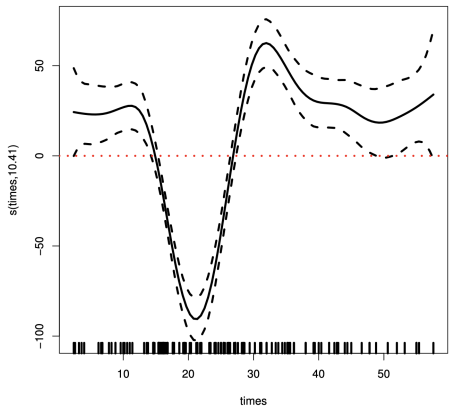


Figure 4.4: P-splines - optimal $\tilde{\lambda}$

Concluding remarks

- Beyond polynomials and splines, many other functions admit linear basis expansion representations
- Penalized estimation works with any nonlinear model based on linear basis expansions
- These techniques generalize to multivariate models:

Additive models:

$$\mathbb{E}[Y_i \mid x_{1i}, \ldots, x_{pi}] = f_1(x_{1i}; \gamma_1) + \cdots + f_p(x_{pi}; \gamma_k)$$

Generalized Linear Model

Part V

Binary regression

1. Beyond Gaussian models

Gaussian linear models are the basis of many regression analysis: amazing theoretical properties, simplicity in the interpretation and easy to fit!

Although we can use a Gaussian linear model almost always without impunity, sometimes it is not appropriate due to the 'nature' of the dependent variable.
For example, what about:

- continuous but only positive variables? *(i.e., speed, acceleration, time...)*
- variables counting items or events or occurrences? *(i.e., number of extreme rainfalls, insurance claims, births, deaths...)*
- categorical variables, binary or not. *(i.e., client creditworthiness, bankruptcy, elder patient falling...)*

A general modelling framework

Generalized Linear Models (GLMs): framework that unifies models where

- an appropriate distribution, not necessarily Gaussian, is adopted for Y; accounts for domain and characteristics of response variable; *hence, generalized*
- the linear combination of covariates and coefficients is correctly 'connected' to the response variable. *hence, linear*

$$\mathbb{E}[Y_i] = \mu_i$$

$$\boldsymbol{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \longleftarrow \quad \boldsymbol{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \qquad n \times (k+1)$$

Using covariates, you model a vector given covariates
(do not touch the intercept)

2. Model Structure

You're aiming at $\mathbb{E}[y_i]$

$$\mathbb{E}[y_i] = \mu_i \Rightarrow \begin{cases} \pi_i = h(\boldsymbol{x}_i^\top \boldsymbol{\beta}) = h(\eta_i) & \text{(I)} \\ \hat{\lambda}_i = h(\boldsymbol{x}_i^\top \boldsymbol{\beta}) = h(\hat{\eta}_i) & \text{(II)} \end{cases}$$

(I) Since you need π_i to be in $[0, 1]$, you introduce the **response function** when dealing with Binary responses.

(II) Here the restriction is more about **non negativity constraints**, so you need a **response function** to transform the linear prediction to accommodate that.

$$\begin{pmatrix} \pi_i \\ \lambda_i \end{pmatrix} \quad \rightarrow \text{depending on which distribution you are using}$$

The point is: when Gaussian does not work (given the nature of the response), you need to adjust, choosing the correct parameters of the distribution.

2.1 Binary regression

Our parameter describes the Probability mass function of the response.

$$Y \sim Bern(\pi)$$
$$\mathbb{P}(\boldsymbol{Y} = y) = \pi^y (1 - \pi)^{1-y}$$

Our aim is to model
and impact of covariates $x_{i1}, \ldots, x_{ik}$ on the (conditional) probability:

$$\pi_i = \mathbb{P}(y_i = 1) = \mathbb{E}(y_i).$$

Naive approach: *linear probability model*

$$\pi_i = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

But $\pi \neq$the linear predictor as it was in gaussian models

$\rightarrow \pi_i$ is a probability: no guarantee, however, that η_i will be in $[0, 1]$.

The response function$h(\cdot)$, defined as:

- strictly monotonically increasing on $\mathbb{R}$,
- ensuring $h(\eta_i) \in [0, 1]$

$$\pi_i = h(\eta_i) = h(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik})$$

It is the transformation on the linear predictor that ensure that η_i stays in the domain of the response, its inverse $g = h^{-1}$ is the **link function**

$$\eta_i = g(\pi_i).$$

Specific response/link functions identify specific (binary) regression models.

2.2 Logit model

Identified by the usage of **logistic response function:** $\rightarrow \quad \frac{\exp(x)}{1+\exp(x)}$

$$\pi_i = h(\eta_i) = \frac{\exp(\eta_i)}{1+\exp(\eta_i)}$$

or equivalently the **logit link function:** $\rightarrow \quad \log\left(\frac{x}{1-x}\right)$

$$g(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.]Estimation of \sigma^2$$

$$\hat{\sigma}^2 = \frac{\sum_i \left(y_i - \hat{f}_{\tilde{\lambda}}(z_i)\right)^2}{n - \mathrm{df}(\mathbf{S}_{\tilde{\lambda}})}$$

$\rightarrow \quad n - \mathrm{df}(\mathbf{S}_{\tilde{\lambda}})$ the equivalent of the degrees of freedome

Some authors suggest replacing the equivalent degrees of freedom $\mathrm{df}(\mathbf{S}_{\tilde{\lambda}})$

$$\mathrm{df}_{\mathrm{res}}(\mathbf{S}_{\tilde{\lambda}}) = \mathrm{trace}(2\mathbf{S}_{\tilde{\lambda}} - \mathbf{S}_{\tilde{\lambda}}\mathbf{S}_{\tilde{\lambda}}) \quad \text{or} \quad \mathrm{df}_{\mathrm{var}}(\mathbf{S}_{\tilde{\lambda}}) = \mathrm{trace}(\mathbf{S}_{\tilde{\lambda}}\mathbf{S}_{\tilde{\lambda}})$$

2.3 Choice of the smoothing parameter

In the P-spline approach the problem of choosing the number and location of knots is replaced by selecting the smoothing parameter $\tilde{\lambda}$ which is crucial for balancing goodness of fit vs. complexity.

The selection of the optimal value for $\tilde{\lambda}$ can be based on a model selection criterion

- **Cross Validation** Leave one out:

$$CV = \frac{1}{n}\sum_i \left(\frac{y_i - \hat{f}_{\tilde{\lambda}}(x_i)}{1 - s_{\tilde{\lambda},ii}}\right)^2$$

  where $s_{\tilde{\lambda},ii}$ is the ith diagonal element of $\boldsymbol{S}_{\tilde{\lambda}}$ it is not necessary to refit the model n times

- **Generalized Cross Validation (GCV):**

$$GCV = \frac{1}{n}\sum_i \left(\frac{y_i - \hat{f}_{\tilde{\lambda}}(x_i)}{1 - \mathrm{df}(\mathbf{S}_{\tilde{\lambda}})/n}\right)^2$$

- **AIC or BIC:**
  - the maximum likelihood estimate for σ^2 is needed $\rightarrow \hat{\sigma}^2_{ML} = \frac{1}{n}\sum_i \left(y_i - \hat{f}_{\tilde{\lambda}}(x_i)\right)^2$
  - and number of parameters: $\mathrm{df}(\mathbf{S}_{\tilde{\lambda}}) + 1$

Note: the number of parameter estimated trough $\mathrm{df}(\mathbf{S}_{\tilde{\lambda}})$ it's almost the same. Also, the actual value of γ is not that interesting, the linear smoothed estimates, these are interesting.

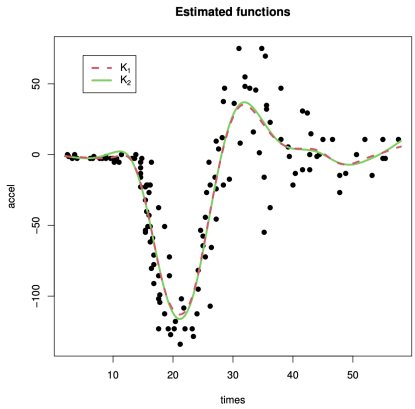


Figure 2.1: P-splines - optimal $\tilde{\lambda}$

Degree	Penalty	m	$\tilde{\lambda}$	$\mathbf{df}(\mathbf{S}_{\tilde{\lambda}})$	GCV	AIC
cubic	first-order diff.	22	3.369	12.275	569.540	1222.092
cubic	second-order diff.	22	12.182	11.414	562.227	1220.542

Table 2.1: Comparison of penalties and model criteria for cubic splines

2.4 Inference for P-splines

Model assumptions:

$$\mathbf{Y} \mid \mathbf{Z} \sim \mathcal{N}_n(\mathbf{f}, \sigma^2 \mathbf{I}_n) \quad \text{with } \mathbf{f} = (f(z_1), \ldots, f(z_n))^\top$$

where $\mathbf{f} = (f(z_1), \ldots, f(z_n))^\top$ is n-dimensional vector containing the (unknown) true conditional expected values $\mathbb{E}[Y_i \mid z_i]$

Note: here we are assuming conditional homoscedasticity and independence

Usually, the interest is in the function $f(\cdot)$, not in the individual parameters γ_j of the expansion

$$f(\cdot) \approx \sum_j \gamma_j B_j(\cdot)$$

Then:

$\hat{\mathbf{f}}_{\tilde{\lambda}} \mid \mathbf{Z} = \mathbf{S}_{\tilde{\lambda}}\mathbf{Y} \mid \mathbf{X} \sim \mathcal{N}_n(\mathbf{S}_{\tilde{\lambda}}\mathbf{f}, \sigma^2 \mathbf{S}_{\tilde{\lambda}}\mathbf{S}_{\tilde{\lambda}}^\top) \rightarrow$ *here we do have the true vector multiplied by the smoothing m*

This introduce a BIAS:

$$\mathbb{E}\left[\hat{\mathbf{f}}_{\tilde{\lambda}} \mid \mathbf{Z}\right] = \mathbf{S}_{\tilde{\lambda}}\mathbf{f} \neq \mathbf{f}$$

- P-splines are biased estimators for the unknown function $f(\cdot)$: $\mathbf{S}_{\tilde{\lambda}}\mathbf{f} - \mathbf{f} \neq \mathbf{0}_n$
- This bias comes from penalty-induced constraints, and the fact that splines approximate $f(\cdot)$
- The distributional result can still be used for approximate inference on $f(\cdot)$

2.5 Hypothesis testing for P-splines

The approximate distributional result for $\hat{\mathbf{f}}_{\tilde{\lambda}}$ allows us to test hypotheses on $f(\cdot)$:

- **Independence assumption**:

$$H_0 : f(\cdot) \text{ is a constant function}$$

- **Linearity assumption**:

$$H_0 : f(\cdot) \text{ is a linear function}$$

Nested models test statistics: Since constants and linear functions are nested in cubic splines, one can use an approximate likelihood ratio or Wald test (typically approximated with an F-test).after expressing these hypothesis in terms of linear restrictions on the cubic spline coefficients (omitted)

Warning:he resulting p-values rely on several approximations and do not take into account the uncertainty related to the choice of the smoothing parameter. In particular, they are anticonservative, in the sense that they tend to underestimate the "true" p-values. → those p-values do not consider sample variability of the smoothing parameter

It is also possible to derive approximate pointwise confidence bands for $f(\cdot)$.

2.5.1 Crash test data – P-splines – R output

`gam` function (package: `mgcv`) – $l = 3, m = 22$, second-order difference penalty

Parametric coefficients:

```
Approximate significance of smooth terms:
edf Ref.df F p-value s(times) 10.414 12.438 37.198 0.000

R-sq.(adj) = 0.78, Deviance explained = 79.7GCV = 562.23, Scale est. =
513.98, n = 133
```

The estimated function $\hat{f}_{\tilde{\lambda}}$ is decomposed into:

- the intercept ⇒ 1 degree of freedom
- a (nonconstant) function centred around 0⇒ df($\mathbf{S}_{\tilde{\lambda}}$) − 1equivalent degrees of freedom

⇒ *p-value suggests that* $\hat{f}_{\tilde{\lambda}}$ *according to the approximate p-value, one may conclude that the estimated function differs significantly from a constant one*

Approximate pointwise confidence bands can be associated with the centered estimated function.

Note:

- The horizontal line at 0 is not included in the band ⇒ Consistently confirms rejection of the constant function null hypothesis

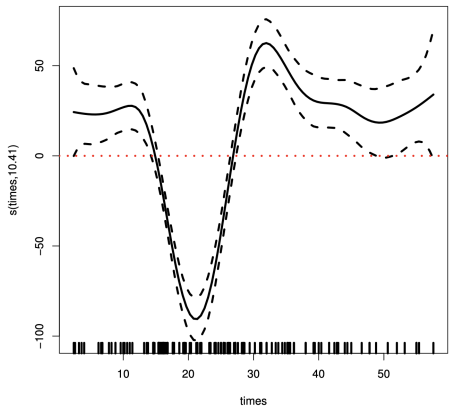


Figure 2.2: P-splines - optimal $\tilde{\lambda}$

Concluding remarks

- Beyond polynomials and splines, many other functions admit linear basis expansion representations
- Penalized estimation works with any nonlinear model based on linear basis expansions
- These techniques generalize to multivariate models:

Additive models:

$$\mathbb{E}[Y_i \mid x_{1i}, \ldots, x_{pi}] = f_1(x_{1i}; \gamma_1) + \cdots + f_p(x_{pi}; \gamma_k)$$

Generalized Linear Model

Part VI

Binary regression

1. Beyond Gaussian models

Gaussian linear models are the basis of many regression analysis: amazing theoretical properties, simplicity in the interpretation and easy to fit!

Although we can use a Gaussian linear model almost always without impunity, sometimes it is not appropriate due to the 'nature' of the dependent variable.

For example, what about:

- continuous but only positive variables? *(i.e., speed, acceleration, time...)*
- variables counting items or events or occurrences? *(i.e., number of extreme rainfalls, insurance claims, births, deaths...)*
- categorical variables, binary or not. *(i.e., client creditworthiness, bankruptcy, elder patient falling...)*

A general modelling framework

Generalized Linear Models (GLMs): framework that unifies models where

- an appropriate distribution, not necessarily Gaussian, is adopted for Y; accounts for domain and characteristics of response variable; *hence, generalized*
- the linear combination of covariates and coefficients is correctly 'connected' to the response variable. *hence, linear*

$$\mathbb{E}[Y_i] = \mu_i$$

$$\boldsymbol{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \longleftarrow \quad \boldsymbol{X} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix} \qquad n \times (k+1)$$

Using covariates, you model a vector given covariates
(do not touch the intercept)

2. Model Structure

You're aiming at $\mathbb{E}[y_i]$

$$\mathbb{E}[y_i] = \mu_i \Rightarrow \begin{cases} \pi_i = h(\boldsymbol{x}_i^\top \boldsymbol{\beta}) = h(\eta_i) & \text{(I)} \\ \hat{\lambda}_i = h(\boldsymbol{x}_i^\top \boldsymbol{\beta}) = h(\hat{\eta}_i) & \text{(II)} \end{cases}$$

(I) Since you need π_i to be in $[0, 1]$, you introduce the **response function** when dealing with Binary responses.

(II) Here the restriction is more about **non negativity constraints**, so you need a **response function** to transform the linear prediction to accommodate that.

$$\begin{pmatrix} \pi_i \\ \lambda_i \end{pmatrix} \quad \rightarrow \text{depending on which distribution you are using}$$

The point is: when Gaussian does not work (given the nature of the response), you need to adjust, choosing the correct parameters of the distribution.

2.1 Binary regression

Our parameter describes the Probability mass function of the response.

$$Y \sim Bern(\pi)$$
$$\mathbb{P}(\boldsymbol{Y} = y) = \pi^y (1 - \pi)^{1-y}$$

Our aim is to model

and impact of covariates $x_{i1}, \ldots, x_{ik}$ on the (conditional) probability:

$$\pi_i = \mathbb{P}(y_i = 1) = \mathbb{E}(y_i).$$

Naive approach: *linear probability model*

$$\pi_i = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

But $\pi \neq$the linear predictor as it was in gaussian models

$\rightarrow \pi_i$ is a probability: no guarantee, however, that η_i will be in $[0, 1]$.

The response function$h(\cdot)$, defined as:

- strictly monotonically increasing on $\mathbb{R}$,
- ensuring $h(\eta_i) \in [0, 1]$

$$\pi_i = h(\eta_i) = h(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik})$$

It is the transformation on the linear predictor that ensure that η_i stays in the domain of the response, its inverse $g = h^{-1}$ is the **link function**

$$\eta_i = g(\pi_i).$$

Specific response/link functions identify specific (binary) regression models.

2.2 Logit model

Identified by the usage of **logistic response function:** $\rightarrow \quad \frac{\exp(x)}{1+\exp(x)}$

$$\pi_i = h(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

or equivalently the **logit link function:** $\rightarrow \quad \log\left(\frac{x}{1-x}\right)$

$$g(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

$\Rightarrow$*when the response is logistic the link is logit*

Note: in the scale of the logit the effect of the coefficient is linear, here the functional form of the response function returns a predictor which is linear in the parameter

Interpretation of β_j

β_j is the relative change **in the log of the odds (**$= \eta_i$**)** after increasing the regressor by 1 unit.

In this model, covariates affect odds in exponential-multiplicative form:

$$\textbf{odds} \quad \frac{\pi_i}{1-\pi_i} = \frac{\mathbb{P}(y_i = 1 \mid \mathbf{x}_i)}{\mathbb{P}(y_i = 0 \mid \mathbf{x}_i)} = \exp(\beta_0) \cdot \exp(\beta_1 x_{i1}) \cdots \exp(\beta_k x_{ik})$$

$$\textbf{Log-odds} \quad \log\left(\frac{\pi_i}{1-\pi_i}\right) = \log\left(\frac{\mathbb{P}(y_i = 1 \mid \mathbf{x}_i)}{\mathbb{P}(y_i = 0 \mid \mathbf{x}_i)}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

which means if we increase for example x_1 by 1 unit we have:

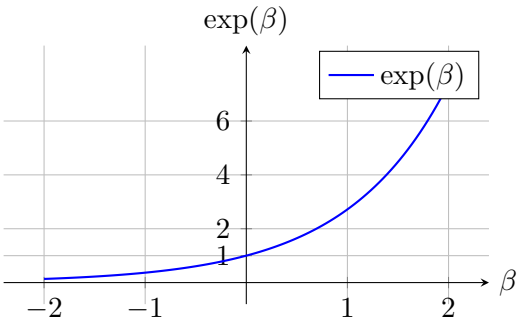
$$\frac{\mathbb{P}(y_i = 1 \mid x_{i1} + 1, \ldots)}{\mathbb{P}(y_i = 0 \mid x_{i1} + 1, \ldots)} \Bigg/ \frac{\mathbb{P}(y_i = 1 \mid x_{i1}, \ldots)}{\mathbb{P}(y_i = 0 \mid x_{i1}, \ldots)} = \exp(\beta_1),$$

$\rightarrow$ when x increase by one unit then the odds of $y = 1$ is $exp(\beta_j)$ times higher (or smaller). and the odds ratio:

- increase when $\beta_1 > 0$
- decrease if $\beta_1 < 0$

– stays the same if $\beta_1 = 0$

→ **in the scale of the response the effect of the coefficient is non linear**



– $\beta_j < 0$ the exponential is still positive but takes vale between 0 and 1,

– $\beta_j = 0$ there is no effect

– $\beta_j > 0$ odds increase and so does the probability

2.3 Probit model

With the standard ($\sigma^2 = 1$) Gaussian CDF as the response function we obtain

$$\pi_i = h(\eta_i) = \Phi(\mathbf{x}_i'\boldsymbol{\beta})$$

Here the response function is the CDF of the stadard normal

problem need numerical evaluation of $\Phi(\cdot)$

good (easier) conjugate in a Bayesian setting

→ σ in Φ is arbitrary but it has to be fixed for *identifiability* of the model

$$\pi_i = \Phi(\mathbf{x}_i'\boldsymbol{\beta}/\sigma) = \Phi(\mathbf{x}_i'\tilde{\boldsymbol{\beta}})$$

2.4 Complementary log-log model

Adapt in nice situations, obtained by choosing

$$\pi_i = h(\eta_i) = 1 - \exp(-\exp(\eta_i))$$

or, alternatively,

$$\eta_i = g(\pi_i) = \log(-\log(1 - \pi_i))$$

– Probit/Logit give similar results; c-log-log is usually different.

The best model for interpretability is the Logit model

Dataset example: Forbes100

Comparison of coefficients estimates (adjusted columns account for different variances of distributions)

To compare model you have to rescale them (recall: σ^2 is one for the probit)

	Probit	cLog-Log	Logit	Probit (adj.)	cLog-Log (adj.)
(Intercept)	-4.5694	-6.0742	-7.9167	-8.2880	-7.2133
sales	0.0222	0.0214	0.0402	0.0420	0.0309
profits	0.5279	0.6921	0.9212	0.9575	0.8876
assets	0.0046	0.0064	0.0080	0.0084	0.0077
marketvalue	0.0061	0.0054	0.0094	0.0110	0.0070

Comparison of coefficients estimates (with standardized predictors)

you have $\beta = \Delta y \quad when x_i \to x_i + 1$
but you need $\beta = \Delta y \quad when x_i \to x_i + 1$(standard error)

	Probit	cLog-Log	Logit	Probit (adj.)	cLog-Log (adj.)
(Intercept)	0.8189	0.4749	1.4384	1.4854	1.1863
sales	1.0507	1.1396	1.8960	1.9057	1.4616
profits	1.7987	2.3580	3.1389	3.2624	3.0032
assets	1.2515	1.6251	2.1608	2.2699	2.0843
marketvalue	0.3978	0.3545	0.6156	0.7215	0.4547

2.5 A latent utility formulation

Latent is something you do not observe. to which it ocrrespond what you see ($y_i = 0$ and $y_i = 1$)

Some binary regression models can be formulated as **latent linear models** with unobserved utilities as response variables.
→ then you should maximise latent utilities

- $i = 1, \ldots, n$ statistical units
- two alternatives, $y_i = 0$ and $y_i = 1$
- each alternative has an associated utility: u_{i0} and u_{i1} (individual specific)
- in practice, we only observe $(y_i, \mathbf{x}_i)$

Units select the alternatives maximizing **utility differences**

$$\tilde{y}_i = u_{i1} - u_{i0} = \mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i$$

with a thresholding mechanism:

$$\begin{aligned} y_i &= 1 \quad \text{if } \tilde{y}_i = u_{i1} - u_{i0} > 0 \\ y_i &= 0 \quad \text{if } \tilde{y}_i = u_{i1} - u_{i0} \leq 0 \end{aligned}$$

Thus:

$$\pi_i = \mathbb{P}(y_i = 1) = \mathbb{P}(\tilde{y}_i > 0) = \mathbb{P}(\mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i > 0) = \int \mathbb{I}(\mathbf{x}_i'\boldsymbol{\beta} + \varepsilon_i > 0) f(\varepsilon_i) d\varepsilon_i$$

If $\varepsilon_i \sim$ Logistic we have logit, if $\varepsilon_i \sim \mathcal{N}(0, 1)$ we have probit.

3. Grouped data

If there are duplicated rows (called *covariate patterns*) in X, data can be aggregated into $i = 1, \dots, G$ groups. It is very difficult when p is large or x_i are continous.
For each i-th group,

- n_i, number of original units
- $\bar{y}_i$, relative frequency
- $\mathbf{x}_i$, values of covariate pattern

$$y_i, \cdots, y_n \sim Bern(\pi) \quad \rightarrow \frac{\sum_{i=1}^{n_i} y_i}{n_i} \sim \frac{Bin(n_i, \pi)}{n_i}$$

and we now model:

$$\bar{y}_i \sim \text{Bin}(n_i; \pi_i)/n_i$$

which is a (scaled) Binomial distribution, with

$$\mathbb{E}(\bar{y}_i) = \pi_i \qquad \text{and} \qquad \text{Var}(\bar{y}_i) = \frac{\pi_i(1-\pi_i)}{n_i}$$

Grouped data formulation provides:

- data identification protection
- decreased computational efforts (due to the reduction of the dataset dimension)
- some inferential tools otherwise not properly available
- arise usually when most covariates are categorical
- when **all** covariates are categorical, might be presented in table format.

3.0.1 Dataset example: Forbes100

Contingency Table

	Not in US		US	
Top50	Others	Oil & Gas	Others	Oil & Gas
No	28	3	19	0
Yes	18	4	25	3

Note: here you do have counts instead of frequency

Model Output (glm)

```
Call:
glm(formula = cbind(Top50_Yes, Top50_No) ~ OilorNot + USorNot,
    family = binomial(link = logit), data = df)

Deviance Residuals:
   1     2     3     4
 0.1726 -0.4447 -0.1736  1.1232

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.4941     0.2992  -1.651  0.0987 .
OilorNotOil    1.1250     0.7400   1.520  0.1285
USorNotUS      0.8214     0.4173   1.968  0.0491 *

Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7.3143  on 3  degrees of freedom
Residual deviance: 1.5194  on 1  degrees of freedom
AIC: 18.441

Number of Fisher Scoring iterations: 4
```

`family = binomial(link = logit)` → needed to specify what type of mdel we are fitting

`(link = logit)` without any specification R will assume the canonical link fuction:

`gamma` → **inverse** that we need to change in `log`

`normale`

`binomial` → logit

`probit` →

`cbind` → means that you are probably dealing with grouped data

`Null deviance: on 3 degrees of freedom ;1 degrees of freedom` → means that you are probably dealing with grouped data,

`(Dispersion parameter for binomial family taken to be 1)` → means that we are not accounting for over dispersion, 1 is the default value

`Number of Fisher Scoring iterations: 4` → means that it cannot be solved in a close form, in binary regression there is no $X^\top Y X$, but instead an iteration algorithm is needed

`cbind` the response you need to set when dealing with grouped data

3.0.2 su R

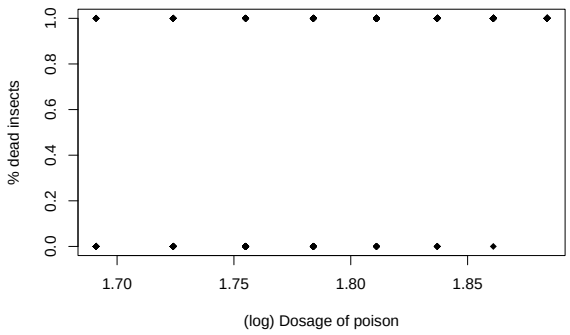
```r
library("car")  #for residuals and QQplots
library(AER)  #contains data "SwissLabor"
library(ResourceSelection)  #if you want Hosmer-Lemeshow test statistic
alpha <- 0.05 #Significance level
```

```r
ds2 <- read.table("Beetles2.txt",header=TRUE)   #che contiene indivisual data
str(ds2)
```

```
 'data.frame':    481 obs. of  2 variables:
  $ x: num  1.69 1.69 1.69 1.69 1.69 ...
  $ y: int  1 1 1 1 1 1 0 0 0 0 ... #indicator binary variable (dead = 1, alive = 0)
```

Uninformative plot

```r
plot(ds2$x,ds2$y, pch=18, col=1, ylab="% dead insects",
     xlab="(log) Dosage of poison")
```



Find a covariate pattern

Per capire quali sono le covariate pattern ci occupiamo della design matrix, **ignorando la response!**le covariate pattern sono unique values nella design matrix.

```r
unique(ds2$x)   #ci trova righe ripetute
```

```
 [1] 1.691 1.724 1.755 1.784 1.811 1.837 1.861 1.884
```

```r
table(ds2$x,ds2$y)
```

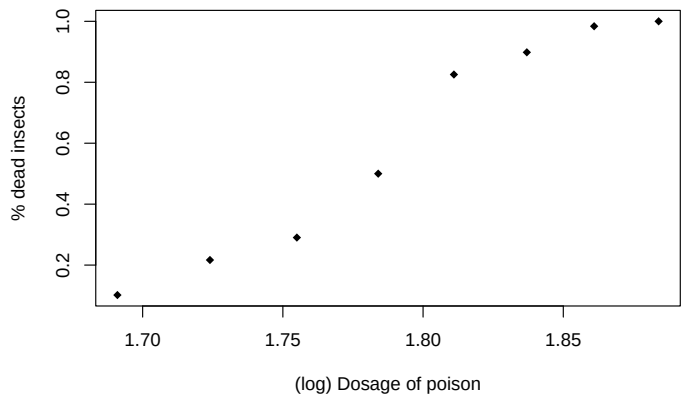
```
        0  1
  1.691 53  6
  1.724 47 13
  1.755 44 18
  1.784 28 28
  1.811 11 52
  1.837  6 53
  1.861  1 61
  1.884  0 60
```

```
'data.frame':    8 obs. of  3 variables:
 $ logdose: num  1.69 1.72 1.75 1.78 1.81 ...  # logarithm of poison dosage
 $ n      : int  59 60 62 56 63 59 62 60       # number of exposed insects to poison
 $ dead   : int  6 13 18 28 52 53 61 60        # number of dead insects
```

8 obs. of 3 variables→ questo ci fa capire che a meno di cose strane questi sono groped data ds abbiamo 8 covariate pattern che sono i repeated values del logdose

Anche se vedi dei numeri e non 0 e 1 questa è una binary regression, non sono $(0, 1)$ perchè si tratta di grouped data, quindi abbiamo le frequenza, non successo insuccesso.

```
plot(ds$logdose,ds$dead/ds$n, pch=18, col=1, ylab="% dead insects",
     xlab="(log) Dosage of poison")
```



Questo grafico ha senso solo se si lavora con grouped data, notiamo che esiste una relazione tra la x e la y(`n`, `log-dose`), questo però non è approssimabile con una linea retta perchè come si nota da un certo dosaggio in poi questo ha un effetto decresccente.

```
mod_logistic<-glm(cbind(dead,n-dead) ~ logdose,
                  data=ds,
                  family=binomial)
```

Note:

1. Questi non hanno close form solutions → sono fitted da Fisher scoring o IWLS
2. in case of grouped data, formula of glm requires on the LHS a matrix with two columns: 1st, number of successes; 2nd, number of failures (è cbind(dead,n-dead), che è una matrice);

```
cbind(ds$dead,ds$n - ds$dead)
```

```
     [,1] [,2]
[1,]    6   53
[2,]   13   47
[3,]   18   44
```

```
[4,]   28   28
[5,]   52   11
[6,]   53    6
[7,]   61    1
[8,]   60    0
```

```
summary(mod_logistic)
```

```
Call:
glm(formula = cbind(dead, n - dead) ~ logdose, family = binomial,
    data = ds)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -60.740      5.182  -11.72   <2e-16 ***
logdose       34.286      2.913   11.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 284.202  on 7  degrees of freedom
Residual deviance:  11.116  on 6  degrees of freedom
AIC: 41.314

Number of Fisher Scoring iterations: 4
```

Intercept interpretation
Dal momento che **usiamo la logit** *trasformazione logaritmica*, cambia l'interpretazione dell'intercetta → non è più il livello medio della y quando la covariate è 0, ma è l'effetto sui *log-odds* quando .... la covariata è uguale a 1?

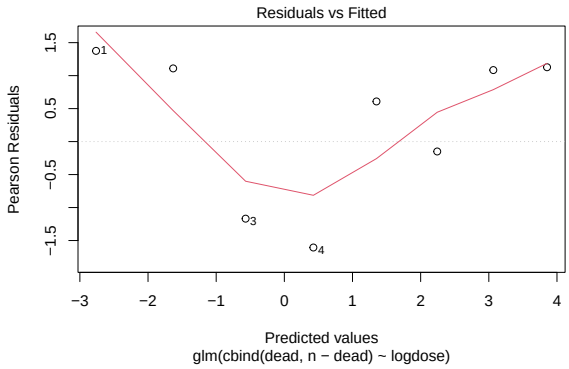
Cosa ci fa capire che è grouped data

1. `cbind(dead, n - dead)`
2. `8 obs.  of 3 variable`
3. `Null deviance:  284.202 on 7 degrees of freedom` $7 = G - 1$ (the intercept), il Null model predice lo stesso valore per tutti
4. `Residual deviance:  284.202 on 5 degrees of freedom` $5 = G - p$

3.0.3 Residuals: visual inspection

Questo tipo di analisi è valida e si può fare solo sui grouped data

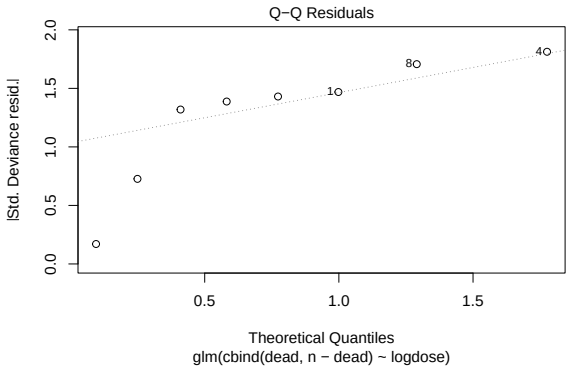
```
plot(mod_logistic)
```



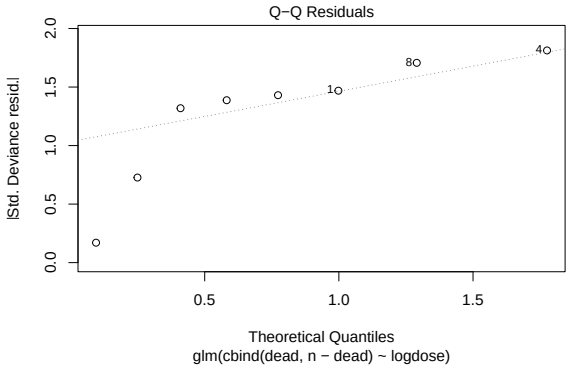
Here we observe predicted residual values vs pearson residuals (a type of residuals).

Possimao notare che la linea rossa non è retta, il che significa che non stiamo ignorando qualcosa nello spiegare i dati, per esempio un effetto non lineare della covariata

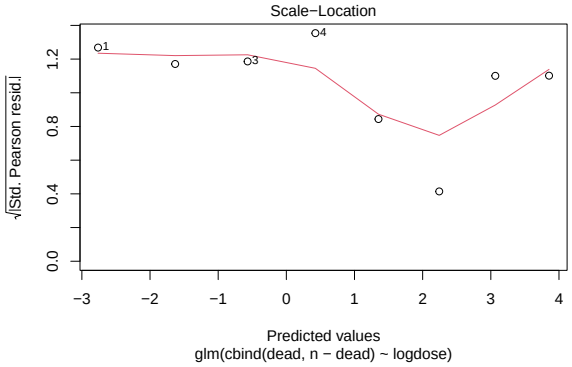
```
plot(mod_logistic, which=2)
```

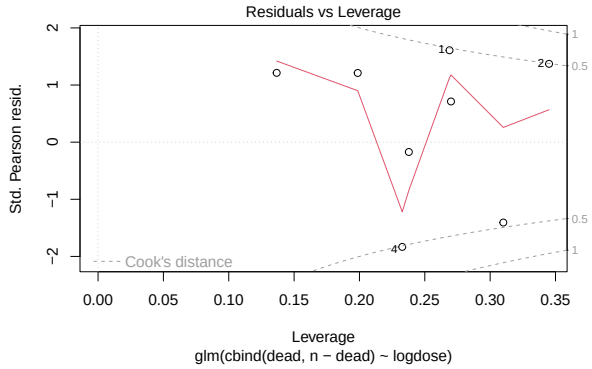


Standardize deviance residuals sono 8, che è la grandezza della nostra popolazione → ci sono troppi pochi punti per ricostruire una quantile per la distribuzione



La cosa più utile che possiamo fare è attenzionare lo scale location plot, dal quale possimao desumere che è necessario introdurre un termine quadratico nel modello


Scale−Location
√|Std. Pearson resid.|
0.0
0.4
0.8
1.2
1
4
3
−3
−2
−1
0
1
2
3
4
Predicted values
glm(cbind(dead, n − dead) ~ logdose)


Residuals vs Leverage
Std. Pearson resid.
2
1
0
−1
−2
1
2
4
1
0.5
0.5
1
Cook's distance
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Leverage
glm(cbind(dead, n − dead) ~ logdose)

3.1 Overdispersion

Empirical variance > theoretical variance $= \hat{\pi}_i(1-\hat{\pi}_i)/n_i, \quad \hat{\pi}_i = h(\mathbf{x}_i'\boldsymbol{\beta})$

Which means that you observe more variability than what you would expect from the model, the problem is that in the logit model thhere is only one parameter for the variance and the means.

Why there is overdispersion:

- unobserved heterogeneity
- positive correlations between individual y_i

Quick fix: introduce overdispersion parameter ϕ

With grouped data we can estimate the variance within each group: $\bar{y}_i(1-\bar{y}_i)/n_i$ with $\bar{y}_i$ being ML estimator for π_i. then you correct the estimated theoretical variance for a **scaling parameter**

$$\text{Var}(y_i) = \phi \cdot \frac{\pi_i(1-\pi_i)}{n_i}$$

- ϕ can be estimated
- **affects the variance of the estimator of $\boldsymbol{\beta}$**

Note: Tutti i test funzionano sui grouped data se hai almeno 5 osservazioni.

output

Contingency Table

	Not in US		US	
	Others	Oil & Gas	Others	Oil & Gas
Top50 = No	28	3	19	0
Top50 = Yes	18	4	25	3

Model Output (glm with quasibinomial)

```
Call:
glm(formula = cbind(Top50_Yes, Top50_No) ~ OilorNot + USorNot,
    family = quasibinomial(link = logit), data = df)

Deviance Residuals:
   1     2     3     4
 0.1726 -0.4447 -0.1736  1.1232

Coefficients:
```

```
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    -0.4941     0.2991  -1.650  0.302
OilorNotOil     1.1250     0.7471   1.507  0.365
USorNotUS       0.8214     0.4102   2.002  0.295

(Dispersion parameter for quasibinomial family taken to be 0.9661612)

    Null deviance: 7.3143  on 3  degrees of freedom
Residual deviance: 1.5194  on 1  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 4
```

`family =quasibinomial(link = logit)` → Solution for the overdispersion

`Dispersion parameter for quasibinomial family taken to be 0.9661612` → it is almost one which means the theoretical variance was nice

`AIC: NA` due to accounting for overdispersion you do not have the likelihood anymore and so you, cannot compute the AIC.

3.2 Model Comparison

3.2.1 Akaike criterion

→ per comparare modelli diversi

```
AIC(mod_logistic)
 [1] 41.31361

extractAIC(mod_logistic)

 [1]  2.00000 41.31361
```

predict

```
predict(mod_logistic)
```

predict in generale da le predizioni nella linear predictor scale (estimated betas) se vuoi cambiarli: è possibile specificare nella link scale (scala della link function) o nella scala della response inserendo `type =‘‘response’’` adesso abbiamo probabilità.

Ungroped data è la situazione estrema in cui ogni osservazione è un gruppo con se stessa

```
mod_logistic2<-glm(cbind(y,1-y)~x,data=ds2,family=binomial)
```

Puoi trattare gli ungrouped data come grouped data inserendo la matrice di successi e insuccessi che in questo caso sarà $(0, 1)$ o viceversa, percè in questo caso consideri una singola osservazione per gruppo.

```
mod_logistic3<-glm(y~x,data=ds2,family=binomial)
```

questi due sono ungroped esattamente uguali

```
summary(mod_logistic) #grouped data
```

```
Call:
glm(formula = cbind(dead, n - dead) ~ logdose, family = binomial,
    data = ds)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -60.740      5.182  -11.72   <2e-16 ***
logdose       34.286      2.913   11.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 284.202  on 7  degrees of freedom
Residual deviance:  11.116  on 6  degrees of freedom
AIC: 41.314

Number of Fisher Scoring iterations: 4
```

```
summary(mod_logistic2) #fake grouped data
```

```
Call:
glm(formula = cbind(y, 1 - y) ~ x, family = binomial, data = ds2)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -60.740      5.182  -11.72   <2e-16 ***
x             34.286      2.913   11.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 645.44  on 480  degrees of freedom
Residual deviance: 372.35  on 479  degrees of freedom
AIC: 376.35

Number of Fisher Scoring iterations: 5
```

```
summary(mod_logistic3) #ungrouped data
```

```
Call:
glm(formula = y ~ x, family = binomial, data = ds2)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -60.740      5.182  -11.72   <2e-16 ***
x             34.286      2.913   11.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 645.44  on 480  degrees of freedom
Residual deviance: 372.35  on 479  degrees of freedom
AIC: 376.35

Number of Fisher Scoring iterations: 5
```

nota bene (potenziale domanda esame
Se cambi la scala dei dati non puoi più comparare l'AIC!, un'altra situazione analoga è la situazione nella quale si vuole comparare Gamma Regression con la gaussian Linear regression in un caso la response è y, nell'altra è $\log(y)$, comparare l'AIC NON ha senso

3.2.2 Regarding Test and statistics

Tutti i test funzionano sui grouped data se hai almeno 5 osservazioni. Se fai un test su questa situazione invece non c'è garanzia che la distribuzione si chi2.

Inference and estimates are equivalent... however, deviances and information criterion are different! The 'appropriate' results for deviance and gof quantities are those coming from grouped data

```
plot(mod_logistic2)
```

4. Maximum likelihood estimation

Assuming conditional independence of the responses

$$\mathcal{L}(\boldsymbol{\beta}) = \prod_{i=1}^{n} f(y_i|\boldsymbol{\beta}) \quad \rightarrow \quad \ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \log(f(y_i|\boldsymbol{\beta})) \quad \text{(to be maximized)}$$

No closed form solution: we use iterative methods (mainly, Fisher scoring)

4.1 Individual data

$$\mathcal{L}(\boldsymbol{\beta}) = \prod_{i=1}^{n} \pi_i^{y_i}(1-\pi_i)^{1-y_i} \quad \rightarrow \quad \ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \ell_i(\boldsymbol{\beta}) \quad \rightarrow \quad \left[y_i \log\left(\frac{\pi_i}{1-\pi_i}\right) + \log(1-\pi_i) \right]$$

Recall:

$$g(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

which means, for the logit model:

$$\ell_i(\boldsymbol{\beta}) = y_i(\mathbf{x}_i^\top \boldsymbol{\beta}) - \log\left(1 + \exp(\mathbf{x}_i^\top \boldsymbol{\beta})\right)$$

Score function

$$\mathbf{s}(\boldsymbol{\beta}) = \frac{\partial \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{n} \frac{\partial \ell_i(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \sum_{i=1}^{n} \mathbf{s}_i(\boldsymbol{\beta})$$

Score and information matrix The best models in term of goodness of fit is the **saturated model** where the parameter is gorup specific $(y_i - \pi_i)$

For the logistic regression:

$$\mathbf{s}_i(\boldsymbol{\beta}) = \mathbf{x}_i(y_i - \pi_i) \quad \Rightarrow \quad \mathbf{s}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i(y_i - \pi_i)$$

ML equations:

$$\mathbf{s}(\hat{\boldsymbol{\beta}}) = \sum_{i=1}^{n} \mathbf{x}_i \left(y_i - \frac{\exp(\mathbf{x}_i\hat{\boldsymbol{\beta}})}{1 + \exp(\mathbf{x}_i\hat{\boldsymbol{\beta}})} \right) = 0 \quad \text{with } \hat{\boldsymbol{\beta}} \in \mathbb{R}^p$$

Since it's a nonlinear system of equations, we solve with Fisher scoring or Newton-Raphson, which need either the

– **Observed information matrix**

$$H(\boldsymbol{\beta}) = -\frac{\partial^2 \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'}$$

– **or Fisher matrix** (a.k.a., expected information matrix)

$$F(\boldsymbol{\beta}) = \mathbb{E}\left(-\frac{\partial^2 \ell(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'}\right) = \mathrm{Cov}\left(\mathbf{s}(\boldsymbol{\beta})\right) = \mathbb{E}(\mathbf{s}(\boldsymbol{\beta})\mathbf{s}(\boldsymbol{\beta})')$$

(since $\mathbb{E}(\mathbf{s}(\boldsymbol{\beta})) = 0$ *due to* $\mathbb{E}(y_i) = \pi_i$*)*

4.1.1 Iterative computation for MLE

Fisher scoring algorithm

$$\hat{\boldsymbol{\beta}}^{(t+1)} = \hat{\boldsymbol{\beta}}^{(t)} + F^{-1}(\hat{\boldsymbol{\beta}})\mathbf{s}(\hat{\boldsymbol{\beta}}), \quad t = 0, 1, 2, \ldots$$

until convergence criterion is met (usually, relative change below threshold).

– Invertibility of $F(\hat{\boldsymbol{\beta}})$ is crucial for convergence to ML estimate
  ◦ For our regression settings, it means full rank of design matrix: $\mathrm{rank}(\mathbf{X}) = p$
  ◦ When n is small and p large, divergence issues of Fisher scoring algorithm
  ◦ For logit, $F(\boldsymbol{\beta}) = H(\boldsymbol{\beta})$, so Fisher = Newton algorithm
  ◦ For probit / c-log-log, $F(\boldsymbol{\beta})$ is used (easier to compute)
  ◦ For grouped data, derivations are analogous
– $F^{-1}(\hat{\boldsymbol{\beta}})$ tell the procedure how much to move, the larger the variance of β of which $F^{-1}(\hat{\boldsymbol{\beta}})$ is an estimate
– $\mathbf{s}(\hat{\boldsymbol{\beta}})$ the updating factor, gives the direction of the updating

We use MLE estimates for:

4.1.2 Asymptotic properties of ML estimator

1. $\tilde{\boldsymbol{\beta}} \approx \mathcal{N}(\boldsymbol{\beta}, A)$ can be approximated with a gaussian
2. asymptotically consistent
3. centered on the true parameter due to SLLN
4. $A = \mathrm{Cov}(\tilde{\boldsymbol{\beta}}) = F^{-1}(\tilde{\boldsymbol{\beta}}), \quad a_{jj} = \widehat{\mathrm{Var}}(\tilde{\beta}_j)$

5. Hypotheses testing

Linear hypotheses

(same as gaussian linear neted models)

$$H_0 : C\boldsymbol{\beta} = d \quad \text{vs.} \quad H_1 : C\boldsymbol{\beta} \neq d \quad \text{with row rank}(C) = r \leq p$$

Three options:

1. **Likelihood Ratio (LR) test**:

$$\text{lr} = -2[\ell(\hat{\boldsymbol{\beta}}_{H0}) - \ell(\hat{\boldsymbol{\beta}})]$$

→ here you need to fit **both model**

2. **Wald statistic**: no need to fit the restricted (sub)model

$$W = (C\hat{\boldsymbol{\beta}} - d)^\top [CF^{-1}(\hat{\boldsymbol{\beta}})C^\top]^{-1}(C\hat{\boldsymbol{\beta}} - d)$$

→ here you need only the **unrestricted model**

3. **Score statistic**: need to fit only simpler/restricted model

$$u = \mathbf{s}^\top(\hat{\boldsymbol{\beta}}_{H0})F^{-1}(\hat{\boldsymbol{\beta}}_{H0})\mathbf{s}(\hat{\boldsymbol{\beta}}_{H0})$$

→ here you need to fit just the **restricted model**

Under mild conditions, asymptotically the three tests: $\text{lr}, w, u \sim \chi^2_r$

5.0.1 R implementation

Quale test usare?

- Se non ci sono problemi con fittare il modelli con i dati usiamo o LRT o il χ^2
- Se hai **quasimodels** e gaussian (dove devi stimare il dispersion parameter) puoi usare l'F test, è il più appropriato.(Per il gaussian sono tutti corretti e la differenza tra guesti modelli è praticamente 0)
- Quando hai problemi nel fittare modelli perché sono troppo grandi → l'optimize function è lo score test, questa è per le situazioni con grandi modelli perchè ha bisogno solo del modello restricted (che è quello più semplice)

Anova function

`anova` function → Compute an analysis of deviance table for one or more generalized linear model fits. Anche in questo caso la comparazione è con il `Null Model`

- **senza specifiche:** avrai solo residual deviance.
- χ^2 o Pearson statistics

```r
anova(mod_logistic, test="Chisq")
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(dead, n - dead)

Terms added sequentially (first to last)


        Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                        7    284.202
logdose  1   273.09         6     11.116 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

– Likelihood Ratio test

```r
anova(mod_logistic, test="LRT")   #log lik ratio test
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(dead, n - dead)

Terms added sequentially (first to last)


        Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                        7    284.202
logdose  1   273.09         6     11.116 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

– score test

```r
anova(mod_logistic, test="Rao")   #guarda su help che cosa è
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(dead, n - dead)

Terms added sequentially (first to last)


        Df Deviance Resid. Df Resid. Dev   Rao  Pr(>Chi)
NULL                        7    284.202
logdose  1   273.09         6     11.116 227.6 < 2.2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

If we are testing a quasi-model the F-test is the best one.

```
anova(mod_logistic, test="F")  #check the warning!

 Warning in anova.glm(mod_logistic, test = "F"): utilizzare il test F con una
 famiglia 'binomial' è inappropriato


 Analysis of Deviance Table

 Model: binomial, link: logit

 Response: cbind(dead, n - dead)

 Terms added sequentially (first to last)


         Df Deviance Resid. Df Resid. Dev      F    Pr(>F)
 NULL                        7    284.202
 logdose  1   273.09         6     11.116 273.09 < 2.2e-16 ***
 ---
 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA limitation

When there is a large number of predictors the sequential nature of `anova()` becomes inadequate. Specifically:

In such cases, it is recommended to compare models by directly computing the Deviance, and assess statistical significance using the χ^2 distribution with degrees of freedom equal to the difference in the number of estimated parameters.

Dataset example: Forbes100

5.0.2 Tests: LRT, Wald, Score

Analysis of Deviance Table

```
Model 1: cbind(Top50_Yes, Top50_No) ~ USorNot
Model 2: cbind(Top50_Yes, Top50_No) ~ OilorNot + USorNot

  Res.Df Resid. Dev Df Deviance Pr(>Chi)
1      2     4.0445
2      1     1.5194  1   2.5251   0.112
```

Linear hypothesis test

```
Hypothesis:
OilorNotOil & Gas = 0

Model 1: restricted model
Model 2: cbind(Top50_Yes, Top50_No) ~ OilorNot + USorNot

  Res.DF Df Chisq Pr(>Chisq)
1      2
2      1  1 2.311   0.1285
```

Analysis of Deviance Table (Score test)

```
Model 1: cbind(Top50_Yes, Top50_No) ~ USorNot
Model 2: cbind(Top50_Yes, Top50_No) ~ OilorNot + USorNot

  Res.Df Resid. Dev Df Deviance   Rao Pr(>Chi)
1      2     4.0445
2      1     1.5194  1   2.5251  2.474   0.1157
```

5.1 Grouped data

5.1.1 Model fit

Saturated model: "best fit" to the data because each group has specific parameter $\rightarrow \pi_i$ estimated with $\bar{y}_i$

Two options to measure 'departure' between fit model (π_i) vs. saturated ($\bar{y}_i$):

- **Pearson statistic**
$$\chi^2 = \sum_{i=1}^{G} \frac{(\bar{y}_i - \hat{\pi}_i)^2}{\hat{\pi}_i(1-\hat{\pi}_i)/n_i}$$
- **Deviance** , a more general approach that compares your model fit with the best fit possible (the one from the saturated model)
$$D = -2\sum_{i=1}^{G} [\ell_i(\hat{\pi}_i) - \ell_i(\bar{y}_i)]$$
- χ^2 and D are different in finite samples but equivalent in asymptotics and for grouped data
- If n_i is large enough for every group, both are distributed as Chi-square distribution with $G - p$ degrees of freedom

To balance goodness-of-fit and model complexity we can use instead:

$$\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}) + 2p$$

- Generalization of R^2 from linear models is available as Nagelkerke's R^2

5.1.1.1 su R

Può essere testato usando il **Deviance test** che considera la residual deviance distrbuita come χ^2_r con r gradi di libertà. $r = G - p$ in questo caso $8 - 2 = 6$. Va implementato a mano

```
deviance(mod_logistic)
 [1] 11.11558  quella del summary di sopra

summary(mod_logistic)$df.residual
 [1] 6

qchisq(alpha,summary(mod_logistic)$df.residual,lower.tail=FALSE)
 [1] 12.59159  valore del test

pchisq(deviance(mod_logistic),summary(mod_logistic)$df.residual,lower.tail=FALSE)
 [1] 0.08486944
```

Essendo che il numero ottenuto è superiore a α allora accettiamo $H_0 \rightarrow$ preferisco il `Null Model` la riduzione nella devianza non è significativa.

è possibile usare anche la funzione anova, which (without specification) will provides same results as the Deviance measure.

```
anova(mod_logistic)
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(dead, n - dead)

Terms added sequentially (first to last)


        Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                        7    284.202
logdose  1   273.09         6     11.116 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.1.2 Estimating overdispersion

For grouped data, allowing overdispersion means:

$$\text{Var}(\bar{y}_i) = \phi \cdot \frac{\pi_i(1-\pi_i)}{n_i}$$

where $\text{Var}(\bar{y}_i)$ is the empirical variance.

which can be estimated with the two goodness-of-fit statistics:

$$\hat{\phi}_P = \frac{1}{n-p}\chi^2 \qquad \text{or} \qquad \hat{\phi}_D = \frac{1}{n-p}D$$

similar to linear models using average $\frac{RSS}{n-p}$.

– We need to adjust estimated covariance matrix of coefficients:

$$\widehat{\text{Cov}}(\boldsymbol{\beta}) = \hat{\phi}F^{-1}(\boldsymbol{\beta})$$

In general, this is a **quasi-likelihood** approach (not using the "true" likelihood).

When you account for over dispersion you change the var-cov matrix of $\boldsymbol{\beta}$, this affect confidence intervals, for example making them bigger and so it will contain the 0 even if the coefficient was significative.

5.2 Individual data

```
 Another dataset with 'sparser' data
```

```
# #?SwissLabor
data(SwissLabor)
str(SwissLabor)
```

```
 'data.frame':    872 obs. of  7 variables:
  $ participation: Factor w/ 2 levels "no","yes": 1 2 1 1 1 2 1 2 1 1 ...
  $ income       : num  10.8 10.5 11 11.1 11.1 ...
  $ age          : num  3 4.5 4.6 3.1 4.4 4.2 5.1 3.2 3.9 4.3 ...
  $ education    : num  8 8 9 11 12 12 8 8 12 11 ...
  $ youngkids    : num  1 0 0 2 0 0 0 0 0 0 ...
  $ oldkids      : num  1 1 0 0 2 1 0 2 0 2 ...
  $ foreign      : Factor w/ 2 levels "no","yes": 1 1 1 1 1 1 1 1 1 1 ...
```

Studiamo la correlazione tra i predittori continui.

```
predictors <- SwissLabor[,-1]   #senza response variable
round(cor(predictors[,-6]), 2)   #correlation matrix
```

```
           income   age education youngkids oldkids
 income      1.00  0.01      0.33     -0.02    0.14
 age         0.01  1.00     -0.15     -0.52   -0.12
 education   0.33 -0.15      1.00      0.10   -0.04
 youngkids  -0.02 -0.52      0.10      1.00   -0.24
 oldkids     0.14 -0.12     -0.04     -0.24    1.00
```

Si tratta di correlazioni basse in questo caso quindi vanno bene

```
cov.patterns<-unique(predictors)
nrow(cov.patterns)
 [1] 872

nrow(SwissLabor)
 [1] 872
```

Ungroupable data!

```
mod <- glm(participation~.,data=SwissLabor,family=binomial) #usiamo tutte le colonne
summary(mod)
```

```
 Call:
 glm(formula = participation ~ ., family = binomial, data = SwissLabor)

 Coefficients:
             Estimate Std. Error z value Pr(>|z|)
 (Intercept) 10.37435    2.16685   4.788 1.69e-06 ***
```

```
income       -0.81504    0.20550  -3.966 7.31e-05 ***
age          -0.51033    0.09052  -5.638 1.72e-08 ***
education     0.03173    0.02904   1.093    0.275
youngkids    -1.33072    0.18017  -7.386 1.51e-13 ***
oldkids      -0.02199    0.07377  -0.298    0.766
foreignyes    1.31040    0.19976   6.560 5.38e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

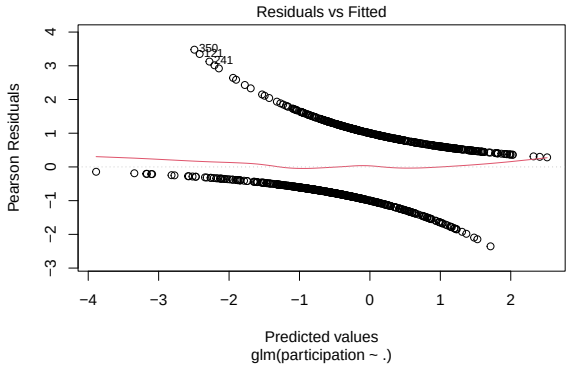
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 1203.2  on 871  degrees of freedom
Residual deviance: 1052.8  on 865  degrees of freedom
AIC: 1066.8

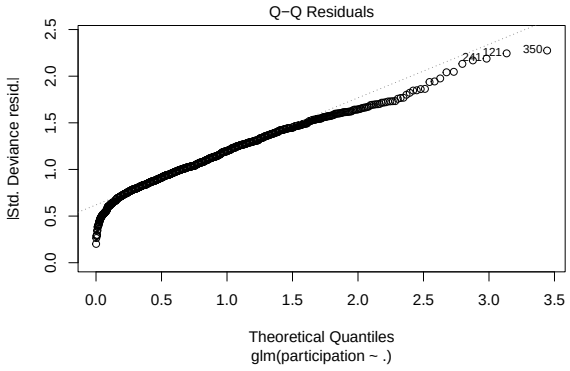
Number of Fisher Scoring iterations: 4
```

5.2.1 Residuals: visual inspection

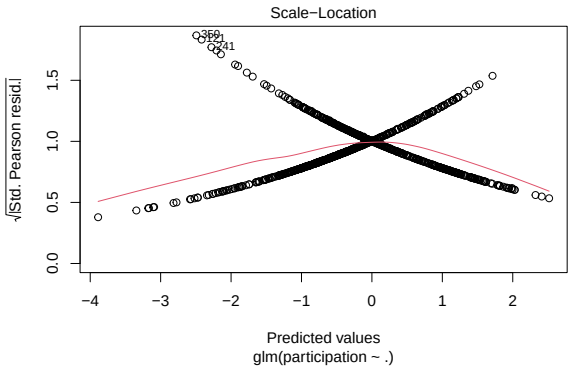
```
plot(mod, which=1)
```




```
plot(mod, which=2)    #non attendibili
```



```
plot(mod, which=3)
```



le linee si attraversano Quindi residual anaylisis sono davvero poco utili

5.2.2 Model fit

```
deviance(mod)
 [1] 1052.798

summary(mod)$df.residual
 [1] 865

qchisq(0.05,summary(mod)$df.residual,lower.tail=FALSE) #p-value for Deviance test
 [1] 934.5329

pchisq(deviance(mod),summary(mod)$df.residual,lower.tail=FALSE) #non è attendibile!
 [1] 1.117042e-05
```

5.2.3 On the deviance and Pearson statistic: Hosmer-Lemeshow Test

Grouping data is almost impossible when predictors are continuous/too many $\rightarrow \chi^2$ and D not reliable

Hosmer-Lemeshow Test: group according to fitted values (it actually find a way to build groups):

1. order y_i according to $\hat{\pi}_i$ trough structuring the model we find similarity among covariates
2. form N (usually 10) equally sized groups
3. first group contains $\approx n/N$ observations with smallest predicted probabilities
4. ...same for all the others!

$$\chi^2_{HL} = \sum_{i=1}^{N} n_i \frac{(\bar{y}_i - \hat{\pi}_i)^2}{\hat{\pi}_i(1-\hat{\pi}_i)} \overset{\text{asymp.}}{\approx} \chi^2 \text{ distribution with df} = N - 2$$

where $\hat{\pi}_i$ denotes average estimated probability in group i

– Still has issues: unstable results, moderate power.

Using HL (hosmer-lesmeshov)

```
# #?hoslem.test
#?predict.glm
```

```
testHL<-hoslem.test(SwissLabor$participation=="yes",
                    predict(mod,type="response"))    #prediction                    nel
testHL
```

```
  Hosmer and Lemeshow goodness of fit (GOF) test

 data:  SwissLabor$participation == "yes", predict(mod, type = "response")
 X-squared = 4.2006, df = 8, p-value = 0.8386
```

non dobbiamo rifiutare ho, ma qual è l'ipotesi nulla #? (guarda help)

```
testHL$observed
```

```
 cutyhat          y0 y1
   [0.0201,0.222] 74 14
   (0.222,0.291]  64 23
   (0.291,0.337]  64 23
   (0.337,0.383]  57 30
   (0.383,0.431]  52 35
   (0.431,0.484]  43 44
   (0.484,0.564]  40 47
   (0.564,0.658]  35 52
   (0.658,0.758]  29 58
   (0.758,0.925]  13 75
```

```
testHL$expected
```

```
 cutyhat             yhat0    yhat1
   [0.0201,0.222] 75.56766 12.43234
   (0.222,0.291]  64.17984 22.82016
   (0.291,0.337]  59.68108 27.31892
   (0.337,0.383]  55.66075 31.33925
   (0.383,0.431]  51.65704 35.34296
   (0.431,0.484]  47.44441 39.55559
   (0.484,0.564]  41.39471 45.60529
   (0.564,0.658]  33.80097 53.19903
   (0.658,0.758]  25.06687 61.93313
   (0.758,0.925]  16.54667 71.45333
```

sono i fake groups che abbiamo creato → [0.0201,0.222] sono 10 fake gropus hosmer - lesmeshov è chi-2 test quando compari predictive unobserved la corrispondente test statistic è: X-squared = 4.2006, df = 8, p-value = 0.8386

5.3 Binary regression issues

- **unbalanced data:**
  The same β does predict both 0s and 1s, but when the proportion is unbalanced it is hard that the vector of parameter $\boldsymbol{\beta}$ succesfully represebt both situation **solution:**
  - reducing 0s $\rightarrow$ subsample and reducing the sample size
  - increasing 1s$\rightarrow$ resample and increasing the sample size (adding fake information)
- perfect separability
- if $\hat{\pi}_i$ are close to 0 or to 1, than you will have an inflated standard error because you get closer to the saturated model

Part VII

Count data regression

6. Models for count data

- For each individual row, how many times the events of reference happen with respect to the covariates?
- Is there some covariate that influences the number of events?

We will model the expected counts of events, i.e.,

When we model the number of times an event happens in a fixed time frame or any count of objects, we need to deal with **count data**(use Poisson or Negative Binomial).

Very important: if you chose the Poisson then you are assuming $\lambda_i = \mathbb{E}(y_i) = \text{Var}(y_i)$ with $y_i \in \{0, 1, 2, \dots\}$

- **Linear Poisson model:**

  $$\lambda_i = \eta_i = \mathbf{x}_i'\boldsymbol{\beta}$$

  (same limitation as linear probability model): does not ensure that η_i stays in the domanain of the response.

- **Log-linear Poisson model:**

  $$\lambda_i = \exp(\eta_i) = \exp(\beta_0)\exp(\beta_1 x_{i1}) \cdots \exp(\beta_k x_{ik}) \quad \text{or} \quad \log(\lambda_i) = \eta_i = \mathbf{x}_i'\boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}$$

  usefull for modelling rates, here the effect become multiplicative on λ, not on the odds

Overdispersion: frequently, empirical variance $\gg$ expected variance.

Two options:

- introduce ϕ to have $\text{Var}(y_i) = \phi\lambda_i$ (quasi-likelihood) *with issue: no AIC, inflating standard errors, increasing var-cov matrix of $\boldsymbol{\beta}$*
- using **negative Binomial** $\rightarrow$ it has one parameter for variance and one for expectation, so you can model directly in the variance.
- **Random Effects** $\rightarrow$ In general, come from data with **extra variability** which might come from grouped data or unobserved extra effects. $\Rightarrow$ So including **R.E.** could help in handling **overdispersion**.

6.1 Maximum Likelihood

We assume (conditional) independence between observed responses y_i,

$$f(y_i \mid \mathbf{x}_i, \boldsymbol{\beta}) = \frac{\lambda_i^{y_i}\exp(-\lambda_i)}{y_i!}$$

Log-likelihood (log-linear Poisson model):

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} \ell_i(\boldsymbol{\beta}) \propto \sum_{i=1}^{n}(y_i \log(\lambda_i) - \lambda_i) = \sum_{i=1}^{n}(y_i\eta_i - \exp(\eta_i))$$

(we ignore additive constant $\sum_{i=1}^{n} \log(y_i!)$*)*

Score function:

$$\mathbf{s}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i(y_i - \lambda_i) = \sum_{i=1}^{n} \mathbf{x}_i(y_i - \exp(\eta_i))$$

we are summing the difference between prediction and observation weighted by the regressor, (recall the saturated model :$(y_i = \lambda_i$ *)* **Fisher information**:

$$F(\boldsymbol{\beta}) = \mathbb{E}(\mathbf{s}(\boldsymbol{\beta})\mathbf{s}(\boldsymbol{\beta})') = \sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i'\lambda_i$$

Estimates computed with Fisher scoring; asymptotically $\hat{\boldsymbol{\beta}} \sim \mathcal{N}(\boldsymbol{\beta}, F^{-1}(\hat{\boldsymbol{\beta}}))$

6.2 Model fit, model choice, quasi-likelihood

- **Linear hypothesis:** same approach as binary regression (*LR, Wald, Score statistics*)
- **Goodness-of-fit:** Pearson statistic (needs grouped data)

$$\chi^2 = \sum_{i=1}^{G} \frac{(y_i - \hat{\lambda}_i)^2}{\hat{\lambda}_i / n_i}$$

  same as for Binary regression , but here the theoretical variance is $\hat{\lambda}_i$
- **Model choice:** AIC (with $0 \cdot \log(0) = 0$ when $y_i = 0$)
- **Overdispersion:** $\text{Var}(y_i \mid \mathbf{x}_i) = \phi\lambda_i$ using a quasi-likelihood approach:

$$\hat{\phi}_P = \frac{1}{n-p}\chi^2 \qquad \text{or} \qquad \hat{\phi}_D = \frac{1}{n-p}D$$

  and adjusting

$$\widehat{\text{Cov}}(\boldsymbol{\beta}) = \hat{\phi}F^{-1}(\boldsymbol{\beta})$$

sectionPoisson data with offset

6.2.1 Modello con dist: rural categoria di riferimento

```
# ?glm
```

```
m1<-glm(c~offset(log(n))+car+dist+age, data=claims, family=poisson)
# non c'è overdispersion perchè non c'è quasipossion
```

Usare `offset(log(n))` è equivalent a dividere ogni each `c` (numero di claims) per il corrispondente `n` (numero di policy-holders). Dal momento che è una trasformazione logaritmica, per includerla come offset devi condiderare il logaritmo del numero di osservazioni in modo da farlo diventare **denominatore del linear predictor**.

```r
summary(m1)
```

```
Call:
glm(formula = c ~ offset(log(n)) + car + dist + age, family = poisson,
    data = claims)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.58753    0.09276 -17.115  < 2e-16 ***
car1-1.5     0.16134    0.05053   3.193 0.001409 **
car1.5-2     0.39281    0.05500   7.142 9.18e-13 ***
car>2        0.56341    0.07232   7.791 6.65e-15 ***
dist1       -0.23421    0.06167  -3.798 0.000146 ***
dist2       -0.20834    0.06476  -3.217 0.001294 **
dist3       -0.19568    0.06984  -2.802 0.005081 **
age25-29    -0.19101    0.08286  -2.305 0.021149 *
age30-35    -0.34495    0.08137  -4.239 2.24e-05 ***
age>35      -0.53667    0.06996  -7.672 1.70e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 236.26  on 63  degrees of freedom
Residual deviance:  51.42  on 54  degrees of freedom
AIC: 388.74

Number of Fisher Scoring iterations: 4
```

Coefficient Interpretation
Per interpretare correttamente i coefficienti di un modello che ha come link function il logaritmo (per esempio Poisson), è necessario prendere l'esponenziale dei coefficienti. Il segno del coefficiente ci fornisce un'indicazione sull'effetto della variabile: un segno positivo implica un aumento del tasso di eventi, uno negativo una diminuzione.

Nel nostro caso, solo una delle categorie della variabile dist risulta significativa. Questo NON implica automaticamente che possiamo eliminare i livelli non significativi: potremmo considerare di collassare alcune categorie, ma ciò richiede cautela e una motivazione chiara, quindi non è sempre consigliabile.

Se almeno un livello di una variabile categorica è significativo, è buona norma mantenere tutta la variabile nel modello. Un'alternativa utile può essere modificare la categoria di riferimento, perché la **significatività** dei coefficienti **può dipendere** da quale livello è utilizzato come **categoria di riferimento**.

Nota: i coefficienti dei singoli livelli variano (in grandezza, segno e significatività) a seconda della categoria di riferimento scelta.

6.3 Goodness of fit

Deviance residuals

```
res.m1.dev<-residuals(m1,type="deviance")
sum(res.m1.dev^2)
```

```
 [1] 51.42003
```

Pearson's residuals

```
res.m1.p<-residuals(m1,type="pearson")
sum(res.m1.p^2)
```

```
 [1] 48.62934
```

Nota: se non eleviamo al quadrato, sommando solamente i residui otterremo 0 (per costruzione). Una volta ottenute queste statistiche inoltre siamo in grado di ricostruire l'overdispersion parameter.

6.3.1 Deviance

Queste misure hanno senso quando solamente quando abbiamo un buon numero di osservazioni in ogni gruppo, altrimenti non sono molto attendibili

```
# summary(m1)$deviance  # 51.42003
# summary(m1.b)$deviance # 51.42003
# summary(m1)$df.residual # 54
# summary(m1.b)$df.residual # 54
```

Possiamo notare come tra i due modelli che hanno categoria di riferimento diversa queste siano uguali, cosa che ci fa capire che il fit totale non è cambiato.

Dobbiamo assicurarci che l'offset sia grande abbastanza da assicurare che le proprietà asintotiche della devianza siano rispettate. Per farlo guardiamo quante unità abbiamo in ogni gruppo della variabile `n`.

```
sort(claims$n)
```

```
  [1]    3    7    9   16   18   20   24   24   25   29   31   33   35   39   40
 [16]   43   48   53   66   68   71   72   73   78   81   85   89   99  114  121
 [31]  122  133  139  149  151  155  175  197  221  240  245  246  264  284  286
 [46]  313  316  322  344  355  419  452  536  648  692  696  724  931 1110 1635
 [61] 1640 1680 2443 3582
```

Apparte i primi 4 gruppi tutti gli altri vanno bene per avere una buona approssimazione dei risultati teorici asintotici.

```
pchisq(summary(m1)$deviance, summary(m1)$df.residual, lower.tail=FALSE)
```

```
 [1] 0.5745071
```

Dobbiamo calcolare noi il p-value perché il summary del modello non restituisce quello della deviance statistic, ma quello della pearson statistic.

6.3.2 Pearson

```
pchisq(sum(res.m1.p^2),summary(m1)$df.residual, lower.tail=FALSE)
 [1] 0.6809085
```

Il modello quindi sembra appropriato ai dati. Vediamo che il p-value ottenuto non è uguale a quello della deviance. Questi infatti sono uguali asintoticamente, ma ovviamente noi abbiamo un campione di unità finito. Può succedere allora in casi particolari che uno sia significativo e l'altro no (di solito quando numerosità campionaria è bassa). Quando capita riporta *entrambi* e comunica il "problema".

6.3.3 Graphical inspection of the Residual

Linearity of the (log) ExpectedValue

```
plot(m1,which=1) # la linea rossa è molto buona
```



Gaussianity of the standardized residuals

```
qqPlot(rstandard(m1)) # abbiamo anche i confidence bands perché è qqplot da car
```



Figure 6.1: Second count plot

Non dobbiamo credere troppo a questi intervalli, li usiamo solo per capire cosa potrebbe essere fuori da quello che mi aspetto dai residui. Ricorda che questi sono grouped data (abbiamo tanti gruppi), quindi ogni punto che vediamo nel grafico è un gruppo.

6.4 Pratical Session

6.4.1 Fitting a `quasipoisson`

Dataset example: `epilepsy`

Call:

```
glm(formula = seizure.rate ~ treatment + age + base,
    family = quasipoisson(), data = epilepsy)

Deviance Residuals:
    Min      1Q  Median      3Q     Max
-4.4360 -1.4034 -0.5029  0.4842 12.3223

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)         0.562532   0.305959   1.839  0.0673 .
treatmentProgabide -0.152701   0.107849  -1.416  0.1582
age                 0.022740   0.009078   2.505  0.0129 *
base                0.022652   0.001149  19.714  <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for quasipoisson family taken to be 5.08961)

    Null deviance:     2521.75  on 235  degrees of freedom
Residual deviance:  958.46  on 232  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

$\mathbb{E}[y_i^2] = \text{Var}(y_i) \quad \Longleftrightarrow \quad$ marginal that does not guarantee this: $\mathbb{E}[y_i \mid \boldsymbol{X}] = \text{Var}(y_i \mid \boldsymbol{X})$

If you see the distribution is a little skewed, $\Rightarrow$ go with Negative Binomial.

Look at ϕ: if $\phi \approx 1$, it's okay otherwise $\Rightarrow$ go with Negative Binomial.

7. Offset

Offset ⇒ Differentiating situation

If you do not take account for the offset, you might estimate incorrectly the average level of your count data.

> ⇒ If you do not have the same exposure for each unit, then the expectation of each unit cannot be the same.

Assuming i.i.d. data then you are expecting to obtain same $\hat{\boldsymbol{\beta}}$ when you observe same covariates values.
⇒ But if the units are exposed differently, you cannot expect the same parameters.

Example

Number of cigarettes here the **offset** can be time frame that you are measuring.
It counts the **exposure** of the phenomenon (e.g., how long).

By adding a new regressor implies:

- loosing 1 (or more if categorical) degree of freedom
- Using a parameter ⇒ more uncertainty

By using it as an offset you earn one degree of freedome

For grouped data, the **offset** ⇒ number of each group.

7.1 Modelling with an offset

In many applications, the counting process and the *rate* at which events happen are affected, for example, by the length of the time frame considered.

Poisson process with intensity rate λ produces counts y in interval of length Δ, distributed as:

$$\text{Pois}(\lambda\Delta) \quad \rightarrow \quad \mathbb{E}(y) = \mu = \lambda\Delta \quad \text{depends on } \Delta$$

when working with Poisson be careful on the time window Δ

If we use log-linear Poisson model, then you are rescalinig for Δ:

$$\log(\mu_i) = \log(\Delta_i) + \mathbf{x}_i'\boldsymbol{\beta}$$

$$\mu_i = \Delta_i + \exp(\mathbf{x}_i'\boldsymbol{\beta})$$

$$\log(\frac{\mu_i}{\Delta_i}) = \mathbf{x}_i'\boldsymbol{\beta}$$

For grouped data:

$$\log(\mu_i) = \log(n_i) + \mathbf{x}_i'\boldsymbol{\beta}$$

in each group i of size n_i.

The fixed parameter γ_i (i.e., $\log(\Delta_i)$ or $\log(n_i)$) is called **offset**.

Derivations show that:

$$F(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i\mathbf{x}_i'\lambda_i \exp(\gamma_i)$$

which means standard errors of $\hat{\boldsymbol{\beta}}$ *decrease with larger offset values.*

7.2 Interpretation of the Offset Term

$\Delta_i \rightarrow$ accounts exposure.

- Unit-specific, accounts for that information.
- Does not have a parameter, it is an additional constant (known).

You obtain new $\boldsymbol{\beta}$, but the input remains the same. This has consequences on the Fisher information matrix: you are less unsure about the estimation.

Allows you to model data with more information. This additional information helps reduce the entropy.

7.3 Warnings and Limitations

Be careful with the offset – it gets weird when each y_i has very different Δ_i,

$y_1, y_2, \ldots, y_n \Rightarrow$ counts

$\Delta_1, \Delta_2, \ldots, \Delta_n \Rightarrow$ offsets

this means that more information is hosted by Δ_i.

Including the offset might lead to:

- $\hat{\beta}_j$ become not significant
- since smaller standard errors $\Rightarrow$ less likely to reject null hypothesis (inflated Type I error)
- If Δ_i values vary too much across observations, the model does not work properly

7.4 Offset Exclusion

```
m4<-glm(c~car+age+distcity,data=claims,family=poisson)
summary(m4)
```

```
Call:
glm(formula = c ~ car + age + distcity, family = poisson, data = claims)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)        2.46014    0.07708  31.917  < 2e-16 ***
car1-1.5           0.98960    0.05045  19.617  < 2e-16 ***
car1.5-2           0.47070    0.05490   8.574  < 2e-16 ***
car>2             -0.58927    0.07211  -8.172 3.04e-16 ***
age25-29           0.56769    0.08272   6.863 6.74e-12 ***
age30-35           0.68217    0.08108   8.413  < 2e-16 ***
age>35             2.19916    0.06965  31.575  < 2e-16 ***
distcityMajorCity -1.06075    0.05849 -18.135  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 4236.68  on 63  degrees of freedom
Residual deviance:  491.64  on 56  degrees of freedom
AIC: 824.96

Number of Fisher Scoring iterations: 5
```

```
cbind(coefficients(m3), coefficients(m4))
```

```
                       [,1]       [,2]
(Intercept)       -1.8102073  2.4601386
car1-1.5           0.1622910  0.9896033
car1.5-2           0.3935180  0.4706991
car>2              0.5653953 -0.5892720
age25-29          -0.1890172  0.5676929
age30-35          -0.3421109  0.6821701
age>35            -0.5327488  2.1991635
distcityMajorCity  0.2184953 -1.0607540
```

Guarda i segni dei modelli, sono diversi! scegliamo il modello con l' offset.

```
pchisq(summary(m4)$deviance, summary(m4)$df.residual, lower.tail=FALSE)
```

```
[1] 6.330768e-71
```

Confrontando un modello senza offset con uno con offset, possiamo osservare che:

- La **devianza** è maggiore (peggiore), segnalando un peggior adattamento del modello.

– Alcuni coefficienti cambiano **segno**, a causa della correlazione tra i predittori e la variabile di offset.

Questo dimostra che, in assenza di offset, il modello può attribuire erroneamente l'aumento del numero di eventi (es. sinistri) alla variabile esplicativa, quando in realtà è dovuto semplicemente a una maggiore esposizione (più persone nel gruppo).

Per ottenere quindi risultati corretti, è fondamentale bilanciare per l'esposizione usando un offset. Senza offset, le stime possono essere distorte e portare a conclusioni sbagliate.

```
# ?by
by(claims$c,claims$distcity,sum)
```

```
claims$distcity: Other
[1] 2825
------------------------------------------------------------
claims$distcity: MajorCity
[1] 326
```

```
by(claims$n,claims$distcity,sum)
```

```
claims$distcity: Other
[1] 21365
------------------------------------------------------------
claims$distcity: MajorCity
[1] 1994
```

Il numero inferiore di sinistri osservati nelle `Major Cities` non implica necessariamente un rischio più basso. Infatti, il motivo potrebbe essere semplicemente che il numero di assicurati in queste aree è minore, e quindi anche l'esposizione complessiva è più bassa.

```
by(claims$c,claims$distcity,sum)/by(claims$n,claims$distcity,sum)
```

```
claims$distcity: Other
[1] 0.1322256
------------------------------------------------------------
claims$distcity: MajorCity
[1] 0.1634905
```

Il tasso grezzo di sinistri (raw rate), calcolato come rapporto tra numero di sinistri e numero di assicurati, è in realtà più alto nelle `Major Cities` rispetto alle altre aree.

7.5 Using offset as a regressor

Attenzione: è una possibile domanda esame! Un altro modo per vedere effect dell'offeset è insierirlo dentro il modello come predittore (sempre in scala logaritmica). Per farlo dobbiamo quindi solo specificare `log(n)`, senza `offset`.

```r
mod.offreg <- glm(c~log(n)+car+age+distcity,data=claims,family=poisson)
summary(mod.offreg) # ci sono 9 parametri
```

```
Call:
glm(formula = c ~ log(n) + car + age + distcity, family = poisson,
    data = claims)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      -1.74398    0.23182  -7.523 5.36e-14 ***
log(n)            0.98526    0.04881  20.185  < 2e-16 ***
car1-1.5          0.17407    0.06384   2.727 0.006398 **
car1.5-2          0.39435    0.05505   7.163 7.88e-13 ***
car>2             0.54781    0.09281   5.902 3.58e-09 ***
age25-29         -0.17924    0.08894  -2.015 0.043880 *
age30-35         -0.32872    0.09265  -3.548 0.000388 ***
age>35           -0.49465    0.14430  -3.428 0.000608 ***
distcityMajorCity  0.19870    0.08788   2.261 0.023754 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 4236.679  on 63  degrees of freedom
Residual deviance:   52.044  on 55  degrees of freedom
AIC: 387.37

Number of Fisher Scoring iterations: 4
```

7.5.1 Hypothesis testing su H0: $\beta_{offset} = 1$

Nel `summary` di un modello GLM, tutti i p-value associati ai coefficienti testano l'ipotesi nulla secondo cui il parametro è uguale a 0.

Tuttavia, quando includiamo una variabile come offset, il test corretto dovrebbe verificare se il coefficiente è uguale a 1, non a 0. Infatti, un offset rappresenta una quantità fissa che entra nel modello come denominatore del predittore lineare: se il coefficiente è 1, la variabile può essere trattata come offset.

Se il coefficiente si discosta significativamente da 1, allora non è un offset valido, e la variabile va trattata come un normale predittore.

Attenzione: Di conseguenza, il p-value nella tabella del summary non è informativo per valutare la correttezza dell'uso come offset: è necessario condurre un test specifico sull'ipotesi $H_0 : \beta_{offset} = 1$ ad esempio con un test lineare.

```r
K.offreg <- rep(0,9) # vettore di nove 9 (numero di parametri)
K.offreg[1+1] <- 1 # l'offset è il secondo parametro (dopo l'intercetta)
K.offreg
```

```
[1] 0 1 0 0 0 0 0 0 0
```

```
t.offreg <- 1
t.offreg
```

```
[1] 1
```

```
lht(mod.offreg,K.offreg,rhs=t.offreg)
```

```
Linear hypothesis test:
log(n) = 1

Model 1: restricted model
Model 2: c ~ log(n) + car + age + distcity

  Res.Df Df  Chisq Pr(>Chisq)
1     56
2     55  1 0.0911     0.7627
```

Siccome è maggiore di 0.05 non rigettiamo che il parametro log(n) sia uguale a 1, quindi è un offset!

Procedura per capire se una variabile può essere usata come offset o no:

- testala come offset
- testala come predittore
- controlla se dovrebbe essere un offset

Nota: se non usi la variabile come offset, ma come predittore, perdi un grado di libertà perché stai usando un altro parametro. Questa cosa se hai una numerosità campionaria piccola potrebbe non aiutarti. Ricorda che quando hai una numerosità campionaria piccola devi sempre pensare al numero di paramteri del tuo modello.

Changing link function

Se cambi il link puoi avere problemi con la convergenza dell'algortimo. Per questo motivo il modello può essere peggiore rispetto agli altri, soprattutto in termini di fit (per esempio di AIC)

`link = identity`

```
# mod.id<-glm(c~offset(log(n))+car+age+distcity, data=claims, start=rep(1,8),family=po
# extractAIC(mod.id)
```

Se lanciassimo il comando con link="identity" vedremmo un lungo warning. Nota: si può calcolare l'AIC, questo è però molto più grande dei precedenti (1552.313)

`link = sqrt`

```
# mod.quadr<-glm(c~offset(log(n))+car+age+distcity,data=claims,start=rep(100,8),family
# extractAIC(mod.quadr)
```

Stesso discorso con link="sqrt". Qua l'AIC si abbassa un po'(973.0204), ma è ancora molto alto.

Part VIII

Nonnegative continuous data

8. Intro

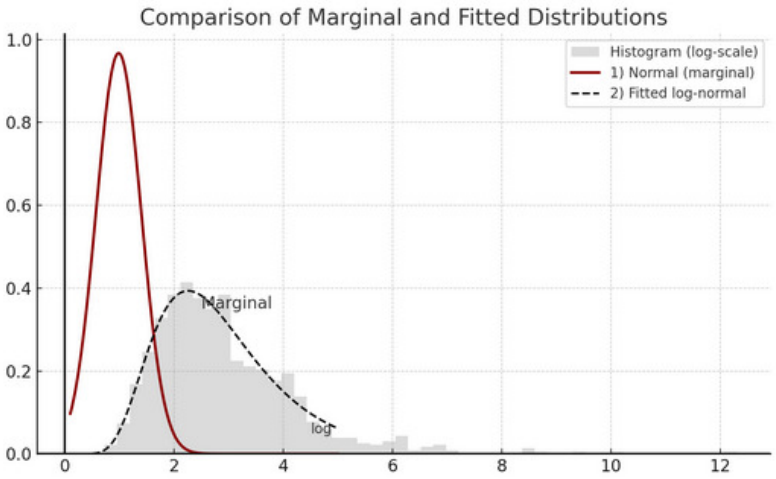


Figure 8.1: Marginal distribution (red) vs fitted distribution (dashed). Histogram represents the empirical log-scale distribution.

1. **Gaussian** If values are far from 0, you can use a Gaussian model (normal), which looks at the variability.
   However, you will still have a problem if the distribution is **asymmetric.**
2. **Log(data)**: this transformation stretches the data such that:
   - Smaller values ⇒ larger
   - Larger values ⇒ smaller
   - It is more symmetrical (marginally)
   - $\log(y_i)$ belongs to $\mathbb{R}$, so it removes nonnegativity constraint
   - Good for rates

   Problem: when values are too close to 0. But you are protected because:

   $$y_i \in (0,1) \quad \Rightarrow \quad \log(y_i) < 0$$

   This solves the problem of forbidden negativity in models, because after the transformation, you can use a Gaussian model.

   Example: interest rate $\in [0,1]$

   $$\log\left(\frac{\text{interest}}{\text{rate}}\right) = \log(\cdot) - \log(\cdot)$$

   (You model differences, an example of simplification using the logarithm)
3. Use a **GLM**

8.1 Log-Normal Model

In many applications, responses y_i are only measured as **nonnegative**, which means they are usually highly *non-normal* and often (right) *skewed*.

Lognormal model: if we transform the response $\tilde{y} = \log(y)$ and use

$$\tilde{y}_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + \varepsilon_i \quad \text{or equivalently} \quad \tilde{y}_i \sim \mathcal{N}(\boldsymbol{x}_i^\top \boldsymbol{\beta}, \sigma^2)$$

we implicitly assume y_i has **log-normal** distribution with

$$\mathbb{E}(y_i) = \exp(\boldsymbol{x}_i^\top \boldsymbol{\beta} + \sigma^2/2) \qquad \text{Var}(y_i) = \exp(2\boldsymbol{x}_i^\top \boldsymbol{\beta} + \sigma^2)(\exp(\sigma^2) - 1)$$

If we estimate $(\boldsymbol{\beta}, \sigma^2)$ from $\tilde{y}$ and plug-in into $\mu = \mathbb{E}(y_i)$ we introduce **considerable bias** (nonlinear back-transformation).

8.1.1 Problems

When

$$\log(y_i) \sim \mathcal{N} \Leftrightarrow y_i \sim \log \mathcal{N}$$

To go back from $\tilde{y}_i \to y_i$, suppose:

$$\mathbb{E}[\tilde{y}_i] = \boldsymbol{x}_i^\top \boldsymbol{\beta} \Rightarrow \mathbb{E}[\exp(\tilde{y}_i)] = \exp(\boldsymbol{x}_i^\top \boldsymbol{\beta}) \stackrel{?}{=} \mathbb{E}[y_i]$$

But this is **not true**, because if we assume $\log(y_i) \sim \mathcal{N}$, then $y_i \sim$ logNormal :

$$\Rightarrow \mathbb{E}[y_i] = \exp(\boldsymbol{x}_i^\top \boldsymbol{\beta} + \sigma^2/2)$$

Just plugging in $\exp(\boldsymbol{x}_i^\top \boldsymbol{\beta})$ induces a **bias**, because we are not working directly on $\log \mathcal{N}$. We also need to look at the variance.

$$\text{Bias} = \exp(\sigma^2/2) \quad \text{(which also needs to be estimated to correct the bias)}$$

$\Rightarrow$ The log is used for symmetry reasons.

But using *log + Gaussian* leads to problems when you go back to the original scale.

However, if you rework on the log scale directly $\Rightarrow$ it's OK.

9. Gamma regression

GLM framework with proper distribution for y.

$$y_i \sim \text{Gamma}\,(\mu_i, \nu) \rightarrow \text{(shape, scale)}$$

with $\mathbb{E}(y_i) = \mu_i$ and $\text{Var}(y_i) = \sigma_i^2 = \mu_i^2/\nu$ *(units specific variance → this means you can model overdispersion)*

Linear link $\mu_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$ not feasible (usual reason why)

Multiplicative exponential model: most used, with $g(\mu) = \log(\mu)$

$$\mu_i = \exp(\eta_i) = \exp(\boldsymbol{x}_i^\top \boldsymbol{\beta}) = \exp(\beta_0)\exp(\beta_1 x_{i1}) \cdots \exp(\beta_k x_{ik})$$

Another option is the **canonical (or natural)** response function

$$\mu_i = -\frac{1}{\eta_i} = -\frac{1}{\boldsymbol{x}_i^\top \boldsymbol{\beta}}$$

which comes from theory of GLMs but same problem of linear link (cannot ensure non-negativity).

9.0.1 Gamma GLM Output Example

```
Call:
glm(formula = recovtime ~ logdose + bloodp,
    family = Gamma(link = log), data = bp)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.31726 -0.61486 -0.07994  0.26537  1.10078

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.64599    0.75892   4.804 2.21e-05 ***
logdose       0.69059    0.30900   2.235   0.0311 *
bloodp       -0.02951    0.01242  -2.375   0.0224 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Gamma family taken to be 0.3432751)

    Null deviance: 17.481  on 42  degrees of freedom
Residual deviance: 14.785  on 40  degrees of freedom
(10 osservazioni eliminate a causa di valori mancanti)
AIC: 336.46

Number of Fisher Scoring iterations: 6
```

- `family = Gamma(link = log)` you need to specify the link function due to the problems with the canonical
- `Estimate` you need to take the exponetial to have the effect
- `Dispersion parameter for Gamma family taken to be 0.3432751` here it refers to ν
- `AIC: 336.46` no problems, because no quasi-gamma

10. R nonneg

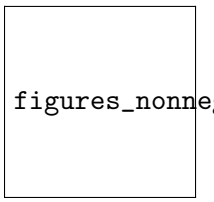
```
library("car")
library(HSAUR3)
```

vogliamo fittare un modello che ci dice l'effetto della density sull'hardness

```
#Load timber.txt
timber <- read.table("timber.txt",header=TRUE)
str(timber)
```

```
'data.frame':    36 obs. of  2 variables:
 $ hardness: int  484 427 413 517 549 648 587 704 979 914 ...
 $ density : num  24.7 24.8 27.3 28.4 28.4 29 30.3 32.7 35.6 38.5 ...
```

```
par(mfrow=c(1,2))
plot(timber$density,timber$hardness,xlab="density", ylab="hardness",
ylim=c(0,3500),xlim=c(0,80),pch=19)
plot(timber$density,timber$hardness,xlab="density", ylab="hardness",
ylim=c(0,3500),xlim=c(0,80),pch=19)
m.normal.id <- lm(hardness ~ density, data = timber)
abline(m.normal.id,lwd=3,col=4)
```




```
[1] 413
```

Dal momento che le nostre osservazioni sono lontane da 0, proviamo con un:

Gaussian Linear Model

```
summary(m.normal.id)
```

```
Call:
lm(formula = hardness ~ density, data = timber)

Residuals:
    Min      1Q  Median      3Q     Max
-338.40  -96.98  -15.71   92.71  625.06
```

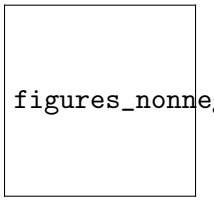
```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -1160.500    108.580  -10.69 2.07e-12 ***
density        57.507      2.279   25.24  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 183.1 on 34 degrees of freedom
Multiple R-squared:  0.9493, Adjusted R-squared:  0.9478
F-statistic:   637 on 1 and 34 DF,  p-value: < 2.2e-16
```

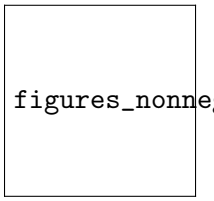
questa volta stiamo usando la funzione `lm`, al posto della funzione `glm` → Notiamo una prima differenza nell'output: è presenta l' `F-statistic` e una stima della varianza, che non abbiamo nel `glm`.

Graphical check of the residual

```
plot(m.normal.id,which=1,main=" ",sub=" ",pch=19)
```

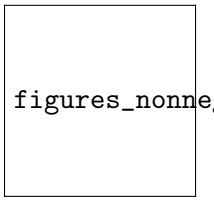



```
plot(m.normal.id,which=2,main=" ",sub=" ",pch=19)
```



ci sono dei punti che sono molto lontani senza però essere outlier.

```
plot(density(timber$hardness))
```



Questo grafico ci dice una cosa importante → i dati non sono simmetrici. Quindi andiamo a usare la gamma sul logaritmo delle y

Fit GLM gamma model

```
m.gamma.log1 <- glm(hardness ~ density,
                    data = timber,
                    family = Gamma(link = log)) #dobbiamo specificare
```

Note: we need to change the canonical function

```
summary(m.gamma.log1)
```

```
Call:
glm(formula = hardness ~ density, family = Gamma(link = log),
    data = timber)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 5.222967   0.079497   65.70   <2e-16 ***
density     0.042022   0.001668   25.19   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

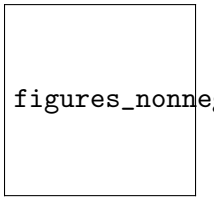
(Dispersion parameter for Gamma family taken to be 0.01796336)

    Null deviance: 11.26788  on 35  degrees of freedom
Residual deviance:  0.61977  on 34  degrees of freedom
AIC: 475.84

Number of Fisher Scoring iterations: 4
```

Ricorda che R usa la pearson chi2 statistic per calcolare l'overdispersion parameter, che corrisponde allo shape paramter della gamma (ν)

```
plot(timber$density,timber$hardness,xlab="density",ylab="hardness",
       ylim=c(0,3500),xlim=c(0,80),pch=19)
x<-seq(0,80,length.out=100)
ey.x<-exp(m.gamma.log1$coeff[1]+m.gamma.log1$coeff[2]*x)
lines(x,ey.x,lwd=3,col=2) #gamma regression model
abline(m.normal.id,lwd=3,col=4) #gaussian model
legend("topleft",
       legend = c("Observed data",
             "Gamma regression",
            "Gaussian regression"),
       col = c("black", "red", "blue"),
       lwd = c(NA, 3, 3),
       pch = c(19, NA, NA),
       bty = "n")
```



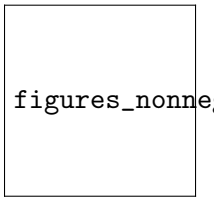
Note: è il predittore a essere lineare, non l'effetto delle covariate sulla response → attraverso il passaggio dall'esponenziale.

Graphical check of the residual

```
x<-seq(0,80,length.out=100)
ey.x<-exp(m.gamma.log1$coeff[1]+m.gamma.log1$coeff[2]*x)
```

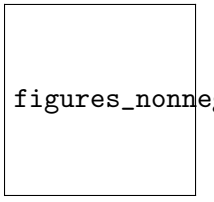
check residuals from Gamma model considerazioni se l'effetto è lineare del linear predictor → posso anche introdurre quadratic polinomial delle covariate vediamo qualcosa di strano in termini di residuals → non sono allineati##? (se c'è un S shape il modello non va bene, non c'è good fit)

```
plot(m.gamma.log1,which=1,main=" ",sub=" ",pch=19) #è tipo scale-location plot
```



vediamo che la linera rossa non è dritta, probabilmente manca un quadratic term dobbiamo inserirlo

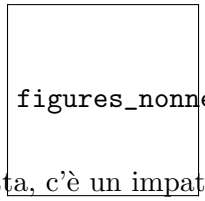
```
plot(m.gamma.log1,which=2,main=" ",sub=" ",pch=19)
```



include a quadratic term

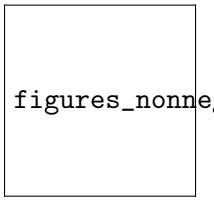
```
m.gamma.log2 <- glm(hardness ~ density + I(density^2),
                    data = timber, family = Gamma(link = log))

par(mfrow=c(1,2))
plot(m.gamma.log1,which=1,main=" ",sub=" ",pch=19) #prima
plot(m.gamma.log2,which=1,main=" ",sub=" ",pch=19) #ora
```



è meglio di prima, la linea è più dritta, c'è un impatto non sostanziale sui residui

```
par(mfrow=c(1,2))
plot(m.gamma.log1,which=2,main=" ",sub=" ",pch=19) #prima
plot(m.gamma.log2,which=2,main=" ",sub=" ",pch=19) #ora
```




```
summary(m.gamma.log2)
```

```
Call:
glm(formula = hardness ~ density + I(density^2), family = Gamma(link = log),
    data = timber)

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.147e+00  2.089e-01  19.848  < 2e-16 ***
density       9.133e-02  9.316e-03   9.803 2.66e-11 ***
I(density^2) -5.210e-04  9.776e-05  -5.329 6.98e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.01017894)

    Null deviance: 11.26788  on 35  degrees of freedom
Residual deviance:  0.33499  on 33  degrees of freedom
AIC: 455.64

Number of Fisher Scoring iterations: 4
```

Coefficient interpretation

Il coefficiente del terminne quadratico è di segno opposto, il che significa che la curva raggiungerà un plateau e poi smetterà di aumentare.

Model selection

Se vogliamo fare una selezione del modello (model selection), ad esempio decidere se rimuovere o meno il predittore X, non possiamo più affidarci ai test sui singoli coefficienti (come i singoli test t o z presenti nella tabella dei coefficienti).

Questo perché rimuovere X vuol dire rimuovere due variabili insieme. Quindi non possiamo guardare a una sola riga della tabella dei coefficienti (che mostra test univariati per ogni parametro).

In questo caso, dobbiamo usare un test globale, come:

- un'ANOVA (analisi della varianza tra modelli annidati),
- oppure un test per la differenza delle devianze (deviance difference test), che confronta i due modelli usando:

Overdispersion

Quando si calcola il parametro ν nella gamma, è importante fare attenzione a come il software parametrizza la distribuzione.

A seconda della convenzione adottata, in due modi diversi:

$$\nu = \frac{\sum(\texttt{pearson residual})^2}{\texttt{degrees of freedom}} \tag{1}$$

$$\nu = \frac{\texttt{degrees of freedom}}{\sum(\texttt{pearson residual})^2} \tag{2}$$

In questo caso è la (1)

```
summary(m.gamma.log2)$df[2]/sum(residuals(m.gamma.log2,type="pearson")^2)
```

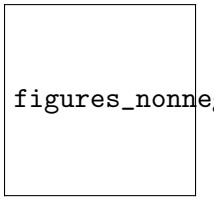
```
  [1] 98.24201
```

```
1/summary(m.gamma.log2)$dispersion
```

```
  [1] 98.24201
```

alcuni software non usano questa parametrizzazione e quindi restituiscono 98.24201 come risulato poi li ti domandi se devi moltiplicare per 98 la varaiza per aggiustarla, NO! É solo una diversa parametrizzazione, quindi sarà 1/98 l'actual dispersion parameter

```
x<-seq(0,80,length.out=100)
ey.x<-exp(m.gamma.log2$coeff[1]+m.gamma.log2$coeff[2]*x+m.gamma.log2$coeff[3]*x^2)
plot(timber$density,timber$hardness,xlab="density",ylab="hardness",
ylim=c(0,3500),xlim=c(0,80),pch=19)
lines(x,ey.x,lwd=3,col=5)
x<-seq(0,80,length.out=100)
ey.x<-exp(m.gamma.log1$coeff[1]+m.gamma.log1$coeff[2]*x)
# è un gamma model -> è l'exponential del linear predictor
# intercetta + beta1 * x
# prendiamo l'esponenziale perché la link function è il logaritmo, se vuoi andare indi
lines(x,ey.x,lwd=3,col=2) #gamma regression model
abline(m.normal.id,lwd=3,col=4) # gaussian model
```



Model comparison

Se vogliamo testare i modelli che abbiamo fittato con un singolo predittore, è sufficiente osservare il *p-value* associato al coefficiente nella tabella dei coefficienti.

Tuttavia, quando il modello include **più termini legati a uno stesso predittore** (ad esempio x e x^2), oppure in presenza di interazioni o trasformazioni, **non è corretto considerare i test sui singoli coefficienti in modo isolato**.

In questi casi, per valutare l'effetto complessivo del predittore, è preferibile utilizzare **altri tipi di test**, come ad esempio la funzione `anova()` in R, che confronta modelli annidati sulla base della devianza o di altri criteri globali.

Null model

```
m.gamma.log0 <- glm(hardness ~ 1, data = timber,family = Gamma(link = log))
#solo intercetta
summary(m.gamma.log0)
```

```
Call:
glm(formula = hardness ~ 1, family = Gamma(link = log), data = timber)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.29266    0.09091   80.22   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.2975109)

    Null deviance: 11.268  on 35  degrees of freedom
Residual deviance: 11.268  on 35  degrees of freedom
AIC: 580.02

Number of Fisher Scoring iterations: 5
```

Note: per comparare due modelli questi devono avere la stessa link function!

```
anova(m.gamma.log0,m.gamma.log2,test="F")
```

```
Analysis of Deviance Table
```

```
Model 1: hardness ~ 1
Model 2: hardness ~ density + I(density^2)
  Resid. Df Resid. Dev Df Deviance      F    Pr(>F)
1        35     11.268
2        33      0.335  2   10.933 537.03 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Test

- `F-test` appropriato quando dobbiamo stimare il dispersion parameter, qui siamo nella situazione in cui stiamo comparando il `gaussian model` e `gamma model` ed entrambi stimano un overdspersion parameter
- likelihood ratio test
- χ^2

```
anova(m.gamma.log0,m.gamma.log2,test="LRT")
```

```
Analysis of Deviance Table

Model 1: hardness ~ 1
Model 2: hardness ~ density + I(density^2)
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        35     11.268
2        33      0.335  2   10.933 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(m.gamma.log0,m.gamma.log2,test="Chisq")
```

```
Analysis of Deviance Table

Model 1: hardness ~ 1
Model 2: hardness ~ density + I(density^2)
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        35     11.268
2        33      0.335  2   10.933 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

tutti i test dicono di rifiutare l'ipotesi nulla

prova con link = inversa → fa vedere che è sbagliato

```
mod.inv <- glm(hardness ~ density + I(density^2),
                    data = timber, family = Gamma)
summary(mod.inv)
```

```
Call:
glm(formula = hardness ~ density + I(density^2), family = Gamma,
    data = timber)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.279e-03  2.773e-04  15.431  < 2e-16 ***
density      -1.164e-04  1.063e-05 -10.953 1.58e-12 ***
I(density^2)  8.636e-07  9.915e-08   8.711 4.56e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.01670585)

    Null deviance: 11.268  on 35  degrees of freedom
Residual deviance:  0.563  on 33  degrees of freedom
AIC: 474.37

Number of Fisher Scoring iterations: 4
```

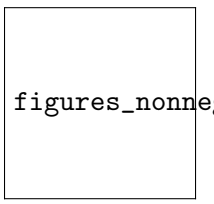
Usando la canonical link function tutti i segni dei coefficienti vengono invertiti.

Note:se vuoi comparare sue modelli con stessi dati e stessi predittori, ma con link diversa, non puoi usare l'anova dobbiamo guardare l'AIC

è meglio quello con link = log, con x e x^2

Try to model log-transformed data

```
m.lognormal <- glm(log(hardness) ~ density + I(density^2), data = timber)
plot(timber$density,log(timber$hardness),xlab="density",ylab="ln(hardness)",
    xlim=c(0,80),pch=19)
elny.x<-(m.lognormal$coeff[1]+m.lognormal$coeff[2]*x+m.lognormal$coeff[3]*x^2)
lines(x,elny.x,lwd=3,col=2) #curva risultante
```



Note:
Se ci fossero stati 0 nei dati avremmo avuto un errore. R dice: `Some values are been forced to be missing values` → problema perché non puoi usarli nel tuo modello, stai perdendo informazioni e diminuendo la sample size

La relazione sembra lineare quinsi sembra un approccio buono da seguire.

coefficients
I coefficienti non sono nella scala della risposta per via del bias

→ $\mathbb{E}[\text{log-normal}] = \exp(\eta + \sigma^2/2)$ quindi se vuoi tornare indietro prendendo $\exp(\beta_i)$ ti manca $(\sigma^2/2)$. Ma σ^2 va stimato.

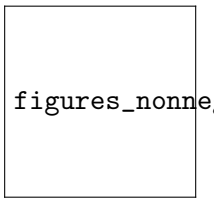
É sbagliato quindi fittare una guassian sui dati log e poi prendere exp dei coefficienti, nel farlo perdi informazioni

```
m.lognormal <- glm(log(hardness) ~ density + I(density^2), data = timber)
elny.x<-(m.lognormal$coeff[1]+m.lognormal$coeff[2]*x+m.lognormal$coeff[3]*x^2)
lines(x,elny.x,lwd=3,col=2) #curva risultante
```

Note:
`glm` senza `link = gaussian` equivale al comando `lm`.

Il predittore è lineare in questo caso, ma siccome usiamo un effeto quadratico nelle x

```
par(mfrow=c(1,2))
plot(m.lognormal,which=1,main=" ",sub=" ",pch=19)
plot(m.lognormal,which=2,main=" ",sub=" ",pch=19)
```




```
exp(coefficients(m.lognormal))
```

```
  (Intercept)       density I(density^2)
   62.6783332     1.0958337     0.9994773
```

Note:

Non sono gli stessi paramteri che avresti in un gamma regression model manca $\sigma^2/2$, che va stimato, quindi anche la correzione per il bias non sarà il proper bias adjustment.

Possibile domanda di esame: *BE CAREFUL: AIC from model on* $\log(Y)$ *is NOT comparable with AIC from model on Y AIC può comparare solo modelli che sono sulla stessa scale variable!!!* → un modello fittato su y e un modello fittato su $\log(y)$ NON sono comparabili! perché nella formula di AIC c'è una parte che viene dalla penalità $(\log(n) \cdot p)$ → questa non è impattata dalla trasformazione, quindi la penalità è la stessa in entrambi i casi ma l'altra parte della formula è -2loglik dipende dalla scala di y.

```
summary(timber$hardness)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  413.0   962.8  1195.0  1469.5  1980.0  3260.0
```

```
summary(log(timber$hardness))
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  6.023   6.869   7.086   7.136   7.591   8.089 
```

Quindi in questa situazione dobbiamo calcolare l'AIC a mano:

```
#?dlnorm
lik.lnorm <- rep(0,nrow(timber))
lik.lnorm #vettore di 36 posizioni
```

```
 [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

ognugna è il contribution sulla likelihood per ogni osservazione che vengono da una log-normal

```
for (i in 1:nrow(timber)) lik.lnorm[i] <-
                      dlnorm(timber$hardness[i],
                             meanlog=predict(m.lognormal)[i],
                            sdlog=sqrt(summary(m.lognormal)$dispersion))
```

calcoliamo densità di una lognormal dove ci sono osservazioni e la media della distribuzione è quella che il modello predice è la definizione del modello, con una varianza specifica che nella nostra situazione è sqrt(disp.par)

```
lik.lnorm
```

```
 [1] 0.0048718936 0.0088882351 0.0008226970 0.0061246197 0.0071922681
 [6] 0.0029502446 0.0057811263 0.0055632889 0.0012746805 0.0034348842
```

```
-2*sum(log(lik.lnorm)) + 2*4
```

```
[1] 455.7336
```

```
AIC(m.gamma.log2) #455.6448
```

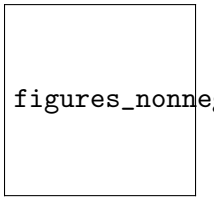
```
[1] 455.6448
```

adesso questo è comparabile e preferiamo la gamma regression

plot degli original data

```
plot(timber$density,timber$hardness,xlab="density",ylab="hardness",
        xlim=c(0,80),ylim=c(0,3500),pch=19)
lines(x,exp(elny.x+summary(m.lognormal)$dispersion/2),lwd=3,lty=2,col=3)

x<-seq(0,80,length.out=100)
ey.x<-exp(m.gamma.log2$coeff[1]+m.gamma.log2$coeff[2]*x+m.gamma.log2$coeff[3]*x^2)
lines(x,ey.x,lwd=3,col=2,lty=3)
legend(0,3500,legend=c("Gamma","Lognormal"),col=c(2,3),lwd=3,lty=c(3,2))
```




```
summary(m.lognormal)$dispersion / 2
```

```
  [1] 0.005077338
```

non è molto, ma dobbiamo prendere l'exp() quindi questo valore può avere un impatto sulle tue stime queste ti servono per non calcolare la gamma sbagliata.

Lui preferisce gamma regression perché è una proper regression sui original variable specialmente perché nella vita vera avremo di 0, problema e quando abbiamo 0 non possiamo prendere il logaritmo

Part IX

A general model definition

Since we do not assume Gaussianity, we write:

$$y_i \sim \mathcal{F}(\theta)$$

where $\mathcal{F}(\cdot)$ is any probabilistic distribution in the **exponential family**.

All of the models seen until now fall under the **GLM** framework. Unified framework for model seen so far (GLMs):

- expectation/mean $\mu = \mathbb{E}(y)$ of response connected to linear predictor through response (or link) function

$$\mu = h(\eta) = g^{-1}(\eta) \qquad g(\mu) = \eta = h^{-1}(\mu)$$

- distribution of y can be written as univariate exponential family

Linearity

The effects are injected **linearly** into the expected value (as part of the fitting procedure): **linear in the parameter**.

$$\boldsymbol{x}_i^\top \boldsymbol{\beta} = \beta_0 + x_{1i}\beta_1 + x_{2i}\beta_2 = \eta_i$$

or even with higher-order terms:

$$\boldsymbol{x}_i^\top \boldsymbol{\beta} = \beta_0 + x_{1i}\beta_1 + x_{1i}^2\beta_2 + x_{2i}\beta_3 = \eta_i \quad \text{(still linear in the parameter)}$$

Since we do not assume Gaussianity, we write:

$$y_i \sim \mathcal{F}(\theta)$$

where $\mathcal{F}(\cdot)$ is any probabilistic distribution in the **exponential family**.

All of the models seen until now fall under the **GLM** framework. Unified framework for model seen so far (GLMs):

- expectation/mean $\mu = \mathbb{E}(y)$ of response connected to linear predictor through response (or link) function

$$\mu = h(\eta) = g^{-1}(\eta) \qquad g(\mu) = \eta = h^{-1}(\mu)$$

- distribution of y can be written as univariate exponential family

10.1 Univariate exponential family (EF)

if you can write the density this way then it belongs to the EF:

$$f(y|\theta) = \exp\left(\frac{y\theta - b(\theta)}{\phi} w + c(y, \phi, w)\right)$$

where:

- θ: canonical (or *natural*) parameter
- ϕ: dispersion (or nuisance) parameter
- w: known value, usually weight (e.g. for grouped data w_i is size of group i)
- support of $f(\cdot)$ should **not** depend on θ

10.1.1 GLMs and EFs

$$f(y|\theta) = \exp\left(\frac{y\theta - b(\theta)}{\phi} w + c(y, \phi, w)\right)$$

Function $b(\theta)$ should be such that:

- $f(\cdot)$ can be normalized (finite integral)
- twice differentiable (at least), so both $b'(\theta)$ and $b''(\theta)$ exist

It can be shown:

$$\mathbb{E}(y) = \mu = b'(\theta) \qquad \text{Var}(y) = \phi b''(\theta)/w$$

The larger ϕ the larger σ^2 (this is why it is called disperse parameter), w refers to thedata amount in eac

A specific GLM is completely determined by:

- member of EF as the distribution of y
- chosen link/response function (and selected covariates)

Each EF has unique canonical link function: if chosen, it guarantees concavity of log-likelihood, (unique) existence of ML estimate, and $F(\boldsymbol{\beta}) = H(\boldsymbol{\beta})$.

- **Gaussian** is the only one where $g(\mu) = \mu$ and $\phi = \sigma^2$.
- **Bernoulli** and **Poisson** $\Rightarrow \phi = 1$.
  - This is why we use *quasi-likelihood* methods to account for overdispersion.
- $\theta(\mu)$ **for binomial models:** $\rightarrow$ **logit** link, which corresponds to the **log-odds**.
  Do not need to memorize these for the exam.

Main point: The maximum of the log-likelihood always exists and is unique.
Warnings or convergence issues are due to features of the problem:

- Perfect separability
- Unbalanced data

10.2 Likelihood inference in GLMs

ML estimator $\hat{\boldsymbol{\beta}} \rightarrow$ solution to $s(\hat{\boldsymbol{\beta}}) = 0$ with score function (general formula for EF):

$$\mathbf{s}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i \frac{h'(\eta_i)}{\sigma_i^2}(y_i - \mu_i) = \mathbf{X}'\mathbf{D}\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$$

where:

- $\mathbf{D} = \text{diag}(h'(\eta_1), \ldots, h'(\eta_n))$
- $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)$
- $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_n)$

Fisher matrix:

$$F(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i' \tilde{w}_i = \mathbf{X}'\mathbf{W}\mathbf{X}$$

with $\mathbf{W} = \text{diag}(\tilde{w}_1, \ldots, \tilde{w}_n)$ and

$$\tilde{w}_i = \frac{(h'(\eta_i))^2}{\sigma_i^2}$$

Matrix form:

$$\text{covariates} \times \text{derivative matrix} \times \boldsymbol{\Sigma}$$

where $\boldsymbol{\Sigma}$ collects variability in the data.

This becomes:

$$(\text{fixed value}) \times (y - \mu)$$

$\equiv$ For the **Gaussian model**, this corresponds to the **residual**, i.e., anything not captured by the model.

10.2.1 Fisher scoring algorithm as IWLS

In this setting, Fisher scoring algorithm has alternative form as **iterative weighted least squares (IWLS)** *a step method to find the estimates* :

$$\hat{\boldsymbol{\beta}}^{(t+1)} = (\mathbf{X}'\mathbf{W}^{(t)}\mathbf{X})^{-1}\mathbf{W}^{(t)}\tilde{\mathbf{y}}^{(t)} \quad t = 0, 1, 2, \ldots$$

with working observations:

$$\tilde{\mathbf{y}}^{(t)} = (\tilde{y}_1^{(t)}, \ldots, \tilde{y}_n^{(t)}), \quad \tilde{y}_i^{(t)} = \hat{\eta}_i^{(t)} + \frac{y_i - h(\hat{\eta}_i^{(t)})}{h'(\hat{\eta}_i^{(t)})} \quad \hat{\eta}_i^{(t)} = \mathbf{x}_i'\hat{\boldsymbol{\beta}}^{(t)}$$

and working weights:

$$\tilde{w}_i^{(t)} = \frac{(h'(\hat{\eta}_i^{(t)}))^2}{\sigma_i^2(\hat{\eta}_i^{(t)})} \quad \text{stored into } \mathbf{W}^{(t)} = \text{diag}(\tilde{w}_1^{(t)}, \ldots, \tilde{w}_n^{(t)})$$

Computational Problems

- when large dataset
- when the matrix is dense
- when covariates are higly correlated $\mathbf{X}'\mathbf{W}^{(t)}\mathbf{X})^{-1}$ might not have full rank

Requirement: $(\mathbf{X}'\mathbf{W}^{(t)}\mathbf{X})$ invertible!

If $h(\cdot)$ is canonical, then IWLS maximum (solution) is unique.

Otherwise, multiple starting values.

Note

The idea is to minimize the distance between a point and a straight line.

$\Rightarrow$ We can abstract from distributional assumptions and reformulate the problem as a least squares problem.
Do the notation and insert weights (same result as Fisher scores), but easier.

$$\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{Y} \quad \Rightarrow \quad \text{solution for the Gaussian model}$$

10.2.2 Asymptotic properties of ML estimates

Ef function are nice for these asymptotical properties

If $h(\cdot)$ is canonical, then:

$$(\mathbf{X}_n'\mathbf{X}_n)^{-1} \xrightarrow{n\to\infty} 0 \quad \text{or} \quad \lambda_{\min}(\mathbf{X}_n'\mathbf{X}_n) \xrightarrow{n\to\infty} \infty$$

is sufficient for β to have:

- (i) asymptotic gaussianity
- (ii) weak consistency

Asymptotic variance-covariance matrix of the estimator is $F^{-1}(\hat{\boldsymbol{\beta}})$

10.2.3 Overdispersion parameter

For EFs:

$$\text{Var}(y_i) = \phi b''(\theta_i)/w_i \quad \text{where } v(\mu_i) \text{ is variance function, and } \mu_i = b'(\theta_i)$$

A consistent estimator is:

$$\hat{\phi} = \frac{1}{G-p}\sum_{i=1}^{G}\frac{(y_i-\hat{\mu}_i)^2}{v(\hat{\mu}_i)/n_i} \quad \text{with } \hat{\mu}_i = h(\mathbf{x}_i'\hat{\boldsymbol{\beta}})$$

All quantities concerning ϕ are adjusted using $\hat{\phi}$.

The inverse of the Fisher information matrix $\mathcal{F}^{-1}(\hat{\boldsymbol{\beta}})$ is going to dictate the result of the t-test of each coefficient.

Eigenvalues

Eigenvalue: just a measure of the true number of dimensions that you have in the data. For example, if $\boldsymbol{X}$ is $n \times p$ and has small eigenvalues, then $\boldsymbol{X}$ is structured in a way that does not require full dimensionality.

$\Rightarrow$ It measures the variability or strength of that dimension, but also other things.

Conclusion

Using the **canonical link function** has a lot of good properties:

- For **Binomial** $\rightarrow$ `logit`
- For **Poisson** $\rightarrow$ `log`
- For **Gamma** (if variance is not appropriate), canonical link might not be suitable

10.3 Model fit and selection for GLMs

Hypothesis testing for GLMs

- definition of $\ell(\boldsymbol{\beta})$, $s(\boldsymbol{\beta})$, and $F(\boldsymbol{\beta})$ specific to the GLM
- LR, Wald, and score statistics are asymptotically χ^2-distributed

Pearson and Deviance statistics

$$\chi^2 = \sum_{i=1}^{G} \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)/w_i} \qquad D = -2\sum_{i=1}^{G} [\ell_i(\hat{\mu}_i) - \ell_i(\bar{y}_i)]$$

Both measure model fit relative to saturated model (where $\mu_i = \bar{y}_i$).
When n_i large in each group, both are asymptotically χ^2 distributed.
Akaike Information Criterion

$$\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}) + 2p$$

10.4 Quasi-likelihood for EFs

In EFs/GLMs, assuming $\mathbb{E}(y) = \mu = h(\mathbf{x}_i'\boldsymbol{\beta})$ implies:

$$\text{Var}(y_i) = \phi b''(\mu_i)/w$$

However, if empirical variance $\neq$ estimated variance $\rightarrow$ **misspecified model!**
Quasi-likelihood approach: specify $\mathbb{E}(y) = h(\mathbf{x}'\boldsymbol{\beta}) = \mu_i$, and:

$$\text{Var}(y) = \sigma_i^2$$

even if resulting likelihood is not proper.
Estimates are the solution to **generalized estimating equation (GEE)**:

$$\mathbf{s}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i \frac{h'(\eta_i)}{\sigma_i^2}(y_i - \mu_i) \qquad \text{with } \mathbb{E}[\mathbf{s}(\boldsymbol{\beta})] = 0$$

and working variance:

$$\sigma^2 = \sigma^2(\mu) = \phi v(\mu)/w \quad \text{with } v(\mu) \text{ quasi-variance function.}$$

10.4.1 Quasi-ML estimator and asymptotics

Quasi-ML estimator: $\hat{\boldsymbol{\beta}}$ obtained iteratively using quasi-Fisher information matrix.
Quasi-ML Fisher information matrix:

$$F(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i' \tilde{w}_i, \quad \text{with } \tilde{w}_i = h'(\eta_i)\sigma_i^2$$

In quasi-likelihood approaches, usually $F(\boldsymbol{\beta}) \neq \text{Cov}(\boldsymbol{\beta})$

Asymptotically, quasi-ML estimators are consistent and Gaussian:

$$\boldsymbol{\beta} \sim \mathcal{N}\left(\boldsymbol{\beta}, \hat{F}^{-1}\hat{V}\hat{F}^{-1}\right)$$

Variance-covariance matrix is a "**sandwich**" adjusted ML matrix.

Important using a quasi-likelihood approach an GEE you loose connection between fisher matrix and Cov($\boldsymbol{\beta}$)

10.5 Exam possible question

Estimation of Overdispersion (Might be on the exam)

- Requires **group data**; the formula has a G index.
- You need a minimum amount of information in each group to leverage the χ^2 asymptotics.

Any linear regression model is about how y_i is affected by $\boldsymbol{x}$:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \varepsilon_i \quad \text{where } g = 1, \ldots, G$$

In GLM if you assume:

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2) \quad \Longleftrightarrow \quad y_i \sim \mathcal{N}(\eta_i, \sigma^2)$$

If y_i is a Bernoulli variable:

$$\mathbb{E}[y_i|\boldsymbol{X}] = h(\eta_i) = \pi_i$$

$$\text{Var}(y_i) = \pi_i(1 - \pi_i)$$

But the **empirical variance** $\frac{\hat{\pi}_i(1-\hat{\pi}_i)}{m_i}$ (from group data) does not match the theoretical one

You rescale the variance by adjusting with a **rescaling factor**.
ϕ is estimated using either:

$$\hat{\phi}_P \quad \text{or} \quad \hat{\phi}_D \quad \text{based on deviance or Pearson}$$

$\Rightarrow \chi^2_{\text{stat}}$

Both statistics measure the **variability left after fitting**, like in Gaussian models:

$$\sigma^2 = \frac{\text{RSS}}{n - p} \quad \text{(residual sum of squares)}$$

Interpretation: what is left *outside* the model ⇒ variability of the residuals.

$$\phi < 1 \Rightarrow \text{data are underdispersed}$$

This happens when you do not have parameters for μ or σ^2 (variance).

AIC is also useful to choose the link function (when there is no $\hat{\phi}$).

How to Choose Between Poisson and Quasi-Poisson?

Group data implies you are measuring correlations.

$$\text{Underestimating } \hat{\phi} \quad \Rightarrow \quad \text{Var}(\cdot,\cdot) = \text{Var}(\cdot) + \text{Var}(\cdot) + \text{Cov}(\cdot,\cdot)$$

Part X

Categorical regression

1. Model definition

When the response y has more than 2 levels/categories/modalities .

$$\pi_r = \mathbb{P}(Y = r), \quad Y \in \{1, \dots, r, \dots, c+1\}$$

- $\boldsymbol{Y} = (Y_1, \dots, Y_n)^\top$: n-dimensional random variable for the dependent variable
- $\boldsymbol{y} = (y_1, \dots, y_n)^\top$: observed sample values it remains the IID assumption

Type of categorical variable:

- ordinal but no sequential (severity of injure in a car crush)
- ordinal and sequential (instruction level, you need a diplome to have a bachelor etc..)
- nominal (not ordinal)

Note: do note consider it as a discrete random variable!

1.1 From categorical to dummy

$$\pi_{ir} = \mathbb{P}(y_{ir} = r \mid \boldsymbol{x_i})$$

Categorical regression attempts to model the frequency of the category , which in the population is the probability of observing exact this category (π_{ir}).
You need to model $c+1$ probabilities, since $\pi_{i,c+1}$ can be obtain as the difference between 1 - the others. Since one event must be observed .

$$\Rightarrow \text{ we can model } \boldsymbol{y} = (y_1, \dots, y_c)' \text{ as a vector of dummy variables.}$$

Reference category
With $c+1$ levels we only need c dummies: one category becomes the *reference*.
If ordinal, we use the last one (or the first one) $c+1$ s.t. $y_{c+1} = 1 - y_1 - \cdots - y_c$.
When they are not ordered you have to decide which one, since all the others parameter will be interpreted with respect to it.
You cannot, fit a models that deal with all $c+1$ categories

For Example

$$\boldsymbol{Y} = \{\text{"HS"}, \text{"Bach"}, \text{"Ms"}, \text{"Phd"}\}$$

To model this set of probability the best you can do is code the categorical variable as binary . This way you have a vector of random variables.

Label	Level	y_1	y_2	y_3	y_4
HS	1	1	0	0	0
BACH	2	0	1	0	0
MS	3	0	0	1	0
PHD	4	0	0	0	1

In this table you do not need the last one. so we remain with

$$y_r = \begin{cases} 1, & Y = r \\ 0, & \text{otherwise} \end{cases} \qquad r = 1, \dots, c$$

Such that the **mapping** is the following

$$Y = r \iff \boldsymbol{y} = (0, \dots, \underbrace{1}_{r\text{-th element}}, \dots, 0)'$$

Modelling the probability of observing the level r, now this is equivalent of the probability that the binary random variable, among all the others, is gonna be the active one. This transform the problem in a **binary regression** Problem.

Why we do not model the separately
Fitting c Binary regression models is not an option. The problem is that they are not independent, you cannot model each proportion seperatly from each others. The data have a proportion. Since each model has its own procedure it might happen that the sum of the proportion from each independent model might not sum up to one.

$$\pi_r = \mathbb{P}(Y = r) = \mathbb{P}(y_r = 1), \quad r = 1, \dots, c$$

1.2 Distributional assumption

- m independent trials
- $\boldsymbol{y} = (y_1, \dots, y_c)'$ where each y_r is the number of times category r is observed.
- $\boldsymbol{\pi}$ the vector containing$(\pi_i, \cdots, \pi_c)$, with $\pi_{c+1} = 1 - \sum_j^c \pi_{ij}$

$$\boldsymbol{y} \sim \text{MultiNom}(m, \boldsymbol{\pi})$$
$$f(\boldsymbol{y} \mid \boldsymbol{\pi}) \propto \pi_1^{y_1} \cdots \pi_c^{y_c} (1 - \pi_1 - \cdots - \pi_c)^{m - y_1 - \cdots - y_c}$$

Basically the Multinomial is a product of the probability of the c category eleveted to the numer of observed units presenting this category, with $(1 - \pi_1 - \cdots - \pi_c)^{m - y_1 - \cdots - y_c}$ this represent the reference category

Note:it is proportional to it, it is not the actual distribution. The combinatorial term is missing but it does not depends the parameter, so we are not interested.

with

$$\mathbb{E}(\boldsymbol{y}) = m\boldsymbol{\pi} = \begin{pmatrix} m\pi_1 \\ \vdots \\ m\pi_c \end{pmatrix}, \quad \text{Cov}(\boldsymbol{y}) = \begin{pmatrix} \pi_1(1-\pi_1) & \cdots & -\pi_1\pi_c \\ \vdots & \ddots & \vdots \\ -\pi_c\pi_1 & \cdots & \pi_c(1-\pi_c) \end{pmatrix}$$

The covariance matrix is not diagonal, and present a negative correlation in nono diagonal elements, which is reasonable: the categories are competing on each observation, the larger the proportion for the r category the lower the proportion of the others (since they have to sum up to 1, increasing one proportion means decreasing another.

We wanted to model **frequencies**, which are probabilities on the population, but accounting for dependence. Where π_1m represent the number of people from category 1 (in our example HS). We can use scaled version $\bar{\boldsymbol{y}} = (\bar{y}_1, \ldots, \bar{y}_c)' \sim \text{MultiNom}(m, \boldsymbol{\pi})/m$ with relative frequencies $\bar{y}_r = y_r/m$, so that you have the expecation is a proportion.

$$\mathbb{E}(\bar{\boldsymbol{y}}) = \boldsymbol{\pi}, \quad \text{Cov}(\bar{\boldsymbol{y}}) = \frac{1}{m}\text{Cov}(\boldsymbol{y})$$

1.3 Group data

Ungrouped data: n units with $\boldsymbol{y}_i \sim \text{MultiNom}(1, \boldsymbol{\pi}_i)$, $\quad \boldsymbol{\pi}_i = (\pi_{i1}, \dots, \pi_{ic})'$

$$\begin{pmatrix} \boldsymbol{y}_1' = (y_{11}, \dots, y_{1c})' \\ \vdots \\ \boldsymbol{y}_i' = (y_{i1}, \dots, y_{ic})' \\ \vdots \\ \boldsymbol{y}_n' = (y_{n1}, \dots, y_{nc})' \end{pmatrix} \quad \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{i1} & \cdots & x_{ik} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix} \qquad i = 1, \dots, n$$

Each observation has his own vector, so you do considere the entire vector, if they ask you which is the value for that observation you respond $(0, 1, 0)$ (in our example), not just 1, it is still one *single phenomenon* that has been expanded.

Grouped data: G groups with $\boldsymbol{y}_i \sim \text{MultiNom}(n_i, \boldsymbol{\pi}_i)$

You just collapse them and you consider the frequency of each group, the only thing you have to remember is the number of people in each group (vector $\boldsymbol{n}$)

$$\begin{pmatrix} n_1 \\ \vdots \\ n_G \end{pmatrix} \quad \begin{pmatrix} \bar{\boldsymbol{y}}_1^\top = (\bar{y}_{11}, \dots, \bar{y}_{1c})^\top \\ \vdots \\ \boldsymbol{y}_G' = (\bar{y}_{G1}, \dots, \bar{y}_{Gc})^\top \end{pmatrix}, \quad \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{G1} & \cdots & x_{Gk} \end{pmatrix} \quad i = 1, \dots, G$$

We want to model/estimate $\pi_{ir} = \mathbb{P}(Y_i = r) = \mathbb{P}(y_{ir} = 1)$ given covariates $\boldsymbol{x}_i$.

Index of the probabilities: because depending on the covariates, the index tell us that each individual has different features ($\boldsymbol{x}_i$).

Here being 2 indeces means that your output represent the **probability that the indiviual i is gonna respond r to the question**

You can have information also on the level (*Example:*amount of exam to obtain a degree), covariates specific to the category, (*Example:*for each level the age)

Example It gets more interesting for groups (for example belonging to a state) you want to estimates 3 sets of probability (vote for democrats, republican, not gonna vote).

2. Models for unordered categories

2.1 Generalization of logit model to *categorical (multinomial)* logit

$$\mathbb{P}(Y_i = r) = \pi_{ir} = \frac{\exp(\boldsymbol{x}_i'\boldsymbol{\beta}_r)}{1 + \sum_{s=1}^{c} \exp(\boldsymbol{x}_i'\boldsymbol{\beta}_s)}, \qquad r = 1, \ldots, c$$

Each individual probability has a probability of presenting category r (π_{ir}) is the effect of it's own parameter (β_r)with respect to all the other potential categories you could see ($1 + \sum_{s=1}^{c} \exp(\boldsymbol{x}_i'\boldsymbol{\beta}_s)$) because each category is competing.

For the reference category : we have that $\beta_{c+1} = \mathbf{0}_r$ and so $\exp(\boldsymbol{x}_i'\mathbf{0}_r) = 1 \rightarrow$ setting a category as the reference means setting its coefficient $= 0$

$$\pi_{i,c+1} = \frac{1}{1 + \sum_{s=1}^{c} \exp(\boldsymbol{x}_i'\boldsymbol{\beta}_s)}$$

2.1.1 Linear predictor

$$\log\left(\frac{\pi_{ir}}{\pi_{i,c+1}}\right) = \eta_{ir} = \boldsymbol{x}_i'\boldsymbol{\beta}_r \quad \text{or} \quad \frac{\pi_{ir}}{\pi_{i,c+1}} = \exp(\boldsymbol{x}_i'\boldsymbol{\beta}_r)$$

Regarding $\boldsymbol{\beta}_r$ and η_{ir}:

- model (log)odds or relative risk between category r and **reference category**
- are **specific for each category**
- $\boldsymbol{\beta}$ coefficients explain increase/decrease due to change in covariates but always **relative to reference category**.

$$\begin{bmatrix} \beta_{11} & \cdots & \beta_{1c} \\ \vdots & \ddots & \vdots \\ \beta_{p1} & \cdots & \beta_{pc} \end{bmatrix}$$

Each regressor has c coefficients, for a total of $c \cdot p$ coefficients, since each y's category needs its own propability (given $\boldsymbol{X}$), same as fitting c interdependent binary regressions.

2.2 Latent utility formulation

Latent is Assumed, (something you can't see) and so you can model it as you want.
Means that there is a continous underline process and what you observe is just the discretization.

In categorical framework you assume that each observed categories is in association with a latent utilities (non observed).

For each category $r = 1, \dots, c+1$ we assume $u_r = \bar{\eta}_r + \varepsilon_r$ with $\bar{\eta}_r$ constant and ε_r a r.v. with F as its CDF. Then,

$$Y = r \iff u_r = \max_{1,\dots,c+1} u_s$$

The linear predictor from the latent formulation is correlated with the one in the logit. This transform the Problem into a linear formulation (without a response function). $\Rightarrow$ you can always right down a logit model using a latent representation.

If $F(\varepsilon_r) = \exp(-\exp(-\varepsilon_r))$ and $c+1$ is reference category (for identifiability)

$$\mathbb{P}(Y = r) = \frac{\exp(\bar{\eta}_r - \bar{\eta}_{c+1})}{1 + \sum_{s=1}^{c} \exp(\bar{\eta}_s - \bar{\eta}_{c+1})} = \frac{\exp(\eta_r)}{1 + \sum_{s=1}^{c} \exp(\eta_s)}$$

Note: this is actually the multinomial logit

Extended multinomial logit: category-specific covariates $\boldsymbol{w}_r$ can be added to the subject-specific (global) covariates $\boldsymbol{x}_i$ to obtain:

Other models: if each ε_r is a gaussian $\varepsilon_r \sim \mathcal{N}(\mu_r, \sigma^2)$we get independent probit model; if the joint vector is Gaussian $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mu, \Sigma)$ we get multivariate probit model.

2.3 Level-specific effects

$$\pi_{ir} = \frac{\exp(\boldsymbol{x}_i'\boldsymbol{\beta}_r + (\boldsymbol{w}_{ir} - \boldsymbol{w}_{i,c+1})'\boldsymbol{\gamma})}{1 + \sum_{s=1}^{c} \exp(\boldsymbol{x}_i'\boldsymbol{\beta}_s + (\boldsymbol{w}_{is} - \boldsymbol{w}_{i,c+1})'\boldsymbol{\gamma})}$$

γ is category-specific, related to category-specific covariates.
π_{ir} is gonna depende on a set of parameter β which are levele specific and a set of covariates which are individual specific x_i, you can then introduce covariates $w_{i,r}$that are both units-specific and also category-specific.

Note: you have to maintain the same reference category for all the parameters, to account for the reference in the w you subtract the reference category $(\boldsymbol{w}_{ir} - \boldsymbol{w}_{i,c+1})$.

2.4 Dataset example -output

```
Call:
multinom(formula = Y ~ NPLAN + RISK + ANTIB, data = csectionlong)

Coefficients:
          (Intercept)     NPLAN      RISK     ANTIB
Type I     -2.621012    1.174251   1.829241  -3.520249
Type II    -2.559917    0.995976   2.195458  -3.087160

Std. Errors:
          (Intercept)     NPLAN      RISK     ANTIB
Type I     0.5567209    0.5213014  0.6023322  0.6717416
Type II    0.5462995    0.4813634  0.5869601  0.5498675

Residual Deviance: 321.8743
AIC: 337.8743
```

RISK	NPLAN	ANTIB	no infection	Type I	Type II	
1	1	0	1	0.957	0.013	0.030
2	0	1	0	0.994	0.002	0.004
3	1	0	0	0.466	0.211	0.323
4	0	1	0	0.870	0.063	0.067
5	1	0	1	0.886	0.038	0.076
6	1	1	0	0.230	0.337	0.433
7	0	1	0	0.692	0.163	0.145

Interpretation

negative coefficient means that i have a decrease of the risk of developing an infection with respect to developing no infection.
With another function you obtain a the table displays the predicted probabilities of each categories per group.

3. R

```
library("car")
library("faraway")
library(nnet) #neural network (c'è collegamento tra NN e categorical regression)
library(MASS)
library(mlogit)
```

Descrizione delle variabili del dataset.

- `incomegp`: Gruppo di reddito (1 = più basso, 5 = più alto).
- `house`: Sicurezza della condizione abitativa (1 = affitto, 2 = mutuo, 3 = proprietà senza mutuo).
- `children`: Numero di figli nel nucleo familiare.
- `singpar`: Genitore single (1 = sì, 0 = no).
- `agegp`: Gruppo di età del rispondente (1 = più giovane).
- `bankacc`: Il rispondente ha un conto bancario? (1 = sì, 0 = no).
- `bsocacc`: Il rispondente ha un conto in una società edilizia? (1 = sì, 0 = no).
- `manage`: Valutazione soggettiva delle abilità nella gestione del denaro (valori alti = abilità elevata).
- `ccarduse`: Frequenza d'uso della carta di credito (1 = mai, 2 = occasionalmente, 3 = regolarmente).
- `cigbuy`: Il rispondente acquista sigarette? (1 = sì, 0 = no).
- `xmasbuy`: Il rispondente acquista regali di Natale per i bambini? (1 = sì, 0 = no).
- `locintrn`: Punteggio sulla scala del locus of control (valori alti = orientamento più interno).
- `prodebt`: Punteggio sulla scala di atteggiamento verso il debito (valori alti = atteggiamento favorevole verso il debito).

Descriptive analysis

```
tab1 <- prop.table(table(Num.Children=debt$children,
    CreditCardUsage=debt$ccarduse),1)
tab1
round(tab1,2)
```

```
             CreditCardUsage
 Num.Children    1    2    3
            0 0.49 0.27 0.25
            1 0.59 0.23 0.19
```

```
2 0.61 0.22 0.17
3 0.53 0.19 0.28
4 0.50 0.40 0.10
5 1.00 0.00 0.00
6 1.00 0.00 0.00
```

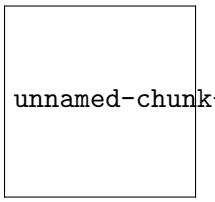
prende table di due categorical quantities e poi prende le proporzioni

quando $x = 0$(0 bambini) → CreditCardUsage è distribuito in modo empirico così: 0.49 0.27 0.25

```
#?matplot
```

→ modo per plottare categorical quantities

```
matplot(0:6,tab1, type="l", xlab="Number of Childred", ylab="Proportions")
```

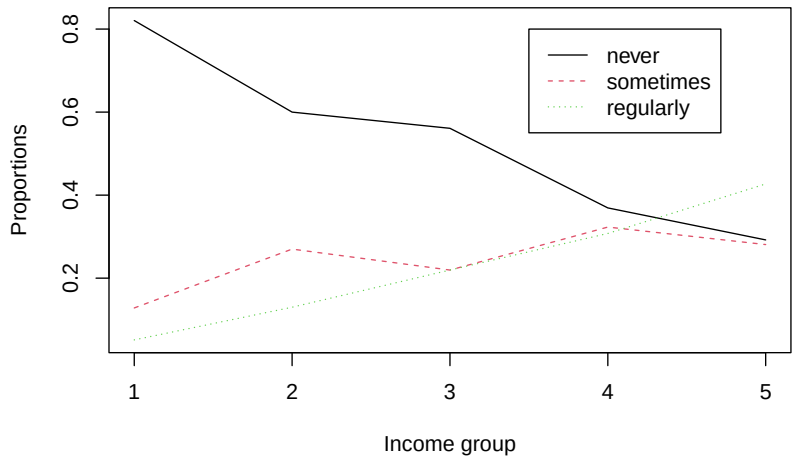


Abbiamo un set di parametri per ogni livello di output, per questo ci sono tre rette (interdipendenti)

```
summary(debt$incomegp)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  1.000   2.000   3.000   2.894   4.000   5.000      19
```

```
tab2 <- prop.table(table(IncomeClass=debt$incomegp,
    CreditCardUsage=debt$ccarduse),1)
matplot(1:5,tab2, type="l", xlab="Income group", ylab="Proportions")
legend(3.5,0.8,legend=c("never","sometimes","regularly"),col=1:3,lty=1:3)
```



Multinomial logistic regression model: fit

```
#?multinom #multinomial distribution -> categorical regression (vedremo logit model)

mod <- multinom(ccarduse ~ incomegp + children, debt)
    #la formula è simile a quella vista fino a adesso
```

```
 weights:  12 (6 variable)
initial  value 454.825488
iter  10 value 387.067521
final  value 387.067005
converged
```

`weights:  12 (6 variable)` da il numero totale di parametri nel modello.
Se l'algoritmo non converge vuol dire che il modello non fitta.

```
mod
```

```
Call:
multinom(formula = ccarduse ~ incomegp + children, data = debt)

Coefficients:
  (Intercept)  incomegp   children
2   -1.814161 0.4074938 -0.1307578
3   -2.993671 0.7281990 -0.1706964

Residual Deviance: 774.134
AIC: 786.134
```

```
summary(mod) #aggiunge gli standard errors
```

```
Call:
multinom(formula = ccarduse ~ incomegp + children, data = debt)

Coefficients:
  (Intercept)  incomegp   children
2   -1.814161 0.4074938 -0.1307578
3   -2.993671 0.7281990 -0.1706964

Std. Errors:
  (Intercept)   incomegp  children
2   0.3122300 0.09110833 0.1074471
3   0.3867818 0.10266386 0.1194816

Residual Deviance: 774.134
AIC: 786.134
```

I p-value non sono inclusi nel summary (abbiamo solo i coefficienti e gl standard error).

Coefficient interpretation

Coefficients ***Domanda esame:*** *dato l'output è un order model o cumulative model?* Se abbiamo un set di coefficienti che diverso per ogni livello significa che tratti la risposta come unordered come unordered É l'unico caso in cui hai vettori di coefficienti per ogni livello.

categorical predictor interpretation effetto di incomegp → consideri il rischio relativo della categoria 2 rispetto alla categoria 1 (se non trasformi sei in scala logaritmica.)

`0.4074938` (l'effetto sul log(odds)) dove gli odds sono essere in categoria 2 rispetto alla probabilità di essere in categoria 1 –quando muovo il mio predittore da x a $x+1$.

Quindi quando muovo da una classe di income alla successiva 0.4074938 è l'effetto sul rischio di essere nella categoria 2: usare la carta di credito qualche volta rispetto a non usarla mai.

```
exp(coefficients(mod)) #odds scale -> ci può dare l'usal interpretation
```

```
  (Intercept) incomegp  children
2  0.16297466 1.503046 0.8774303
3  0.05010315 2.071347 0.8430775
```

Interpretazione sugli odds `1.503046` → aumento del 50 % del rischio relativo da never a sometimes Se vuoi avere lo stesso impatto di income quando considere l'ultima categoria ma `2.071347` quindi c'è un aumento del 100%, quindi muoversi nella categoria seconda di income produce un 100% di aumento quando consideri da never a regularly. Abbiamo un increase amount di

Hypothesis testing-nested models

```
summary(mod) #non abbiamo i p-value
```

```
Call:
multinom(formula = ccarduse ~ incomegp + children, data = debt)

Coefficients:
  (Intercept)  incomegp   children
2   -1.814161 0.4074938 -0.1307578
3   -2.993671 0.7281990 -0.1706964

Std. Errors:
  (Intercept)   incomegp  children
2   0.3122300 0.09110833 0.1074471
3   0.3867818 0.10266386 0.1194816

Residual Deviance: 774.134
AIC: 786.134
```

`incomegp`

Cosa succede quando il predittore `incomegp` è significativo per il secondo livello (es. gruppo 2 vs riferimento), ma non per l'ultimo livello (es. gruppo 5 vs riferimento)?

poiché la variabile income ha coefficienti distinti per ciascun livello, è possibile che influenzi alcuni livelli ma non altri. Per decidere se mantenerla nel modello, è opportuno valutare la sua significatività complessiva tramite un test globale, e non basarsi solo sui p-value dei singoli coefficienti.

- Deviance test
- AIC
- **NO** linear hp test , è difficile per il setup qua.

`children`

```
mod1 <- multinom(ccarduse ~ incomegp, debt)
```

```
 # weights:  9 (4 variable)
 initial  value 454.825488
 iter  10 value 388.435868
 iter  10 value 388.435865
 final  value 388.435865
 converged
```

```
mod1
```

```
 Call:
 multinom(formula = ccarduse ~ incomegp, data = debt)

 Coefficients:
   (Intercept)   incomegp
 2    1.935531 -0.4050475
 3   -1.216433  0.3210794

 Residual Deviance: 776.8717
 AIC: 784.8717
```

```
summary(mod1)
```

```
 Call:
 multinom(formula = ccarduse ~ incomegp, data = debt)

 Coefficients:
   (Intercept)   incomegp
 2    1.935531 -0.4050475
 3   -1.216433  0.3210794
```

```
Std. Errors:
  (Intercept)   incomegp
2   0.2978532 0.09091127
3   0.4192959 0.11036030

Residual Deviance: 776.8717
AIC: 784.8717
```

Confrontiamo le differenze tra i due modelli.

Note: la differenza deve essere fatta in modo tale da non essere negativa (sempre prima il modello con meno predittori che avrà sempre deviance maggiore).

→ É questa differenza significativamente piccola?
→ Posso considerare questa differenza nella deviance qualcosa che i predittori portano alla tabella o è solo un problema di randomness?

```
dev <- mod1$dev - mod$dev
dev1 <- mod$dev - mod1$dev
dev

 [1] 2.73772
dev1 #hai usato il wrong order dei modelli

 [1] -2.73772
```

```
gdl <- mod$edf - mod1$edf #differenza tra gradi di libertà
 [1] 2
gdl
```

Nel contesto della regressione multinomiale, i gradi di libertà sono determinati dal numero di parametri del modello, quindi sono noti e fissi. Il termine *estimated degrees of freedom (edf)* compare solo per motivi terminologici, legati all'uso di pacchetti generalizzati (ad es. reti neurali o smoothing).

```
p_val <- 1 - pchisq(dev, gdl) # p-value del test del deviance
 [1] 0.2543968
```

→ non è significativo In questo caso, non si rigetta l'ipotesi nulla, che ricordiamo essere: *"il modello più semplice è sufficiente"*. La riduzione nella devianza pari a 2.73772 non è sufficientemente grande da giustificare l'inclusione del predittore. Pertanto, si preferisce il modello più semplice, senza includere la variabile aggiuntiva.

```
#plot(mod)
# Errore in xy.coords(x, y, xlabel, ylabel, log) :
#  'x' è una lista, ma non ha componenti 'x' e 'y'
# Questo per mostrare che le categorical reegression si focalizzano di più sulle
# predizioni che sull'inferenza
```

4. Models for ordered categories

With the orders comes more information, it means that you are able to leverage this information:

- making the models more accurate
- reducing the number of parameter (which might lead to overfitting)

Ordered category can be

- sequential
- not sequential, *you can jump from one to another/ cumulative*

Most models for ordered categories are (i) parsimonious (less parameters) (ii) and use a thresholding mechanism.

4.1 Cumulative model

$$y_r = \begin{cases} 1, & u_r > 0 \\ 0, & u_r < 0 \end{cases}$$

thresholding mechanism → the threshold is zero when the latent reach it $y_i = 1$. It is simple when binary, when categorical you have more than one threshold.

4.1.1 Definition

Let u be unobserved

$$u_i = -x_i'\beta + \varepsilon_i$$
$$\text{or} \quad u_i = -\eta_i + \varepsilon_i$$

Note: we do not include the intercept (otherwise you cannot identify model's parameter).

Then

$$Y_i = r \iff \theta_{r-1} < u_i < \theta_r,$$

with

- $r = 1, \ldots, c+1$
- and latent thresholds $-\infty = \theta_0 < \theta_1 < \cdots < \theta_c < \theta_{c+1} = +\infty$

Which mean that there is a one to one correspondence, and also that there are needed $c+1$ thresholds.

$\tilde{\eta_r} = u_r - \varepsilon_r \Rightarrow \quad \tilde{\eta}_{c+1} = u_{c+1} - \varepsilon_{c+1}$

$\eta_r = \tilde{\eta_r} - \tilde{\eta}_{c+1}$ (always considering with respect to the reference category).

$\Rightarrow \eta_r = u_r - \varepsilon_r - u_{c+1} - \varepsilon_{c+1}$ → to this we do apply the threshold meccanism

Recall that $\eta_{c+1} = 0$

Note: you are not modelling $\pi_r = \mathbb{P}(Y_i = r)$, you are modelling the probability of staying up to r ($\mathbb{P}(Y_i \leq r)$), so that the latent u_i is below the threshold $\rightarrow$, which means that the category that you reach is at least r

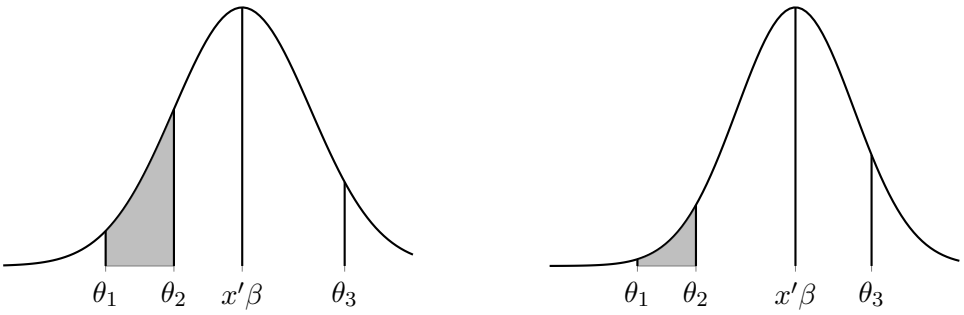
$$\mathbb{P}(Y_i \leq r) = \mathbb{P}(u_i \leq \theta_r) = \mathbb{P}(-x_i'\beta + \varepsilon_i \leq \theta_r) = \mathbb{P}(\varepsilon_i \leq x_i'\beta + \theta_r) = F(\theta_r + x_i'\beta)$$

When saying that the above is the CDF, that we are interested in, it follows that the random variable of interest is ε_i. so the distribution of ε determines the formulation of the cumulative logistic model (or cumulative logit model).

Since y is not continous to evaluate the precise probability you just compute:

$$\mathbb{P}(Y_i = r) = F(\theta_r + x_i'\beta) - F(\theta_{r-1} + x_i'\beta)$$

4.1.2 The effect of β



A positive shift of the linear predictor, decrease the probability of belonging to the previous category. The density is mooving while the threshold is fixed.

Probability of staying in the second category is the probability of staying up to θ_2- Probability of staying below θ_1. The effect of β if you slide the linear predictor to be larger (when $\beta_r > 0$) suddenly the lower category has lower probability

4.2 Example Dataset

Ungrouped data, the response is the severity of the damage (categorical and ordered), effect of the predictors of staying up to a category, there are plenty predictor (20), but due to being two category you should have 40 parameter, but there is 20.

You do not now if the coefficient is significant, anyway a negative β means a dicrease (but it depends on the parametrization of your software)

Note: when there is written intercept it actually refers to threshold, when data are balanced usually are evenly spread out.

```
Call:
polr(formula = defol3 ~ ageo + ageo2 + ageo3 + yearo + yearo2 +
    yearo3 + gradient + canopyd + alt + depth + type + fert +
    humus0 + humus2 + humus3 + humus4 + watermoisture1 + watermoisture3 +
    alkali1 + alkali3 + alkali4, data = forest)

Coefficients:
                  Value      Std. Error  t value
ageo              1.137020    0.087969   12.9252
ageo2            -0.476667    0.081710   -5.8336
ageo3             0.138459    0.068650    2.0169
yearo            -0.064551    0.060147   -1.0732
yearo2           -0.113025    0.059724   -1.8924
yearo3           -0.319380    0.064156   -4.9782
gradient          0.002722    0.006703    0.4061
canopyd          -0.031549    0.003029  -10.4167
alt              -0.000233    0.001358   -0.1718
depth            -0.020476    0.006863   -2.9836
type             -0.506335    0.072742   -6.9580
fert             -0.478945    0.124341   -3.8519
humus0           -0.250414    0.127071   -1.9707
humus2            0.324633    0.115472    2.8114
humus3            0.218513    0.132473    1.6495
humus4           -0.184750    0.176823   -1.0448
watermoisture1   -0.566765   0.149370   -3.7944
watermoisture3    0.450174   0.106808    4.2148
alkali1           0.542298   0.150486    3.6036
alkali3          -0.493799   0.153862   -3.2094
alkali4           0.540431   0.190051    2.8436

Intercepts:
                                 Value   Std. Error t value
no damage|weak damage           -2.0117    0.6415   -3.1361
weak damage|severe damage        2.0259    0.6449    3.1415
```

4.3 Proportional Odds Linear Regression

It is a cumulative logit Model, The structure is the same of the binary model (logit in particular).
The effect of the linear predictor is on the logartithm of the odds that actually compared the probability of staying below the threshold with the probability of staying above.
Note: no intercept

By choosing the logistic distribution for F we get *cumulative logit model*

$$\mathbb{P}(Y_i \leq r) = \frac{\exp(\theta_r + x_i'\beta)}{1 + \exp(\theta_r + x_i'\beta)} \quad \text{or, equivalently} \quad \log\left(\frac{\mathbb{P}(Y_i \leq r)}{\mathbb{P}(Y_i > r)}\right) = \theta_r + x_i'\beta$$

also called *proportional odds model* because the odds ratio does not depend on the category

(r) so the effect of β is constant trough the categories.

$$\frac{\mathbb{P}(Y_i \leq r \mid x_i)}{\mathbb{P}(Y_i > r \mid x_i)} \Big/ \frac{\mathbb{P}(Y_i \leq r \mid \tilde{x}_i)}{\mathbb{P}(Y_i > r \mid \tilde{x}_i)} = \exp\left((x_i - \tilde{x}_i)'\beta\right)$$

4.3.0.1 Threshold interpretation

Here you do not observe neither the reference category nor intercept, since thresholds became intercept.

```
library(MASS)
low <- read.table("lowbwt.csv", header=TRUE, sep=",")
str(low)
```

```
'data.frame':    189 obs. of  11 variables:
 $ ID   : int  85 86 87 88 89 91 92 93 94 95 ...
 $ LOW  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ AGE  : int  19 33 20 21 18 21 22 17 29 26 ...
 $ LWT  : int  182 155 105 108 107 124 118 103 123 113 ...
 $ RACE : int  2 3 1 1 1 3 1 3 1 1 ...
 $ SMOKE: int  0 0 1 1 1 0 0 0 1 1 ...
 $ PTL  : int  0 0 0 0 0 0 0 0 0 0 ...
 $ HT   : int  0 0 0 0 0 0 0 0 0 0 ...
 $ UI   : int  1 0 0 1 1 0 0 0 0 0 ...
 $ FTV  : int  0 3 1 2 0 0 1 1 1 0 ...
 $ BWT  : int  2523 2551 2557 2594 2600 2622 2637 2637 2663 2665 ...
```

let'scategorize/discretize variable 'BWT'
levBWT = 0 if BWT > 3500
levBWT = 1 if 3000 < BWT <= 3500
levBWT = 2 if 2500 < BWT <= 3000
levBWT = 3 if BWT <= 2500

```
levBWT <- cut(low$BWT, c(0,2500, 3000, 3500, 10000), right=TRUE, label=FALSE)

levBWT <- as.factor(abs(levBWT - 4))
levels(levBWT)
```

```
[1] "0" "1" "2" "3"
```

```
table(levBWT) #proportion nel dataset
```

```
levBWT
 0  1  2  3
46 46 38 59
```

Ricorda che unbalanced data è un problema anche in questi casi

Multinomial ordinal logistic regression 1 regressor

```
#?polr #proportional odds logistic regression
```

```
mod11 <- polr(levBWT ~ LWT, data=low) #response ~ predictor
mod11
```

```
Call:
polr(formula = levBWT ~ LWT, data = low)

Coefficients:
        LWT
-0.01273745

Intercepts:
       0|1        1|2        2|3
-2.8311127 -1.7069495 -0.8316029

Residual Deviance: 510.2954
AIC: 518.2954
```

```
summary(mod11)
```

```
Ri-adattamento per ottenere Hessian

Call:
polr(formula = levBWT ~ LWT, data = low)

Coefficients:
       Value Std. Error t value
LWT -0.01274   0.004325  -2.945

Intercepts:
    Value   Std. Error t value
0|1 -2.8311  0.6034    -4.6916
1|2 -1.7069  0.5789    -2.9488
2|3 -0.8316  0.5694    -1.4606

Residual Deviance: 510.2954
AIC: 518.2954
```

4.3.1 Coeffients interpretation

Il modello ordinal logit stimato con `polr` è simile a un modello lineare gaussiano in quanto:

- Non prevede un set distinto di coefficienti per ciascuna categoria dell'output;
- Ciascun predittore ha un **effetto unico** che agisce cumulativamente su tutte le soglie.

- l'introducione delle **soglie** $(\theta_0, \theta_1, \ldots)$ che rappresenta i punti di passaggio tra le categorie ordinate.

$$\text{logit}(\mathbb{P}(Y \leq j)) = \theta_j - \eta_i, \quad \text{dove } \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$$

- La presenza del segno negativo davanti al predittore lineare implica un'importante conseguenza interpretativa:
  - Un coefficiente $\beta_k > 0$ **aumenta** il valore del predittore lineare η_i.
    $\rightarrow$ Poiché η_i viene **sottratto** nella formula del logit, questo comporta una **diminuzione della probabilità** di essere nelle categorie più basse $(Y \leq j)$.
    $\rightarrow$ Di conseguenza, un $\beta_k > 0$ implica una **maggiore probabilità di appartenere a una categoria più alta**.
  - Viceversa, un $\beta_k < 0$ aumenta la probabilità di rimanere nelle categorie basse.
- Questa interpretazione è opposta rispetto alla regressione logistica classica, in cui un $\beta > 0$ è direttamente associato a un aumento della probabilità di successo (categoria 1).

Quando si utilizza la funzione `polr` per stimare un modello ordinal logit (proportional odds), è importante ricordare che l'output riportato nella sezione `Coefficients` non rappresenta direttamente i coefficienti $\boldsymbol{\beta}$, ma piuttosto i valori di $\boldsymbol{\eta} = -\boldsymbol{\beta}$.

4.3.1.1 Different paramterization on R

- La parametrizzazione adottata da `polr` segue un segno opposto rispetto alla formulazione canonica.
- Pertanto:
  - Un **coefficiente positivo** nell'output indica un β **negativo**, e quindi un aumento della probabilità di appartenere a **categorie più basse**.
  - Un **coefficiente negativo** nell'output indica un β **positivo**, e quindi un aumento della probabilità di appartenere a **categorie più alte**.
- In sintesi: i segni dei coefficienti vanno interpretati all'opposto rispetto a quelli che comparirebbero nella parametrizzazione standard.

se usaimo la funzione polr l'output nei coefficients: non è beta, ma eta -> è quindi - original effect possiamo interpretarlo così: negativo → più probabilità alla lower category positive -> più probabilità alla larger category

Check function help: it is parametrized with the alternative parametrization! every coefficient in the summary is $\eta = -\beta$ which means that positive coefficient increases probability for higher categories and negative coefficient increases probability for lower categories

For Example

```
Coefficients:
         Value   Std. Error t value
LWT   -0.01274     0.004325  -2.945
```

indica che:

- Il predittore `LWT` ha un singolo effetto cumulativo;

- Essendo il modello cumulativo, questo effetto **shifta le probabilità** di appartenere a tutte le categorie:
  - un coefficiente negativo aumenta la probabilità di appartenere a categorie più basse;
  - un coefficiente positivo aumenta la probabilità di appartenere a categorie più alte.
- Il coefficiente si riferisce al logaritmo delle **odds cumulative**, ovvero le odds di essere al di sopra di una soglia rispetto a essere al di sotto.

Le soglie stimate (intercette) sono:

```
Intercepts:
            Value    Std. Error t value
0|1        -2.8311     0.6034   -4.6916
1|2        -1.7069     0.5789   -2.9488
2|3        -0.8316     0.5694   -1.4606
```

Queste soglie $(\theta_1, \theta_2, \theta_3)$ rappresentano i punti di separazione stimati tra le categorie. Ad esempio, per passare dalla categoria 0 alla 1 è necessario superare la soglia $\theta_1 = -2.8311$.

```
exp(mod11$coef)
```

```
      LWT
0.9873433
```

L'esponenziale del coefficiente associato a `LWT` ($\exp(-0.0127) \approx 0.987$) indica una **riduzione di circa il 2% nelle odds cumulative** per ogni aumento di una unità (pound) nel peso corporeo. Trattandosi di un **modello proportional odds**, questo effetto è costante attraverso tutte le soglie del modello:

- la riduzione del 2% nelle odds si applica allo stesso modo nel passaggio dalla categoria 3 alla 2, dalla 2 alla 1 e dalla 1 alla 0;
- si tratta quindi di uno **shift costante** nella distribuzione della probabilità verso le categorie più basse perchè il coefficiente era negativo.

Di conseguenza, se si vuole stimare l'effetto di un aumento di 10 unità in `LWT` (ad esempio un incremento di 10 pounds nel peso corporeo), è sufficiente moltiplicare il coefficiente all'interno dell'esponenziale per 10:

```
exp(mod11$coef*10)
```

```
      LWT
0.8804039
```

La riduzione è del 12%

Questo consente di interpretare facilmente l'effetto di variazioni di entità diversa nel predittore, sfruttando la natura log-lineare del modello.

More than one regressor

```
mod12 <- polr(levBWT ~ SMOKE + LWT, data=low)
mod12
```

```
Call:
polr(formula = levBWT ~ SMOKE + LWT, data = low)

Coefficients:
      SMOKE         LWT
 0.72022303 -0.01221501

Intercepts:
       0|1        1|2        2|3
-2.5092652 -1.3558199 -0.4568955

Residual Deviance: 503.253
AIC: 513.253
```

```
summary(mod12)
```

```
Ri-adattamento per ottenere Hessian

Call:
polr(formula = levBWT ~ SMOKE + LWT, data = low)

Coefficients:
         Value Std. Error t value
SMOKE  0.72022   0.273295   2.635
LWT   -0.01222   0.004345  -2.811

Intercepts:
    Value   Std. Error t value
0|1 -2.5093  0.6169    -4.0676
1|2 -1.3558  0.5956    -2.2763
2|3 -0.4569  0.5886    -0.7763

Residual Deviance: 503.253
AIC: 513.253
```

Nel modello ordinal logit hai comunque una **soglia specifica per ogni coppia di categorie** (es. da 0 a 1, da 1 a 2, da 2 a 3), ma l'effetto delle covariate sulla variabile risposta è rappresentato da un **singolo coefficiente**.

Per interpretare l'effetto di una covariata sulla **scala originale delle odds**, è sufficiente applicare l'esponenziale al coefficiente stimato:

```
exp(mod12$coef)
```

```
    SMOKE       LWT
2.0548915 0.9878593
```

Se vogliamo stimare l'effetto di un aumento di 10 unità nel predittore (ad esempio, 10 pounds in più nel peso corporeo), dobbiamo **moltiplicare il coefficiente** corrispondente per 10 *prima* di applicare l'esponenziale:

```
exp(mod12$coef[2]*10)
```

```
      LWT
0.8850155
```

4.4 Model comparison (come prima)

```
dev1 <- mod11$dev - mod12$dev #piccolo - grande
dev1
```

```
[1] 7.042398
```

la differenza tra i modelli riguarda un solo predittore, quindi la differenza nei parametri è pari a 1.

```
gdl1 <- mod12$edf - mod11$edf
gdl1
```

```
[1] 1
```

H_0: smaller model is better

```
1-pchisq(dev1, 1) #il secondo 1 sono i dof
```

```
[1] 0.007960236
```

Poiché si rigetta l'ipotesi nulla, il modello più ampio presenta una riduzione significativa della devianza. Pertanto, è opportuno includere il predittore `SMOKE` nel modello.

4.5 proportional hazards model

Where the shift its proportional on the definition of hazard By choosing the extreme minimum-value distribution for F we get *grouped Cox model*

$$\mathbb{P}(Y_i \leq r) = 1 - \exp\left(-\exp(\theta_r + x_i'\beta)\right)$$

5. Appendix

(not on the exam)

Sometimes, categories can only be obtained sequentially, i.e. only gradually achievable $\Rightarrow$ models based on binary transitions from r to $r+1$.

- **Sequential mechanism:**

$$\mathbb{P}(Y_i = r \mid Y_i \geq r) = F(\theta_r + x_i'\beta), \quad r = 1, \ldots, c$$

- **Actual observations:** NO transitions, only realized category. We formulate the model with marginal probabilities

$$\mathbb{P}(Y_i = r) = F(\theta_r + x_i'\beta) \prod_{s=1}^{r-1} \big(1 - F(\theta_s + x_i'\beta)\big)$$

- **Choosing F:** depending on the distribution we have logistic, extreme value, or probit sequential models.

  - If $c+1$ is small, we can consider category specific coefficients β_r and global parameters γ.

Same likelihood approach as binary regression. For grouped data,

$$y_i \sim \text{MultiNom}(n_i, \boldsymbol{\pi}_i) \quad \text{with log-likelihood}$$

$$\ell(\beta) \propto \sum_{i=1}^{G} \left(y_{i1}\log(\pi_{i1}) + \cdots + y_{ic}\log(\pi_{ic}) + \cdots + y_{i,c+1}\log(\pi_{i,c+1})\right)$$

with $y_{i,c+1} = n_i - \sum_{s=1}^{c} y_{is}$ and $\pi_{i,c+1} = 1 - \sum_{s=1}^{c} \pi_{is}$.

In every categorical regression model:

$$\mathbb{P}(Y_i = r) = \mathbb{P}(y_{ir} = 1) = \pi_{ir} = h_r(\eta_{i1}, \ldots, \eta_{ic})$$

with $h_r(\cdot)$ defining the specific model.

• If we combine predictors into $\boldsymbol{\eta}_i = (\eta_{i1}, \ldots, \eta_{ic})' = X_i\beta$ we can write compactly $\boldsymbol{\pi}_i = \boldsymbol{h}(\boldsymbol{\eta}_i)$, with design matrix and response/link function depending on the selected model.

5.1 ML estimator

It has not a close form, so you go with Fisher iteration. it also coincide with OLS.
If you have to adjust for the standard error you are going to have to write the $\boldsymbol{X}$ matrix yourself (not for the exam).
Extended multinomial logit model:

$$\beta = (\beta_1, \dots, \beta_c, \gamma) \quad \text{and} \quad X_i = \begin{pmatrix} x_i & & & \\ & x_i & & \\ & & \ddots & \\ & & & x_i \end{pmatrix}, \quad W_i = \begin{pmatrix} w_{i1} = w_{i,c+1} \\ w_{i2} = w_{i,c+1} \\ \vdots \\ w_{ic} = w_{i,c+1} \end{pmatrix}$$

Ordinal cumulative model:

$$\beta = (\theta_1, \dots, \theta_c, \beta_1, \dots, \beta_k) \quad \text{and} \quad X_i = \begin{pmatrix} 1 & x_i \\ 1 & x_i \\ \vdots & \vdots \\ 1 & x_i \end{pmatrix}$$

ML estimator: $\hat{\beta}$ such that $s(\hat{\beta}) = 0$ with

$$s(\beta) = \sum_{i=1}^{G} X_i' D_i \Sigma_i^{-1} (y_i - n_i \pi_i) \qquad F(\beta) = \sum_{i=1}^{G} X_i' W_i X_i'$$

$$D_i = \frac{\partial h(\eta_i)}{\partial \eta_i}, \quad W_i = D_i \Sigma_i^{-1} D_i$$

5.2 Numerical computation of $\hat{\beta}$

By stacking all relevant quantities:

$$y = (y_1', \dots, y_G'), \quad \pi = (\pi_1', \dots, \pi_G'), \quad X = \begin{pmatrix} X_1 \\ \vdots \\ X_G \end{pmatrix}$$

$$\Sigma = \text{block-diag}(\Sigma_1, \dots, \Sigma_G), \quad D = \text{block-diag}(D_1, \dots, D_G), \quad W = \text{block-diag}(W_1, \dots, W_G)$$

We can use iterative Fisher scoring (be carefull because it is not said in the output):

$$\hat{\beta}^{(t+1)} = \hat{\beta}^{(t)} + F^{-1}(\hat{\beta}^{(t)}) s(\hat{\beta}^{(t)})$$

or its iteratively (re)weighted least squares formulation.

- **Asymptotics:** as $n \to \infty$, larger $\quad \hat{\beta} \sim \mathcal{N}(\beta, F^{-1}(\hat{\beta}))$
- **Hypothesis testing:** same as univariate case, replace all relevant quantities.

NON VUOLE PARLARNE, NON É IN ESAME

Changing reference category

recoding the dependent variable. Utile quando fittiamo un **unordered categorical regression model** e quando fittare il modello richiede tempo.

Tutti i parametri cambiano e l'interpretazione cambia.

Esempio Vogliamo sapere qual è l'impatto di passare da sometimes alle altre due In questo esempio prendiamo "somentimes" come reference recalling.

```
fact1 <- which(debt$ccarduse==1)
fact2 <- which(debt$ccarduse==2)
```

abbiamo switchato 1 e 2 category

```
debt$ccarduse[fact1] <- 2
debt$ccarduse[fact2] <- 1

mod2 <- multinom(ccarduse ~ incomegp + children, debt)
```

```
 # weights:  12 (6 variable)
 initial  value 454.825488
 iter  10 value 387.068864
 final  value 387.067005
 converged
```

```
mod2
```

```
 Call:
 multinom(formula = ccarduse ~ incomegp + children, data = debt)

 Coefficients:
   (Intercept)   incomegp   children
 2    1.814161 -0.4074893  0.1307437
 3   -1.179491  0.3207063 -0.0399505

 Residual Deviance: 774.134
 AIC: 786.134
```

paragone con il modello di prima (mod)

```
summary(mod)$coefficients
```

```
   (Intercept)  incomegp   children
 2   -1.814161 0.4074938 -0.1307578
 3   -2.993671 0.7281990 -0.1706964
```

```
summary(mod2)$coefficients
```

```
  (Intercept)   incomegp   children
2    1.814161 -0.4074893  0.1307437
3   -1.179491  0.3207063 -0.0399505
```

5.2.1 Model connection

La prima riga (della tabella o del vettore dei coefficienti) mostra che i coefficienti dei due modelli sono gli stessi, ma con segno opposto. Questo suggerisce che esiste una connessione tra i due modelli.

Quindi possiamo passare da un modello all'altro senza rifare il secondo fit, cioè senza rifittare il secondo modello. Questo è un punto importante.

Possiamo farlo semplicemente calcolando la differenza tra i vettori dei parametri dei due modelli.

Evaluation of the "change of category level" effect model reparametrization

```
b1 <- -summary(mod2)$coefficient[1,2]
b2 <- summary(mod2)$coefficient[2,2]-summary(mod2)$coefficient[1,2]
b3 <- -summary(mod2)$coefficient[1,3]
b4 <- summary(mod2)$coefficient[2,3]-summary(mod2)$coefficient[1,3]

temp <- cbind(rbind(b1,b2),rbind(b3,b4))
colnames(temp)=c("incomegp","children")
temp #coefficienti dal nuovo modello ma trasformate
```

```
    incomegp   children
b1 0.4074893 -0.1307437
b2 0.7281956 -0.1706942
```

```
summary(mod)$coefficient[,2:3] #coefficienti dal modello originale
```

```
   incomegp   children
2 0.4074938 -0.1307578
3 0.7281990 -0.1706964
```

se guardaimo ai due insiemi di coefficienti vediamo che sono uguali

hanno la stessa deviance, $\Rightarrow$ è è lo stesso modello con una diversa reference category

$\rightarrow$ cambiando il denominatore (la reference) del log-odds i parametri cambiano il complete set di predittori da però la stessa deviance (quindi incomgp e childre hanno sempre lo stesso impatto)

Part XI

Generalized Linear Mixed Models

1. Generalized Linear Mixed Models

Mixed since we have both fixed effect and random effect, if in bayesian contest everything random so it would be a nosense, here parameters hare fixed. The easiest part is that the random effect acccounts for grouping, data are clustered and groups are known.

Random effect: try to capture variability that comes from a grouping structured regardless if longitudinal (repeated measurement), or clusters.

Fixed or random?

All the levels of the categorical value that groups the observation are exaustive or there are some of them that are unknown?

1.1 Model structure:

We need to account for the clustered structure of the data.

- conditional density $f(y \mid \cdot)$ belongs to EF
- we model $\mathbb{E}(y_{ij}) = \mu_{ij} = h(\eta_{ij})$
- response variables y_{ij} have two 'levels':
  - $i = 1, \dots, m$: clusters or subjects
  - $j = 1, \dots, n_i$: repeated measurements or units within cluster
- u' usually sub-vector of x'
- if $u'_{ij} = 1$, then random intercept GLM with $\eta_{ij} = x'_{ij}\beta + \gamma_i$ **Note:** here we do not add ε_i to the linear predictor.
- introduce random effects $\Rightarrow \mu_{ij} = \mathbb{E}(y_{ij} \mid \gamma_i) = h(\eta_{ij})$ with

$$\eta_{ij} = x'_{ij}\beta + u'_{ij}\gamma_i$$

  The random part is given by γ_j since $u_{i,j}$ is another set of predictors which is given as fixed quantities

- we assume $\gamma_i \sim \mathcal{N}(0, Q)$ and i.i.d.

$$\text{special case is } Q = \text{diag}(\tau_0^2, \dots, \tau_q^2)$$

Note: decide the distribution of the random effect, it is the most important part. But generally the random effects are Gaussian centered on 0, the center means that on average we are not introducing a Bias in the linear predictor, in fact the random effect does not enter in the mean.

Intraclass Correlation Coefficient (ICC)

A percentage of the variability of the response can be explained by the grouping of the data, which means that the larger the ICC the larger the evidence, in a sense, that those groups are actually different, that give a structer to the data. (you should look at it when you fit the model).

For a Gaussian Linear Mixed Model:

$$y_{i,j} = \beta_0 + \gamma_{0,i} + \varepsilon_{0,i}$$

It is straight forward to compute the ICC, as soon as you change something, you will need a package or to do it manually. *Example:* Binary case $\rightarrow$ you need the variance of the logistic distribution

1.2 Matrics Formulation

Extending LMMs to distributions other than Gaussian does not impact the general structure of linear mixed models, which have linear predictor:

$$\eta = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{U}\boldsymbol{\gamma}$$

- $\boldsymbol{X}$and $\boldsymbol{U}$ are design matrices
- $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_r)'$ is the vector of all random effects
- i.i.d. assumption $\gamma_i \sim \mathcal{N}(0, \boldsymbol{Q})$ can be relaxed into $\boldsymbol{\gamma} \sim \mathcal{N}(0, \boldsymbol{G})$, where $\boldsymbol{Q}$ is diagonal, $\boldsymbol{G}$ is not recommended.

2. Mixed binary and count

2.1 Mixed Logit/Probit models

For binary response variables simply takes the previous linear predictor (with one more index, due to hierarchical structure),

$$\log\left(\frac{\mathbb{P}(y_{ij}=1 \mid \gamma_i)}{\mathbb{P}(y_{ij}=0 \mid \gamma_i)}\right) = \eta_{ij} = x'_{ij}\beta + u'_{ij}\gamma_i \quad \text{or} \quad \mathbb{P}(y_{ij}=1 \mid \gamma_i) = \Phi(\eta_{ij}) = x'_{ij}\beta + u'_{ij}\gamma_i$$

with $n_i > 1$ for identifiability in random intercept models.

When you work with binary data and you want a random intercept you must ensure that in each group there are at least 2 observation, otherwise you cannot fit the model.

For example: take the 20 region if you use them as fixed effect you will need 19 β 's, or you can threat it as random effects (u_{ij}) you take the region predictors, then you only need the variance of that predictor, and then with a fixed variance you are gonna generate 20 values one for each region, each of them will adjust for the difference in each region. This way you need 1 parameter (the variance) nor 19.

2.2 Mixed Poisson models

For count responses $y_{ij} \mid \gamma_i \sim \text{Pois}(\lambda_{ij})$

$$\log(\lambda_{ij}) = x'_{ij}\beta + u'_{ij}\gamma_i$$

Random intercept model:

$$\lambda_{ij} = \exp(x'_{ij}\beta + \gamma_{0i}), \quad \text{where } \gamma_{0i} \sim \mathcal{N}(0, \tau_0^2)$$

Since the exponential of a sum is a product of exponential and since for the poisson model:

$$\mathbb{E}[y_i] = \lambda_{ij} = \mathbb{V}[y_i]$$

$$\lambda_{ij} = \exp(x'_{ij}\beta) \cdot \exp(\gamma_{01})$$

You obtain an over disperse model, because you are adjusting the variance for a factor, this si way better then doing a quasi-poisson model (when you want to adjust the variance).

connection with the log-normal
Saying that γ_i is gaussian the expnonetial of Gaussian is normal, (same for the logarithm). Also called log-linear random intercept or Poisson-normal or Poisson-lognormal model because

$$\lambda_i = \nu_i \exp(x'_i\beta) \quad \text{with } \nu_i = \exp(\gamma_{0i}) \text{ has lognormal distribution}$$

3. Mixed models for categorical data

You just have to increase your linear predictor by adding buy adding random coefficient $(u'_{ij}\gamma_{ir})$.
the professor advice is to stay away from Multinomial Logit Model with random effects, evenif it is easy to fit, the interpretation of the result is not that easy.
Since for each level of your outcome you have a vector of coefficient, (4 levels and 3 predictors = 9 parameters) .. very messy very fast.
When you add the random intercept it's just another parameter, no problems.

3.1 Mixed multinomial logit model

Considering both global covariates and category-specific fixed effects:

$$\mathbb{P}(y_{ij} = r \mid \gamma_r) = \frac{\exp(\eta_{ijr})}{1 + \sum_{s=1}^{c} \exp(\eta_{ijs})}$$

with linear predictors

$$\eta_{ijr} = x'_{ij}\beta_r + u'_{ij}\gamma_{ir}$$

The random effects $\gamma_{i1}, \ldots, \gamma_{ic}$ are assumed to be i.i.d. from a Gaussian

$$\gamma_{ir} \sim \mathcal{N}(0, \boldsymbol{Q}_r)$$

Analogous arguments inject random effects into linear predictors of cumulative and sequential models.

3.2 Conditional and marginal models

Always recall that you are obtaining a conditional model,

$$\mathbb{E}(y_{ij} \mid \boldsymbol{X}, \boldsymbol{\gamma}) = h(\eta_{ij}) = h(x'_{ij}\beta + u_{i,j}\gamma_i)$$

Models we build are **conditional** on random effects, i.e. we use $p(y \mid \gamma)$.

Real focus is actually on **marginal** density which can be obtained by integrating out – marginalizing – the random effects:

$$p(y) = \int p(y \mid \gamma)p(\gamma)d\gamma$$

The marginal model does not consider the variability introduced by γ_i to account for the grouping structure.

Usually $p(\gamma)$ is Gaussian, which means no problem; outside of Gaussian LMM and few special cases, the integral does not have a closed form solution.

The important thing is: In general, parameters β in conditional and $\beta^\star$ marginal models are different from each other.
Marginalization means that you take inito account for all the possible values of $\gamma_{i,j}$ could have assumed, usually the effect of marginalising a model is an increase in variability (beacuse you are considering all the variability that the random effects could accounts for).

Note: fitting a model without considering the random effects does not mean that you have obtained the marginalized model, because the data are still structured, ignoring it is not enough.

3.2.0.1 In log-linear Poisson random intercept model ($n_i = 1$)

We have analytical marginal means, variances and covariances:

- because $\mu_i \mid \gamma_{0i} = \nu_i \exp(x_i'\beta)$, then $\nu_i = \exp(\gamma_{0i})$ is log-normal
- so $\mathbb{E}(y_i) = \exp(\tau_0^2/2 + x_i'\beta)$ which means marginal and conditional model have same fixed effects parameters
- $\mathrm{Var}(y_i) = \mathbb{E}(y_i) \cdot \kappa$ with κ always greater than 1, so the marginal density is not Poisson as the conditional but overdispersed

4. Maximum Likelihood Estimation

Inference is performed within a likelihood framework:

- if we know $\boldsymbol{G}$ or $\boldsymbol{Q}$ we use *penalized likelihood*;
- if we don't, we use *approximate penalized likelihood.*

4.1 Known variance-covariance parameters

Joint likelihood

$$L(\boldsymbol{\beta}, \boldsymbol{\gamma}) = p(y \mid \boldsymbol{\beta}, \boldsymbol{\gamma}) p(\boldsymbol{\gamma})$$

The Penalized log-likelihood is just the original minus (or plus) a penalization therm, to penalised for the complexity of the model

$$\ell_{\text{pen}}(\boldsymbol{\beta}, \boldsymbol{\gamma}) = \ell(\boldsymbol{\beta}, \boldsymbol{\gamma}) - \frac{1}{2} \boldsymbol{\gamma}' \boldsymbol{G}^{-1} \boldsymbol{\gamma}$$

This is actually the prior for $\boldsymbol{\gamma}$, recall that if gamma is gaussian this is the gaussian density for $\boldsymbol{\gamma} \sim \mathcal{N}(0, \boldsymbol{Q})$.

Also GLMM can be written as iterative weighted least squares.

$$\begin{pmatrix} \hat{\beta}^{(t+1)} \\ \hat{\gamma}^{(t+1)} \end{pmatrix} = \left(\boldsymbol{C}^\top \boldsymbol{W}^{(t)} \boldsymbol{C} + \boldsymbol{B} \right)^{-1} \boldsymbol{C}^\top \boldsymbol{W}^{(t)} \boldsymbol{z}^{(t)}$$

you can find a recursive estimation by taking the previous estimation and adjusting by $\left(\boldsymbol{C}^\top \boldsymbol{W}^{(t)} \boldsymbol{C} + \boldsymbol{B} \right)^{-1} \boldsymbol{C}^\top \boldsymbol{W}^T \boldsymbol{z}^T$, the inverse of a matrix times the weighted version of the observation.

Recall the OLS ($(\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y}$) it is the same structure. $\rightarrow$ consider ($\boldsymbol{X}^T \boldsymbol{X}$ as a measure of the variability of the predictors and $\boldsymbol{X}' \boldsymbol{y}$ the covariance between $\boldsymbol{X}$ and y. So at the end it is a correlation measured rescaled by the variability of $\boldsymbol{X}$. More or less this is the idea.

with (non important)

$$B = \begin{pmatrix} 0 & 0 \\ 0 & G^{-1} \end{pmatrix}, \quad W = D \Sigma^{-1} D, \quad \tilde{y} = X\beta + U\gamma + D^{-1}(y - \mu)$$

and

$$D = \operatorname{diag}\left(\frac{\partial h(\eta)}{\partial \eta}, \dots \right), \quad \Sigma = \operatorname{diag}(\dots, \sigma_i^2, \dots)$$

4.2 Unknown variance-covariance parameters

(Non l'ha nemmeno spiegata, the important thing is that the covariates of the parameter are estimated via the fisher information matrix)

Unknown parameters $\boldsymbol{\theta}$ in $\boldsymbol{G}$ are estimated with approximate marginal likelihood.

- ◇ Conditional log-likelihood written in quadratic form via Laplace approximation:

  $$\ell(\beta, \gamma; \theta) = \log(p(y \mid \beta, \gamma)) \propto -\frac{1}{2}(y-\mu)'\boldsymbol{\Sigma}^{-1}(y-\mu) \approx (y-\boldsymbol{X}\hat{\beta}-\boldsymbol{U}\hat{\gamma})'\boldsymbol{W}(y-\boldsymbol{X}\hat{\beta}-\boldsymbol{U}\hat{\gamma})$$

  means for **fixed** $\boldsymbol{\theta}$ it leads to log-likelihood of LMM:

  $$\tilde{y} \mid \gamma \sim \mathcal{N}(\boldsymbol{X}\beta + \boldsymbol{U}\gamma, \boldsymbol{W}^{-1})$$

If we set $\boldsymbol{V} = \boldsymbol{W} + \boldsymbol{U}\boldsymbol{G}^{-1}\boldsymbol{U}'$, we obtain the approximate marginal (or restricted) log-likelihood $\ell_R(\theta)$ [check main reference book for full definition].

- estimation algorithm alternates between (β, γ) and $\boldsymbol{\theta}$;
- estimated covariance matrix

  $$\operatorname{Cov}\left(\begin{pmatrix}\hat{\beta}\\ \hat{\gamma}\end{pmatrix}\right) \approx \hat{\boldsymbol{F}}^{-1}$$

5. R mixed models

Consiglio: quando usi un pacchetto, leggi le sue **vignettes**, ti danno un'idea di come usarlo.

```
library("lme4")
library("glmm")
library("mgcv")
```

In questo contesto, è importante non confondere *clustered data* con *grouped data*, poiché rappresentano concetti differenti:

- **Grouped data**: le osservazioni condividono lo stesso *covariate pattern*, ovvero presentano esattamente gli stessi valori nelle covariate. Il raggruppamento può essere identificato direttamente tramite la design matrix (ad esempio, usando `unique()` sulle righe)
- **Clustered data**: le osservazioni appartengono a gruppi (cluster) in cui alcune caratteristiche sono condivise tra le unità, ma **non necessariamente** hanno lo stesso covariate pattern. Le unità sono simili tra loro, ma non identiche.
  Un esempio tipico è costituito da individui appartenenti alla stessa famiglia: potrebbero avere comportamenti d'acquisto simili, pur non avendo esattamente gli stessi valori delle covariate.

Quelli che usiamo in questo contesto sono **clustered data**!

5.1 Dati longitudinali (conteggi) individuali

```
set.seed(42) # lo mettiamo per la riproducibilità dei risultati
I <- 25  # numero di soggetti (persone)
n <- 4  # numero di time points per soggetto
N <- n * I  # numero totale di osservazioni (100)
X <- runif(N)  # la nostra X è un Uniforme in (0,1)
# vettore con 100 osservazioni tra 0 e 1
i <- rep(1:I, each = n) # colonna id del soggetto:
# vettore lungo 100 con dove ognuno dei 25 id è ripetuto 4 volte
```

```
sig.alpha.DGP <- 2
```

É la deviazione standard del random effect, che chiamiamo τ e non τ^2. É quindi la variabilità aggiuntiva che viene introdotto nei dati solamente perché sono longitudinali.

```
alpha <- rnorm(I, mean = 0, sd = sig.alpha.DGP)  # notiamo come servano solamente M va
```

Generiamo random effect normali centrati in 0 con deviazione standard `sig.alpha.DGP`, decisa prima. Questo sarà un vettore di lunghezza 25. Le osservazioni infatti per ogni soggetto non sono indipendenti tra loro, ma condividono una componente di variabilità individuale. Questi random effcet quindi rappresentano la variabilità intra-soggetto.

```r
int.coef <- 0.5 # intercetta
x.coef <- 1.25 # coefficiente di regressione
```

```r
lin.pred <- int.coef + x.coef * X + alpha[i] # predittore lineare
# int.coef -> intercetta (beta_0)
# x.coef -> (beta_1)
# alpha[i] -> random effect
```

Il predittore lineare è definito come: $\beta_o + \beta_1 X + \alpha_i$, dove:

- β_0 è l'intercetta comune a tutti,
- $\beta_1 X$ rappresenta l'effetto della covariata specifico per ogni osservazione
- α_i è il random effect associato al soggetto i

Poiché ogni soggetto è osservato 4 volte, le prime 4 osservazioni condividono lo stesso valore di α_i, ma differiscono nei valori della covariata X.
Otteniamo quindi un vettore di 100 valori del predittore lineare, ciascuno influenzato sia dalla covariata X che dall'effetto casuale del soggetto corrispondente. Le osservazioni dello stesso soggetto risultano pertanto correlate, in quanto condividono la stessa componente di variabilità α_i.

Nota: Differenza tra modelli marginali e modelli condizionati
Se si ripetesse l'esempio senza fissare il seed, i random effects generati cambierebbero, poiché sono estratti da una distribuzione casuale. Questo comporta che i risultati di un modello condizionato dipendono dallo specifico insieme di effetti casuali utilizzato nella stima. Tuttavia, spesso l'interesse è rivolto all'effetto delle covariate X attraverso i coefficienti β, indipendentemente dalla particolare realizzazione dei random effects. Questo tipo di informazione si ottiene da un modello marginale, in cui la variabilità indotta dai termini casuali α viene "marginalizzata", ovvero calcolata tramite integrale.
Poiché l'integrazione analitica non è sempre disponibile in forma chiusa, nella pratica si ricorre frequentemente a modelli condizionati. In questi casi, l'inferenza è fatta condizionatamente al particolare set di effetti casuali stimato dal modello.

```r
Y <- rpois(N, lambda=exp(lin.pred)) # generiamo i dati
```

Per generare la variabile risposta Y, utilizziamo la distribuzione di Poisson.
Poiché i valori del predittore lineare possono essere sia positivi che negativi, applichiamo la funzione di risposta e per garantire che il parametro di intensità λ sia sempre positivo, come richiesto dalla distribuzione di Poisson: $Y \sim \text{Poisson}(\lambda)$ con $\lambda = \exp(\text{lin.pred})$

```r
# Y # conteggi
# X # matrice di disegno
head(cbind(Y, X)) # matrice dei dati
```

```
      Y         X
[1,] 12 0.9148060
[2,]  8 0.9370754
[3,]  4 0.2861395
[4,] 10 0.8304476
[5,]  1 0.6417455
[6,]  0 0.5190959
```

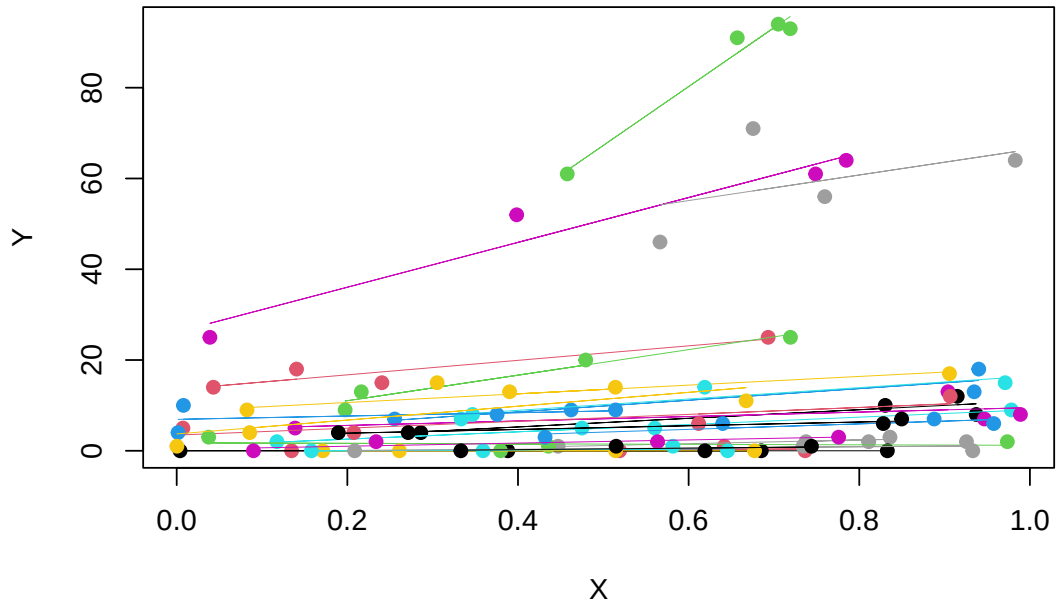
Nel contesto reale non osserviamo direttamente il termine casuale α ma cerchiamo di stimarne la variabilità attraverso la sua deviazione standard. Questa rappresenta l'effetto non osservato che introduce dipendenza tra le osservazioni ripetute per lo stesso individuo. In un'analisi simulata, come in questo caso, conosciamo il valore vero di α, ma nel mondo reale dobbiamo stimarlo dal campione.

```
df <- data.frame(time = factor(rep(1:n,I),ord=F), subjID = factor(i),
                 X = X, Y = Y)
head(df)
```

```
  time subjID         X  Y
1    1      1 0.9148060 12
2    2      1 0.9370754  8
3    3      1 0.2861395  4
4    4      1 0.8304476 10
5    1      2 0.6417455  1
6    2      2 0.5190959  0
```

Abbiamo aggiunto al dataset le informazioni relative al soggetto e al time point, così da rappresentare i dati in una struttura tipica delle analisi longitudinali. Il dataset risultante include le seguenti variabili: **tempo di osservazione**, **identificativo del soggetto**, **covariata** X e **variabile risposta** Y. Nella pratica, questa struttura è molto comune nei dati reali, con l'unica differenza che spesso si dispone di più covariate.

```
# empty plot
plot(NA, xlab="X", ylab="Y", xlim=range(df$X), ylim=range(df$Y))
for(j in 1:length(unique(df$subjID))){
    points(df$X[df$subjID==j], df$Y[df$subjID==j], col=j, pch=19)
    lines(df$X[df$subjID==j],
          predict(lm(df$Y[df$subjID==j]~df$X[df$subjID==j])),
          col=j, lwd=0.5)
}
```



Nel grafico ogni soggetto è rappresentato da punti dello stesso colore, corrispondenti alle quattro osservazioni longitudinali (baseline, 1°, 2° e 3° time point).
Per ciascun individuo, tracciamo una semplice retta di regressione che collega le osservazioni, permettendo di visualizzare l'andamento della risposta nel tempo all'interno dello stesso soggetto.

Nota: per creare i grafici usa `ggplot2` è meglio. Uno dei problemi dei plot base di R è la scarsità di colori, cosa che alla lunga fa confondere.

5.1.0.1 Dati longitudinali (conteggi) clustered

```
set.seed(42) # per riproducibilità
J <- 5  # numero di cluster
n <- 25  # numero medio di individui per gruppo
N <- n * J # totale osservazioni (125) osservazioni
X <- runif(N) # simuliamo da un'uniforme in (0,1)
```

```
j <- as.numeric(cut(X, J)) # label dei cluster
head(j) # i primi due individui sono nel cluster 5, il terzo nel 2, ...
```

```
[1] 5 5 2 5 4 3
```

```
table(j) # vediamo quanti individui ci sono in ogni cluster
```

```
j
 1  2  3  4  5
21 23 24 30 27
```

Da qua abbiamo tutto è uguale a prima

```
sig.alpha.DGP <- 2
alpha <- rnorm(J, mean = 0, sd = sig.alpha.DGP)
alpha
```

```
[1] -0.5327612  3.5626494  1.1925103  0.5246994  3.0663862
```

`alpha` è un random effect specifico per i cluster specific, quindi individui dello stesso cluster condividono lo stesso random effect. Per questo motivo il valore del predittore lineare di questi individui aumenta o diminuisce in base al cluster a cui appartengono.

Per determinare gli effetti specifici del cluster (cluster effects), è sufficiente stimare un valore per ciascun cluster (in questo caso 5). Anche se due individui appartenenti a cluster diversi hanno lo stesso valore della covariata X, il valore del predittore lineare sarà diverso, perché ciascun individuo eredita la propria intercetta casuale (random intercept) in base al cluster di appartenenza. Questo comporta uno spostamento verticale nel valore previsto della risposta, indipendentemente da X.

```
int.coef <- 0.5
x.coef <- 1.25
lin.pred <- int.coef + x.coef * X + alpha[j]
```

Una volta calcolato il linear predictor, possiamo generare la variabile risposta. Il termine casuale specifico per ciascun cluster è rappresentato da α_j, mentre la parte deterministica del predittore lineare, data da `int.coef + x.coef * X` è coerente con quella già vista in modelli precedenti. La componente α_j introduce quindi variabilità dovuta alla struttura di cluster nei dati.

```
Y <- rpois(N, lambda=exp(lin.pred))
```

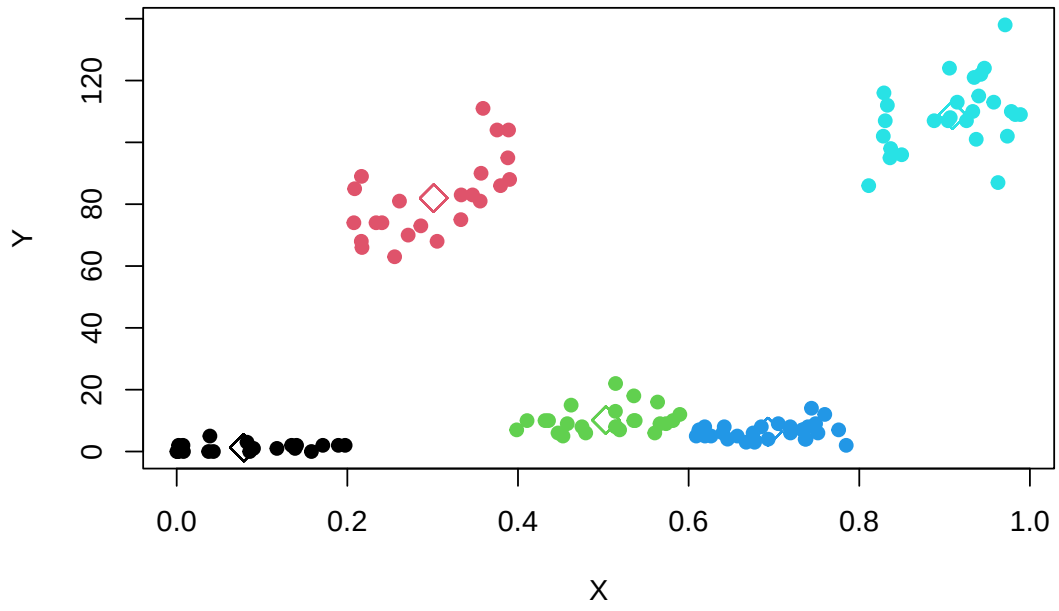
```
meanX <- tapply(X, j, mean)[j]
meanY <- tapply(Y, j, mean)[j]
# le calcoliamo per creare dopo i grafici
# sono le medie di X e Y in ogni cluster per ogni individuo
```

```
df1 <- data.frame(cluster = factor(j), X = X, Y = Y)
head(df1)
```

```
  cluster         X   Y
1       5 0.9148060 113
2       5 0.9370754 101
3       2 0.2861395  73
4       5 0.8304476 107
5       4 0.6417455   8
6       3 0.5190959   7
```

```
# è il  dataset finale, abbiamo il cluster, X e Y
```

```
plot(df1$X, df1$Y, xlab="X", ylab="Y", col=df1$cluster,pch=19)
points(meanX, meanY, col=as.numeric(df1$cluster), pch=5, cex=1.5)
```




```
# Vediamo che persone di cluster diversi hanno un comportamento diverso di X e Y
```

5.1.0.2 Dati Binari clustered

```
set.seed(42) # riproducibilità
J <- 4  # numero di gruppi
n <- 50  # numero medio di individui per gruppo
N <- n * J  # numero totale di osservazioni (200)
X <- runif(N,-0.5,0.5) # simuliamo X da una normale
j <- as.numeric(cut(X, J))  # label dei cluster
sig.alpha.DGP <- 1.5
alpha <- rnorm(J, mean = 0, sd = sig.alpha.DGP)  # random effect
int.coef <- -0.75
x.coef <- 1
lin.pred <- int.coef + x.coef * X + alpha[j]
Y <- rbinom(N,1,prob=(exp(lin.pred)/(1+exp(lin.pred))))
```

Per generare dati binari utilizziamo la funzione `rbinom`, poiché la variabile risposta segue una distribuzione binomiale. In questo esempio, generiamo un vettore di 200 osservazioni, dove la probabilità di successo è modellata tramite la funzione logit inversa: $p = \frac{\exp(\text{lin.pred})}{1+\exp(\text{lin.pred})}$. Questa trasformazione assicura che i valori predetti siano compresi tra 0 e 1, come richiesto per una probabilità.

```
df2 <- data.frame(cluster = factor(j), X = X, Y = Y)
head(df2)
```

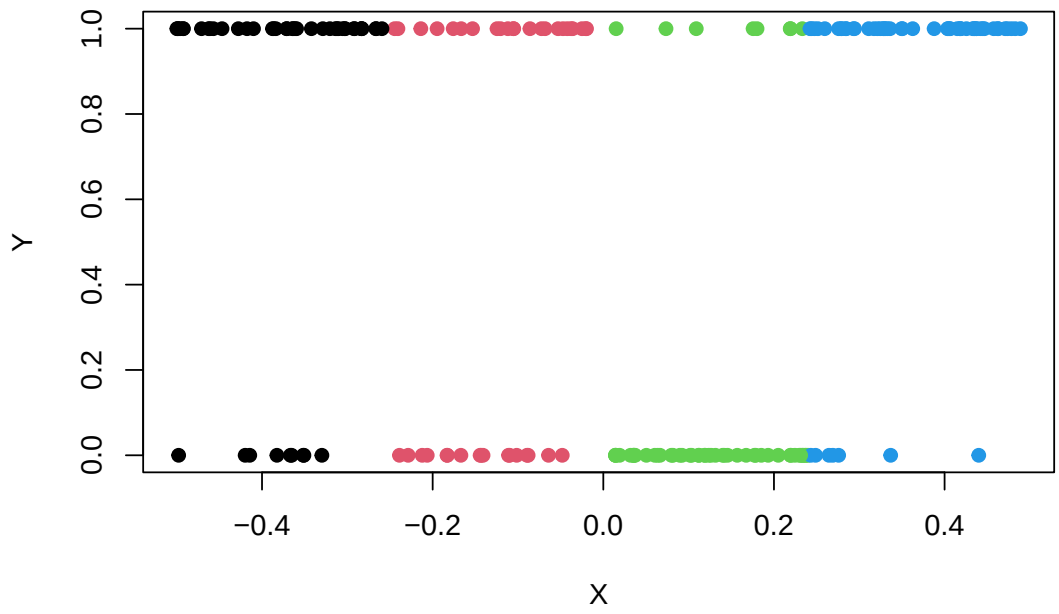
```
  cluster           X Y
1       4  0.41480604 1
2       4  0.43707541 1
3       2 -0.21386047 1
4       4  0.33044763 1
5       3  0.14174552 0
6       3  0.01909595 0
```

```
dim(unique(df2))
```

```
[1] 200   3
```

Vediamo che sono abbiamo 200 osservazioni, quindi il nostro dataset è clustered, non grouped.

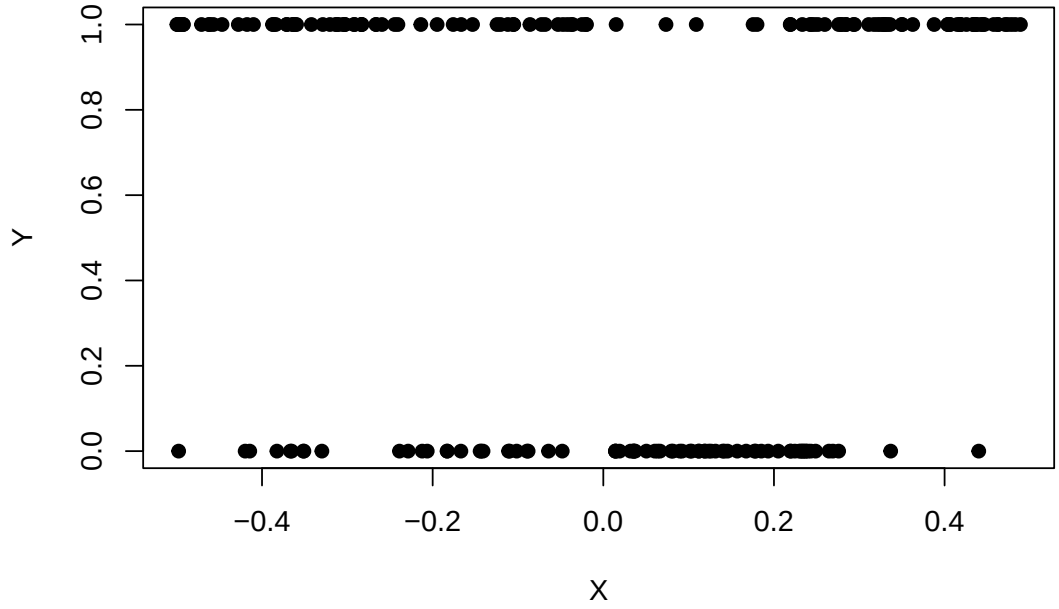
```
plot(df2$X, df2$Y, xlab="X", ylab="Y", col=df2$cluster,pch=19)
```




```
# con gruppi e colori
```

Vediamo la divisione dei punti nei cluster grazie ai colori. Se li togliessimo non capiremmo più nulla:

```r
plot(df2$X, df2$Y, xlab="X", ylab="Y",pch=19) # senza colori
```



Adesso quindi possiamo fittare i modelli. Potremmo fittarne migliaia diversi tra loro, dobbiamo quindi capire dove mettere il random effect e perché.
Attenzione: va poi interpretato.

5.1.1 Count Data

5.1.1.1 Modello con `time`

```r
fit.glmm <- lme4::glmer(Y ~ time + X + (1 | subjID), family = poisson, data = df)
```

La struttura della formula in `glmer` è simile a quella dei modelli `glm`, ma include un termine per i random effects. Questi sono specificati tra parentesi tonde con la sintassi:

(termine casuale | variabile di raggruppamento)

Nel nostro caso quindi: (1 | `subjID`)

Questo indica che ogni soggetto ha una propria intercetta casuale, cioè esiste una variabilità tra soggetti che non è spiegata dalle covariate osservate.

Non utilizziamo `time` come random effect, perché la variabilità condivisa nelle osservazioni deriva dal fatto che provengono dallo **stesso soggetto**, non dallo stesso istante temporale. Tuttavia, in altri contesti si potrebbe modellare anche `time` come random effect, ad esempio se le misurazioni fossero prese dallo stesso macchinario o dallo stesso operatore.

Nel nostro modello invece, trattiamo `time` e `X` come **effetti fissi**, e `subjID` come **fonte di variabilità**, modellata tramite una intercetta casuale.

In sintesi, abbiamo fittato un modello che spiega il comportamento della variabile di risposta `Y` (conteggio), in funzione dei time point, degli effetti delle covariate, e di una intercetta casuale associata al soggetto.

```
summary(fit.glmm)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: Y ~ time + X + (1 | subjID)
   Data: df

     AIC      BIC   logLik deviance df.resid
   508.7    524.4   -248.4    496.7       94

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9279 -0.5598 -0.1725  0.4206  2.5975

Random effects:
 Groups Name        Variance Std.Dev.
 subjID (Intercept) 3.017    1.737
Number of obs: 100, groups:  subjID, 25

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.95771    0.37009   2.588  0.00966 **
time2       -0.10564    0.07623  -1.386  0.16580
time3       -0.17360    0.08306  -2.090  0.03662 *
time4       -0.13031    0.07668  -1.699  0.08927 .
X            1.02764    0.13546   7.586 3.29e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
      (Intr) time2  time3  time4
time2 -0.066
time3 -0.136  0.417
time4 -0.075  0.500  0.420
X     -0.214 -0.146  0.197 -0.111
```

L'output del modello `glmer` è simile a quello dei modelli `glm`, ma contiene alcune informazioni aggiuntive a causa della presenza dei random effects. Anche se l'ordine con cui le informazioni sono presentate può variare, il contenuto rimane concettualmente coerente.

Vengono riportati **AIC** e **BIC**, che ci servono per confrontare modelli, ma misurano concetti differenti: - L'AIC è orientato alla **capacità predittiva** del modello. - Il BIC mira a selezionare il modello **più vicino alla verità**, assumendo che questa sia tra i modelli candidati.

La differenza fondamentale sta quindi nella penalizzazione: - AIC penalizza i parametri con un fattore costante: $2 \times$ numero di parametri - BIC usa una penalizzazione più severa basata su $\log(n)$, dove n è il numero di osservazioni: $\log(n) \times$ numero di parametri

Per quanto riguarda gli **effetti casuali**, l'output mostra la **varianza** e la **deviazione standard** dell'intercetta casuale per ciascun gruppo (es. `subjID`). Questo indica la variabilità non spiegata dalle covariate e associata alla struttura a gruppi del campione.

Gli **effetti fissi** includono le stime dei coefficienti $\hat{\beta}$, gli standard errore, i valori z e i p-value. Tuttavia, essendo un modello di regressione di Poisson, i coefficienti sono espressi nella **scala del predittore lineare**. Per interpretare l'effetto sulla variabile risposta, bisogna trasformare i coefficienti tramite l'esponenziale:

$$\hat{y} = \exp(\eta) = \exp(\beta_0 + \beta_1 X + \dots)$$

Infine, viene riportata la **matrice di correlazione tra gli stimatori** $\hat{\beta}$, ottenuta come riscalamento della matrice di Fisher. Questa rappresenta la correlazione tra gli stimatori dei coefficienti (e non tra i coefficienti veri, che per noi sono fissi).
In pratica: I parametri β sono **fissi**, mentre gli stimatori $\hat{\beta}$ sono **variabili aleatorie** con media, varianza e covarianza definite.

5.1.1.2 Modello senza `time`

```
fit.glmm2 <- lme4::glmer(Y ~ X + (1 | subjID), family = poisson, data = df)
```

La formula che abbiamo usato rappresenta il **modello vero** con cui sono stati generati i dati, ma con una differenza importante: durante la simulazione, abbiamo incluso un **effetto casuale** (random effect) chiamato `alpha`, mentre nel modello stimato questo effetto non è esplicitamente presente.

Pertanto, nel modello stimato dobbiamo **recuperare indirettamente** questa variabilità stimando la **deviazione standard** corrispondente che ha originato i dati.

Quindi, il modello $Y \sim$ `time` $+ X +$ (1 | `subjID`) non è tecnicamente identico al vero modello di generazione dei dati, perché non conosciamo esattamente i valori dei random effects. Tuttavia, anche se non stiamo usando esattamente la stessa struttura, osserviamo che la covariata `time` risulta **significativa** nel modello stimato. Questo succede perché:

- Quando abbiamo generato i dati, la variabilità associata ai soggetti era esplicitamente modellata con `alpha`
- Durante la stima, questa variabilità viene recuperata indirettamente, e può capitare che covariate come `time` risultino significative anche se **non erano state utilizzate** nella generazione dei dati.

Questo evidenzia il fatto che anche in presenza di un modello leggermente misspecificato, alcune covariate possono risultare comunque significative.

```
summary(fit.glmm2)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: poisson  ( log )
Formula: Y ~ X + (1 | subjID)
   Data: df
```

```
      AIC      BIC   logLik deviance df.resid
    507.9    515.7   -250.9    501.9       97

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9410 -0.5496 -0.2401  0.3709  2.6259

Random effects:
 Groups Name        Variance Std.Dev.
 subjID (Intercept) 3.019    1.738
Number of obs: 100, groups:  subjID, 25

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.8418     0.3667   2.295   0.0217 *
X             1.0548     0.1269   8.311   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
  (Intr)
X -0.206
```

Dal modello stimato osserviamo che la deviazione standard dell'intercetta casuale associata a `subjID` è pari a 1.738, che è **più piccola** rispetto al valore utilizzato nella generazione dei dati $\tau = 2$. Questo è atteso, poiché nella stima cerchiamo di approssimare la variabilità indotta dal random effect.

Poiché il modello include un solo effetto fisso (`X`), l'unica correlazione presente tra i coefficienti stimati riguarda l'intercetta e il coefficiente associato a `X`. In questo caso, la correlazione stimata è moderata e negativa (circa -0.206), e riflette la dipendenza statistica tra questi stimatori nel modello.

Per quanto riguarda il confronto tra modelli `GLMM`, possiamo utilizzare sia l'**AIC** che la funzione `anova()` (con test basati sulla likelihood). Tuttavia, è importante ricordare che:

- La funzione `anova()` può essere usata solo per confrontare **modelli dello stesso tipo**, ad esempio due GLMM stimati con la stessa funzione (`glmer`).
- Non è corretto utilizzare `anova()` per confrontare un GLMM con un modello `GLM` classico: in tal caso, i risultati del test potrebbero essere fuorvianti o non validi, perché i modelli non sono confrontabili direttamente a livello di likelihood.

In conclusione, per confronti tra modelli misti, è preferibile restare coerenti nella struttura del modello e nello strumento di stima.

```
c(AIC(fit.glmm),AIC(fit.glmm2))
```

```
[1] 508.7431 507.8647
```

```
anova(fit.glmm, fit.glmm2, test="Chisq")
```

```
Data: df
Models:
fit.glmm2: Y ~ X + (1 | subjID)
fit.glmm: Y ~ time + X + (1 | subjID)
          npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
fit.glmm2    3 507.86 515.68 -250.93   501.86
fit.glmm     6 508.74 524.37 -248.37   496.74 5.1216  3     0.1631
```

Dal confronto tra modelli tramite la funzione `anova()`, otteniamo una **riduzione nella deviance** pari a circa 5.12. Tuttavia, questa differenza **non è statisticamente significativa**.

Nel dettaglio:

- Il **modello semplice** H_0 è `fit.glmm2`: $Y \sim X + (1 \mid \texttt{subjID})$
- Il **modello complesso** H_1 è `fit.glmm`: $Y \sim \texttt{time} + X + (1 \mid \texttt{subjID})$

La differenza tra i due modelli è data dall'inclusione del predittore `time`, che introduce **3 parametri addizionali**.

La **riduzione della deviance** tra H_0 e H_1 è pari a 5.1216. Questo valore viene confrontato con una distribuzione chi-quadro con 3 gradi di libertà (pari alla differenza di parametri tra i due modelli). Tuttavia, il **p-value** risultante non è inferiore al livello di significatività α (ad esempio, 0.05), quindi **non rifiutiamo** H_0. In altre parole, l'aggiunta del predittore `time` non porta un miglioramento sufficiente da giustificare l'aumento di complessità del modello. Pertanto, il **modello più semplice** risulta preferibile.

5.1.1.3 Modello semplice senza random effects

Ignoriamo la variabile che ci restituisce i gruppi

```
fit.glm <- glm(Y ~ X, family = poisson, data = df)
```

```
summary(fit.glm)
```

```
Call:
glm(formula = Y ~ X, family = poisson, data = df)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.89995    0.06707   28.33   <2e-16 ***
X            1.16720    0.09781   11.93   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2305.2  on 99  degrees of freedom
Residual deviance: 2155.1  on 98  degrees of freedom
AIC: 2475.3

Number of Fisher Scoring iterations: 6
```

L'output del modello mostra coefficienti associati a **standard error molto piccoli** e **valori di z molto grandi**. In teoria, questo potrebbe farci credere di essere estremamente confidenti sulle stime ottenute.

Tuttavia, questa è una **situazione tipica in cui esiste una fonte di variabilità aggiuntiva (e.g., effetto di raggruppamento)** che **non è stata modellata esplicitamente**. Quando si ignora un *grouping factor*, succede spesso che:

- gli **standard error risultano sottostimati** (deflated);
- la **confidenza nelle stime è eccessiva** (overconfident), anche se in realtà non è giustificata.

In altre parole, si ignora una parte della variabilità presente nei dati, e ciò porta a una falsa sensazione di precisione.

Anche il valore dell'AIC è indicativo:

- **AIC: 2475.3**, molto più alto rispetto ai modelli che includono la struttura di dipendenza.
- Questo segnala che il **modello ignora la struttura di raggruppamento** (es. `subjID`, `time`) che invece è chiaramente presente nei dati.
- Dato che tali informazioni sono disponibili, è **opportuno includerle nel modello**, per una rappresentazione più corretta della variabilità nei dati.

5.1.1.4 Modello con `subjID` come fattore

```
fit.glm1 <- glm(Y ~ X + subjID, family = poisson, data = df)
```

```
summary(fit.glm1)
```

```
Call:
glm(formula = Y ~ X + subjID, family = poisson, data = df)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.32324    0.20000   6.616 3.68e-11 ***
X             1.05317    0.12779   8.242  < 2e-16 ***
subjID2      -3.27170    1.01509  -3.223 0.001268 **
subjID3       2.44213    0.18105  13.489  < 2e-16 ***
subjID4       0.41026    0.22531   1.821 0.068625 .
subjID5      -0.27841    0.27843  -1.000 0.317335
subjID6      -0.05588    0.24442  -0.229 0.819159
subjID7       0.71359    0.22184   3.217 0.001297 **
subjID8      -1.38822    0.39307  -3.532 0.000413 ***
subjID9     -20.13451 2792.68114  -0.007 0.994248
subjID10      0.06231    0.25967   0.240 0.810365
subjID11     -1.46329    0.44373  -3.298 0.000975 ***
subjID12     -0.40994    0.27365  -1.498 0.134119
subjID13      0.43798    0.22928   1.910 0.056104 .
subjID14      2.03218    0.18759  10.833  < 2e-16 ***
subjID15    -20.05596 2830.37167  -0.007 0.994346
```

```
subjID16       1.95947    0.18344  10.682  < 2e-16 ***
subjID17      -0.27894    0.27845  -1.002 0.316458
subjID18       1.23616    0.21602   5.722 1.05e-08 ***
subjID19       1.04456    0.21532   4.851 1.23e-06 ***
subjID20       0.45939    0.25730   1.785 0.074193 .
subjID21      -3.18877    1.01548  -3.140 0.001689 **
subjID22      -1.24185    0.41698  -2.978 0.002899 **
subjID23       0.40721    0.25538   1.595 0.110815
subjID24      -2.39191    0.60229  -3.971 7.15e-05 ***
subjID25      -2.61097    0.72820  -3.586 0.000336 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2305.248  on 99  degrees of freedom
Residual deviance:   58.123  on 74  degrees of freedom
AIC: 426.26

Number of Fisher Scoring iterations: 16
```

Nel modello considerato abbiamo **24 variabili aggiuntive**, motivo per cui **non è opportuno trattare questa variabile come un effetto fisso**. Anche se il valore dell'AIC risulta **più basso** rispetto al modello con effetti casuali (AIC = 426.26), **una scelta puramente meccanica** basata sull'AIC può essere fuorviante.

Infatti, il modello potrebbe **non spiegare realmente nulla** (come suggeriscono alcuni standard error anomali), pur senza generare alcun messaggio di warning da parte di R.

Conclusione: è necessario usare il buon senso e la conoscenza del contesto, perché **un AIC più basso non sempre giustifica la scelta del modello**, specialmente quando si ignorano strutture gerarchiche o si includono troppe variabili come effetti fissi.

5.1.2 Cluster Data

Fittiamo il modello esattamente come prima, ma sul dataframe `df2`

```
str(df2)
```

```
'data.frame':    200 obs. of  3 variables:
 $ cluster: Factor w/ 4 levels "1","2","3","4": 4 4 2 4 3 3 3 1 3 3 ...
 $ X      : num  0.415 0.437 -0.214 0.33 0.142 ...
 $ Y      : int  1 1 1 1 0 0 0 0 0 0 ...
```

5.1.2.1 Modello con intercetta sia fissa che random e X fisse

```
fit2.glmm <- lme4::glmer(Y ~ 1 + X + (1 | cluster), family = binomial, data = df2)
```

Utilizziamo una random intercept. Quello che ci da un'idea del grouping è dentro la colonna cluster. Individui nello stesso gruppo condividono effetto simile, cioè una forma simile di fonte di variabilità.

```
summary(fit2.glmm)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: Y ~ 1 + X + (1 | cluster)
   Data: df2

     AIC      BIC   logLik deviance df.resid
   205.9    215.8   -100.0    199.9      197

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.2848 -0.4472  0.3258  0.4810  3.0444

Random effects:
 Groups  Name        Variance Std.Dev.
 cluster (Intercept) 4.022    2.006
Number of obs: 200, groups:  cluster, 4

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.5379     1.0199   0.527   0.5979
X             4.6245     2.5366   1.823   0.0683 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
  (Intr)
X 0.017
```

Stiamo stimando un **modello logit**. Di conseguenza, applicando `exp()` ai coefficienti otteniamo l'effetto sull'**odds** (rapporto tra la probabilità di esito 1 rispetto a 0).
Nel caso analizzato, il coefficiente **non è significativo** al livello di significatività $\alpha = 0{,}05$. Tuttavia, stiamo effettivamente utilizzando **la formula con cui abbiamo generato i dati** (il "true model"). La differenza cruciale è che nel processo di generazione dei dati il **random effect era noto**, mentre ora deve essere **stimato**.

I random effects servono a **stimare la variabilità all'interno dei cluster**, ossia all'interno dei gruppi identificati dal **grouping factor**. Se non abbiamo abbastanza osservazioni per ciascun cluster, la stima della variabilità sarà poco affidabile.

In questo esempio, la deviazione standard stimata è Std.Dev = 2.006, che è **superiore** al valore reale utilizzato nella simulazione (1.5). Inoltre, si nota una **discrepanza tra i coefficienti stimati e quelli reali** (intercetta e pendenza), che possono risultare molto diversi anche se il modello è formalmente corretto. Questo mismatch è comune nei **modelli binari**, dove la variabilità non osservata (random effect) viene stimata indirettamente, causando una distorsione anche se si conosce il modello strutturale corretto.

5.1.2.2 Modello con solamente l'intercetta random

```
fit2.glmm2 <- lme4::glmer(Y ~ (1 | cluster), family = binomial, data = df2)
```

É il modello più semplice che possiamo usare in questi casi. Lo fittiamo sempre per primo per farci un'idea della situazione.

```
summary(fit2.glmm2)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: Y ~ (1 | cluster)
   Data: df2

     AIC      BIC   logLik deviance df.resid
   208.1    214.7   -102.1    204.1      198

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.5355 -0.4489  0.3944  0.4891  2.2277

Random effects:
 Groups  Name        Variance Std.Dev.
 cluster (Intercept) 2.003    1.415
Number of obs: 200, groups:  cluster, 4

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.5133     0.7304   0.703    0.482
```

Se, una volta fittato il modello, osserviamo che la **deviazione standard associata ai random effects è molto piccola**, ciò indica che il **grouping factor non aggiunge informazione rilevante** sulla variabile risposta. In tal caso, l'uso di un modello a effetti misti potrebbe non essere necessario: è uno **starting point diagnostico** utile per decidere se includere o meno componenti random.

Inoltre, il **p-value associato alla stima della varianza** dei random effects può risultare **non significativo** anche in presenza di differenze apparenti tra i gruppi. Questo accade quando gran parte della variabilità in Y è già spiegata dalla struttura di raggruppamento e i gruppi non si discostano molto dalla **media complessiva**.

```
anova(fit2.glmm,fit2.glmm2)
```

```
Data: df2
Models:
fit2.glmm2: Y ~ (1 | cluster)
fit2.glmm: Y ~ 1 + X + (1 | cluster)
           npar    AIC    BIC   logLik deviance  Chisq Df Pr(>Chisq)
fit2.glmm2    2 208.14 214.73 -102.069   204.14
```

```
fit2.glmm     3 205.91 215.81  -99.956   199.91 4.2265  1     0.0398 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.1.2.3 Modello con anche random slope

```
fit2.glmm4 <- lme4::glmer(Y ~ 1 + X + (1 + X | cluster), family = binomial, data = df2
```

Modello con singola barra verticale: effetti casuali correlati

```
boundary (singular) fit: see help('isSingular')
```

Possiamo introdurre una **random slope** nei modelli misti specificando che entrambi gli effetti, intercetta e coefficiente della covariata, varino tra i gruppi. Questo consente di modellare una maggiore flessibilità nella relazione tra la variabile predittiva e la risposta all'interno di ciascun cluster. In pratica, due individui con lo stesso valore di una covariata `X`, ma appartenenti a cluster diversi, possono avere effetti differenti su Y, poiché sia l'intercetta che la pendenza sono specifiche per ciascun gruppo.

Tuttavia, se specifichiamo una struttura che consente la correlazione tra intercetta e pendenza (random slope), potrebbe comparire un **warning** legato alla **matrice di correlazione degli effetti casuali**, con una correlazione stimata pari a `-1`. Questo indica **una correlazione perfetta negativa**, spesso dovuta a **scarso supporto informativo nei dati**, ovvero non ci sono abbastanza osservazioni per stimare in modo affidabile la covarianza tra gli effetti.

Per evitare questo problema, possiamo specificare che gli effetti casuali siano **indipendenti**. In termini statistici, si assume che la matrice di covarianza degli effetti casuali sia **diagonale** (matrice **Q**), escludendo la covarianza tra intercetta e pendenza. Questo si ottiene usando la notazione con **doppia barra verticale**, che forza l'indipendenza tra gli effetti.

In sintesi, l'introduzione di una random slope permette di modellare variazioni nell'effetto di una covariata tra gruppi, ma bisogna fare attenzione alla struttura della matrice di covarianza degli effetti casuali per evitare stime instabili o non interpretabili.

```
summary(fit2.glmm4)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: Y ~ 1 + X + (1 + X | cluster)
   Data: df2

     AIC      BIC   logLik deviance df.resid
   209.5    226.0    -99.7    199.5      195

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.3972 -0.4495  0.3199  0.4926  3.7298
```

```
Random effects:
 Groups  Name        Variance Std.Dev. Corr
 cluster (Intercept) 4.214    2.053
         X           2.491    1.578    -1.00
Number of obs: 200, groups:  cluster, 4

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.2297     1.1531   0.199   0.8421
X             5.4544     2.7624   1.974   0.0483 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
  (Intr)
X -0.501
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')
```

```
fit2.glmm3 <- lme4::glmer(Y ~ 1 + X + (1+X || cluster), family = binomial, data = df2)
```

Modello con doppia barra verticale: effetti casuali incorrelati Quando specifichiamo un modello misto utilizzando la **doppia linea verticale** ||, stiamo dicendo a R di assumere che gli effetti casuali (intercetta e pendenza) siano **incorrelati**. Questo equivale a dire che la **matrice di covarianza degli effetti casuali è diagonale**, ovvero non contiene termini di covarianza tra gli effetti.

Questa scelta è particolarmente utile quando non ci sono abbastanza dati per stimare in modo affidabile la correlazione tra intercetta e slope, evitando così problemi numerici. In questo caso, infatti, **non compare più il warning** che indica che siamo sul **boundary dello spazio dei parametri**, il che solitamente significa che la correlazione stimata tra gli effetti era esattamente ± 1, una situazione sospetta.

In sintesi, l'uso della doppia barra verticale è una buona pratica in presenza di pochi dati o per evitare overfitting strutturale legato alla correlazione tra gli effetti casuali.

```
summary(fit2.glmm3)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: Y ~ 1 + X + (1 + X || cluster)
   Data: df2

     AIC      BIC   logLik deviance df.resid
   207.9    221.1   -100.0    199.9      196

Scaled residuals:
    Min      1Q  Median      3Q     Max
```

```
-3.3103 -0.4473  0.3240  0.4823  3.0288

Random effects:
 Groups    Name        Variance Std.Dev.
 cluster   (Intercept) 3.9429   1.9857
 cluster.1 X           0.5994   0.7742
Number of obs: 200, groups:  cluster, 4

Fixed effects:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.5095     1.2525   0.407   0.6842
X             4.6098     2.5985   1.774   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
  (Intr)
X 0.099
```

Nel modello con effetti casuali specificati tramite ||, non viene più stimata la correlazione tra intercetta casuale (τ_0) e slope casuale (τ_1). In questo caso, il modello include sia la random intercept che la random slope, ma assume che siano incorrelate.

Il problema della correlazione perfetta tra effetti casuali si presenta frequentemente quando abbiamo **molti cluster con poche osservazioni ciascuno** e si cerca di stimare la **matrice di covarianza generale G** degli effetti casuali. In queste situazioni, la mancanza di informazione sufficiente porta l'algoritmo di stima a raggiungere il **boundary dello spazio dei parametri**, restituendo un **warning** o una **matrice singolare**.

Questo significa che la matrice di covarianza non è invertibile, e il modello non riesce a stimare in modo affidabile la **covarianza tra** τ_0 e τ_1. Non si tratta di un problema esclusivo dei dati binari: può verificarsi anche in modelli per conteggi o in altri contesti con struttura a effetti misti.

```
anova(fit2.glmm,fit2.glmm3)
```

```
Data: df2
Models:
fit2.glmm: Y ~ 1 + X + (1 | cluster)
fit2.glmm3: Y ~ 1 + X + (1 + X || cluster)
           npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
fit2.glmm     3 205.91 215.81 -99.956   199.91
fit2.glmm3    4 207.91 221.10 -99.955   199.91 0.0014  1     0.9702
```

Stiamo confrontando due modelli: uno con sola **random intercept**, e uno che include anche una **random slope**, assumendo che τ_0 e τ_1 siano **indipendenti** (quindi con matrice di covarianza **Q**, diagonale). Dal confronto non emerge una riduzione significativa nella deviance, il che indica che l'introduzione della random slope **non apporta alcun miglioramento** al modello.

Di conseguenza, è preferibile mantenere il modello più semplice. Questo risultato è coerente con il **data generating process**, in cui non era stata introdotta alcuna random slope.

5.1.2.4 Modello con variabile cluster come predittore categorico

```
fit2.glm <- glm(Y ~ 1 + X + cluster, family = binomial, data = df2)
```

Un'alternativa che si potrebbe considerare, soprattutto in presenza di un **numero limitato di cluster** (come in questo caso, in cui i cluster sono solo 5), è trattare il cluster come un **predittore categorico** ed inserirlo come **effetto fisso** nel modello. In questo modo non modelliamo la variabilità con un termine casuale, ma stimiamo un parametro distinto per ciascun cluster.

```
summary(fit2.glm)
```

```
Call:
glm(formula = Y ~ 1 + X + cluster, family = binomial, data = df2)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   4.2714     1.1256   3.795 0.000148 ***
X             7.1647     2.6542   2.699 0.006947 **
cluster2     -3.0716     0.8793  -3.493 0.000477 ***
cluster3     -7.0590     1.5760  -4.479  7.5e-06 ***
cluster4     -4.7908     2.0184  -2.374 0.017618 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 268.37  on 199  degrees of freedom
Residual deviance: 181.32  on 195  degrees of freedom
AIC: 191.32

Number of Fisher Scoring iterations: 4
```

Vediamo che tutti i coefficienti sono significativi. Se vogliamo comparare i modelli dobbiamo guardare guardare l'AIC

```
AIC(fit2.glmm)
```

```
[1] 205.9111
```

```
AIC(fit2.glm)
```

```
[1] 191.3206
```

Questo è uno dei dilemmi principali nell'uso dei **GLMM**: quale modello scegliere?

Nel nostro caso, con **200 osservazioni**, il costo di stimare 4 gradi di libertà in più (per i 5 cluster) è accettabile. Ma la scelta tra **effetti fissi** e **effetti casuali** dipende dall'interpretazione del contesto. Per esempio se questi 5 fossero degli ospedali:

- Se si pensasse che questi fossero **gli unici rilevanti**, allora avrebbe senso trattarli come **effetti fissi**.
- Se invece si considerasse che questi siano un **sottogruppo di una popolazione più ampia**, allora è preferibile usare un **effetto casuale**. In tal caso, si assume che ci sia una distribuzione sottostante degli effetti di cluster, e che i cluster osservati rappresentino solo un campione da tale distribuzione.

In sintesi:

- **Effetto fisso**: voglio stimare specificamente l'effetto di ciascun cluster osservato.
- **Effetto casuale**: voglio modellare la variabilità tra cluster come proveniente da una popolazione più ampia.

Part XII

Generalized Additive Models

1. Generalized Additive Models

In order to fit a model you can apply transformation to the predictors, as long as the resulting transformation (the way this enter in the linear predictor) keeps the linear predictor linear in the coefficient. The power of this models is the fact that it is non-paramteric, so can represent non-linear relation. (chapter) of the book geo-spatial model) . it is good for flexibility because you can do a lot of things. But poor diagnostic.

1.1 Model definition

- $i = 1, \cdots, n$ observation of $(y_i, x_{i1}, \cdots, x_{iq})$ a continous response and generic covariates (categorical, binary...)
- $\boldsymbol{z_i} = (z_{i1}, \cdots, z_{iq})$ and additional set of **continous** predictors

The idea is to model the effect of $\boldsymbol{z}$ on the linear predictor trough $f(\mathbf{z}) = f(z_1, \ldots, z_q)$, it is a function of q variable but your are going to split it in order to be a univariate function, here comes the **additive part**:

→ *you break down the effect of all continuous covariates into individual effect that are additive.*

$$f(\mathbf{z}) = f(z_1, \ldots, z_q) = f_1(z_1) + \cdots + f_q(z_q)$$

- Each $f_j(z_j)$ *is a nonlinear smooth effect to be* ***estimated***, no parametrically (usually with splines and polynomial),

$$\begin{aligned} y_i &= f_1(z_{1i}) + \cdots + f_q(z_{qi}) + \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i \\ y_i &= f_1(z_{1i}) + \cdots + f_q(z_{qi}) + \eta_i^{\text{lin}} + \varepsilon_i \\ y_i &= \eta_i^{\text{add}} + \varepsilon_i \end{aligned}$$

- y_i is given by the contribution of these additive predictor, but they enter the model non parametrically + the usual linear predictor.
- model above also called *partial* or *semiparametric* model, due to a parametric part (η_i^{lin}) and a nono parametric part ($f_1(z_{1i}) + \cdots + f_q(z_{qi})$)

A model with no η_i^{lin} is simply called just **additive model**.

$$\begin{aligned} y_i &= f_1(z_{1i}) + \cdots + f_q(z_{qi}) + \varepsilon_i \\ y_i &= \eta_i^{\text{add}} + \varepsilon_i \end{aligned}$$

Note: this construction does not determine the additive part is the one above it is not the construction of the response which determines the additive part of the model but,

the splitting of $f(\mathbf{z})$.

Note: suitable when you have large n due to the increase of needed estimates.
Note: be carefull because there is no interaction. You cannot use interaction!

Note: Sometimes the distinction between linear predictor and non parametrical is not needed, you can use a definition of f tha also correspond to the linear predictors

1.2 Estimation of $f_j(z_j)$

Each function $f_j(z_j)$ is approximated through Thin Plate splines or B-Splines (basic) or P-splines (penalized), you fit models and then you guess (no test) which one is better the another:

$$f_j(z_j) = \sum_{l=1}^{d_j} \gamma_{jl} B_l(z_j)$$

The basis $(B_l(z_j))$ are your predictor,whith $\boldsymbol{\gamma}_j$ regression coefficients, at the end this are just Large Linear Models te complication is that the design matrix $\boldsymbol{X}$ contains also basis computed on observed value, but still $\boldsymbol{X\beta}$.

The vector of function values $\boldsymbol{f}_j = (f_j(z_{1j}), \ldots, f_j(z_{nj}))^\top$ at observed points $(z_{1j}, \ldots, z_{nj})$ is expressed as:

$$\boldsymbol{f}_j = Z_j \boldsymbol{\gamma}_j$$

A vector $(n \times 1)$ of values computed using the first predictor computed with the splines.
$\rightarrow$ each observation is decomposed additively in to q effect.

Note: there is no index because the vector contains all n observations.

Now the the full additive model (without η_i^{lin}) can be expresed express as:

$$y = f_1 + f_2 + \cdots + f_q + \boldsymbol{\varepsilon}$$

1.2.1 Problem of Identifiability

If you have two function $f_1(\cdot)$ $f_2(\cdot)$ and a constant k then you add a k to $f_1(\cdot)$ and subtract the same constant to $f_2(\cdot) \Rightarrow (f_1(\cdot) + k) + (f_2(\cdot) - k) = f_1(\cdot) + (f_2(\cdot)$ this is a problem of **identifiability** (non ho capito) after summing them the two function are now the same.

Solution: Center the functions around zero:

$$\sum_{i=1}^{n} f_1(z_{i1}) = \sum_{i=1}^{n} f_2(z_{i2}) = \cdots = \sum_{i=1}^{n} f_q(z_{iq}) = 0$$

1.2.2 Smoothness

The $f_1, \cdots, f_q$ that you are trying to estimate they cannot be any function, all this function sould be **smooth function**, it means the existence of at least the second derivative. A smooth function has ∞ amounuts of point, but it comes from the data (which are not ∞), so to reconstruct a smooth function from a finite set of points you need to impose constraint:

- the ones of the cubic spline
- the presence of a penalization term ($\lambda_j \gamma_j' \boldsymbol{K_j} \gamma_j$) on the estimation. Choosinig on $\boldsymbol{K_j}$ determines the type of splines

If we had no restrictions on γ_j this would be just a large linear model, fittable with OLS. Instead, we estimate with regularized approaches using penalties of the type $\lambda_j \gamma_j' K_j \gamma_j$, with penalty matrix K depending on the basis functions used for $f_j()$.

1.3 Matrix representation

Additive models can be written in matrix form. Here you have the *additive-nonparametrical* part anD then the usual linear predictor.

$$y = \boldsymbol{Z_1 \gamma_1} + \cdots + \boldsymbol{Z_q \gamma_q} + \boldsymbol{X \beta} + \boldsymbol{\varepsilon}$$

(he also write)

$$y = f_1 + \cdots + f_q + \eta_i^{\text{lin}} + \boldsymbol{\varepsilon}$$

1.4 STAR Models

Model used for Geo-spatial regressor, this is a class of model that allows you to put barely anything inside the linear predictor. STAR stands for Structured (because you can put whatever you want, infact all the compnents list below do have different structure) Additive (because whatever the effect you put it has to be a sunm) Regression Models.

For example you can use:

- nonlinear effect of covariate,
- spatial effects (specific maps),
- random effects,
- interaction terms,
- varying coefficients and so on

Additive $\rightarrow$ the non parametric part must have additive effect.

The general formulation is:

$$y = f_1(v_1) + \cdots + f_q(v_q) + \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \varepsilon$$

which is still a linear predictor.

2. Generalized STAR and GAMs

Generalized because the effect is not continous. Previous were not generalized models, because we are assuming a continous response, to make it a generalized star models it sufficies to: $\mathbb{E}[\boldsymbol{y}] = \boldsymbol{\eta}^{\text{add}} \to \mathbb{E}[\boldsymbol{y}] = g(\boldsymbol{\eta}^{\text{add}})$ with: $\boldsymbol{\eta} = f_1 + \cdots + f_q + \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$ you simply transform linear predictors.

Note: the notation changes

2.1 Model definition

Terms $v_1, \ldots, v_q$ are one- or multi-dimensional covariates built from the original predictors, i.e. consider the model

$$y = f_0(v_0) + f_1(v_1) + f_2(v_2) + \varepsilon$$
$$y = \eta^{\text{add}} + \varepsilon$$

- $f_0(v_0) = x'\beta, \quad v_0 = (1, x_1, x_2, \ldots, x_k)'$, the linear predictor η^{lin}
- $f_1(v_1) = f_1(z_1), \quad v_1 = z_1$, nonlinear smooth term for z_1
- $f_2(v_2) = \gamma_i, \quad v_2 = i$, individual-specific random intercept

Due to the presence of an additional *non-linear* term ($f_1(v_1)$), we can't fit a linear mixed effects model, but an additive model is needed

But in principle if we use basis functions to model $f_1, \ldots, f_q$ we obtain a large linear model

$$y = f_1 + \cdots + f_q + X\beta + \varepsilon = V_1\gamma_1 + \cdots + V_q\gamma_q + X\beta + \varepsilon$$

STAR models can be generalized to allow non-normal regressions

$$E(y_i) = \mu_i = h(\eta_i^{\text{struct}}) = h(f_1(v_1) + \cdots + f_q(v_q) + x'\beta)$$

and if we only consider smooth additive terms of continuous covariates

$$\eta_i^{\text{struct}} = \eta_i^{\text{add}} = f_1(z_1) + \cdots + f_q(z_q) + x'\beta$$

we obtain the class of *Generalized Additive Models (GAMs).*

2.2 Inference in STAR models

(do only the important things)

Inference in STAR models is particularly challenging. Main two avenues: *penalized least squares* or *likelihood estimation.*

Consider model for continuous variable y to be

$$y = f_1 + \cdots + f_q + X\beta + \varepsilon = V_1\gamma_1 + \cdots + V_q\gamma_q + X\beta + \varepsilon,$$

we want to minimize the criterion (important)

$$LS_{\text{pen}}(\gamma_1, \ldots, \gamma_q, \beta) = LS(\gamma_1, \ldots, \gamma_q, \beta) + \sum_{j=1}^{q} \lambda_j \gamma_j' K_j \gamma_j$$

Penalization due to smoothness, in random effect models $\sum_{j=1}^{q} \lambda_j \gamma_j' K_j \gamma_j$ is the prior of the coefficient, here it is the soomthness of the function.
with

$$LS(\gamma_1, \ldots, \gamma_q, \beta) = (y - V_1\gamma_1 - \cdots - V_q\gamma_q - X\beta)'(y - V_1\gamma_1 - \cdots - V_q\gamma_q - X\beta)$$

Smoothing parameters λ_j
Balance the trade-off between fit and smoothness, in particular this are *tuning parameter*, needs to be adjusted in orther to improve the fit. These parameter are found trough cross validation, or AIC. (this are the tricky parameter) .

- if the function is too smooth ⇒ does not provide a good fit to the data, bbut gives good prediction
- if it is not smooth enough ⇒ overfitting on the data, but will do poorly on prediction.

(important)

To minimize $LS_{\text{pen}}(\gamma_1, \ldots, \gamma_q, \beta)$ we have : backfitting algorithm (iterative smoothing of partial residuals) direct optimization

2.3 Inference in Generalized STAR models and GAMs

For generalized STAR models and GAMs, we maximize penalized log-likelihood

$$\ell_{\text{pen}}(\gamma_1, \ldots, \gamma_q, \beta) = \ell(\gamma_1, \ldots, \gamma_q, \beta) - \frac{1}{2} \sum_{j=1}^{q} \lambda_j \gamma_j' K_j \gamma_j$$

which means setting first derivatives to zero and solve system of equations: similar to GLMs, Fisher scoring and/or backfitting.

This is important
Heree you do not have statistics or test, nothing. What you can do is fit the model and compare AIC or use CV.

- ◇ **Choosing smoothing parameters** λ_j: model choice criterion like AIC or generalized cross-validation (GCV)

- ◇ **Choosing predictors and comparing models**: model choice criterion like AIC or GCV

- ◇ **Goodness-of-fit:** model choice criterion like AIC or GCV

(IMPORTANT)
There is no residual qq-plot, etc you fit a GAM model you look at the effect of the covariates, look above, and that's it.

Diagnostics for STAR, generalized STAR and GAMs is complex and an evolving field of research/software implementations.

2.4 Summary

Response type	**Model characteristics and classification**
y **continuous**	– If there is a *nonlinear effect* $f(z_1, \ldots, z_q)$, the model may be: ∗ nonparametric ∗ semiparametric – If $f_1(z_1) + \cdots + f_q(z_q)$, it is an additive model. – If all functions $f_1(z_1), \ldots, f_q(z_q)$ are smooth and *nonlinear*, it is an additive model. – If some $f_1(z_1), \ldots, f_q(z_q)$ are random effects or spatial terms, it is a **Structured Additive Regression** (STAR) model.
y **not continuous**	– If there is a *nonlinear effect* $f(z_1, \ldots, z_q)$, the model may be: ∗ nonparametric ∗ semiparametric – If $f_1(z_1)+\cdots+f_q(z_q)$, it is a **Generalized Additive Model** (GAM). – If all functions $f_1(z_1), \ldots, f_q(z_q)$ are smooth and *nonlinear*, it is a GAM. – If some $f_1(z_1), \ldots, f_q(z_q)$ are random effects or spatial terms, it is a **Generalized Structured Additive Regression** (GSTAR) model.

Comparison: Gaussian Models vs. Additive Models (GAM/STAR)

	Gaussian Model (GLM)	Additive Model (GAM/STAR)
Theoretical	Strong theoretical guarantees, well-established inference	Few theoretical guarantees, especially in nonparametric parts
Flexibility	Low flexibility	High flexibility, nonlinear effects allowed
Interpretability	High interpretability due to linearity	Less interpretable due to smooth nonlinear components
Complexity	Simple to specify, fit and analyze	More complex; requires smoothing, tuning, backfitting
Estimation	Ordinary Least Squares (OLS) or MLE	Penalized estimation, smoothing splines, backfitting algorithms

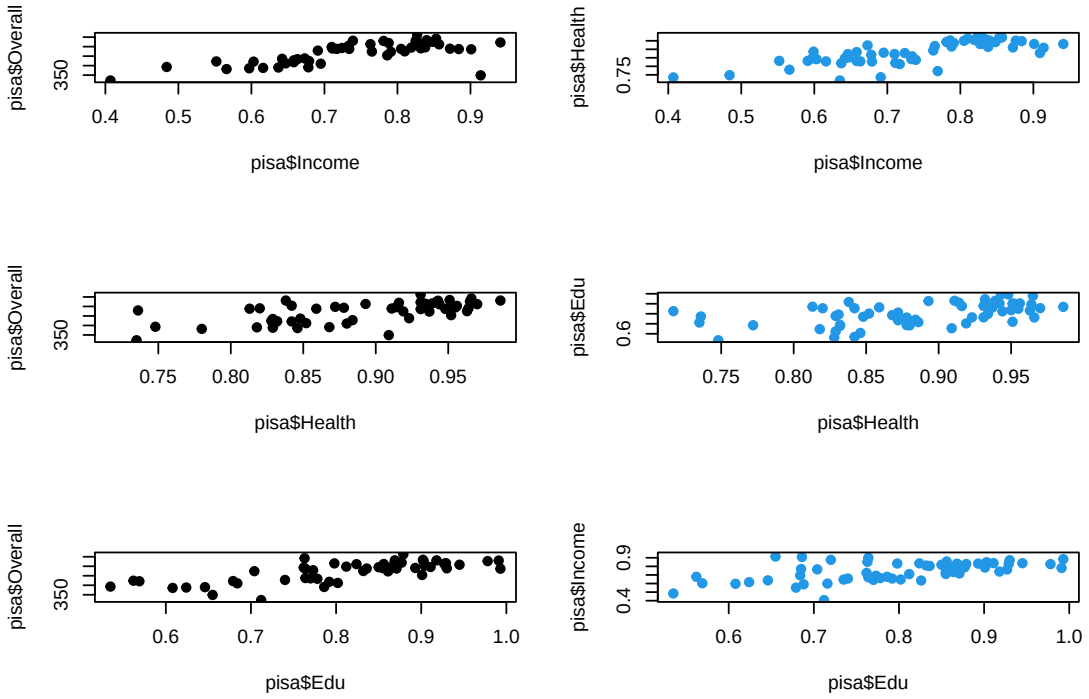
2.5 Generalised Additive Models

Fittare modelli generalizzati additivi (GAM) non è sempre così trasparente come accade nei GLM o in altri modelli parametrici classici. Sebbene la flessibilità dei GAM consenta di inserire effetti non lineari o effetti strutturati (come nei modelli di regressione additiva strutturata), è fondamentale essere molto cauti. Infatti, è facile cadere nell'illusione di "avere tutto sotto controllo": la sintassi semplice e la possibilità di rappresentare graficamente gli effetti li rende strumenti potenti e accattivanti. Tuttavia, se non si presta attenzione, si rischia di perdere il controllo su ciò che si sta effettivamente stimando, sulla natura degli effetti (fissi, random, smooth), e sull'interpretabilità complessiva del modello.

> In sintesi: è facile fittarli, è bello plottarli, ma è altrettanto facile smarrirsi. Serve sempre consapevolezza teorica e attenzione nella costruzione del modello.

```
pisa <- read.csv("pisa2006.csv", stringsAsFactors = T)
```

```
par(mfrow=c(3,2))
plot(pisa$Income, pisa$Overall, pch=19, col=1)
plot(pisa$Income, pisa$Health, pch=19, col=4)
plot(pisa$Health, pisa$Overall, pch=19, col=1)
plot(pisa$Health, pisa$Edu, pch=19, col=4)
plot(pisa$Edu, pisa$Overall, pch=19, col=1)
plot(pisa$Edu, pisa$Income, pch=19, col=4)
```




```
par(mfrow=c(1,1))
```

2.5.0.1 Visualizzazione delle relazioni e introduzione dei tensor product smooth

A sinistra abbiamo dei grafici che mostrano la relazione tra ciascun predittore e la variabile di risposta Y. Questi plot aiutano a comprendere il tipo di associazione (lineare o non lineare) tra singoli predittori e l'output.

A destra, invece, visualizziamo le relazioni **tra coppie di predittori**, per valutarne l'eventuale **correlazione**. Questo passaggio è utile quando si desidera introdurre uno **smooth bidimensionale**, cioè un effetto non lineare che coinvolge **due variabili contemporaneamente**.

In matematica, questo tipo di funzione è chiamato **tensor product smooth**: una funzione che prende in input due (o più) variabili e genera un effetto complessivo, non separabile, sulla risposta.

- La **spline** è la versione univariata: modella un effetto non lineare su una singola variabile.
- Il **tensor** è la generalizzazione multivariata: modella un effetto congiunto di due o più variabili.

Un'analogia con l'informatica:

- **Vettore**: 1 dimensione
- **Matrice**: 2 dimensioni
- **Array**: oggetto multidimensionale (generale)

Allo stesso modo nella modellazione statistica abbiamo:

- **spline = univariate**
- **tensor = multivariate**

2.5.1 Linear regression

La funzione `gam` può fittare un linear model, ma se voglioamo fittare un linear model non ha senso farlo con questa, ma con la funzione `lm`

```
lin.mod <- gam(Overall ~ Income, data=pisa)
summary(lin.mod)
```

```
Family: gaussian
Link function: identity

Formula:
Overall ~ Income

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   204.32      35.37   5.777 4.32e-07 ***
Income        355.85      46.79   7.606 5.36e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


R-sq.(adj) =  0.518   Deviance explained = 52.7%
GCV = 1504.5  Scale est. = 1448.8    n = 54
```

L'unico vantaggio di fittare un modello lineare con `gam` è che ci da anche il `GCV (= 1504.5)` che è una misura del fit del modello. É la cross validation generalizzata

```
# plot(lin.mod)
```

Se si prova a usare `plot()` su un modello lineare classico (ad esempio ottenuto con `lm()`), non è possibile ottenere i grafici dei termini smooth perché **non ci sono smooth terms nel modello**. Nel caso dei modelli `gam`, la funzione `plot()` è progettata per visualizzare le **funzioni smooth** stimate, ovvero i termini non parametrici come $f(X)$. Ma in un modello lineare classico, l'espressione è puramente lineare: $Y = X \cdot \beta + \varepsilon$
Pertanto, quando si invoca `plot(lin.mod)` su un oggetto che **non contiene termini smooth**, la funzione non sa cosa rappresentare e restituisce un output vuoto o un errore. In breve, la funzione di plotting dei GAM **richiede la presenza di componenti non lineari**, altrimenti non ha nulla da plottare.

Per introdurre un effetto non lineare in un modello `gam`, utilizziamo la funzione `s()`. Questa funzione segnala al modello che la variabile deve essere trattata come uno **smooth term**, ovvero una funzione non parametrica stimata dai dati.

La sintassi è molto simile a quella di un modello lineare: $Y \sim s(X)$

In questo caso, invece di stimare un coefficiente lineare per X, si stima una funzione $f(X)$ che cattura in modo flessibile l'associazione tra la variabile X e la risposta Y, lasciando che la forma della relazione emerga dai dati stessi.

```
# ?gam
# ?smooth.terms oppure #s
# s: può essere uno spline, un tensore, un random effect, quello che vogliamo
```

2.5.2 Spline

2.5.2.1 Spline con basi tp

```
mod <- gam(Overall ~ s(Income,bs="tp"), data=pisa)
```

Invece di assumere che l'effetto del reddito (`income`) su una misura di valutazione degli studenti (ad esempio il punteggio medio in scienze) sia lineare, possiamo ipotizzare che tale effetto sia **non lineare**. Per farlo, utilizziamo la funzione `s()` all'interno di un modello GAM, che permette di stimare una relazione flessibile tramite funzioni spline.

Il tipo di spline utilizzata è controllato dal parametro `bs` (basis), e nel caso specifico usiamo `bs = "tp"`, che indica l'uso di **thin plate regression splines**, una delle scelte più comuni per modellare effetti smooth.
Ma, **come scegliere il tipo di spline da usare?**

La scelta può basarsi su diverse considerazioni:

- Proprietà teoriche delle basi spline
- Performance empirica confrontando diversi modelli (ad esempio tramite **cross-validation** o criteri informativi come **AIC** o **GCV**)

L'approccio consigliato è:

- 1. Scegliere un tipo di spline (es. `"tp"`, `"cr"`, `"ps"`, ...)
- 2. Fittare il modello
- 3. Cambiare tipo di spline
- 4. Confrontare le performance

Quindi, stimiamo un effetto non lineare di `income` sulla valutazione degli studenti nei diversi paesi. La struttura del modello sarà: $Y \sim s(\text{income})$

Anche se il fitting del modello è computazionalmente veloce, bisogna tenere conto che viene comunque stimata una funzione complessa, non immediatamente interpretabile come nel caso dei modelli lineari.

```
summary(mod)
```

```
Family: gaussian
Link function: identity

Formula:
Overall ~ s(Income, bs = "tp")

Parametric coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  470.444      3.968   118.6   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
            edf Ref.df    F p-value
s(Income) 7.972  8.729 16.1  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.717   Deviance explained = 75.9%
GCV = 1019.7  Scale est. = 850.26    n = 54
```

Il modello stimato è simile a un modello lineare classico, ma con l'aggiunta di un termine **smooth**.

Nella sezione dei **coefficienti parametrici**, troviamo solo l'intercetta. Questo indica che non abbiamo covariate parametriche nel modello oltre a questa.

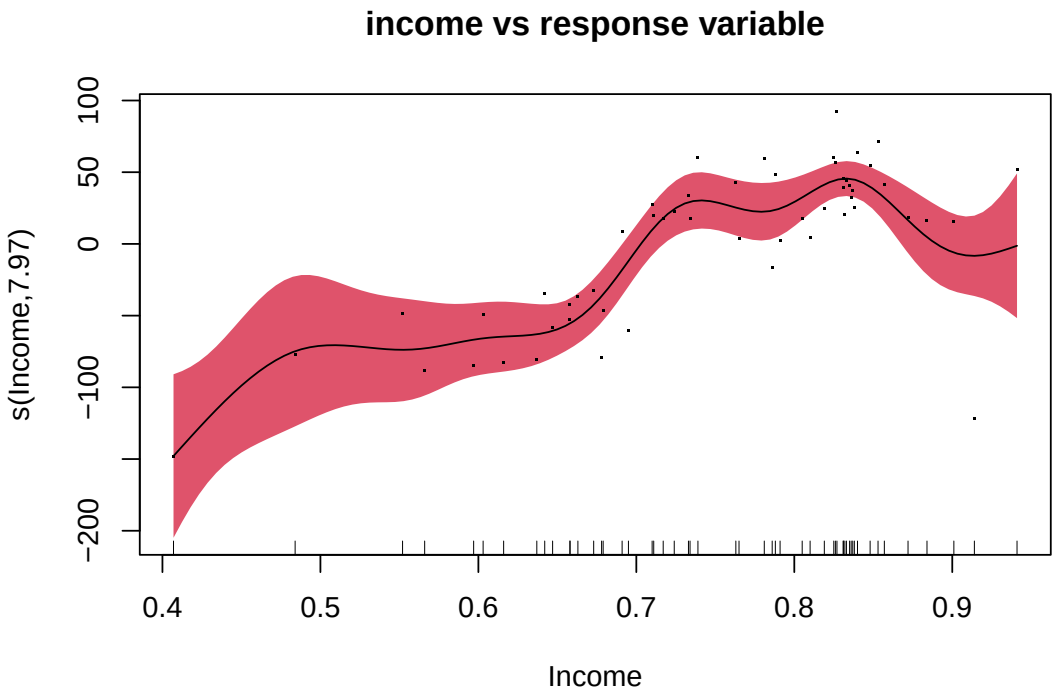
La sezione **Approximate significance of smooth terms** mostra l'effetto del termine non lineare `s(Income)`. Il test non è esatto ma approssimato. L'**edf** vicino a 8 indica che questa funzione non lineare, pur avendo un solo predittore, equivale in complessità a circa 8 parametri. Il p-value significativo conferma che l'effetto di `Income` sia davvero non lineare.

Infine, valutiamo le prestazioni del modello:

- Percentuale di devianza spiegata: **75.9%**
- R-squared (adjusted): **0.717** (contro 0.518 del modello lineare precedente)
- GCV score: **1019.7** (contro 1504.5 del modello lineare)

Tutti gli indicatori suggeriscono che il modello con effetto non lineare di `Income` è migliore rispetto al modello lineare.

```
plot(mod,pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.col=2, , main = "incom
```

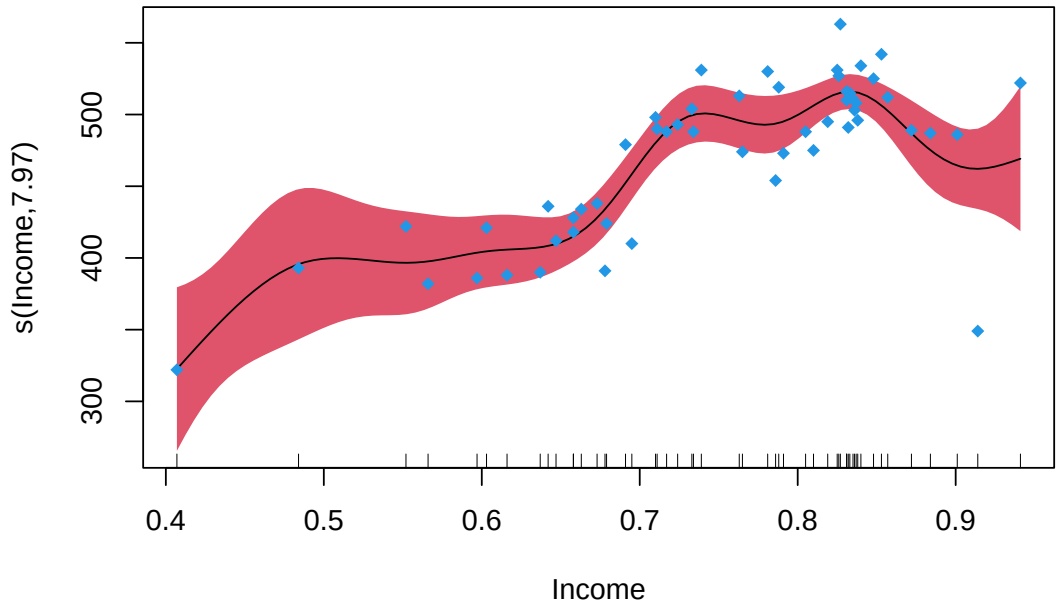


Plot

Poiché il modello contiene un termine non lineare, è possibile visualizzare l'effetto della variabile `Income` sulla risposta. L'effetto stimato risulta chiaramente **non lineare**: cresce per bassi valori di reddito, poi si stabilizza (plateau), riprende a crescere e infine mostra un altro plateau.

Le **fasce rosse** nel grafico rappresentano gli **intervalli di confidenza**, che indicano la variabilità della stima del termine non lineare in ogni intervallo del predittore. I **punti neri** indicano le osservazioni originali. È fondamentale ricordare che l'**asse y del grafico** non rappresenta direttamente i valori della variabile risposta, ma solo il contributo dello smooth term, ovvero la funzione stimata per `Income`. Manca quindi un livello di riferimento, che è dato dall'intercetta del modello. Questo tipo di visualizzazione serve **non tanto a valutare i livelli assoluti della risposta**, quanto piuttosto a comprendere **la forma dell'effetto** (es. lineare, crescente, decrescente, non lineare ecc.). Se si volesse ricostruire il valore atteso della risposta, bisognerebbe **sommare l'intercetta** al termine visualizzato nel grafico.

```
plot(mod,shift=coef(mod)[1],pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.col=
points(pisa$Income,pisa$Overall, pch=18, col=4)
```



Possiamo visualizzare lo smooth term del modello includendo anche l'intercetta, ottenendo così un grafico "shiftato" verso l'alto. Questo cambiamento di scala è dovuto al valore positivo dell'intercetta, che viene sommato all'effetto stimato. Tuttavia, questo shift non è obbligatorio né necessario: nella maggior parte dei casi si preferisce analizzare lo smooth term senza l'intercetta, per cogliere meglio la **forma dell'effetto non lineare**.

Per confrontare le stime con i dati originali, possiamo sovrapporre al grafico anche i punti osservati. In questo modo diventa evidente che:

- Per bassi valori di `Income`, l'effetto stimato mostra un incremento iniziale, indicando che **anche piccoli aumenti del reddito nei paesi più poveri migliorano sensibilmente il rendimento scolastico degli studenti**.
- Segue una fase di **plateau**, dove ulteriori incrementi di reddito non producono grandi cambiamenti nell'assessment degli studenti.
- Successivamente, per valori intermedi di `Income`, si osserva un nuovo aumento dell'effetto, seguito da un altro plateau.

L'aggiunta dei punti osservati evidenzia però un aspetto importante: **nelle regioni a basso reddito ci sono pochissime osservazioni**. Questo rende la stima dello smooth term molto instabile in quella zona. La crescita iniziale osservata potrebbe essere fortemente influenzata da pochi dati (es. due punti), e potrebbe non riflettere un pattern robusto.

La **larghezza delle bande di confidenza** conferma questa incertezza: all'inizio del dominio del predittore l'area è ampia, mentre si restringe nelle zone centrali, dove ci sono più osservazioni. Questo sottolinea che **le stime non parametriche sono fortemente guidate dalla distribuzione dei dati**: più osservazioni significano stime più affidabili, mentre regioni con pochi punti portano a maggiore incertezza.

2.5.2.2 Spline con basi br

```
mod2 <- gam(Overall ~ s(Income,bs="cr"), data=pisa)
summary(mod2)
```

```
Family: gaussian
Link function: identity

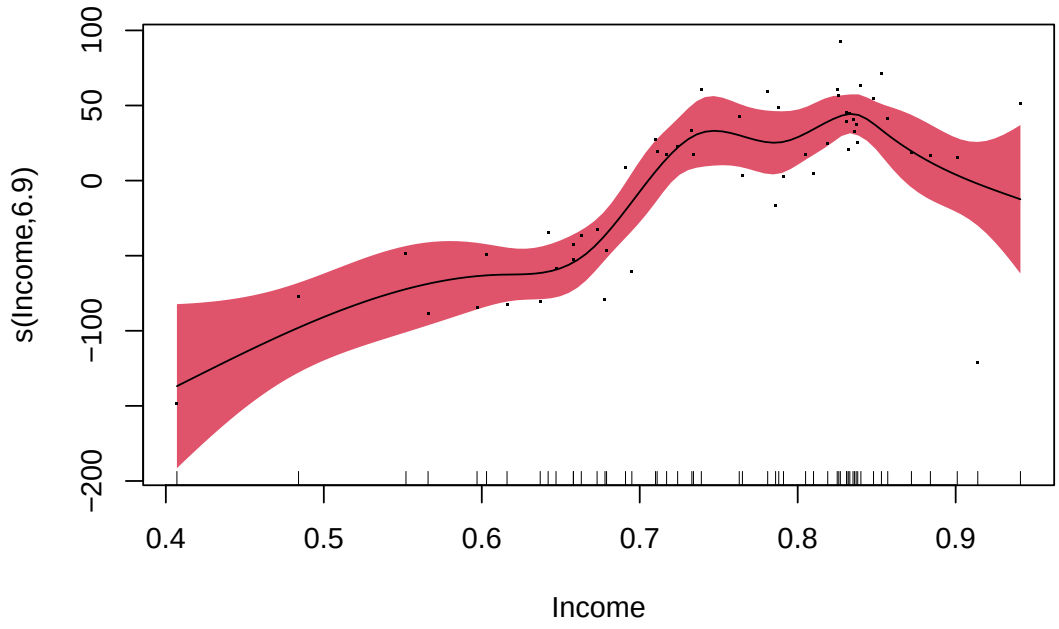
Formula:
Overall ~ s(Income, bs = "cr")

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  470.444      4.082   115.3   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

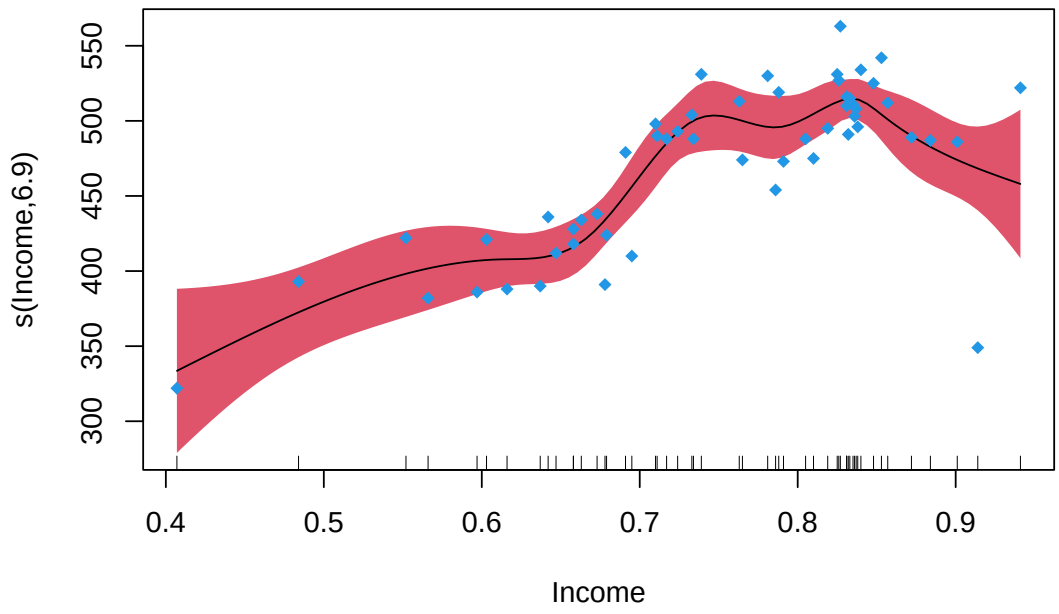
Approximate significance of smooth terms:
            edf Ref.df     F p-value
s(Income) 6.895  7.741 16.67  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =    0.7   Deviance explained = 73.9%
GCV = 1053.7  Scale est. = 899.67    n = 54
```

```
plot(mod2,pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.col=2)
```



```
plot(mod2,shift=coef(mod2)[1],pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.c
points(pisa$Income,pisa$Overall, pch=18, col=4)
```




```
anova(mod,mod2, test="Chisq")
```

```
Analysis of Deviance Table

Model 1: Overall ~ s(Income, bs = "tp")
Model 2: Overall ~ s(Income, bs = "cr")
  Resid. Df Resid. Dev       Df Deviance Pr(>Chi)
1    44.271      38285
2    45.259      41479 -0.98791  -3193.8  0.05168 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2.5.2.3 Spline con Edu lineare ed Income ed Edu con basi tp

```
mod3 <- gam(Overall ~ s(Income,bs="tp")+s(Edu,bs="tp")+Edu, data=pisa)
summary(mod3)
```

```
Family: gaussian
Link function: identity

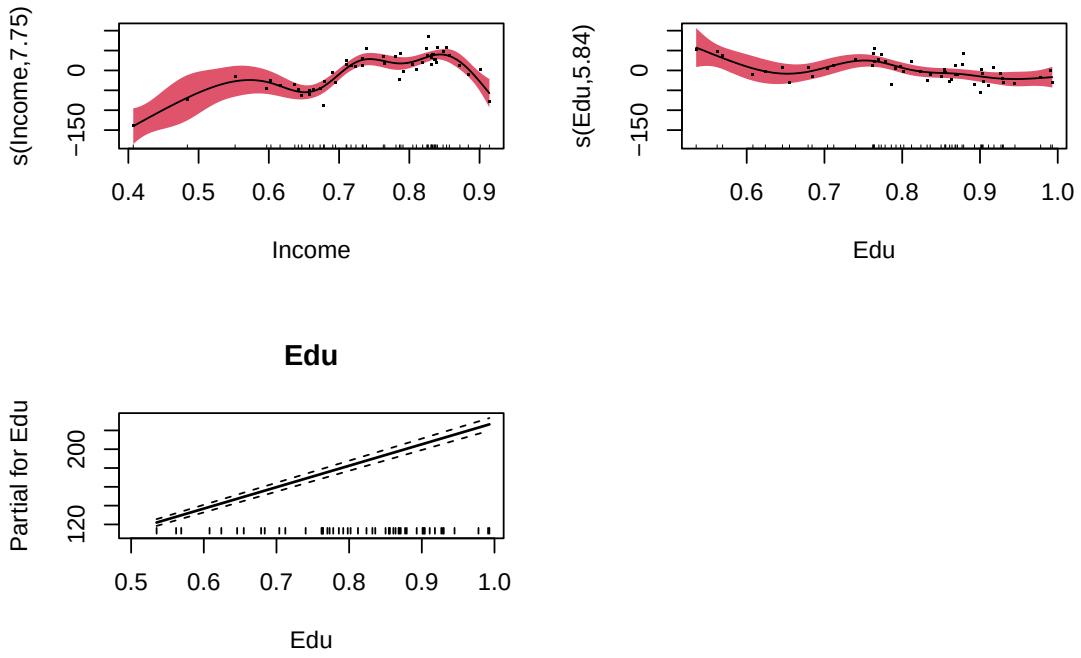
Formula:
Overall ~ s(Income, bs = "tp") + s(Edu, bs = "tp") + Edu
```

```
Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  286.516      3.024   94.76   <2e-16 ***
Edu          228.035      3.360   67.86   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
            edf Ref.df     F  p-value
s(Income) 7.745  8.514 10.32 2.65e-07 ***
s(Edu)    5.839  6.821  2.88   0.0164 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Rank: 19/20
R-sq.(adj) =  0.855   Deviance explained = 89.4%
GCV = 587.69  Scale est. = 422.78    n = 52
```

```
plot(mod3,pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.col=2)
```



2.5.2.4 Spline con Income ed Edu con basi tp

```
mod4 <- gam(Overall ~ s(Income,bs="tp")+s(Edu,bs="tp"), data=pisa)
summary(mod4)
```

```
Family: gaussian
```

```
Link function: identity

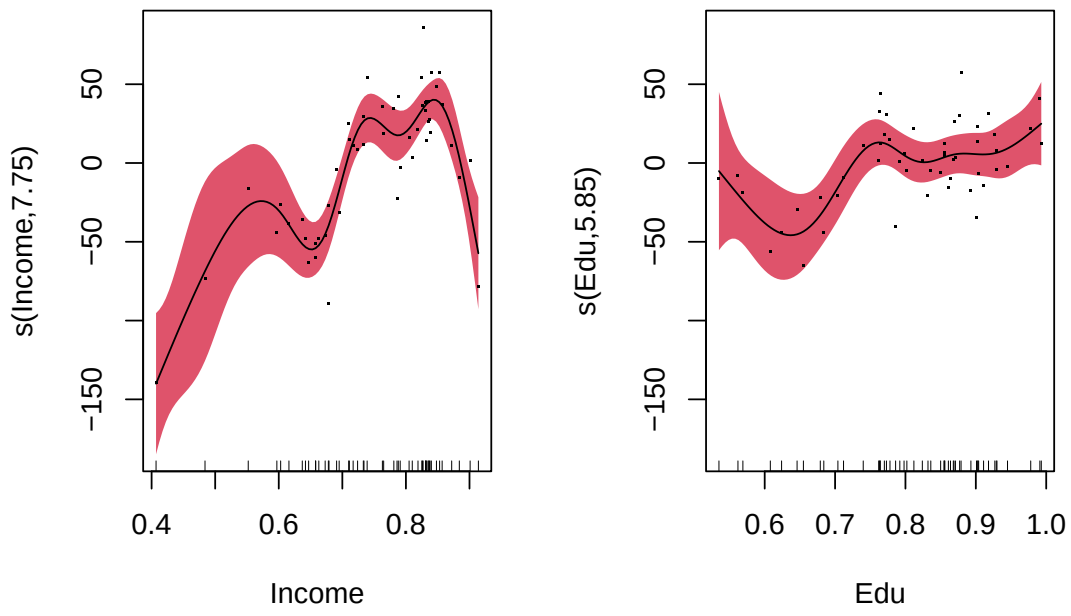
Formula:
Overall ~ s(Income, bs = "tp") + s(Edu, bs = "tp")

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  471.154      2.851   165.2   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

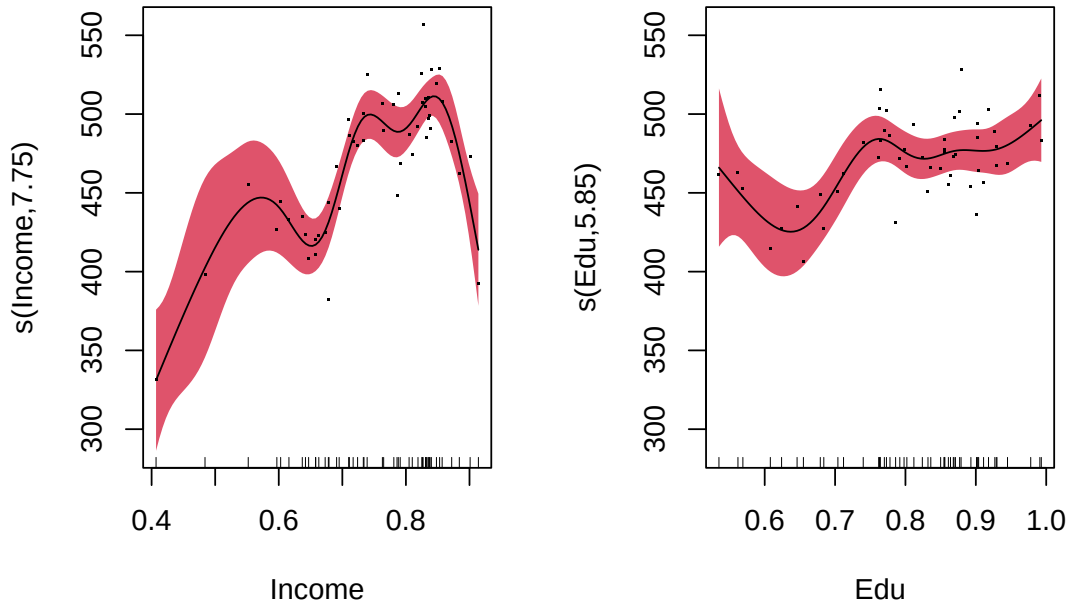
Approximate significance of smooth terms:
            edf Ref.df     F  p-value
s(Income) 7.745  8.514 10.32 2.65e-07 ***
s(Edu)    5.847  6.829  2.84   0.0174 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.855   Deviance explained = 89.4%
GCV = 587.69  Scale est. = 422.78    n = 52
```

```
plot(mod4,pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.col=2)
```




```
plot(mod4,shift=coef(mod4)[1],pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.c
```




```
anova(mod4,mod3,test="Chisq")
```

```
Analysis of Deviance Table

Model 1: Overall ~ s(Income, bs = "tp") + s(Edu, bs = "tp")
Model 2: Overall ~ s(Income, bs = "tp") + s(Edu, bs = "tp") + Edu
  Resid. Df Resid. Dev          Df    Deviance  Pr(>Chi)
1    35.657      15816
2    35.657      15816 -2.7001e-13 -6.5484e-10 3.687e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# attenzione a usare X all'interno di spline E come effetto lineare
```

2.5.3 Tensor

Vediamo cosa succede quando introduciamo un tensore (effetto di 2 covariate allo stesso tempo), che per noi è l'quivalente delle interazioni.

```
mod_tensor <- gam(Overall ~ te(Income, Edu) , data=pisa)
```

Modelliamo l'assessment degli studenti considerando contemporaneamente due predittori: il livello di reddito (`Income`) e il livello di istruzione (`Education`) per ciascun paese. Invece di trattarli separatamente o in modo lineare, introduciamo un **tensor product smooth**, ovvero una funzione che prende entrambe le covariate simultaneamente e ne stima l'**effetto combinato** in modo flessibile e non lineare. Questo approccio permette di catturare eventuali **interazioni complesse** tra `Income` ed `Education`, fornendo una

rappresentazione più realistica dell'impatto congiunto che queste due variabili esercitano sulla performance scolastica degli studenti.

```
summary(mod_tensor)
```

```
Family: gaussian
Link function: identity

Formula:
Overall ~ te(Income, Edu)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  471.154      3.349   140.7   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                edf Ref.df     F p-value
te(Income,Edu) 10.1  12.19 16.93  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =    0.8   Deviance explained =   84%
GCV = 741.42  Scale est. = 583.17    n = 52
```

Il tensor object è un'entità complessa: dal punto di vista computazionale, la matrice di design che implementa questi oggetti è molto grande e presenta una struttura intricata, anche se dal summary non emergono questi dettagli (compare solo una riga riassuntiva).

Nel modello stimato, tra i coefficienti parametrici abbiamo un'unica intercetta, che rappresenta il livello medio dell'assessment degli studenti nei vari paesi.

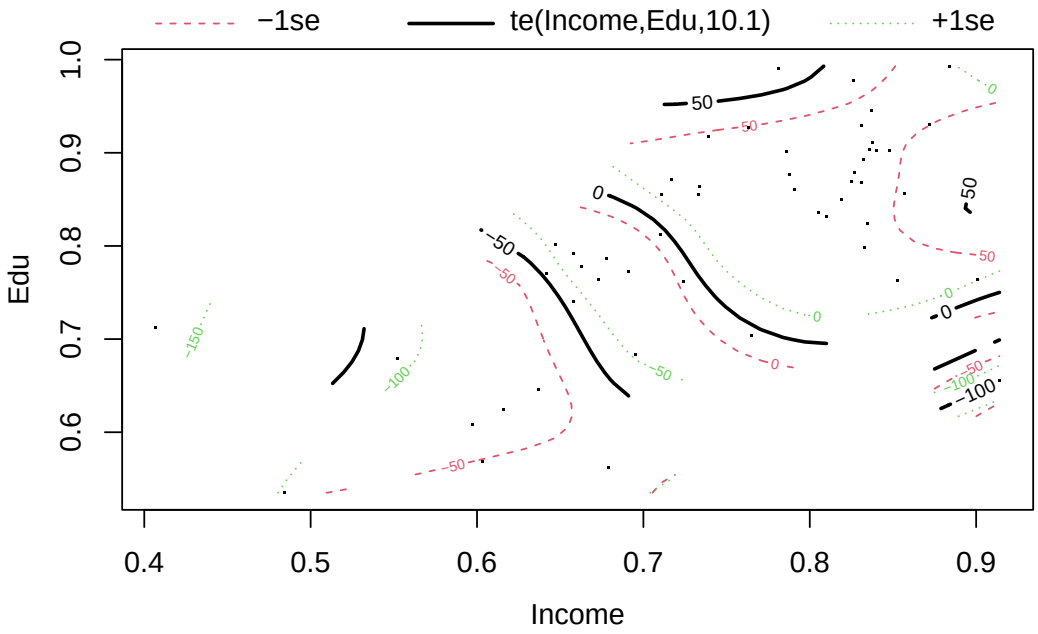
Per quanto riguarda i termini non lineari, il summary riporta:

- `te(Income, Edu)`: effetto combinato non lineare di `Income` e `Education`, stimato con circa 10 gradi di libertà effettivi (edf = 10.1), ed è altamente significativo.

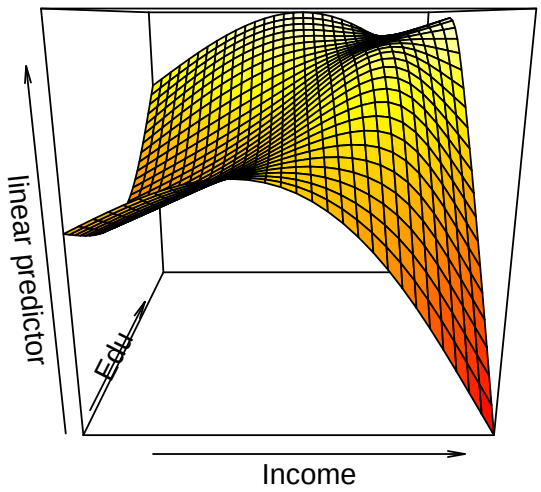
Questo significa che per modellare adeguatamente l'effetto congiunto di `Income` e `Education` sull'assessment degli studenti, sono richiesti almeno 10 parametri, e l'effetto è statisticamente rilevante. Il modello con il tensore spiega circa **l'80% della devianza** e il valore della GCV è sceso a **741.42**, indicando un miglioramento del fit del modello ai dati rispetto ai modelli precedenti.

2.5.3.1 Plot

```
plot(mod_tensor,pages=1,residuals=TRUE,all.terms=TRUE,shade=TRUE,shade.col=2)
```

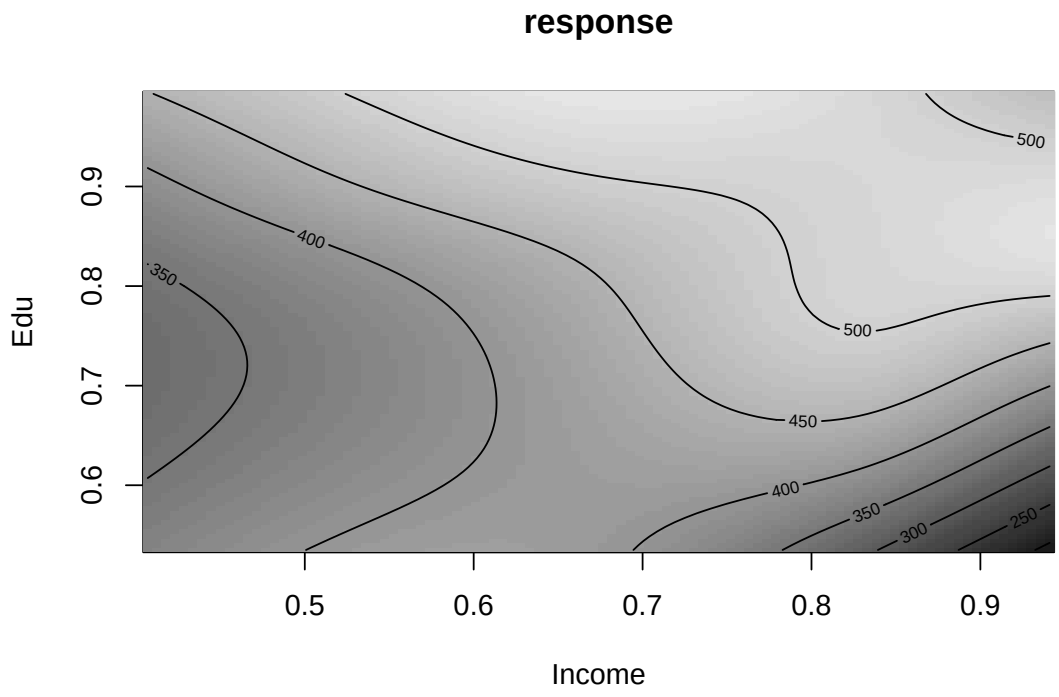



```
# ?vis.gam # funzione che plotta 2 dimensional objects (visualizziamo gam object)
vis.gam(mod_tensor)
```

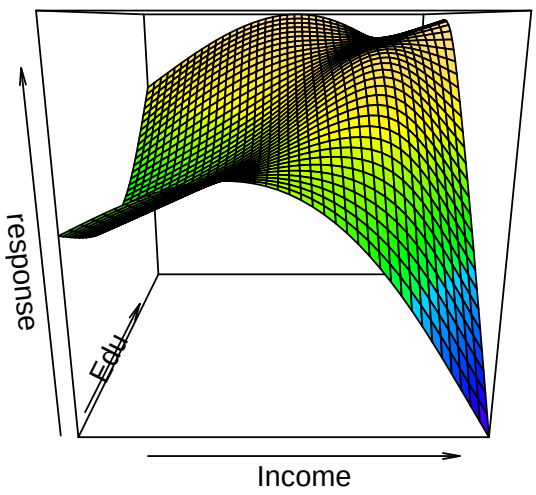



```
# income, education, effect che risulta sul linear predictor
# potremmo anche vederla in versione 3D da ruotare -> uau
```

```
vis.gam(mod_tensor, type = 'response', plot.type = 'contour',
        n.grid=100,color="gray")
```




```
vis.gam(mod_tensor, type = 'response', plot.type = 'persp',
        n.grid=40, color="topo")
```



Valori blu valori più bassi, valori rossi valori più grandi.

Questo effetto è **fortemente non lineare**: non può essere rappresentato tramite una semplice interazione lineare. Nei modelli lineari classici, le interazioni producono **piani paralleli** o inclinati nello spazio predittivo, ma in questo caso la superficie stimata è molto più complessa e non regolare.

Pertanto, **non sarebbe possibile modellare tale effetto con un modello lineare tradizionale**. Solo un **Generalised Additive Model (GAM)**, che consente di stimare relazioni non lineari flessibili tra le variabili tramite smooth functions e tensori, è in grado di cogliere e rappresentare adeguatamente questa complessità.